

As filed with the U.S. Securities and Exchange Commission on March 20, 2026.

**UNITED STATES
SECURITIES AND EXCHANGE COMMISSION**
Washington, D.C. 20549

FORM S-1
REGISTRATION STATEMENT
UNDER
THE SECURITIES ACT OF 1933

X-Energy, Inc.

(Exact name of registrant as specified in its charter)

Delaware
(State or other jurisdiction of
incorporation or organization)

4911
(Primary Standard Industrial
Classification Code Number)

43
(I.R.B. Number)
Identification Number

530 Gaither Road, Suite 700
Rockville, Maryland 20850
(301) 358-5600

(Address, including zip code, and telephone number, including area code, of registrant’s principal executive office)

J. Clay Sell
Chief Executive Officer
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Approximate date of commencement of proposed sale to the public:
As soon as practicable after the effective date of this registration statement.

If any of the securities being registered on this Form are to be offered on a delayed or continuous basis pursuant to Rule 415 under the Securities Act of 1933, as amended, or Securities Act, check the following box: ☐

If this Form is filed to register additional securities for an offering pursuant to Rule 462(b) under the Securities Act of 1933, as amended, check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering: ☐

If this Form is a post-effective amendment filed pursuant to Rule 462(c) under the Securities Act, check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering: ☐

If this Form is a post-effective amendment filed pursuant to Rule 462(d) under the Securities Act, check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering: ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a small business issuer, a company, or an emerging growth company. See the definitions of “large accelerated filer,” “accelerated filer,” “small business issuer,” “company,” and “emerging growth company” in Rule 12b-2 of the Securities Exchange Act of 1934, as amended.

Large accelerated filer	☐	Accelerated filer	☐
Non-accelerated filer	☒	Smaller reporting company	☐
		Emerging growth company	☒

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 7(a)(2)(B) of the Securities Act. ☐

The registrant hereby amends this registration statement on such date or dates as may be necessary to delay its effectiveness until it shall file a further amendment which specifically states that this registration statement shall thereafter become effective in accordance with the provisions of the Securities Act or until the registration statement shall become effective on such date as the Securities and Exchange Commission, to said Section 8(a), may determine.

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SUBJECT TO COMPLETION, DATED _____, 2026

PRELIMINARY PROSPECTUS

Shares

X•ENERGY

X-Energy, Inc.

Class A Common Stock

This is the initial public offering of shares of Class A common stock of X-Energy, Inc. We are offering _____ shares of our Class A common stock.

Prior to this offering, there has been no public market for our Class A common stock. We expect the public offering price per share of our Class A common stock will be between \$ _____ and \$ _____ per share.

We intend to apply to list our Class A common stock on the Nasdaq Stock Market LLC (“Nasdaq”) under the symbol “XE.”

Upon consummation of this offering, we will be a holding company in an organizational structure referred to as an umbrella partnership-C-corporation (or “Up-C”) structure, and our principal asset will be ownership of _____ % of the common units (“Common Units”) of X-Energy Reactor Company, LLC, which owns approximately _____ % of the Common Units if the underwriters exercise in full their option to purchase _____ shares of Class A common stock). See “Risk Factors — Risks Related to Our Capital Structure.” We will not exercise control over all of the business and affairs of XERC and its direct and indirect subsidiaries, and conduct of XERC.

Following this offering, we will have two series of authorized common stock: shares of Class A common stock, having one vote per share and economic rights, and shares of Class B common stock, having one vote per share and economic rights (collectively, the “Common Stock”). Holders of Class A and Class B common stock will vote as a single class on all matters to be presented to our shareholders for their vote or approval, except as required by applicable law or our Bylaws (as defined herein). Our outstanding Class A common stock will represent approximately _____ % and _____ %, respectively, of the total value of our outstanding Common Stock immediately following this offering, assuming no exercise of the underwriters’ option to purchase additional shares of Class A common stock. See “Description of Capital Stock” and “Capital Structure.”

In connection with the offering, we will enter into a Tax Receivable Agreement (“TRA”) (as defined in this prospectus) with XERC and the TRA Holders (as defined elsewhere in this prospectus) that will require cash payments to be made by the Company to such TRA Holders in respect of certain future tax benefits that the Company, Energy, Inc., utilizing cash for the benefit of such unitholders that otherwise would have been available to the Company and for the benefit of all of our shareholders. Although the actual timing and amount of any payments made under the Tax Receivable Agreement will vary, the Company expects such payments under the TRA Agreement are substantial. Any payments made by the Company to the TRA Holders will benefit the Company and will reduce cash that might have otherwise been available to the Company for reinvestment or other purposes. See “Risk Factors — Risks Related to Our Capital Structure” and “Certain Relationships and Related Party Transactions — Tax Receivable Agreement.”

We are an “emerging growth company” as defined under the federal securities laws. As such, we will not have taken advantage of certain reduced disclosure obligations that apply to emerging growth companies, including financial statements and executive compensation arrangements.

Investing in our Class A common stock involves risks. See the section titled “Risk Factors” beginning on page _____ about factors you should consider before buying shares of our Class A common stock.

This prospectus is not complete and may be changed. We may not sell these securities until the registration statement filed with the Securities and Exchange Commission is complete.

Neither the Securities and Exchange Commission nor any other regulatory body has approved or disapproved of these securities or passed upon the accuracy or adequacy of this prospectus. Any representation to the contrary is a criminal offense.

Initial public offering price
Underwriting discounts and commissions ⁽¹⁾
Proceeds, before expenses, to us

(1) See the section titled “Underwriting” for a description of the compensation payable to the underwriters. The underwriters have the option for a period of 30 days from the date of this prospectus to purchase additional shares of our Class A common stock from us to cover over-allotments, if any, offering price less underwriting discounts and commissions. The underwriters expect to deliver the shares of our Class A common stock to purchasers on or about May 1, 2026 through the book-entry facilities of The Depository Trust Company.

Joint Bookrunning Managers

J.P. Morgan	Morgan Stanley	Jefferies	Moelis & Company
Cantor	UBS Investment Bank	TD Securities	
Guggenheim Securities		Wolfe Nomura Alliance	

Prospectus dated May 1, 2026

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X·ENERGY

Building Energy Solutions that Empower the Unprecedented



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Neither we nor the underwriters have authorized anyone to provide you with information or that contained in this prospectus and any free writing prospectus we have prepared. responsibility for, and can provide no assurance as to the reliability of, any other information may give you. We and the underwriters are offering to sell Class A common stock and seeking Class A common stock only under circumstances and in jurisdictions where such offers and sale. The information in this prospectus is accurate only as of the date of this prospectus (or as of any as of which such information is given), regardless of the time of delivery of this prospectus or the Class A common stock. Our business, liquidity position, financial condition, prospects and operations may have changed since the date of this prospectus.

This prospectus contains forward-looking statements that are subject to a number of uncertainties, many of which are beyond our control. See the sections titled “Risk Factors” and “Note Regarding Forward-Looking Statements.”

For investors outside the United States: Neither we nor the underwriters have done anything to permit a public offering of the securities offered hereby or possession or distribution of this prospectus, amendment or supplement to this prospectus, or any applicable free writing prospectus in any jurisdiction where action for that purpose is required, other than in the United States. Persons outside the

who come into possession of this prospectus, any amendment or supplement to this prospectus, any applicable free writing prospectus must inform themselves about, and observe any restrictions on the offering of the securities and the distribution of this prospectus, any amendment or supplement to this prospectus, or any applicable free writing prospectus outside of the United States.

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BASIS OF PRESENTATION AND DEFINITIONS

Organizational Structure

In connection with the closing of this offering, we will effect certain organizational transactions. Unless otherwise stated or the context otherwise requires, all information in this prospectus relates to the consummation of the organizational transactions and this offering, which we refer to collectively as the “Transactions.” See “Organizational Structure” for a description of the Transactions and a diagram of our organizational structure after giving effect to the Transactions, including this offering.

As used in this prospectus, unless the context otherwise requires, references to:

- “we,” “us,” “our,” the “Company,” “X-energy” and similar references refer, from the date of consummation of the Transactions, including this offering, to X-Energy, Inc., and, unless otherwise stated, all of its subsidiaries, including X-Energy Reactor Company, LLC, which we refer to as “XERC” and, unless otherwise stated, all of its subsidiaries. When used in a historical context, these terms collectively refer to X-Energy Reactor Company, LLC, which we refer to as the predecessor for financial reporting purposes, and its operating subsidiaries.
- “ARDP” means the U.S. Department of Energy’s Advanced Reactor Demonstration Program.
- “ARDP Agreement” means the Cooperative Agreement between the U.S. Department of Energy and X-Energy, LLC, Award No. DE-NE0009040, effective as of February 2, 2021.
- “Basis Adjustments” means the tax basis adjustments expected to be obtained by XERC resulting from (a) any future redemptions or exchanges of Common Units from the Transactions; (b) certain distributions (or deemed distributions) by XERC; and (c) payments made under the Receivable Agreement.
- “Blocker Companies” means entities that are owners of Common Units in XERC prior to the Reorganization Transactions and are taxable as corporations for U.S. federal income tax purposes.
- “Blocker Mergers” means the merger of each of the Blocker Companies with X-Energy, Inc. as part of the Reorganization Transactions, pursuant to which the Blocker Shareholders receive shares of Class A common stock as consideration for the applicable Blocker Merger.
- “Blocker Shareholders” means the owners of the Blocker Companies prior to the Reorganization Transactions, who will exchange their interests in the Blocker Companies for shares of Class A common stock in connection with the consummation of the Blocker Mergers.
- “Blocker Tax Attributes” means (i) the share of tax basis (including under Sections 743 and 754 of the Code and Section 1.743-1(h) of the Treasury Regulations and, in the case of non-federal comparable sections of U.S. state and local tax law) of certain of XERC Group’s assets that are amortizable under Section 197 of the Code or that are otherwise amortizable or depreciable for federal income tax purposes, in each case, attributable to the Common Units acquired by the Blocker Companies in the Blocker Mergers and (ii) net operating losses (and carryforwards thereof), capital losses (and carryforwards thereof), disallowed interest expense carryforwards under Section 163(j) of the Code and credit carryforwards of the Blocker Companies relating to tax attributes.

ending on or prior to the date of this offering, but, in the case of clause, (ii) excluding the attribute of a Blocker Company that is used to offset taxes of such Blocker Company, taxes are attributable to taxable periods (or portion thereof) ending on or prior to the Blocker Mergers.

- “*Code*” means the U.S. Internal Revenue Code of 1986, as amended.
- “*Common Units*” means common units of membership interest in X-Energy Reactor Co.
- “*Continuation Application*” means a non-competitive application for an additional block within the contractual award timeline under the ARDP Agreement.
- “*Continuing Equity Owners*” refers collectively to those Original Equity Owners and certain of their affiliates, but excluding Management LLC) that will own Class B common stock in XERC and our Class B common stock after the Reorganization Transactions and following the

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consummation of this offering, redeem their Common Units for cash or shares of common stock as described in “Certain Relationships and Related Party Transactions.”

- “*DGCL*” means the Delaware General Corporation Law, as amended.
- “*DOE*” means the U.S. Department of Energy.
- “*Dow*” means Dow Inc. or its subsidiaries.
- “*Exchange Act*” means the U.S. Securities Exchange Act of 1934, as amended.
- “*Exchange Existing Basis*” means the tax basis in certain assets of XERC and certain of its indirect subsidiaries (including assets that will eventually be subject to depreciation or amortization once placed in service) that is obtained by X-Energy, Inc., in connection with and is attributable to a Common Unit exchanged or redeemed by a TRA Holder, including with respect to the Common Unit contributed to X-Energy, Inc. by a TRA Holder in connection with the consummation of this offering.
- “*Existing Basis*” means the Exchange Existing Basis and IPO Existing Basis.
- “*FOAK*” means first-of-a-kind.
- “*Former Equity Owners*” refers to those Original Equity Owners other than the Blocker Company and the Continuing Equity Owners.
- “*GAAP*” means U.S. generally accepted accounting principles.
- “*GFR*” means Gas-cooled Fast Reactors.
- “*Ghaffarian Enterprises*” means Ghaffarian Enterprises, LLC.
- “*GM Enterprises*” means GM Enterprises, LLC.
- “*HALEU*” means high-assay low-enriched uranium.
- “*HTGR*” means High Temperature Gas-cooled Reactors.
- “*HTF*” means Helium Test Facility.
- “*Incentive Units*” means the Class B-2 Common Units of Management LLC granted under the Management LLC Profits Interest Plan prior to the closing of this offering.

- “*Interest Deductions*” means deductions attributable to imputed interest and other interest by X-Energy, Inc. pursuant to the Tax Receivable Agreement.
- “*IPO Existing Basis*” means the tax basis obtained by X-Energy, Inc. in connection with the offering as a result of existing tax basis in certain assets of XERC and certain of its direct or indirect subsidiaries, including assets that will eventually be subject to depreciation or amortization and placed in service.
- “*LLC Units*” means X-Energy Reactor Company, LLC’s existing Common Units and preferred units.
- “*Management LLC*” means X-Energy Management, LLC, a Delaware limited liability company.
- “*Material Breach*” means, with respect to the Tax Receivable Agreement, means (i) any event or events except those exceptions set forth in the Tax Receivable Agreement (including the exceptions for failure to comply with the terms of the Tax Receivable Agreement, the failure to comply with the terms of the Tax Receivable Agreement to restrictions in X-Energy, Inc.’s debt instruments and similar agreements) X-Energy, Inc. fails to make a payment (along with any applicable interest) within ninety (90) calendar days of the applicable Final Payment Date (as defined in the Tax Receivable Agreement) or (ii) the Tax Receivable Agreement by operation of law in a case commenced in bankruptcy court.
- “*MPa*” means Megapascal.
- “*MSR*” means Molten Salt Reactors.
- “*NOAK*” means Nth-of-a-kind.
- “*NRC*” means the U.S. Nuclear Regulatory Commission.
- “*OPG*” means Ontario Power Generation, Inc.

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- “*Original Equity Owners*” refers to the direct and certain indirect owners of XERC prior to the Transactions.
- “*Person*” means any individual, firm, corporation, partnership, limited liability company, trust, partnership, incorporated or unincorporated association, joint venture, joint stock company, or any other entity with legal authority or instrumentality or other entity of any kind.
- “*Profits Interest Plan*” means the Profits Interest Plan of Management LLC in place at the time of the closing of this offering.
- “*Recapitalization*” means the transactions whereby X-Energy Reactor Company, LLC completed a recapitalization pursuant to which all outstanding equity interests in X-Energy Reactor Company, LLC will be converted or exchanged into Common Units.
- “*Reorganization Transactions*” has the meaning ascribed to it in “Organizational Structure.”
- “*Sarbanes-Oxley Act*” means the Sarbanes-Oxley Act of 2002.
- “*SEC*” means the U.S. Securities and Exchange Commission.
- “*Securities Act*” means the U.S. Securities Act of 1933, as amended.
- “*Series C-1 Investment*” means the purchase by the Series C-1 Investors of the Series C-1 Units.
- “*Series C-1 Investors*” means those certain investors participating in the Series C-1 Investment.
- “*Series C-1 Notes*” means the Series C-1 Convertible/Exchangeable promissory notes issued by XERC.

- “*Series C-2 Investment*” means the purchase by the Series C-2 Investors of the Series C-2 Securities Purchase Agreements.
- “*Series C-2 Investors*” means those certain investors participating in the Series C-2 Investment.
- “*Series C-2 Notes*” means the Series C-2 Convertible/Exchangeable promissory notes issued by XERC.
- “*Series C-2 Securities Purchase Agreements*” means the Series C-2 Convertible/Exchangeable Securities Purchase Agreements entered into between XERC and the Series C-2 Investors.
- “*SMR*” means Small Modular Reactors.
- “*Tax Receivable Agreement*” means the Tax Receivable Agreement to be entered into between X-energy, XERC and certain Original Equity Owners (the “*TRA Holders*”) at the consummation of this offering, pursuant to which, among other things, X-energy will be required to pay the TRA Holder 85% of the amount of cash tax savings, if any, that it realizes (or in certain cases fails to realize) as a result of the Existing Basis, Basis Adjustments, Blocker Tax Attribute Adjustments and Deductions. A copy of the form of Tax Receivable Agreement is attached to this prospectus.
- “*TRISO-X*” means TRISO-X, LLC, our wholly owned subsidiary.
- “*TRISO-X Fuel*” refers to the HALEU-based TRISO pebble fuel manufactured for the XERC by TRISO-X, LLC subsidiary.
- “*U.K.*” means the United Kingdom.
- “*U.S.*” means the United States of America.
- “*X-energy Founder*” means X-Energy Holdings, IBX Company Opportunity Fund 1, LP, IBX Company Opportunity Fund 2, LP, IBX Opportunity GP, Inc., X-energy KG Parent, LP, and X-energy Enterprises.
- “*X-Energy Holdings*” means X-Energy Holdings, LLC.
- “*XERC Group*” means XERC and each of its direct or indirect subsidiaries that is a partnership or disregarded entity for U.S. federal, and applicable state and local, income tax purposes (but excluding any such subsidiary to the extent it is directly or indirectly held by or for the benefit of an entity treated as a corporation for U.S. federal, and applicable state and local, income tax purposes (other than X-Energy, Inc.)).

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- “*XERC LLC Agreement*” refers to XERC’s Eighth Amended and Restated Operating Agreement, which will become effective upon the consummation of this offering.
- “*XERC*” refers to our predecessor for financial reporting purposes, X-Energy Reactor Company, LLC.
- “*Xe-100 plant*” means the optimized four-reactor configuration of 320 MWe that is expected to be commonly deployed in.

This prospectus includes certain historical condensed consolidated financial and other information of X-Energy Reactor Company, LLC. Immediately following this offering, we will be a holding company and the sole managing member of XERC, and upon completion of this offering and the application of the proceeds therefrom, our principal asset will consist of common units of XERC. We will operate and conduct the business and affairs of X-Energy Reactor Company, LLC and conduct our business through XERC.

Reactor Company, LLC and its subsidiaries. Following this offering, X-Energy Reactor Company, LLC will be the predecessor of X-Energy, Inc. for financial reporting purposes. As a result, the consolidated financial statements of X-Energy, Inc. will recognize the assets and liabilities received in the reorganization on their historical carrying amounts, as reflected in the historical financial statements of X-Energy Reactor Company, LLC. We will consolidate X-Energy Reactor Company, LLC on our consolidated financial statements and record a noncontrolling interest related to the LLC Units (as defined below) owned by the Continuing Equity Owners (as defined below) on our consolidated balance sheet and statement of operations.

Presentation of Financial and Other Information

XERC is the predecessor of the issuer, X-Energy, Inc., for financial reporting purposes. X-Energy, Inc. will be the audited financial reporting entity following this offering. Accordingly, this prospectus presents the following historical financial statements:

- *X-Energy, Inc.:* Other than the balance sheet, dated as of December 31, 2025, no other financial information of X-Energy, Inc. has not been included in this prospectus as X-Energy, Inc. is an unincorporated entity, has no business transactions or activities to date and had no assets or liabilities during the periods presented in this prospectus.
- *X-Energy Reactor Company, LLC:* As we will have no other interest in any operations other than those of X-Energy Reactor Company, LLC and its subsidiaries, the historical consolidated financial information included in this prospectus is that of X-Energy Reactor Company, LLC and its subsidiaries.
- *Unaudited Pro Forma Condensed Consolidated Financial Information:* This prospectus contains unaudited pro forma condensed consolidated financial information for the year ended December 31, 2025. The unaudited pro forma condensed consolidated financial information contained in this prospectus is derived from the “Unaudited Pro Forma Condensed Consolidated Financial Information” section of this prospectus. The unaudited pro forma condensed consolidated balance sheet as of December 31, 2025 contained in this prospectus presents the consolidated balance sheet position of XERC after giving effect to the Transactions as if all such transactions had occurred as of December 31, 2025 and has been prepared in accordance with Article 11 of Regulation S-X. The unaudited pro forma condensed consolidated statement of operations for the year ended December 31, 2025 contained in this prospectus presents the consolidated results of operations of XERC after giving effect to the Transactions as if all such transactions had occurred as of December 31, 2025 and has been prepared in accordance with Article 11 of Regulation S-X. The unaudited pro forma condensed consolidated financial information is presented for informational purposes only and may not be indicative of the results that would have been achieved if the foregoing transactions had taken place on an earlier date or on the dates assumed. In addition, the unaudited pro forma condensed consolidated financial information does not purport to project the future financial condition and results of operations of XERC or X-Energy, Inc. See “Unaudited Pro Forma Condensed Consolidated Financial Information” for a complete description of the adjustments and assumptions underlying the summary unaudited pro forma condensed consolidated financial information.

Certain monetary amounts, percentages and other figures included in this prospectus have been rounded to rounding adjustments. Percentage amounts included in this prospectus have not in all cases been calculated on the basis of such rounded figures, but on the basis of such amounts prior to rounding. For this reason,

percentage amounts in this prospectus may vary from those obtained by performing the same calculations using the figures in our consolidated financial statements included elsewhere in this prospectus. Other amounts that appear in this prospectus may not sum due to rounding. Figures shown in certain tables or charts may not be the arithmetic aggregation of the figures that precede them. Percentages expressed as percentages in the text may not total 100% or, as applicable, when aggregated may not be the arithmetic aggregation of the percentages that precede them.

Our fiscal year begins on January 1 and ends on December 31 of the same year.

TRADEMARKS, TRADE NAMES AND SERVICE MARKS

We use the X-energy logo and other marks as trademarks in the U.S. and other countries. This prospectus contains references to our trademarks and service marks and to those belonging to others. Solely for convenience, trademarks and trade names referred to in this prospectus, including logos and other visual displays, may appear without the ® or ™ symbols, but such references are not intended to indicate in any way that we will not assert, to the fullest extent under applicable law, our rights in these trademarks and trade names or the rights of the applicable licensors to these trademarks and trade names. We do not intend our use or display of these trademarks, trade names, trademarks or service marks to imply a relationship with, or endorsement or sponsorship of us by, any other entity.

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We are responsible for the disclosure contained in this prospectus. Information contained in this prospectus concerning the market and the industry in which X-energy competes, including our market position, general expectations of market opportunity, size and growth rates, is based on information obtained from various third-party sources, X-energy's knowledge of the markets for its services and products, and assumptions made by X-energy based on such sources and knowledge. This information and the estimates and market information provided in this prospectus involve numerous assumptions and limitations, and you are cautioned not to give undue weight to such information. Third-party sources generally state that the information obtained from such source has been obtained from sources believed to be reliable but that there can be no assurance as to the accuracy or completeness of such information. Notwithstanding the foregoing, we are providing the information provided in this prospectus. The industry in which X-energy engages or proposes to engage is subject to a high degree of uncertainty and risk. As a result, the estimates and market information provided in this prospectus are subject to change based on various factors, including those described in the sections of this prospectus entitled "*Cautionary Note Regarding Forward-Looking Statements*" and "*Risk Factors — Risks Related to X-energy — Risks Relating to X-energy's Business*" elsewhere in this prospectus.

PROSPECTUS SUMMARY

This summary highlights selected information contained elsewhere in this prospectus and provides an overview of the Company. This summary does not contain all of the information you should consider before investing in our Class A common stock. For a more complete understanding of our business, you should read this entire prospectus carefully, including the sections titled “Risk Factors,” “Description of Business,” “Discussion and Analysis of Financial Condition and Results of Operations of X-energy,” “Description of our audited consolidated financial statements and the related notes included elsewhere in this prospectus,” and “Other Information,” before making an investment decision. Capitalized terms not otherwise defined in this prospectus have the meanings assigned to them under “Basis of Presentation and Definitions” included elsewhere in this prospectus.

Overview

Our company

X-energy is a leading designer of advanced nuclear reactor technology (commonly referred to as small modular reactors, “SMRs”) and manufacturer of advanced nuclear fuels. We believe these small modular reactor generation technologies help satisfy historically unprecedented electricity demand growth, driven by the development of AI and associated data center infrastructure. Total demand for new electricity is expected to increase globally by 7,626 TWh from 2023 to 2030 and the challenges associated with meeting this demand have led policymakers and industry leaders to recognize nuclear energy, particularly small modular nuclear, as a key component to address this need.

Founded in 2009 by Dr. Kamal “Kam” Ghaffarian to bring clean, safe, secure and advanced nuclear technology to market, X-energy is seeking to redefine the energy industry through its flagship Xe-100, an advanced small modular High Temperature Gas-cooled Reactor (“HTGR”), in development for nearly a decade. The Xe-100 reactor is designed to generate 80 megawatts of electric power, 100 megawatts of thermal output (heat), or a combination thereof. This reactor technology builds on over 50 years of research and development by the global nuclear industry and the operating experience of previous HTGRs including those at Peach Bottom in the U.S., and Dragon in the U.K. in the 1970s and more recently with China’s ongoing deployments of HTGRs in the 21st century.

The Xe-100 has several technological attributes that we believe make it advantaged compared to other sources of baseload generation. These include advanced safety features, virtually no direct greenhouse gas (“GHG”) emissions during generation, high thermal output, load-following capabilities and modularity, all of which allow X-energy to more specifically meet a customer’s power and/or industrial heat needs. X-energy’s simple Xe-100 design directly translates into simplicity of project delivery through reduced supply chain complexity and labor intensity during construction, which we believe will lead to lower costs and faster deployment timelines when compared with conventional nuclear energy sources. X-energy has optimized the deployment of its Xe-100 into a four-reactor format that outputs 320 MWe (or 800 MWt) by deploying four independent reactor modules instead of a single unit, this optimized configuration inherently delivers the high levels of reliability and redundancy required for baseload power and industrial heat applications.

X-energy’s reactors use a tri-structural isotropic (“TRISO”) coated particle fuel in the form of a spherical ‘pebble’, called TRISO-X fuel. This pebble fuel consists of HALEU fuel kernels, which are encapsulated in layers of silicon carbide and pyrolytic carbon, forming miniature containment vessels that trap fission products. These particles are then embedded in a graphite matrix to make fuel

possess exceptional safety margins and compacts, enabling operations at very high temperatures. HALEU fuel used in our TRISO-X pebble fuel is enriched to 15.5%, a higher energy density than the less than 5% low-enriched uranium (“LEU”) fuel used in conventional nuclear reactors. TRISO fuel will be produced at our fuel fabrication facility in Oak Ridge, Tennessee. The first facility, known as the TX-1, began construction in October 2024 and is expected to be completed by the first half of 2026. Upon completion, it is expected to be North America’s first purpose-built commercial advanced nuclear fuel fabrication facility. The TX-1 facility will have sufficient production capacity to support the fuel needs of the first 11 Xe-100 reactors at steady state operations.

In addition to its technology leadership, X-energy has three high-quality customers in D and Centrica, who we expect will underpin the deployment of the initial fleets of Xe-100 reactors together, assuming each customer exercises its contingent rights in full, these three customers with a more than 11 gigawatts electric (“GWe”), 144 reactor pipeline across the U.S. and the advanced development efforts already underway on the first Dow project at its Seadrift Operations Texas and the first Amazon project in connection with Energy Northwest.

X-energy maintains a strong relationship with the DOE and in December 2020 was awarded \$1.2 billion as part of its selection as one of two awardees in the ARDP, the most substantial commitment to deploying advanced nuclear technology. The cooperative agreement for the project in February 2021 (the “ARDP Agreement”), provides 50/50 cost share of \$2.4 billion of eligible (billion reimbursement) through 2027, allowing X-energy to continue work toward design, commercialization and construction of its first-of-a-kind commercial advanced nuclear commercial TRISO-X fuel fabrication facility, while benefiting from decades of nuclear engineering knowledge within the DOE.

As of December 31, 2025, X-energy has been reimbursed approximately \$438 million in the ARDP Agreement. We submit our budgets through an ongoing “budget period” basis to milestones under the ARDP Agreement, and our current budget covers a budget period that begins in 2025 and extends through August 2026. We submit a Continuation Application to the DOE for funding into subsequent periods. Extensions beyond the current budget period are subject to DOE review and approval. Under the terms of the ARDP Agreement that rely on the Office of Management and Budget (OMB) guidance, the total extension of the award may not exceed three years (for a total performance of 10 years). Any additional extension would require an approval within DOE at the discretion of the Contracting Officer. If we are unable to obtain extensions and incur eligible costs beyond the currently approved period of performance, we would forgo reimbursement for such costs and have no obligation of unobligated funds at closeout. There can be no assurance that we will receive additional funding beyond the current budget period or that extensions will be granted.

Our organizational structure following the offering and the Reorganization Transactions is referred to as an umbrella partnership-C corporation (or “Up-C”) structure. Pursuant to the offering, following this offering we will hold a number of Common Units equal to the number of shares of our Class A common stock, and holders of Common Units (each, an “Common Unit”) (other than us) will hold a number of Common Units equal to the number of our issued and outstanding Class B common stock. The Up-C structure was selected in order to (i) provide our Continuing Owners with an option to continue to hold their economic ownership interests in our business in “pass-through” form for U.S. federal income tax purposes through their ownership of Common Units, and (ii) potentially allow our Continuing Equity Owners and us to benefit from certain net cash tax savings that we might realize in the future, as more fully described in the subsection titled “Certain Relationships with Related Party Transactions — Tax Receivable Agreement.”

Our Market Opportunity

Load growth is accelerating globally with the International Energy Agency (the “IEA”) projecting electricity consumption to grow by more than 25% from approximately 30,000 TWh in 2020 to approximately 37,000 TWh by 2030. In the U.S., power demand is being driven by the increase in data center buildout from cloud computing providers, industrial growth and reshoring of manufacturing, and broader electrification (e.g., electric vehicle installed base). Other sources (BNEF New Energy Outlook 2025) also indicate that power demand is being driven by growing load growth demand from industrial use. Policymakers and industry leaders recognize that nuclear, particularly advanced nuclear, will be a key contributor to meeting this load growth and securing America’s energy independence.

stated goal of the current federal administration to expand U.S. nuclear capacity to 400 GWe by 2050 (from approximately 97 GWe at present).

We believe the market for SMRs is vast and that X-energy is well positioned to capitalize on this opportunity. According to a 2024 report from PA Consulting, the U.S., the U.K. and Canada, the three planned core markets, comprise approximately one-third of potentially serviceable global electricity demand. Across these three geographies, there is an estimated total addressable market (“TAM”) of 1,146 GWe in capacity additions of 743 GWe and 1,146 GWe by 2040 and 2050, respectively. Of this TAM, PA Consulting estimates 72 GWe in

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2040 and 158 GWe in 2050 will be best served by SMRs based on needs, siting efficiencies, and co-generation. These capacity additions, across a representative set of SMR use cases including data centers, utility power generation, industrial applications and behind-the-meter (“BTM”) implies a potential need for approximately around 1,975 Xe-100 reactors (the equivalent of 49 deployments), or an estimated \$2.3 trillion market opportunity for X-energy in 2050.

We believe the following secular movements will continue to support the use cases and provide scalable, firm, clean baseload power that SMRs, particularly the Xe-100, can deliver.

- ***Demand for AI is Prompting Investment in Nuclear Power Generation.*** Cloud computing and AI companies are deploying capital across data center infrastructure to keep up with computing demand required by AI. As part of this expansion, electricity demand in data centers is expected to grow from approximately 108 TWh in 2020 to approximately 2030. Since data center facilities must be continuously powered, nuclear generation is a key solution for reliable supply. SMRs are particularly well suited to meet the average 1,000 MW power capacity that data centers require. SMRs have a smaller physical footprint, siting requirements, and have modular scalability which provides the flexibility to address specific capacity needs.
- ***High Carbon-Intensity Industrials Require Replacement for Industrial Heat Production.*** Industrial companies have historically relied on fossil fuel fired boilers for electricity and steam for their industrial processes. The current installed base of industrial boilers operates below capacity and is frequently offline for maintenance. The current fleet is nearing term replacement cycle due to age. For industrial processes requiring consistent steam, X-energy’s HTGR solution can reliably provide industrial steam in addition to onsite power. With an expected capacity factor of 95%, X-energy’s SMR presents a compelling replacement of aging infrastructure to both decarbonize and achieve greater reliability.
- ***Current Alternatives Are Not Well-Suited to Deliver Reliable, Uninterrupted, Clean and Co-Located Power.***
 - ***Renewables with or without energy storage*** — Solar and wind generation have relatively low capacity factors (i.e., the actual energy output in a given period of time relative to maximum output) of 23% and 33% (EIA) respectively, since they only produce power when the sun is shining or the wind is blowing. Addressing this inherent intermittency issue requires the integration of large-scale battery energy storage systems or supplementary fossil fuel generators for the production capacity of these renewables to even be comparable to the anticipated 95% capacity factor of our reactor. Large-scale deployment of battery storage is complicated by global supply chain availability and incremental cost.
 - ***Conventional fossil fuel generation with or without carbon capture*** — Standalone fossil fuel fired generation can deliver capacity factors similar to those of nuclear power but lacks the backup generation when maintenance takes key assets offline. Conversely, SMRs can deliver critical redundancy due to its modular nature which results in a reliable power supply without requiring grid redundancy. Further, many consumers and governments have net-zero targets, giving SMRs (with virtually no direct GHG emissions from energy generation) a clear advantage over carbon-intensive alternatives such as coal, oil, and natural gas generation. Achieving net-zero would require pairing them with expensive carbon capture solutions to try to meet net-zero targets.
 - ***Traditional Nuclear Power*** — Large-scale nuclear, while typically operating at high capacity factors (other than planned outage maintenance and refueling cycles, as compared to SMRs with online refueling), requires a large footprint and has consistently suffered from historical

delays and cost overruns. By contrast, advanced nuclear requires a much smaller footprint, on average about $1/4^{\text{th}}$ to $1/10^{\text{th}}$ that of a traditional nuclear plant. Additional sites in the U.S. will require a significantly smaller safety zone radius (400 meters to 1.6 kilometers) because of built-in passive safety features. These safety characteristics reduce costs and enable more compact designs due to a reduction in the quantity of concrete and associated infrastructure. The smaller overall footprint of SMRs allows co-location with existing power demand hubs, and the modular nature of the Xe-100 enables scalable power output with additional on-site reactors to match specific needs.

- **Ongoing Support from the U.S. Government Beyond the ARDP.** Support for nuclear power has been increasing over the past decade with billions of additional dollars being allocated to nuclear energy through recent energy-related legislation. For instance, the One Big Beautiful Bill Act, signed in July 2025, maintained tax credits that were first introduced as part of the Energy Policy Act of 2005 and later expanded under the Inflation Reduction Act for nuclear projects. Additionally, the U.S. President issued four Executive Orders in May 2025, including support for the advanced nuclear regulatory review for advanced nuclear reactors and promotion of investment in a domestic advanced nuclear supply chain. We believe these broader initiatives, building from U.S. nuclear policy priorities, will provide long-term support for advanced nuclear reactors like the Xe-100.

Competitive Strengths

As pioneers of next generation nuclear reactor and fuel technologies, we believe that our expertise can allow us to capitalize on our competitive strengths. We believe we will establish a strong position in the evolving market, including by leveraging the strengths of our early customer relationships to give us a differentiated path to success.

Existing Customer Base Provides Initial Momentum and Support. Both Dow and Amazon have signed agreements for our initial projects, and we expect will serve as offtake customers in the future. These initial agreements enable X-energy's goal of scaling its design to two different offtake customer applications: Industrial Process Heat Applications and Data Center Projects. Dow and Amazon also provide project development in their own operations portfolios. X-energy expects to learn from and engage with these and other customers as it builds the pipeline. This repeatable project schedule, which is also anticipated for our announced Centrica partnership, is expected to enable X-energy to scale from First-of-a-Kind to a Second-of-a-Kind model that we believe will reduce our project cost and significantly derisk future projects.

Differentiated Design that Provides Attractive Solution for Customers. The characteristics of the Xe-100 design provide several advantages over many other energy production alternatives. First, traditional energy production technologies like wind and solar cannot provide reliable baseload power without being paired with longer-term storage technologies. They are also dependent on exogenous factors, which make them incapable of quickly ramping up to address increases in demand. Second, the Xe-100 has a versatile design that can be adapted to several end markets, including industrial heat and conventional power generation. In addition to these emerging SMR technologies, many of which are based on conventional light water reactor technology, may not be designed to efficiently load-follow or operate at temperatures high enough to provide industrial heat, and are further constrained by the availability of large amounts of water, which the Xe-100 does not have.

Improved Proven Technology. Unlike some other advanced reactor technologies, X-energy's design is based on HTGR technology that has been previously deployed across the globe, including in the U.S. at the Fort St. Vrain Bottom and in the U.K. at Dorset (Dragon), and building on more than 50 years of operational experience and development by the global nuclear industry. Both of these reactors served as proof-of-concept for the technology in the U.S. and, along with other HTGRs in Europe and Asia, provide valuable engineering and operational data that X-energy has used to improve its design, leading to a more efficient, commercial-scale reactor technology.

Intrinsically Safer Based on Physics without Needing Active Safety Systems. Our simplified design is not dependent upon active safety systems, which are susceptible to failure and require redundancy to necessitate the redundancy found in a typical LWR reactor design. Our design relies on physics-based passive safety features, such that in the event of a total loss of power to a Xe-100 reactor, the reactor will automatically require no operator or computer actions, grid connections, emergency backup power or additional cooling to cool the reactor. The reactor has a strong negative temperature coefficient, which means that an increase in temperature (such as from the loss of coolant circulation) slows the fission reaction, causing the reactor to shut itself down.

shut down. Finally, due to the relatively low power density of the core, the remaining heat load is dissipated through passive cooling.

Superior Fuel Based on Decades of Research & Development. We expect our TRISO fuel to demonstrate technical quality and a streamlined fuel qualification pathway, providing us a competitive advantage in the commercial fabrication of TRISO fuel forms. The TRISO-X fuel used in the Xe-100 is a proven technology, a vessel itself and designed not to melt, enabling the technological and safety advantages of the technology to decades of research, development and testing, including the DOE's Advanced Gas Reactor (AGR).

Development and Qualification Program, the TRISO-X particle fuel is relatively well understood. Historical data has established the parameters for TRISO-X fuel testing and qualification. Our pebble fuel qualification methodology is approved by the NRC, and we have a streamlined final fuel qualification by the time of our first Xe-100 deployment.

Attractive Intellectual Property-Driven Business Model. X-energy serves as both a reactor provider and a fuel fabrication provider, offering customers an integrated solution. In our real line, we expect to receive technology fees for licensing use of our proprietary Xe-100 technology, receiving fees for coordinating assembly and construction support with customers and anticipated third-party vendors. We also intend to leverage our knowledge and expertise in regulatory construction, procurement, operations, maintenance and other processes to provide customer suite of services from the development of a project, and ultimately for the operating life of the plant. We do not construct the power plants and do not plan to own or operate the power plants once completed. As a result, we do not incur capital expenditures relating to constructing, maintaining, owning or operating facilities. However, we intend to remain involved with the EPC process throughout the project lifecycle, providing strategic consulting related to the integration of the reactor technology. We believe this property-driven business model will position us to generate attractive free cash flow.

Leading TRISO Fuel Provider. From our TRISO-X fuel manufacturing facilities in Tennessee, X-energy plans to provide customers with initial reactor fuel loads of as well as the delivery of TRISO-X fuel required to refuel plants over the lifetime of a plant. This provision of fuel to customers would generate a strong, recurring, revenue stream. X-energy's fuel can also be used in other advanced reactor technologies, making it a key enabler of the broader advancement of the nuclear industry. X-energy was selected to receive one of the first allocations of HALEU from the DOE. The DOE is required to begin initial fuel production but X-energy does not, and does not intend to, bear any inventory risk associated with uranium or fuel feedstock. We intend to provide only fabrication services (e.g., transformation of HALEU into the final TRISO-X fuel form) for customers, thereby limiting the risks associated with holding the title to uranium or enriched uranium fuel feedstock.

Visionary Management Team and Highly Expert Employee Base. We have an experienced and talented team of leaders and innovators who have been directly involved in the development of advanced nuclear technology and who have led large-scale nuclear projects and operations. As of January 1, 2023, we have a highly educated workforce of 889 employees, of whom 312 have master's degrees and 107 have PhDs. Our executive leadership team has experience in nuclear design, nuclear project delivery, nuclear fuel fabrication, operations, government relations and public companies in organizations such as GE Nuclear Energy, Ltd. ("GEH"), the DOE, the Nuclear Regulatory Commission, Constellation Energy, Hunt Consolidated, BWXT, Westinghouse, Hartree Partners, and Emirates Nuclear Energy Corporation. Our management team is led by our CEO, J. Clay Sell, who is one of the foremost leaders in the nuclear industry. As the former Deputy Secretary of the DOE, he brings the perspective of the U.S. energy sector to X-energy. Further, his experience in renewables development after his time as Deputy Secretary of the DOE gives him key experience in major project development and valuable insight into the limitations of renewable technologies, and in turn the value of nuclear generation.

Strength of Government Relationships. The ARDP significantly derisks the delivery of X-energy's first-of-a-kind reactor deployments through the Dow Project. Because of the 50/50 cost share program, X-energy has had significant ongoing engagement with the DOE and benefits from the DOE's collective resources and support. Further, X-energy has developed high credibility with the Nuclear Regulatory Commission through our ongoing engagement for both our FOAK Fuel Fabrication facility and Dow project. This was demonstrated by the docketing of the first Construction Permit Application for an 18-month review schedule, the shortest CPA timelines ever given.

Our Business Model

We have an intellectual property-driven business model based on our reactor and fuel. We derive revenues from technology licensing, services and fuel operations that span the development and operation of the reactors.

- **Reactors:** The revenue stream from reactors includes technology fees for the use of our property of the Xe-100 technology. We will not own and operate the facilities themselves. We believe this model significantly reduces the amount of capital needed to operate our business.

We anticipate offering site-specific engineering and site characterization, project planning, construction coordination, construction support, regulatory support, procurement support and long-term support to customers. Utilizing our knowledge and expertise in licensing, construction, project management and other processes, we plan to provide customers with a full suite of value-added services to support the development of the nuclear power facilities. At the same time, we expect to generate recurring revenue from services such as the ongoing maintenance and operator training over the anticipated 60-year life of a facility.

- **Fuel:** We intend to provide manufacturing services to customers, including producing and delivering a full load of both TRISO-X fuel and an LEU-based TRISO fuel at commissioning of a plant. We expect to generate additional long-term recurring revenue from our own proprietary TRISO-X fuel. We expect to be required to refuel plants during the anticipated 60-year life of each facility. We expect to manage the inventory risks related to uranium or enriched uranium fuel feedstock. We intend to provide fuel fabrication supply services (e.g., transformation of HALEU into the final TRISO-X fuel) to our customers and assume limited risks associated with holding the uranium or enriched uranium feedstock. Additionally, we will not have any responsibility for spent fuel management. The design of such facilities to adequately handle spent fuel during the life of the facility and operations, spent fuel remains the responsibility of the plant operator. Thereafter, spent fuel management remains the responsibility of the DOE.

Growth Strategies

We intend to grow our business by leveraging the competitive advantages of our differentiated, reliable Xe-100 technology and our strong customer and government relationships to achieve our growth objectives. We believe we have several avenues to achieve our growth objectives:

Continue to Develop our Next Generation Technology. We intend to continue developing our advanced SMR fuel technology with the goal of achieving commercial delivery of our first fleets of reactors in the early 2030s. We have substantially advanced detailed design for the Xe-100 and are working with our customers and construction partners to complete that process. Furthermore, we will work with our customers to complete the site-specific environmental studies required for licensing and progress the submission of the construction permit application, including the preliminary safety analysis report. We have produced TRISO-X pebble fuel in kilogram batch quantities in our fuel fabrication pilot facility. We have begun construction on North America's first purpose-built commercial advanced nuclear fuel facility in Oak Ridge, Tennessee.

Execute on Attractive Business Development Pipeline. We believe the market for our Xe-100 advanced SMR X fuel technologies is wherever non-intermittent, reliable power is needed. We are initially targeting deploying our advanced SMRs to both industrial (including large-scale chemical manufacturing) and data center-based service provider customers (e.g. data centers) who have needs for both electric power and the production of high-temperature steam with high reliability needs. We also plan to serve traditional utility and independent power producers ("IPPs") seeking to replace carbon-intensive fossil-fueled power in their jurisdictions.

Leverage Repeated Project Execution Learnings to Scale From FOAK to NOAK. Traditionally, large-scale nuclear power plants have been plagued by cost and schedule overruns. Vogtle units 3 & 4 cost more than \$16 .0 billion and took over seven years over schedule in part because they were the first and second AP1000 reactors deployed in the U.S. While the fourth unit was reportedly approximately 20% less expensive than the third, it still took over seven years over schedule.

were applied, achieving expected Nth-of-a-Kind scale with standard costs and schedule timelines only through repeated project delivery over a large order book. The DOE and industry expect Nth-of-a-Kind delivery can unlock potential savings of more than 30%. Starting with Dow and energy is already working to deliver eight reactors across its first announced sites at Seadrift, Richland, Washington, respectively. With substantial potential for further targeted pipeline and Centrica, we expect to be able to achieve reductions in costs and the acceleration of schedule as we deliver more reactors in

the future. We believe that NOAK will be achieved through repetitive project execution and risking, including early supply chain engagement, construction timeline compression, and streamlining. As we apply lessons learned from each deployment, we expect these steps will reduce overall project execution costs and risks for subsequent Xe-100 projects.

Continue Geographic Expansion. Our initial core markets are the U.S., Canada and the U.K. We have sophisticated regulators that are capable of licensing our technology. Beyond these initial markets, which together represent only one-third of the serviceable energy consumption, we anticipate entering new markets where we already have customer engagement, including countries in Eastern Europe, Middle East and East Asia.

Drive Technology Advancements. Using our innovative technology platform, we believe we are well-positioned to continue making technology advancements over time. These improvements include optimizing the Xe-100 for industrial heat applications and innovations in the design to support higher temperatures greater than 800° C to support more efficient hydrogen production processes.

Develop New Products. We continue to explore the development of innovative new products using our core technology and varied use cases for our proprietary Xe-100 reactor, TRISO-X fuel and microreactor technologies. TRISO-X is currently performing fuel fabrication work for others. Other technologies also use TRISO-based fuel, including space applications. Similarly, our Technologies team is involved in U.S. government-funded studies (such as those from the DOE) for innovative remote powering and lunar applications that may involve our microreactor technologies.

Our Technology

X-energy was founded by Kam Ghaffarian. The technology used today was matured through the Bed Modular Reactor program, a South African public-private partnership which advanced high-temperature gas-cooled reactor technology. X-energy initially focused on the conceptual design of the Xe-100, shown below, with an emphasis on top-level requirements designed to meet the broadest set of use cases. The company has continued to mature the Xe-100 design and has expanded into the production of other reactor designs. Finally, X-energy has developed the design for the XENITH microreactor and we continue to invest in research and development efforts.

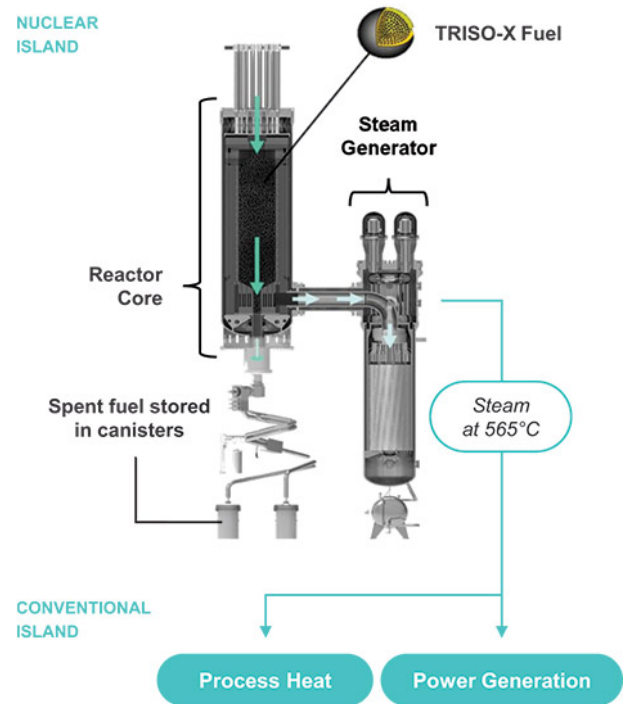
Xe-100 — Our Advanced Reactor Technology

The Xe-100 is an HTGR which is designed to have advantages in economics, reliability and safety over conventional LWRs and other advanced SMR designs. The Xe-100 is a Generation IV advanced nuclear technology that has been designed to remedy weaknesses associated with traditional LWRs. The Xe-100 can efficiently produce electricity or process heat. When configured for electricity generation, the reactor can power an 80 MW turbine generator. Deployment of these reactors is scalable and is designed to form a four-reactor configuration to form a 320 MWe power plant.

The Xe-100 has made substantial progress in the NRC licensing process in the U.S. After completion of pre-application engagement, we supported Dow's subsidiary, Long Mott Energy, LLC, in filing a construction permit application to NRC for the proposed advanced nuclear project utilizing the Xe-100 reactor design in Dow's Seadrift, Texas site. The pre-application engagement was intended to reduce licensing risk, as regulatory challenges or delays can materially affect a project's timing, cost and overall viability. Citing the completeness and quality of the application, and the effectiveness of the

application engagements, the NRC established an 18-month review schedule to complete its technical reviews for the Dow Construction Permit Application.

Xe-100 Schematic

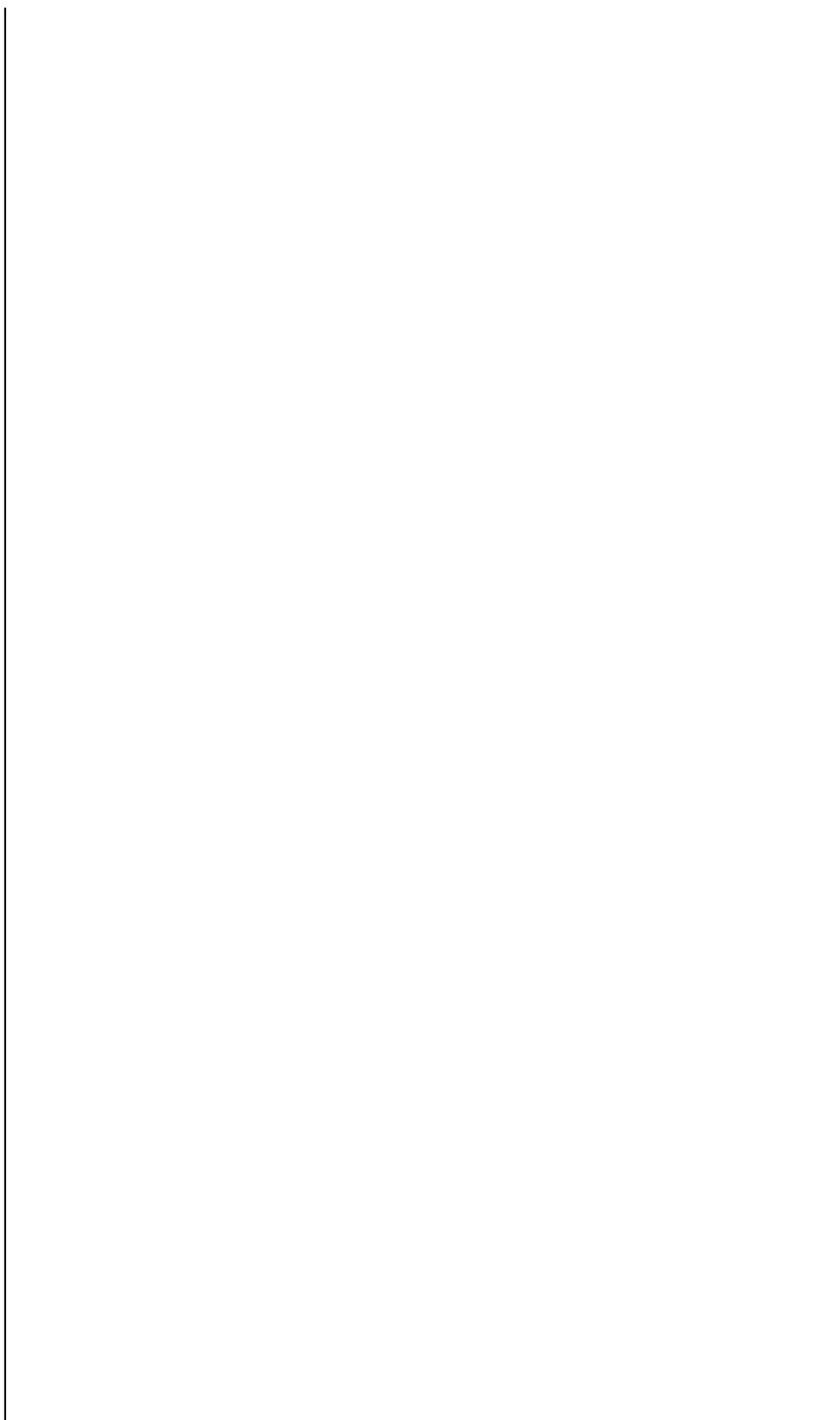


One Xe-100 plant produces 320 MWe¹

Note: Figure is intended to represent an illustrative rendering and is for illustrative purposes only.

The Xe-100 has the following characteristics:

- **Cleaner and more reliable** — Like all nuclear reactors, HTGRs produce virtually zero emissions during energy generation, are “always” on (at 95%+ availability), and provide dispatchable power that is cleaner than most available firm alternatives.
- **Designed with Intrinsic Safety Features** — The Xe-100 HTGR is safer by design and by the intrinsic physics of the reactor, requires fewer mechanical safety systems than traditional reactors, as well as fewer personnel for operations.
- **More Secure** — The TRISO-X pebble fuel is a containment vessel in itself, supporting the inherent safety nature of the Xe-100.
- **Scalable** — The Xe-100 is expected to be commonly deployed in its optimized configuration of 320 MWe, referred to as the “Xe-100 plant,” with the opportunity to scale up to additional reactors as needed.
- **Streamlined** — Due to its superior intrinsic safety attributes, the Xe-100 design needs fewer related specialized materials and therefore utilizes substantially more off-the-shelf components than traditional nuclear reactors, allowing for scalability from commercial vendors.



Versatile — In addition to electricity, the Xe-100 plant can be configured to deliver high-temperature steam at 565°C, providing a solution for difficult-to-decarbonize industrial heat applications such as oil sands operations, mining, chemical production and petroleum refining and other industrial processes.

- Load-following — The Xe-100 is designed to ramp down from full power to 40% power in a short time span and ramp back up in a similar short time span.

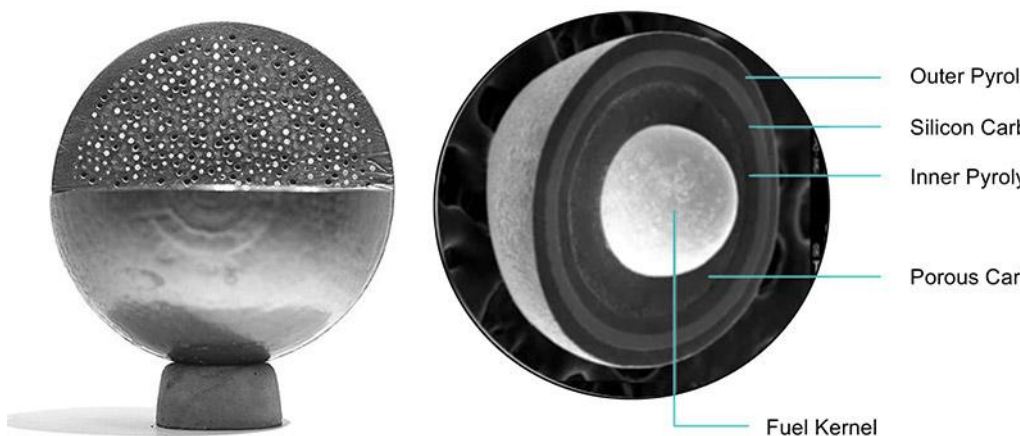
TRISO-X Pebble Fuel—Our Proprietary TRISO Fuel

Our reactors use TRISO particle fuel, a technology first developed in conjunction with the Dragon reactor in the 1960s. The DOE describes TRISO particles as “the most robust nuclear fuel on Earth.” We manufacture our own TRISO fuel using proprietary methods, through our wholly owned subsidiary, TRISO-X, LLC, to ensure supply and quality control. Further, many of the advanced reactor designs in development are expected to be powered by TRISO fuel or other coated-particle fuels. As part of our business model, we intend to not only fabricate and sell TRISO-X fuel to our customers, we also fabricate and sell TRISO-X fuel and other encapsulated fuels to other advanced nuclear customers, including governmental and private entities.

Safe, established fuel form: While TRISO fuel is already an established fuel after decades of use and development, we expect our proprietary fabrication method to further differentiate our TRISO fuel and deliver a competitive edge in the commercial fabrication of TRISO fuel forms. We anticipate our patented and novel process will demonstrate superior quality characteristics as well as improved performance compared to historical TRISO and potential future TRISO competitors. The TRISO fuel is a containment vessel in itself, encased in graphite, pyrolytic carbon and silicon carbide layers that do not melt due to its ability to withstand extreme temperatures. After HALEU has been encapsulated in pebble form, security and proliferation risks are substantially reduced. Its robust nature allows for enhanced safety and, we expect a small emergency planning zone for our reactors, providing additional flexibility in site design and access to non-traditional nuclear markets and customers.

Robust integrity: Our TRISO-X pebble fuel is designed not to melt due to its robust construction. The structure minimizes the requirement for the extensive use of expensive and large concrete containment structures typically required in conventional reactors.

TRISO-X Fuel ***X-energy's Proprietary TRISO Fuel***



Integrated fuel fabrication business: We currently manufacture TRISO-X fuel in our TRISO-X Fuel Facility in Oak Ridge, Tennessee, which has operated since 2016. In February 2026, we received a Nuclear Material License from the NRC that establishes TX-1 as the first-ever Category I nuclear facility licensed in the United States. TX-1 will be a state-of-the-art Category II nuclear facility specifically for handling and processing HALEU feedstock for the fabrication of TRISO-X pebbles.

anticipate constructing TX-2 on the same site, which will allow for the license to extend to b
We do not expect to hold significant inventory of the fuel feedstock, which we expect to be pr
customers. Our business will be fabricating the pebble fuels into their final fuel form for
purchase for operation of the Xe-100 reactor and its refueling needs.

Path to qualification due to extensive R&D: The ability to leverage decades of prior TR
and development, particularly the DOE's Advanced Gas Reactor (AGR) Fuel Developm
provides

X-energy with a well-defined pathway for fuel qualification. X-energy's TRISO-X Qualification Methodology has been approved by the NRC and references the AGR test results qualification. This established framework provides X-energy a streamlined path towards obtaining TRISO-X fuel by the time of its first Xe-100 deployment. Based on the approved fuel methodology, we are currently in the irradiation test phase and in the final stages of qualifying X pebble fuel.

Long-term revenues derisked: We plan to provide our Xe-100 customers with their initial revenue while generating additional long-term and recurring revenue streams by refueling Xe-100 throughout their anticipated 60-plus year lives. We further assume that we will be the initial supplier of proprietary TRISO-X fuel.

Research and Development

In addition to our core Xe-100 reactor, we continue to innovate and develop new products with our Emerging Technologies team. We are expanding our portfolio to address a broader range of applications and use cases beyond utility-scale power generation.

XENITH Microreactor

We have developed the XENITH microreactor, a compact solution designed to deliver power in locations where traditional power infrastructure is unavailable, impractical or vulnerable. XENITH is designed to be deployed in months and operate continuously for 20 years, delivering 3-10 MWe with minimal maintenance requirements.

XENITH is not yet operational but is currently exploring commercialization opportunities. The development of the microreactor began in the Pele program through the U.S. Department of Energy ("Project Pele"), for which X-energy was awarded approximately \$60 million in various contracts. Conceptual design was progressed during Pele Phase 1 and 2, and design work has continued on now the XENITH microreactor. The microreactor is likely to be deployed in a 5 MWe form factor or multiple reactors in a 10 MWe electric format.

Like the Xe-100, XENITH is also a High Temperature Gas-cooled reactor that runs on TRISO fuel. The required fuel is expected to be in the form of compacts rather than pebbles. The expected operating temperature is 750 degrees Celsius. The value proposition is expected to be as follows:

- Scalable Power. Factory-built, transportable nuclear energy solution that prioritizes deployment readiness and compatibility with existing infrastructure to support deployment in a variety of locations.
- Resilient & Reliable. Provides reliable, long-term power with no required refueling over the reactor's lifetime, eliminating vulnerability to supply chain disruptions while providing 24/7 reliable power.
- Tailored to the Target Market. Military, defense, and remote applications, including power for black-starts, and remote communities that require secure, independent power.

Space Applications

Our innovative HTGR design and high-temperature-resistant fuel make a unique and powerful combination for nuclear electric propulsion, nuclear thermal propulsion and lunar surface power.

Our use of robust TRISO fuel in combination with innovative reactor designs and advanced materials, all work together to increase the efficiency of the fission reaction while minimizing weight, keeping the reactor light enough for spaceflight. These characteristics have allowed

viable concepts for a Fission Surface Power (FSP) reactor. This concept was studied through a
award from the DOE and the U.S. National Aeronautics and Space Administration (NASA) in
advance the design of a Fission Surface Power (FSP) solution.

Our Customers

Dow

Dow is X-energy's first customer to receive a reactor and is a global leader in the special
industry, involved in the manufacturing and distribution of a wide range of chemical and plasti

the course of its annual operations, Dow produces over 12 million MWh of electricity and is committed to the ongoing decarbonization of its operations.

X-energy has partnered through a Master Project Development Agreement (“MPDA”) and a Cooperation Agreement (“CCA”) with Dow to provide our services in support of a FOAK of four Xe-100 reactors to provide power with virtually zero direct GHG emissions and industrial heat for Dow’s UCC Seadrift site in Texas. With the support and assistance of X-energy, Long Mott Energy, a wholly owned subsidiary of Dow, filed a Construction Permit Application (“CPA”) with the Texas Department of Licensing and Regulation in March 2025 which was docketed in May 2025 for an 18-month review period with an expected completion by late 2026 and receipt of the CPA expected in the first quarter of 2027. Initial construction will commence after receipt of the CPA.

Dow provides expertise in plant design and industrial heat as well as in the development of Xe-100’s first plant, which will be a first-of-its-kind co-generation facility with unique heat capabilities not available in other technologies. Under the CCA, as between the parties, X-energy will own the intellectual property related to the reactor systems, controls, software, and related nuclear plant technology, while Dow will own all of the conventional island intellectual property related to the non-nuclear island. X-energy will turn reactor steam into electricity and run steam and water for the facility, as well as intellectual property for equipment and processes that sit outside both the nuclear and conventional plant areas, and the parties for this project.

Amazon

After an extensive evaluation of potential carbon free generation solutions for its extensive data center footprint, Amazon made an equity investment in X-energy in 2024 and announced options for more than 5 GWe of new Xe-100 projects online across the U.S. by 2039, which assuming full execution of all options, will represent the largest commercial deployment target of SMRs to date as of August 2025.

The first deployment under this 5 GWe total potential target is a project with Energy Northwest in central Washington. Amazon and Energy Northwest entered into a Carbon Free Development Agreement for an initial deployment of four reactors representing 320 MWe, with the potential to expand to a power capacity to 960 MWe.

Centrica

Centrica is a major provider of energy services across the U.K. and owns a 20% stake in the country of operating nuclear reactors in the country. In September 2025, X-energy and Centrica signed a Joint Development Agreement (“JDA”) dedicated to building and operating Xe-100 reactors in the U.K. This announcement followed a pledge from President Donald Trump and Prime Minister Keir Starmer to work together on nuclear power and constitutes a new strategic commercial alliance to accelerate the deployment of SMRs, bringing together a leading player in delivering the U.K.’s clean energy future and the world’s most advanced nuclear technology.

The Xe-100 was selected after a significant review period by Centrica of advanced Generation IV nuclear technologies. Centrica prioritized partnership with X-energy because of the reactor’s safety and ability to provide industrial heat that Centrica sees as key to expansion in the U.K. market. The U.K. government is currently pursuing a “three-legged” approach to nuclear development, designed to ensure a resilient nuclear energy sector. Under this strategy, the first “leg” focuses on large-scale nuclear power, such as those being developed in partnership with EDF, to provide substantial baseload power. The second “leg” supports the deployment of Gen III+ light water SMRs, with Rolls Royce’s SMR technology.

as the leading solution in this category. The third “leg” is dedicated to other small and advanced reactors, including HTGRs like Xe-100, which are prioritized for their potential to deliver base load and high-grade industrial heat, supporting decarbonization across a range of sectors.

X-energy and Centrica have identified Hartlepool as the preferred site for the first of a fleet of approximately six GWe (representative of 76 reactors likely deployed as 19 configurations). A project at Hartlepool will be comprised of up to twelve 80 MWe reactors, capability to provide

high temperature steam for industrial decarbonization. Subject to securing appropriate permits and licenses, the first electricity generation is expected to be in the mid-2030s.

Other Key Partnerships

Department of Energy

Through the ARDP, we are partnering with DOE and others to build the world's first commercial advanced nuclear reactor. In December 2020, X-energy, initially in collaboration with Energy Transfer, was selected for an award, initially in the amount of \$1.2 billion to deliver a first-of-a-kind advanced nuclear plant and TRISO-X fuel fabrication facility. This project is now being developed by X-energy and Energy Transfer with continued financial support from the DOE on a 50/50 cost share basis. As one of the few companies selected out of many applicants for the ARDP, X-energy was recognized as having an advanced technology of choice.

Ontario Power Generation

In 2022, X-energy and OPG entered into a framework agreement to work exclusively with OPG with respect to advanced nuclear power industrial applications in Ontario, Canada, and to co-develop and advance the Xe-100 as the nuclear technology of choice for industrial applications throughout the world. OPG will be the operator of any Xe-100 facilities that are deployed under this agreement. OPG is a long-term strategic investor in X-energy.

Our Key Supply Chain Partners

We have a network of partners that enable our growth and success. These partners include:

- ***Curtiss-Wright.*** Curtiss-Wright is a leading U.S. designer and supplier of critical nuclear reactor plant systems, equipment, services, and spare parts to the U.S. domestic and global nuclear power industry. Following several competitive bid processes, X-energy selected Curtiss-Wright as the successful bidder for multiple Xe-100 systems. Curtiss-Wright was selected as a preferred supplier to X-energy for the ARDP and subsequent Xe-100 projects in the U.S. Curtiss-Wright is an equity investor in X-energy.
- ***Doosan Enerbility ("Doosan").*** Doosan is a preferred strategic supplier to X-energy. Doosan is a major global manufacturer and supplier of core components of nuclear power plants, such as pressure vessels, steam generators, and steam turbines. Doosan has a vertical manufacturing facility in Changwon, Korea, which is capable of raw material production and assembly of nuclear components. Doosan has manufactured and supplied 34 reactor pressure vessels and 124 steam generators globally. Doosan is an equity investor in X-energy. In December 2022, Doosan signed a Reservation Agreement with X-energy, committing to the construction of a SMR fabrication facility to support the execution of X-energy's more than 11 GW pipeline.
- ***DL E&C.*** DL E&C will work with X-energy to identify opportunities around the world to share best practices to support the development and deployment of Xe-100 plants on a global scale. Founded in 1939, DL E&C has the longest business history among construction companies in Korea and has maintained a top ten Korean engineering and construction company ranking for over 50 years. DL E&C is the flagship company of DL Group, and has a broad range of experience in the mid/downstream energy sector engineering, procurement and construction, providing

and solutions in more than 35 nations, focusing on a more sustainable and better future. X-energy is an equity investor in X-energy.

- *Korea Hydro & Nuclear Power (“KHNP”)*. KHNP is the Korean entity that operates in the hydroelectric sector as a subsidiary of Korea Electric Power Corporation (“KEPCO”), a state-owned utility. Since 1971, KHNP has successfully constructed and operated 30 nuclear power plants still in operation today, 26 in South Korea and four in the United Arab Emirates. KHNP provides significant expertise in the construction of nuclear reactors. In August 2023, KHNP

signed a joint compact with X-energy, Amazon, and Doosan outlining the intention to deploy the Amazon order book in the U.S.

- *SGL Carbon LLC (“SGL”)*. SGL is a developer and manufacturer of advanced carbon capture technology and SGL have collaborated since 2015 on the qualification of NBG-18 graphite for the Xe-100, leveraging SGL’s experience manufacturing graphite for high-temperature reactors. In January 2026, SGL announced a 10-year graphite supply agreement for components of X-energy’s Xe-100 SMRs, and has commenced production for the Dow site as part of an initial three-year award valued at over \$100 million.

SUMMARY RISK FACTORS

Investing in our common stock involves substantial risk. The risks described under the “Risk Factors” immediately following this prospectus summary may cause us to not realize the goals of our objectives or may cause us to be unable to successfully execute all or part of our strategy. The more significant challenges include the following:

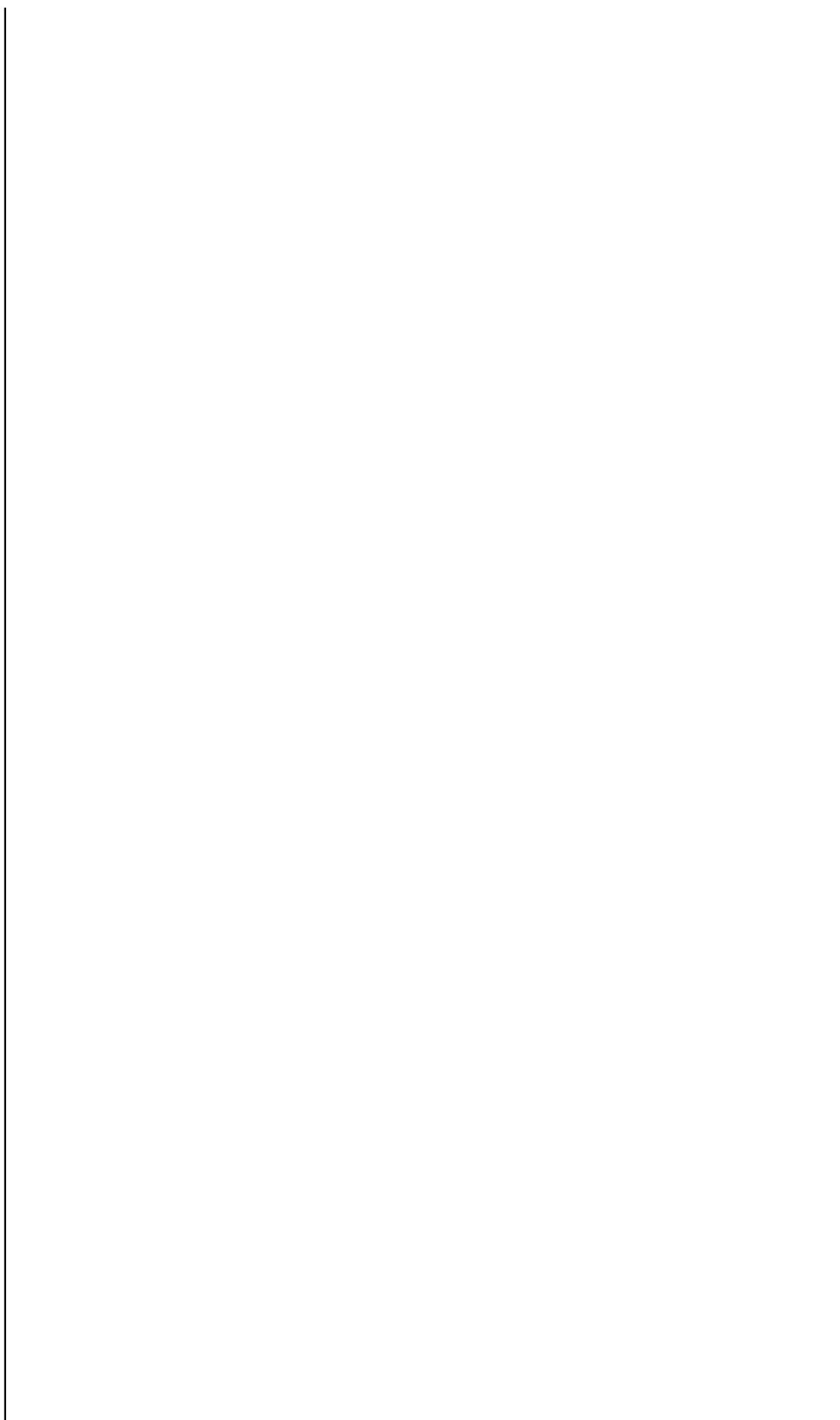
- We have not yet delivered a commercial Xe-100 or achieved final investment decision for any deployments; our FOAK schedule, cost, and performance are uncertain and delays, particularly on initial projects, could materially harm our business, reputation, and financial results.
- We may not generate sufficient liquidity to fund operations and growth; our business is capital intensive and we will require substantial additional financing, which may not be available on acceptable terms and could be dilutive.
- Our business depends heavily on U.S. government support, including the ARDP appropriations are uncertain, and any reduction, delay, or termination, or failure to obtain extensions or modifications under the ARDP Agreement or increases due to inflation or economic growth, could materially and adversely affect our projects, TX-1 fuel facility, and commercialization.
- We face significant regulatory and licensing risks for both our reactors and fuel facilities; other regulatory approvals may be delayed, conditioned, or denied, including as a result of government interventions and hearings, which could increase costs and extend timelines.
- Access to HALEU is limited and failure of commercial HALEU supply to materialize on timelines and at predictable costs could delay fuel fabrication (TX-1 and beyond) and deployments and impair competitiveness.
- Our cost, schedule, and unit economics are subject to substantial uncertainty, including supply chain constraints, labor availability, site-specific factors, and first-of-a-kind costs; reductions from NOAK learning are lower or slower than expected, our products may not be competitive.
- We have limited operating and commercial experience at our intended scale and configuration; design, manufacturing, construction, or operational issues may emerge late and require remediate.
- We rely on a limited number of specialized suppliers (including graphite and other materials and first-of-a-kind components); supply disruptions, quality issues, shipping/logistics, tariffs, or trade policy changes could delay projects and increase costs.
-

Safety, security, and cybersecurity incidents at our facilities, our customers' facilities, or in the nuclear industry, could result in regulatory actions, project delays, increased costs, harm, and reduced demand.

- Market adoption of Gen IV SMRs is nascent and uncertain; demand may grow more slowly than expected, customers may defer or cancel projects, and our products may face competition from lower cost alternatives (including gas, renewables with storage, or other advanced reactors).
- We face intense competition from domestic and state-supported international nuclear competitors; competitors may have greater resources, faster regulatory paths in some jurisdictions, or other advantages which could pressure pricing and margins.

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- Our reliance on key partners and customers, including Dow (ARDP) and cloud computing, exposes us to counterparty and execution risk; changes in partner priorities, funding, or other factors could materially affect our path to commercialization. Our JDA with Centrica is not definitive and may not result in definitive agreements or revenues.
- Customer contractual terms (including Amazon's priority queue slots, rights of first refusal, and favored pricing) may constrain capacity allocation, compress margins, increase complexity, and expose us to payment or performance obligations.
- Our fuel business depends on licensing and scaling TX-1 and subsequent facilities; delays in licensing, materials qualification, or facility construction, or inability to recruit and retain talent, could impair fuel availability and recurring revenue.
- We may not carry insurance covering performance of the Xe-100 or all relevant projects; if not obtained, insurance may be unavailable, insufficient, or costly.
- Changes in laws, regulations, incentives, export controls, or government policies could increase costs, delay approvals, limit market access, restrict technology transfers, or reduce expected incentives.
- Public perception and political support for nuclear energy can shift; adverse events or changes in policy priorities could slow licensing and deployment and increase costs.
- Spent fuel and governmental waste management policies remain unsettled; customer costs for storage or disposal costs and responsibilities could reduce demand for our technology.
- We depend on key personnel and our ability to hire and retain highly specialized talent, immigration constraints, or turnover could delay programs and increase costs.
- We have identified material weaknesses in internal control over financial reporting; failure to remediate the existing weakness or to maintain effective controls could adversely affect our reporting and the market price of our stock.
- We are subject to intellectual property risks; we may be unable to obtain, maintain, or enforce intellectual property rights, or we may face third-party claims that could require costly licensing, redesigns, or limit commercialization; our IP protection is territorial and may not be effective abroad.
- Data privacy and cybersecurity risks could lead to service interruptions, regulatory liability, and reputational harm; increased AI use introduces additional privacy, security, and compliance risks.
- We are exposed to construction, siting, industrial application, severe weather, disaster, and other risks; catastrophic events or unusual siting requirements could increase costs, delay start-up, and strain resources.
- Our illustrative capacity figures and unit economics in this prospectus are estimates based on numerous assumptions (including technology fees, services, and fuel pricing); actual results may differ materially, including lower pre- and post-COD revenues and margins.
- Compliance with government contracting requirements (FAR, False Claims, pricing/cost accounting audits) carries risk of penalties, repayment, suspension, or debarment; U.S. budget cuts, shutdowns, and continuing resolutions may disrupt program funding and payments.
- Export/import controls and sanctions can restrict sales, technology transfers, and commercial operations; violations or policy shifts could result in penalties and lost market access.
-



Energy market rules and oversight (including FERC, NERC, and ISO/RTO market) affect customer project economics and indirectly our demand; noncompliance could result in sanctions on customers.

- Tax law changes and interpretations (including OBBBA) may increase complexity, affect effective tax rates; changes in accounting estimates and quarterly variability could materially affect reported results.
- As a holding company, we will depend on distributions from our operating subsidiary for operating expenses (including under a Tax Receivable Agreement), and any dividends; such distributions may not be sufficient to cover our needs.

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restricted, and TRA payments (including with respect to a change of control (as defined in the Receivable Agreement), Material Breach or early termination) may be substantial and realized tax benefits, constraining liquidity.

- An active trading market for our Class A common stock may not develop or be sustained. The price may be volatile, large sales (including post lock-up) may depress the price, and the market may be limited, and you may lose part or all of your investment.

Please see “Risk Factors” for a more complete discussion of these and other risks.

IMPLICATIONS OF BEING AN EMERGING GROWTH COMPANY

We are an emerging growth company, as defined in Section 2(a) of the Securities Act, and the Jumpstart Our Business Startups Act of 2012 (the “JOBS Act”), and we may take advantage of exemptions from various reporting requirements that are applicable to other public companies that are not emerging growth companies, including relief from the auditor attestation requirements of Section 404 of the Sarbanes-Oxley Act, less extensive disclosure obligations regarding executive compensation, reduced registration statements, periodic reports and proxy statements, exemptions from the requirement to hold a non-binding advisory vote on executive compensation, and exemptions from stockholder approval of golden parachute payments not previously approved. We may also elect to take advantage of certain reporting requirements in future filings. As a result, our stockholders may not have access to certain information that they may deem important and the information that we provide to our stockholders may be different than, and not comparable to, information presented by other public reporting companies. We will remain an emerging growth company until the earliest of (i) the last day of the year following the end of the anniversary of the completion of this offering, (ii) the last day of the year in which we have annual gross revenue of at least \$1.235 billion, (iii) the last day of the year in which we are deemed to be an “accelerated filer” as defined in Rule 12b-2 under the Exchange Act, which would occur if the aggregate market value of our Class A common stock and Class B common stock held by non-affiliates exceeded \$700 million as of the last business day of the second fiscal quarter of such year and (iv) the date on which we have more than \$1.0 billion in non-convertible debt securities during the prior three-year period.

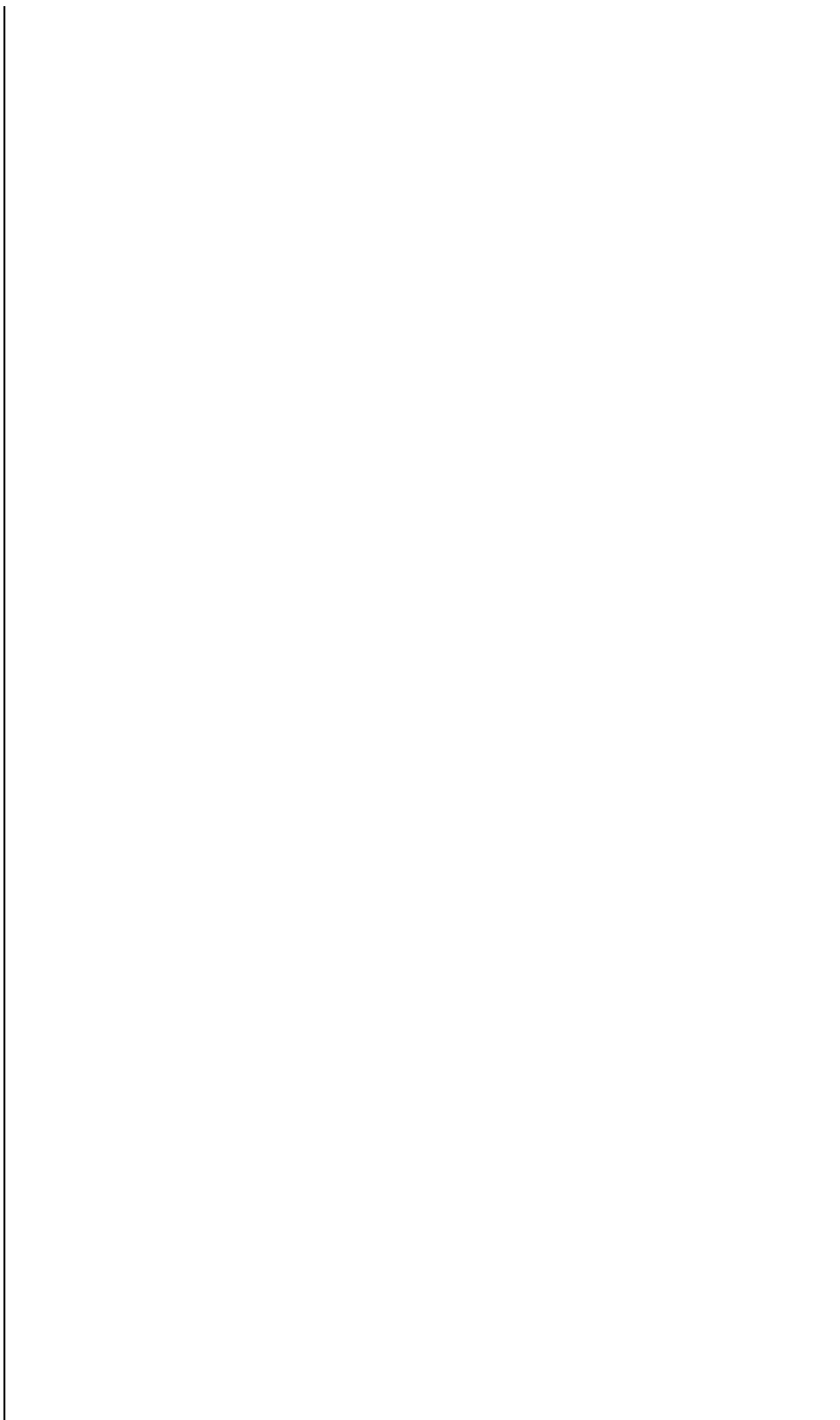
In addition, the JOBS Act also provides that an emerging growth company may take advantage of an extended transition period provided in the Securities Act for complying with new or revised accounting standards. An emerging growth company may therefore delay the adoption of certain accounting standards until those standards would otherwise apply to private companies. We have elected to avail ourselves of this exemption and, as a result, will not be subject to the same implementation timing for new accounting standards as are required of other public companies that are not emerging growth companies, which may make comparison of our financial information to those of other public companies more difficult.

CORPORATE INFORMATION

X-Energy Reactor Company, LLC is a Delaware limited liability company formed on March 1, 2018, and is the predecessor for financial reporting purposes of X-Energy, Inc., a Delaware corporation formed on September 18, 2025.

X-energy’s principal executive offices are located at 530 Gaither Road, Suite 700, Rockville, MD 20850 and our phone number is (301) 358-5600.

Our website address is www.x-energy.com. Information contained in, or accessible through, our website is not a part of this prospectus, and the inclusion of our website address in this prospectus is only an inactive textual reference.



ORGANIZATIONAL STRUCTURE

In connection with the closing of this offering, we will undertake certain organizational transactions subsequent to which we will conduct our business through what is commonly referred to as a partnership structure, which is often used by partnerships and limited liability companies when they decide to raise capital in an initial public offering. Unless otherwise stated or the context otherwise requires, all information in this prospectus reflects the consummation of the Reorganization Transactions.

Following the consummation of the Reorganization Transactions, we will be a holding company. Our sole material asset will be our equity interest in XERC, which, through its direct and indirect ownership, conducts all of our operations. Because X-Energy, Inc. will be the sole managing member of XERC, it will indirectly operate and control all of the business and affairs (and will consolidate the financial statements) of XERC and its subsidiaries.

Prior to the consummation of the Transactions, the capital structure of XERC consists of three classes of membership interests: Common Units, Preferred Units and Incentive Units.

In connection with the consummation of this offering, we will complete a series of organizational transactions, including: (i) the eighth amendment and restatement of the XERC LLC Agreement to, among other things, effect a recapitalization in which all existing ownership interests in XERC are converted into one class of Common Units; (ii) the amendment and restatement of the X-Energy, Inc. Certificate of Incorporation to, among other things, authorize two classes of common stock; (iii) X-Energy, Inc., as managing member of XERC, (iv) X-Energy, Inc.'s acquisition of Common Units held by the Former Equity Owners pursuant to the Blocker Mergers, (v) X-Energy, Inc.'s acquisition of all of the Common Units held by the Former Equity Owners (except for Management LLC, who is addressed in clauses (vi) and (vii) below) and a portion of the Common Units held by the Continuing Equity Owners, in exchange for an equal number of shares of Class A common stock, (vi) the second amendment and restatement of the Management LLC Agreement to, among other things, effect a recapitalization in which all existing ownership interests in Management LLC are converted into one class of Common Units, (vii) Management LLC's contribution of all of its Common Units of XERC to X-Energy, Inc. in exchange for an equal number of shares of Class A common stock, which shares shall remain subject to certain vesting conditions applicable to the corresponding Common Units immediately prior to such contribution, and (viii) X-Energy, Inc.'s issuance to the Continuing Equity Owners of a number of shares of Class A common stock (equal to the number of Common Units held by the Continuing Equity Owners) in exchange for a nominal cash contribution made by such Continuing Equity Owners, resulting in a combined company organized in an umbrella partnership C corporation structure (such structure, an "Up-C"). Substantially all of the assets and the business of the company will be held by X-Energy Receivable Agreement LLC, as more fully described elsewhere in this prospectus (such transactions, the "Reorganization Transactions"). See "Certain Relationships and Related Party Transactions" for additional information.

Prior to the completion of the offering, X-Energy, Inc. will enter into a Tax Receivable Agreement with XERC and the TRA Holders. This Tax Receivable Agreement will provide for the payment by X-Energy, Inc. to the TRA Holders of 85% of the amount of cash tax savings, if any, that X-Energy, Inc. is deemed to realize (calculated using certain assumptions) as a result of the Basis Adjustment. The TRA Agreement, Basis, Blocker Tax Attributes and Interest Deductions. Assuming no material changes in the tax laws and that we earn sufficient taxable income to realize all tax benefits that are subject to the TRA Receivable Agreement, we expect the tax savings associated with the purchase of Common Units in connection with this offering, together with future redemptions or exchanges of all remaining Common Units owned by the TRA Holders pursuant to the XERC LLC Agreement as described above, to aggregate to approximately \$ million over years from the date of this offering, assuming an assumed initial public offering price of \$ per share of our Class A common stock (the maximum estimated price range set forth on the cover page of this prospectus), and assuming all redemptions or exchanges would occur immediately after the initial public offering for the remaining owners of Common Units not acquired by X-Energy, Inc. Under such scenario, assuming future payments are made on

relevant tax return is due, without extensions, we would be required to pay approximately amount, or approximately \$ million over the -year period from the date of this o TRA Holders. The actual amounts we will be required to pay under the Tax Receivable Agre significantly different from the amounts described in the preceding sentence.

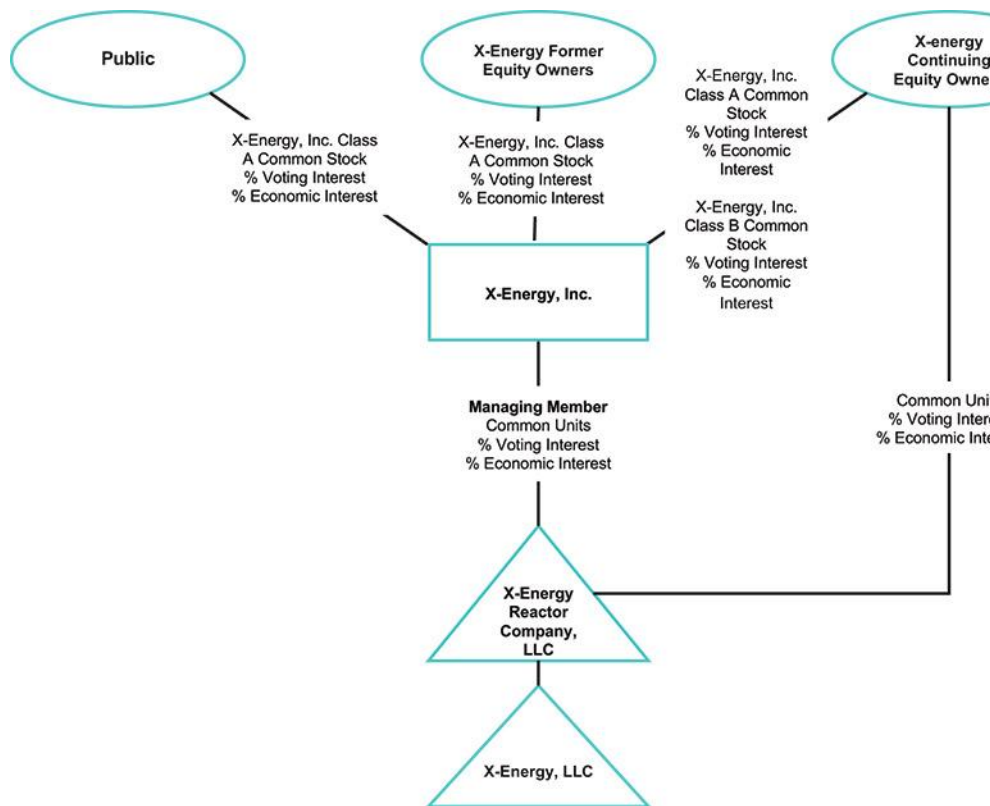
The term of the Tax Receivable Agreement will continue until all such tax benefits have or expired unless we exercise our right to terminate the Tax Receivable Agreement early, in w would

be required to pay to the TRA Holders an amount representing the present value of anticipated tax benefits under the Tax Receivable Agreement (computed using certain assumptions). For example, if we elect to terminate the Tax Receivable Agreement immediately following this offering, and there are no material changes in the relevant tax laws or tax rates and that we earn sufficient taxable income to utilize the potential tax benefits that are subject to the Tax Receivable Agreement, we estimate that the termination payments would be approximately \$ _____ million (at a discount rate of SOFR plus _____ points) based on the assumed initial public offering price of \$ _____ per share of our Class A common stock, which is the midpoint of the range set forth on the cover page of this prospectus, and assuming that the TRA Holders (as defined in the Tax Receivable Agreement) were to be _____ %.

Importantly, upon a change of control or a Material Breach of our obligations under the Tax Receivable Agreement, the Tax Receivable Agreement will not terminate nor will a single, accelerated payment be due. Thus, if we commit a Material Breach under the Tax Receivable Agreement, including a change of control (as defined in the Tax Receivable Agreement) or elect to terminate the Tax Receivable Agreement and do not elect to make a single, lump sum payment, our (or our successor's) future obligations under the Tax Receivable Agreement for each taxable year after any such event would be determined by utilizing certain valuation assumptions, including that (i) any Common Units that have not been redeemed are deemed exchanged for the market value of the shares of our Class A common stock at the time of the change of control and (ii) X-Energy, Inc. will have sufficient taxable income to fully utilize the tax attributes covered by the Tax Receivable Agreement.

For additional information, see “Risk Factors — Risks Related to Our Capital Structure” and “Relationships and Related Party Transactions — Tax Receivable Agreement” for additional information regarding the Tax Receivable Agreement.

The following diagram reflects our simplified organizational structure after giving effect to the Transactions, including this offering:



We intend to use all of the net proceeds we receive from this offering to purchase Common Units from XERC. In the event the underwriters exercise their option to purchase additional shares of Class A common stock, we intend to use any proceeds from such exercise to purchase additional

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issued Common Units from XERC. See “Organizational Structure,” “Use of Proceeds,” “Relationships and Related Party Transactions”.

Subject to the terms and conditions of the XERC LLC Agreement, the Continuing Equity Owners will have the right to have XERC redeem their Common Units for shares of Class A common stock of XERC on a one-for-one basis or, at our election (determined solely by our independent directors (within the meaning of Nasdaq rules) who are disinterested), to the extent there is cash available from a contemporaneous public offering or private sale of Class A common stock by us, a corresponding amount of cash, in either case, to be paid to XERC by X-Energy, Inc., unless X-Energy, Inc. elects, in its sole discretion (determined by our independent directors (within the meaning of Nasdaq rules) who are disinterested), to effect such redemption as a direct exchange with the relevant Continuing Equity Owner. Upon any such redemption of Common Units, the corresponding shares of Class B common stock held by such Continuing Equity Owner will be surrendered and immediately canceled. See “Certain Relationships and Related Party Transactions” — XERC LLC Agreement” for additional information regarding such redemption and exchange.

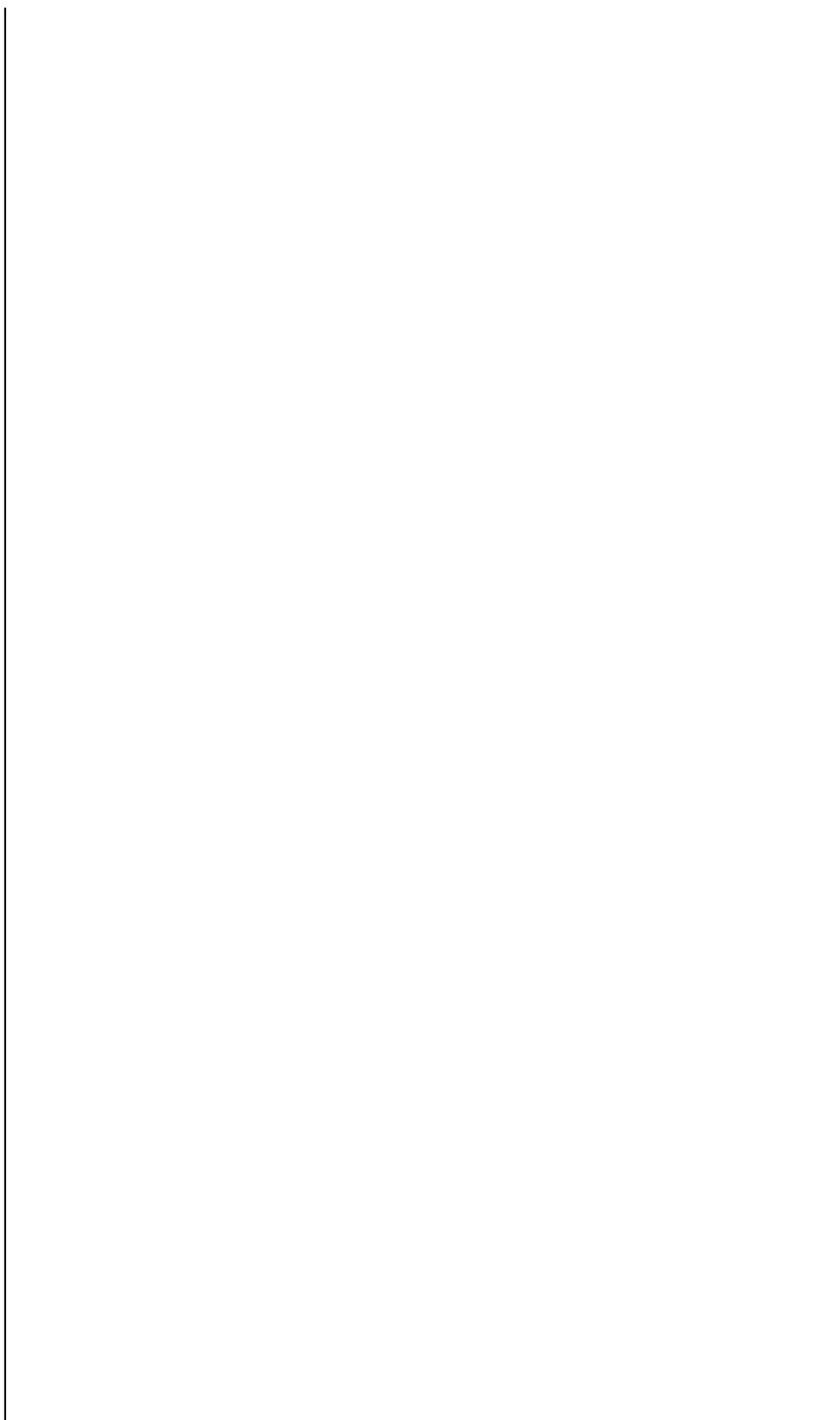


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	THE OFFERING
Common stock offered by us	shares of Class A common stock.
Class A common stock to be outstanding after this offering	shares, or shares if the underwriters exercise their option to purchase additional shares of Class A common stock in full).
Class B common stock to be outstanding after this offering	shares, or shares if the underwriters exercise their option to purchase additional shares of Class A common stock in full).
Common Units to be outstanding after this offering	Common Units, representing an approximate economic interest in XERC (or Common Units, representing approximately % economic interest in XERC if the underwriters exercise their option to purchase additional shares of Class A common stock in full).
Option to purchase additional shares	The underwriters have an option to purchase up to of additional shares of Class A common stock. The underwriters can exercise this option at any time within 90 days from the date of this prospectus.
Use of proceeds	We estimate that the net proceeds to us from the sale of our common stock in this offering will be a \$ million. We intend to use the proceeds from this offering to a portion of the proceeds for future growth projects. Proceeds.”
Voting rights	Holders of shares of our Class A common stock and Class B common stock will vote together as a single class of common stock presented to shareholders for their vote or approval, except as otherwise required by law or our amended and restated certificate of incorporation. Each share of our Class A common stock and Class B common stock entitles its holder to one vote on all matters to be voted on by shareholders generally. See “Description of Capital Stock”.

**Redemption rights for Common
Units**

Prior to this offering, we will amend and restate the Agreement so that the Continuing Equity Owners may, at the terms of such limited liability company agreement, have XERC redeem their Common Units for either shares of Class A common stock on a one-for-one basis or at a price (determined solely by our independent directors (in the meaning of Nasdaq rules) who are disinterested), to the extent there is cash available from a contemporaneous public offering, in the sale of Class A

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Dividend policy

common stock by us, a corresponding amount of common stock, in the case, contributed to XERC by X-Energy, Inc., unless X-Energy, Inc. elects, in its sole discretion (determined solely by our Board of directors (within the meaning of Nasdaq rules and regulations, disinterested), to effect such transaction as a direct exchange with the relevant Continuing Equity Owner. Upon any such exchange of Common Units, the corresponding shares of common stock will be canceled. See “Certain Relationships and Related Party Transactions — XERC LLC Agreement.”

We do not expect to pay any dividends on our common stock in the foreseeable future. See “Dividend Policy.” The declaration and payment of any future dividends will be at the sole discretion of our Board. Our Board may take into account general and specific business conditions, our financial condition and operating performance, our available cash and current and anticipated cash requirements, contractual, legal, tax, and regulatory requirements and implications on the payment of dividends by us to our subsidiaries or by our subsidiaries (including XERC) to us, and any other factors as our Board may deem relevant. Holders of our common stock do not have any economic rights or entitlement to receive dividends, or to receive a distribution upon our liquidation, dissolution, or winding up of X-Energy, Inc., with respect to our Class B common stock.

X-Energy, Inc. is a holding company and has no other assets other than a controlling equity interest in XERC. The TRA Parties agree to an Agreement that will be in effect at the time of this offering, pursuant to which that certain distributions to cover the taxes of the holders of our Common Units will be made based upon assumed cash flow and other assumptions provided in such limited liability partnership agreement. Additionally, in the event X-Energy, Inc. declares a cash dividend, we intend to cause XERC to make distributions to X-Energy, Inc., in an amount sufficient to cover such dividend if declared by us. If XERC makes such distributions to X-Energy, Inc., the other holders of Common Units will also be entitled to receive the respective equivalent pro rata distributions in accordance with their respective ownership of vested Common Units.

Tax Receivable Agreement

Upon the completion of this offering, we will be a party to a Tax Receivable Agreement with XERC and the TRA Parties. Pursuant to the Tax Receivable Agreement, we generally will be required to pay to the TRA Parties 85% of the amount of cash tax savings that we actually realize (or in some circumstances are deemed to realize) as a result of (i) Basis Adjustments, (ii) Existing Basis Adjustments, (iii) Tax Attributes and (iv) Interest Deductions. We will retain the benefit of the remaining 15% of these cash tax savings. See “Certain Relationships and Related Party Transactions — Tax Receivable Agreement.”

Stock exchange symbol

“XE”

Risk factors

Investing in our common stock involves a high degree of risk. For a more detailed discussion of the “Risk Factors” beginning on page [24](#) in this prospectus, you should carefully consider the discussion of factors you should carefully consider before investing in our common stock.

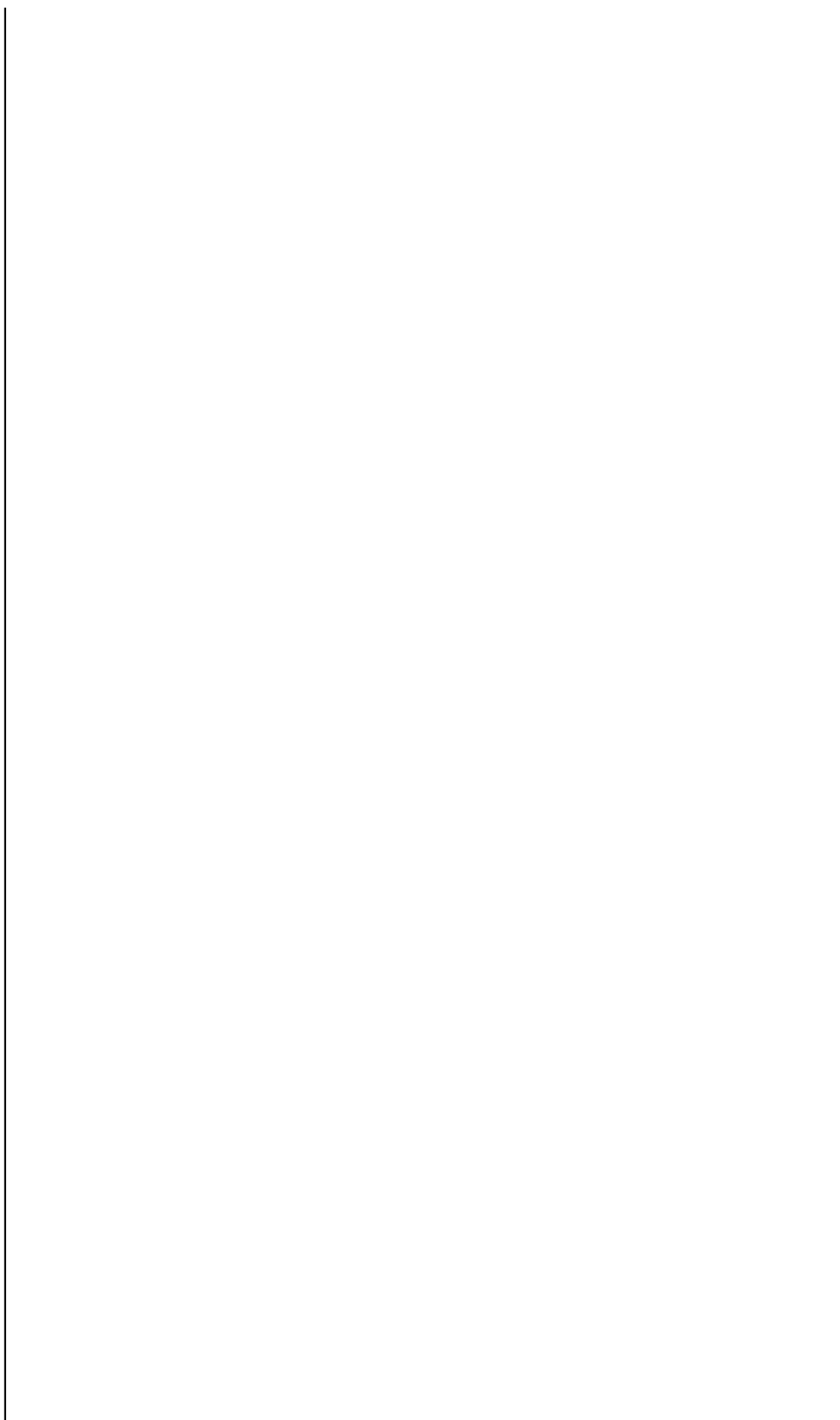
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Except as otherwise indicated, the number of shares of our Class A common stock common stock to be outstanding immediately after this offering is based on an aggregate Class A common stock and Class B common stock (which includes shares of common stock unvested restricted common stock subject to a repurchase option by us) outstanding as excludes:

- shares of Class A common stock reserved for issuance upon redemption of Common Units that will be held by the Continuing Equity Owners on a one-for-one basis;
- shares of Class A common stock reserved for future issuance under our Incentive Plan (the “2026 Plan”), which will become effective upon the effective registration statement of which this prospectus forms a part (which number includes our Class A common stock subject to restricted stock unit awards and stock options granted to certain of our employees and directors pursuant to our 2026 Plan in connection with the consummation of this offering, based upon an assumed initial public offering price per share (the midpoint of the estimated price range set forth on the cover page of this prospectus);
- shares of Class A common stock reserved for future issuance under our 2026 Stock Purchase Plan (the “ESPP”), which will become effective upon the effective registration statement of which this prospectus forms a part.
- In addition, our 2026 Plan and ESPP each provide for annual automatic increases in the number of shares reserved thereunder.

Unless otherwise indicated, this prospectus assumes or gives effect to:

- the completion of the Transactions described under “Organizational Structure,” the amendment and restatement of the XERC LLC Agreement and the reclassification of all equity of XERC into a single class of Common Units, which will occur immediately prior to the consummation of this offering (the Recapitalization);
- no exercise of outstanding stock options referred to above;
- no exercise by the underwriters of their over-allotment option;
- an assumed initial public offering price of \$ per share, which is the midpoint of the estimated price range set forth on the cover page of this prospectus; and
- the filing and effectiveness of our amended and restated certificate of incorporation and the effectiveness of our amended and restated bylaws, each of which will occur immediately prior to the completion of this offering.



SUMMARY HISTORICAL AND PRO FORMA FINANCIAL INFORMATION OF X-ENERGY

The following table shows summary historical and pro forma financial data for each of the periods indicated. The summary historical financial information for X-energy presented below for the periods ended December 31, 2025 and 2024 have been derived from X-energy's audited financial statements included elsewhere in this prospectus. The unaudited summary pro forma consolidated financial data set forth below as of and for the year ended December 31, 2025 have been derived from our unaudited pro forma consolidated financial statements included elsewhere in this prospectus.

The unaudited pro forma condensed statement of operations for the year ended December 31, 2025 gives effect to the pro forma events as if they had been consummated on January 1, 2025. The unaudited pro forma condensed balance sheet as of December 31, 2025 gives effect to the pro forma events as if they had been consummated on December 31, 2025. We have derived the summary unaudited pro forma condensed balance sheet and the unaudited pro forma condensed statement of operations from our unaudited pro forma condensed consolidated financial statements as of and for the year ended December 31, 2025 included elsewhere in this prospectus. The pro forma adjustments are based on available information and are based upon assumptions that management believes are reasonable in order to reflect, on a pro forma basis, the effect of the pro forma events on our historical financial information. The adjustments are described in the notes to the unaudited pro forma condensed balance sheet and the unaudited pro forma condensed statement of operations. The unaudited pro forma condensed financial information is presented for illustrative purposes only and do not purport to represent the results of operations or the financial position that would actually have occurred had the pro forma events been consummated on the dates and periods presented. We do not project the Company's results of operations or financial position for any future date or period.

The summary information in the following tables should be read in conjunction with the following sections of this prospectus: "Presentation — Organizational Structure," "Capitalization," "Management's Discussion and Analysis of Financial Condition and Results of Operations of X-energy," "Unaudited Pro Forma Consolidated Financial Information" and our financial statements and accompanying notes included elsewhere in this prospectus.

	Year ended December 31, 2024	Year ended December 31, 2025
Consolidated Statements of Operations		
Services revenue	\$ 83,986	\$ 94,260
Grant income	36,166	14,830
Total revenues and grant income	\$ 120,152	\$ 109,090
Operating expenses:		
Direct costs	130,115	161,367
Selling, general and administrative	111,887	116,318
Research and development	1,662	1,708
Total operating expenses	243,664	279,393
Operating loss	(123,512)	(170,293)
Other income (expense):		
Interest expense	(16,190)	(47,300)
Interest income	2,833	20,293
Other income (expense), net	10,909	(239,300)
Total other expense, net	(2,448)	(219,483)
Net loss	<u><u>\$ (125,960)</u></u>	<u><u>\$ (389,776)</u></u>

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Balance Sheet Data	As of December 31, 2025	As of December 31, 2024	As of December 31, 2023
Total assets	\$ 1,211,271	\$ 579,510	\$ 579,510
Total liabilities	369,011	125,695	125,695
Total mezzanine equity	2,066,555	1,300,376	1,300,376
Total members' deficit	(1,224,295)	(846,561)	(846,561)

RISK FACTORS

Investing in our Class A common stock involves a high degree of risk. You should consider risks and uncertainties described below, together with all of the other information in this prospectus, including the section titled “Management’s Discussion and Analysis of Financial Condition and Results of Operations of X-energy” and our consolidated financial statements and the accompanying notes thereto, elsewhere in this prospectus before deciding whether to invest in shares of our Class A common stock. The risks and uncertainties described below are not the only ones we face. Additional risks and uncertainties that we are unaware of or that we deem immaterial may also become important factors that affect our business. If any of the following risks occur, our business, financial condition, operating results and future prospects could be materially and adversely affected. In that event, the market price of our Class A common stock could decline, and you could lose part or all of your investment.

Throughout this section, unless otherwise noted, the “Company,” “we,” “us” or “our” refers to X-energy, Inc. and its consolidated subsidiaries.

Risks Relating to X-energy’s Business

Our ability to execute on our business plan and our continued existence are dependent upon our ability to raise additional funding and financing.

While we currently expect that we have sufficient sources of liquidity, taking into account cash on hand and the expected net proceeds from this offering, to continue working on our research and development projects with our key customers and partners, we will need additional sources of funding and financing to support our ongoing operations and to execute on our business plan.

We have had, and expect that we will continue to have, an ongoing need to raise additional capital from outside sources to fund our operations and expand our business. Historically, our primary sources of capital to support our operations have been revenue from the ARDP Agreement, contributions from our investors and members, loans from financial institutions and capital raises.

As of December 31, 2025, we had cash and cash equivalents of \$458.9 million, short-term investments of \$304.9 million and long-term investment balances of \$261.5 million and continuing obligations to customers, vendors, counterparties and employees in the ordinary course of business. Our expected primary uses of cash on a short-term and long-term basis are for working capital requirements, capital expenditures, and other general corporate services. Our primary working capital requirements are for project execution activities including purchases of materials, subcontracted services and personnel, which fluctuate during the year, driven primarily by the timing and extent of activities required for the completion of existing projects. We expect that working capital requirements will need to continue to be funded by a combination of cash on hand, funding awarded under the ARDP Agreement, further issuances of debt or equity securities and capital raises.

In addition, management expects that future operating losses and negative operating cash flow will increase from historical levels because of additional costs and expenses related to the development of our technology and the development of market and strategic relationships with other businesses. In connection with our business plan, management anticipates additional increases in operating expenses and capital expenditures relating to the development and commercialization of the Xe-100. We intend to fund these expenses with further issuances of debt or equity securities. Thereafter, we expect we will raise additional capital and generate revenues to meet long-term operating requirements.

If we raise additional funds through the issuance of equity or convertible debt securities, the ownership of our equity holders could be significantly diluted, and these newly issued securities could have rights, preferences or privileges senior to those of existing equity holders. If we raise additional funds by obtaining loans from third parties, the terms of those financing arrangements may include restrictive covenants or other restrictions on our business that could impair our operating flexibility and ability to incur interest expense.

If we are unable to raise additional capital when desired, we may not be able to construct, test, and deliver the Xe-100 or any other reactor to customers on time or on budget or at all, conduct operations, and

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development, undertake other projects, or fulfill our business plan, and therefore, we may need to abandon these and other projects.

Any of the foregoing would materially and adversely affect our business, financial condition, operating results, and future prospects.

We may not generate sufficient revenues or liquidity to operate our business, and a successful transition to profitable operations depends upon achieving a level of revenue adequate to support us.

We expect that working capital requirements will need to continue to be funded through a combination of cash on hand, funding awarded under the ARDP Agreement, further issuances of securities, and other raises. In addition, we will need to generate increased revenues sufficient to meet long-term capital requirements.

At present, our revenues are generally derived from contract services performed for government and commercial entities from cost-share agreements such as the ARDP and other programs for development, product development, and fuel services provided to other government and commercial entities. In the future, we expect to generate revenue through licensing or technology transfer, the use of the proprietary intellectual property and build-to-print design of the Xe-100 technology, engineering planning, assembly coordination, construction support, regulatory support, procurement support, and other services to customers and the supply of fuel and associated services. However, we have not yet entered into any such licensing, technology or other agreements, and there can be no assurances that customers will accept our anticipated fees and pricing under these agreements or that we will be able to charge the fees and pricing that we anticipate under our business plan. If we are unable to obtain sufficient fees and pricing under these agreements, our business, financial condition, operating results and future prospects will be materially and adversely affected.

Our revenue growth also may be adversely affected by other factors including: our ability to maintain, grow and develop the X-energy products; our ability to continue to receive and perform government contracts; weakness in the industry generally; general economic conditions, including the result of tariffs, high interest rates and inflation; terrorism, sanctions or other geopolitical events; global pandemics and other public health emergencies; increasing competition; and the other risk factors in this “Risk Factors” section.

Our historical revenue growth is not indicative of our future performance. If we are unable to maintain, grow, and develop X-energy’s business and generate sufficient revenue and achieve profitability, our business, financial condition, operating results and future prospects will be materially and adversely affected. We may also achieve business growth in quantity of deployed plants, but achieve less than anticipated revenue on each deployment, including achieving lower than anticipated margin on each deployment. Additionally, our cash needs may increase in the future as we focus on growing and developing

Our future capital requirements will depend on many factors, including the ability to meet (or, as the case may be) the requirements of Nasdaq for listing or any other exchange, losses, the ability to obtain the necessary financing to meet obligations and repay liabilities of business operations when they come due, the ability to generate and maintain sufficient cash, and the ability to generate profitable operations in the future. There can be no assurance that our liquidity will be sufficient to achieve our long-term objectives, grow and develop our products, operate our business, or meet the terms of our indebtedness.

If we are not able to generate sufficient revenues or liquidity to operate our business, and to execute our long-term business plan, it would materially and adversely affect our business, financial condition, operating results, and future prospects.

We have not yet delivered the Xe-100 or any other reactor to customers and have not achieved final investment decisions for the purchase or deployment of any of our reactors, and there is no guarantee that we will be able to do so. Any delays or setbacks we may experience during our first commercial delivery, which is planned for the 2030s, or failure to obtain final investment decisions would have a material adverse effect on our business, financial condition, operating results, and future prospects, and could harm our reputation.

The success of our business will depend in large part on our ability to successfully deliver our reactors to customers on time and on budget at guaranteed performance levels, which would give greater confidence in our business.

our subsequent customers. There is no guarantee that our planned deployments of the Xe-100 will be successful, on schedule, or on budget, or that our customers will elect to exercise their options to purchase Xe-100 units. We are in the design phase of the Xe-100, and as a result our cost and schedule estimates are subject to significant uncertainty and change. Our current cost model and estimates have limitations and may prove inaccurate, which could result in material cost growth, schedule extensions, or schedule changes that adversely affect project economics, final investment decisions and customer commitments.

Moreover, because the Xe-100 will be considered a first-of-a-kind technology, there can be no assurance that we will not experience operational or process failures and other problems during our first commercial deployment or any planned deployment thereafter. The nuclear industry has experienced significant cost overruns, schedule delays and cancellations on first-of-a-kind advanced nuclear projects, which have, in many cases, rendered projects uneconomical; similar dynamics could occur with our projects notwithstanding our planning, risk management and contracting strategies. In the event we fail to develop and successfully commercialize our technology, if we fail to develop such first-of-a-kind technologies before our competitors or if such technologies fail to perform as expected, are inferior to those of our competitors or are perceived as less safe than those of our competitors, our business, financial condition could be materially and adversely impacted. Any failures, delays or setbacks, particularly during our first commercial deployments, would likely harm our reputation and have a material adverse effect on our business prospects, financial condition, results of operation and cash flows.

Any delays in the development and manufacture of our SMRs and related technology may adversely affect our business and financial condition.

We have previously experienced and may experience in the future, delays, cost overruns, and other complications in the design, manufacture, production and delivery of the Xe-100 and related technologies, such as the TRISO-X Fuel Fabrication Facility, that could prevent us from delivering any of our reactors on our current anticipated timeline. For example, as part of our iterative Xe-100 reactor design process, we have phased development cycles and review gates to progress from concept through product development and testing.

timing, scope, and economics of any resulting project may differ materially from our current expectations. If the relationship with Centrica does not progress as anticipated, or if Centrica elects not to move forward with the development or purchase of our reactors, we would not realize any of the potential strategic benefits associated with the arrangement. Any failure to convert the JDA into a commercial agreement, or any delay or cancellation of potential projects under the JDA, could materially affect our business, financial condition, results of operations and cash flows.

Our sales and profitability may be impacted by, and we may incur liabilities as a result of, terms of our contracts with customers, our failure to meet performance guarantees under customer contracts or customer safety

Terms in contracts or agreements we have entered into with existing or future customers may impact our capacity allocation, impact our pricing and margins, and limit our strategic flexibility. Under our agreements with Amazon, we have granted Amazon first-priority allocation of Xe-100 manufacturing slots across 2031 to 2039 to allocate to utilities and independent power producers, primarily commercial nuclear operators, who will ultimately own and operate the SMR facilities. We have also granted Amazon a first refusal on a portion of our scheduled delivery. While Amazon is not obligated to purchase capacity, it could limit our ability to allocate capacity to other customers and may lead to underutilization and inefficiencies if Amazon delays or declines to exercise its allocated capacity. Under our agreements with Amazon, we are also obligated to offer Amazon the most favorable pricing and commercial terms to any non-affiliate customer, including minimum TRISO-X fuel allocations commensurate with those offered to favored customers. If we cannot make a reserved slot available, we must use commercial efforts to provide replacement capacity to honor our targets under the agreement.

These commitments may reduce our flexibility to prioritize higher-margin or strategic customers, which may require us to match the best terms we offer to others, which could compress margins and adversely affect our revenue mix. They may also increase operational complexity and costs (including TRISO-X fuel allocations) and result in disputes. Additionally, successor and transfer provisions require any transferee of manufacturing assets or relevant equity interests to assume queue-slot obligations, which could limit our ability to pursue strategic transactions or restructurings on favorable terms.

Under our agreements with Amazon, beginning on the fifth anniversary of the commercial operation date of the first Amazon-sponsored project, we are required to make payments to Amazon reflecting the efficiencies we have achieved relative to the PPA prices applicable to the initial projects. These payments could be material, and could adversely affect our liquidity, cash flows, and financing flexibility. If our costs decline on other projects, which would increase the premium owed on Amazon-sponsored projects. The timing and magnitude of payments may be volatile due to annual reforecasts of performance, and exogenous variables, reducing cash available for operations, investment, and potentially extending the final maturity beyond our original expectations.

Further, we anticipate that our customers may require performance guarantees as to the performance of the Xe-100s we deliver. Prospective future customers may also require that we comply with their unique requirements relating to their compliance with policies, priorities, regulations, and government mandates, including provision of data and related assurance for environmental, social, and governance-related standards or goals, and such compliance may add cost and timeline uncertainty or risk. Our failure to deliver products to operate properly or to meet specifications of our customers or our failure to meet performance guarantees may increase costs by requiring additional engineering resources, replacement of parts and equipment or monetary reimbursement to a customer, or could otherwise result in liability to our customers. There are significant uncertainties and judgments involved in meeting performance guarantee obligations, including changing product designs, differences in customer requirements, and failure to identify or disclaim certain variables. To the extent that we incur performance guarantee claims in any period, our earnings and ability to obtain future business may be materially adversely affected.

The illustrative capacity figures and unit economics included in this prospectus are illustrative in nature and based on a number of assumptions, and may not reflect our actual future performance.

The illustrative capacity figures presented in this prospectus are based on our current agreements with our customers, including Dow, Amazon, and Centrica, and reflect our internal assumptions regarding the capacity of the Xe-100, the number of units that may be deployed, and the time periods over which we anticipate such units being constructed and placed into service. Numerous risks and uncertainties could cause the actual deployed capacity to differ materially from the illustrative figures, including our customers' ability to successfully construct and deliver Xe-100s (none of which have been built to date) or to do so on a timely basis, obtain required regulatory approvals, secure adequate financing, manage supply chain constraints, labor availability, cost overruns, and construction delays. Our customers such as Dow, Amazon, and Centrica have rights to delay, reduce, or terminate their orders, and any such changes, along with potential modifications in the scope, scale, or timing of customer orders, technological or engineering challenges, or shifts in energy policy, electricity market demand, or public acceptance of nuclear power, could materially impact our ability to achieve the levels of capacity shown in this prospectus. As a result, our actual cumulative deployed capacity may be substantially less than shown, or may not be achieved at all, and investors should not place undue reliance on the capacity figures in evaluating our business, financial condition, operating results or future prospects.

We do not carry insurance coverage for the risks associated with the Xe-100 or its components, and any insurance coverage may not be adequate, or we may not be able to obtain insurance at acceptable rates.

We do not carry insurance coverage for the performance of the Xe-100 or its components. If we were to purchase this kind of insurance, the insurance may not fully protect us from the financial impact of defending against product liability claims that may occur in future. In the event that our technology does not perform as expected, are inferior to those of our competitors or are perceived as less safe than those of our competitors, our business and financial condition could be materially and adversely impacted. In addition to contractual protections provided by our vendors for certain components and systems, we have entered into other risk sharing structures to mitigate all risks associated with the successful delivery and performance of the Xe-100. As we have not yet delivered the Xe-100 or any other reactor to customers, we have not determined that our current insurance coverage is sufficient for our business operations. However, we may choose to purchase additional insurance to operate our business. If we fail to obtain the required insurance, we may be exposed to substantial losses or liabilities due to fire, explosions, floods, other natural disasters, or accidents or business interruption that we are not adequately covered under our existing insurance. Our business and results of operations could be materially and adversely affected.

We depend significantly on U.S. government contracts, which often are only partially funded, subject to termination, and heavily regulated and audited. Continued full funding of, and any upward adjustments to, the ARDP available under, the ARDP are subject to future government appropriations and continued political support. Our ability to access ARDP funding is subject to significant timing and regulatory risks. The termination or non-fulfillment of the ARDP, or our failure to timely secure upward adjustments in the ARDP funding to cover the full cost, could have a material, adverse impact on our business prospects, relationship with our partners, financial results of operations and cash flows and may result in a more limited ARDP fuel facility development project development timelines, or other commercial changes to the ARDP.

In addition to at least \$500 million in funding appropriated in FY2020-FY2024, the Infrastructure Investment and Jobs Act (Public Law 11758) appropriated funds for the ARDP in total of \$2.1 billion. ARDP funding to date, at least \$1.1 billion has been allocated to X-energy's award. Ongoing ARDP funding is provided to X-energy through Continuation Applications for each "budget period", with the latest period through at least August of 2026. Congress has appropriated funding that was allocated toward our award, in total of approximately \$1.1 billion of the \$1.2 billion commitment as part of the Infrastructure Investment and Jobs Act in 2022 to support our ARDP award, as well as recent additional appropriations of \$3.1 billion to ARDP, some portion of which we expect to be allocated to X-energy. In the future, U.S. government appropriations have been affected by larger U.S. government budgetary issues and legislative changes. As of December 31, 2025, we have been reimbursed over \$438 million in funding from the ARDP. We cannot predict the extent to which funding may be increased, if at all, or if total funding will be sufficient to complete the ARDP.

its procurement practices. These initiatives and changes to procurement practices may change government contracts are solicited, negotiated and managed, which may affect whether and how we have opportunities to provide our products and services to the U.S. government, including the conditions under which we do so, which may have an adverse impact on our business, financial condition, operating results and future prospects.

Failure to comply with applicable regulations and requirements could lead to fines, penalties, repayments, or compensatory or treble damages, or suspension or debarment from U.S. government contracting or subcontracting for a period of time. Among the causes for debarment are violations of U.S. laws and regulations, including those related to procurement integrity, export control (International Traffic in Arms Regulations at 22 C.F.R. Parts 120-130), U.S. government employment practices, protection of the environment, accuracy of records, proper recording and reporting of foreign corruption. The termination

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of a U.S. government contract or relationship as a result of any of these acts would have an adverse effect on our operations and could have an adverse effect on our standing and eligibility for future government contracts.

Government funding is subject to the political process, which is inherently unpredictable, competitive and dependent on budgetary limitations, congressional appropriations and the allotment of funds, all of which may be affected by changes in U.S. government policies resulting from various political developments. If political support for the prioritization of the development of ARDP energy decreases, including due to policy changes by the current administration or future administration and changing congressional funding priorities, we may be unable to secure continued government funding under the ARDP or any requested increases to such funding. Our failure to secure upward adjustments to ARDP funding to cover actual costs, should such a need arise, could have a material, adverse effect on our business prospects, which would materially and adversely affect our development timeline, financial condition, relationship with our partners, results of operations and cash flows. In addition, any failure to obtain requested increases to such funding could result in a more limited ARDP fuel facility development or other commercial changes to the ARDP.

In addition, in connection with future requests for increases to funding under the ARDP, or with our entry into U.S. government contracts, or for any other reason, the U.S. government may impose conditions or make certain requests of us in return, including, but not limited to, requests for additional funding on existing U.S. government contracts or requests to acquire and/or purchase shares of our Class A common stock or securities that give the U.S. government the right to acquire shares of our Class A common stock. Any such conditions or requests, whether or not we agree to them, and the terms of any such conditions or requests could have a material adverse impact on our relationship with the U.S. government, could have a material adverse effect on our business prospects, financial condition, results of operations and cash flows, and could result in significant volatility in, or a decline of, our stock price.

Our ability to receive ARDP funding is subject to budget period limitations, extension approvals, and the date, and we may not receive the full amount of the ARDP reimbursement.

Our current ARDP Agreement provides for 50% reimbursement of \$2.4 billion in ARDP costs (\$1.2 billion in total reimbursement). As of December 31, 2025, we have received a total of \$438 million in ARDP funding. We submit our budgets through an ongoing “budget period” process with project milestones under the ARDP Agreement, and our current budget covers a budget period from March of 2025 and extends through August 2026. At the time of the initial award, the estimated total cost was approximately \$2.4 billion, which reflected management’s then-current estimate of the total cost of the ARDP project.

licensing, fuel fabrication facility construction, and demonstrator reactor construction activities during the ARDP program period. Since 2020, the estimated total project costs have increased, and we expect our cost estimates will continue to be revised in connection with each budget period's Continuation Application. As project scope, market conditions, and external factors evolve.

Continuation of funding is contingent on DOE approval of additional budget periods. Extensions beyond the ARDP Agreement are at the DOE's discretion and are subject to criteria including: (1) availability of appropriations; (2) availability of future-year budget authority; (3) substantial progress toward achieving project objectives; (4) submittal of required reports; and (5) compliance with the terms and conditions of the award. We submit a Continuation Application to the DOE to extend funding; however, there is no assurance that extensions will be approved. As part of the Continuation Application process required by the ARDP Agreement, we are required to submit updated budgets for each subsequent budget period, which are subject to review and approval by the DOE. These updated budgets reflect the best information available at the time of submission and may differ materially from prior estimates due to the factors described above. The DOE's current reimbursement share under the ARDP is \$1.2 billion (representing 50% of our \$2.4 billion estimate), meaning any increases in estimated project costs above the original estimate without adjustment of the current award value would require X-energy and Dow to fund the incremental amounts from non-ARDP sources. If actual project costs exceed the amounts eligible for reimbursement we and Dow would be required to bear the full amount of such excess costs, reducing government cost-sharing, which could have a material adverse effect on our business, financial condition, results of operations and cash flows. X-energy and Dow are under no obligation to continue the project or project scope from ARDP.

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Under the ARDP Agreement, the total extension of the period of performance may be up to three years beyond the original 7 year award period, resulting in a maximum period of performance of 10 years. Any additional extension would require an approval within DOE above the Contracting Officer. Based on the original award date in February 2021, the outside date for ARDP funding, assuming all extensions are granted, is expected to be in or around 2030. Dow, as our partner under the ARDP Agreement, does not separately need to apply for ARDP funding or extensions. Reimbursements for the Dow project are tied to project milestones and they are not subject to annual budget periods. If the Dow project, with a target commercial operations date in the early 2030s, is not completed within the ARDP funding period and Dow elects to construct the project, Dow would be responsible for remaining costs related to the construction of the reactor without ARDP reimbursement. If Dow is unable to do so, the Dow project could be delayed or terminated, which would have a material adverse effect on our business, financial condition, results of operations and cash flows. X-energy has no obligation to fund the construction of the Dow project, and is under no obligation to construct the project without a final investment decision from Dow.

If we fail to incur eligible costs within the currently approved period of performance, or a DOE-approved extension, we would forgo reimbursement for such costs and could face depletion of unobligated funds at closeout. Even with an extension granted, we may not receive additional ARDP funding unless and until DOE agrees to obligate further amounts to us. Any adverse award outcomes, performance shortfalls, contractual compliance issues, or denial of extension requests could materially reduce or eliminate ARDP funding available to us, which would have a material adverse effect on our business, financial condition, results of operations and cash flows.

The original ARDP cost estimate was developed in 2020 and has since increased; future cost increases could require us and our partners to fund amounts in excess of available ARDP reimbursement.

The ARDP Agreement was based on management’s reasonable cost estimates developed for X-energy’s ARDP application in 2020. At the time of award, total estimated project costs were approximately \$2.4 billion, with DOE providing \$1.2 billion in 50/50 cost-share reimbursement for project costs under the ARDP Agreement. This reflected management’s then-current estimates and did not account for subsequent inflation, supply chain cost increases, scope evolution, or other factors that have affected project costs.

Since the original award in 2020, our estimated project costs have increased, and we expect our cost estimates will continue to be revised in connection with each budget period’s Continuation Application. As the nuclear industry has historically experienced significant cost overruns on first-of-a-kind projects, and although Xe-100’s modular design is intended to reduce first-of-a-kind risks, we cannot provide assurance that cost increases will not occur. Factors that may contribute to cost increases include:

- Inflation and general price escalation across labor, materials, and services;
- Supply chain disruptions and increased costs for nuclear-grade materials and components;
- Scope refinements and design modifications resulting from ongoing engineering work;
- First-of-a-kind engineering challenges and regulatory requirements;
- Extended project timelines, including potential delays in obtaining regulatory approvals;
- Changes in tariff regimes or trade restrictions affecting procurement costs.

As part of the Continuation Application process, we submit updated budgets to the DOE for each budget period, including cost estimates and supporting justification. These updated estimates are subject to DOE review and approval and may differ materially from prior estimates. In January 2023, DOE appropriated additional funding of \$3.1 billion towards three Advanced Reactor programs, including 1) the Generation III SMR deployments, 2) the Risk Reduction for Future Demonstrations Program, and 3) the Advanced Reactor Demonstration Program, of which X-energy is one of two awardees. We have allocated some portion of this to our project incremental to the current \$1.1B allocated to date. Our business plan does not approve updated budgets or if additional funding is not appropriated by Congress, we may be unable to complete project activities within the ARDP cost-share framework. This could require us to raise additional capital on terms that may not be favorable, which could materially impact our business condition, operating results and future prospects.

Our business plan may also include the development of other configurations of our SMR or other new projects, which makes certain assumptions with respect to learnings, efficiencies, and regulatory approvals as a result of our development approach which may not be accurate or correct. Any adverse change to these assumptions could have a material adverse effect on our business prospects, financial condition and results of operations and cash flows.

We compete in a market characterized by rapid technological advances, evolving regulatory requirements, and frequent new product introductions and enhancements. To succeed, we must continue to rely on the development of other configurations of our SMRs or other new projects we may undertake. The development of other configurations of our SMRs, or the undertaking of other projects, has inherent risks, including, but not limited to:

- higher than expected research and development costs;
- delays or unexpected costs in developing new plant configurations;
- inability or delays in obtaining regulatory approval;
- customers delaying purchase decisions in anticipation of new configurations;

- customer confusion and extended evaluation and negotiation time;
- educating our sales, marketing, and consulting personnel to work with new configurations;
- competition from earlier and more established entrants;
- market acceptance of earlier configurations; and
- the accuracy of assumptions about the nature of customer demand.

If we are unable to successfully introduce, market, and sell other configurations of our products, we may be unable to undertake any other new projects in a timely and cost-effective manner, and properly position our products, our business, financial condition, operating results and future prospects could be impacted.

If we fail to manage our growth effectively, we may be unable to execute our business plan, and our financial condition, operating results and future prospects could be harmed.

In order to achieve the substantial future revenue growth we have projected, we must successfully design, construct, and operate our reactors, receive regulatory approvals, including NRC licensing of additional fuel fabrication facilities, and continue to develop and market new products and services to traditional and nontraditional end users. We intend to expand our operations significantly to meet anticipated demand. To manage our growth, we will need to hire and retain additional personnel, upgrade our existing management and financial and reporting systems, and improve our business processes and controls. Our future expansion will be dependent on:

- hiring and training new personnel with the skill and expertise required;
- completing the licensing and construction of TX-1 and subsequent fuel fabrication facilities and optimizing the production of our TRISO-X fuel;
- finalizing our reactor design and developing new technologies and services (e.g., maintenance, procurement);
- optimizing applications of our reactors to serve both traditional utility and electric power markets and a broad base of nontraditional industrial customers interested in utilizing the high-temperature heat produced by our design;
- controlling expenses and investments in anticipation of expanded operations and rising costs;
- manufacturing and maintaining a sufficient supply of pebbles;
- managing the costs of our reactors;
- managing construction timelines, performance and budgets of our engineering, procurement and construction (“EPC”) partners and other third-party contractors;

- upgrading the existing operational management and financial reporting systems and technology to meet requirements as a public company; and
- implementing and enhancing administrative infrastructure, systems and processes.

If our operations continue to grow as planned, of which there can be no assurance, we will need to expand our sales and marketing, research and development, customer and commercial strategy, operations, services, supply chain, and manufacturing functions, among others. These efforts will require significant financial and other resources, including in industries and sales channels in which we have limited experience.

limited experience to date. We will also need to continue to leverage our manufacturing and systems and processes, and there is no guarantee that we will be able to scale the business planned or within the planned timeframe. The continued expansion of our business may require additional manufacturing and operational facilities, as well as space for administrative support. There is no guarantee that we will be able to find suitable locations for such facilities.

Our continued growth could increase the strain on our resources, and we could experience difficulties, including difficulties in hiring and training employees, finding manufacturing facilities to produce our SMRs and related equipment, delays in production, challenges in scaling component fabrication capacity and difficulty sourcing adequate raw material, such as graphite (which our customers are responsible for procuring), for our reactors. These difficulties may require the attention of management and key employees and impact financial and operational results. If we are unable to drive commensurate growth, these costs, which include lease commitments, headcount and other expenses, could result in decreased margins, which could have a material adverse effect on our business, financial condition, operating results and future prospects.

Further, if our capital needs are greater than anticipated or the timing of expenditures is such that we may need to raise additional debt or equity, reduce scope, defer projects, or pursue alternative financing models and risk-sharing structures; there can be no assurance such capital will be available when needed on acceptable terms. See *Risks Related to X-energy's Capital Resources — In order to fulfill our long-term plan, we will require additional funding. To the extent we require such additional investor funding in the future, such funding may be dilutive to our investors and no assurances can be provided as to the availability of such funding. Any such funding and the associated terms will be highly dependent upon market conditions and the progress of our business at the time we seek such funding. The terms of any financing we pursue may be less favorable than previously anticipated and could become even less favorable if market conditions on the amount of funds we may require.*

There is limited commercial operating experience for our planned SMRs and facilities, configuration of components. This creates risks in cost and schedule estimates and lack of recent domestic commercial experience. Limited labor and supply chain and other factors may result in greater than expected construction costs, delays in timelines, maintenance requirements, differing power output and greater operating expense.

While our SMR design will be actively managed through design reviews, prototyping, testing and the involvement of external partners with subject matter expertise, and application of approaches used in the operation of the Xe-100, we could still fail to identify latent design, manufacturing, construction or operations issues early enough to avoid negative effects on production, fabrication, construction or performance of the Xe-100 and related technologies, or we may encounter unexpected regulatory requirements. Moreover, the cost and time associated with the construction and maintenance of our SMRs may be more than we or our customers expect because we or they may face a lack of a domestic labor force with sufficient commercial experience and an inexperienced or insufficient supply chain for this type of reactor. If these issues arise at later stages of deployment, deployment could be subject to greater delays, which could be significantly delayed, which could materially and adversely affect our business. Although near-term costs to construct the Xe-100 is borne by our customers, such costs incurred by our customers could significantly exceed their and our expectations, including for reasons outside of their and our control, which could make existing or prospective customers less likely to contract with us and use our SMRs in the future, which could have a material adverse effect on our business, results of operations and future prospects.

We have not yet entered into any technology fee agreements with our customers, which could materially impact our revenue model.

A key component of our business model generally and the illustrative unit economics presented in our prospectus is the ability to reach binding agreements with potential customers for use of and access to our SMRs.

Xe-100 technology and designs and to realize the fees associated with such agreements. To date, we have not entered into any such agreements with our customers, and our customers have not paid such fees. Therefore, there can be no assurance that they will enter into such agreements in the future at levels we have anticipated or at all. Potential customers may resist paying technology fees given the substantial expenditures already associated with the construction of SMRs, or the potential development of other technologies that may prove more efficient or effective. If we are unable to enter into these agreements or enforce or collect these fees, our expected revenue streams could be materially reduced.

Any failure to effectively create the design, ensure its commercial viability and implement the construction and operations of our planned SMRs and facilities or ensure cost competitiveness could reduce the marketability of X-energy designs and has the potential to impact deployment schedules.

Creating our designs and ensuring their commercial viability and implementing the construction and operations will be necessary to be competitive and attractive in the market, particularly in the U.S. where the price of power is generally lower than in certain other key markets. If we are not able to maintain cost competitiveness of our fuel or our planned SMRs in the U.S. or elsewhere, our business could be materially and adversely affected.

We may not attract customers to our Xe-100 technology as quickly as we expect, or at all, and acquiring customers may be more expensive than we currently anticipate.

Adoption of the Xe-100 among our potential customers may progress more slowly than we expect, or it may be more expensive to bring potential customers into our pipeline. Any delay or failure to attract potential customers to our reactors or SMR technology may have a material and adverse impact on our business and financial condition.

The amount of time and funding needed to bring our nuclear fuel to market at scale may significantly exceed our expectations. Any material change to our assumptions or expectations, or any material overruns or unexpected increases in costs, could have a material adverse effect on our expected revenues, gross margins, and the other information included in the illustrative unit economics provided in this prospectus or our ability to compete in the market for other coated particle fuels.

The development of our TRISO-X fuel at scale will take a significant amount of time and funding. TRISO-X fuel is produced through a highly specialized, small-batch manufacturing process. Scaling from pilot-scale production to commercial-scale production involves significant technical, regulatory, and economic challenges. Any shortfall in research, development, and testing funds, unexpected increases in costs, any delay in achieving fuel development milestones, and any uncertainty in licensing timelines or adverse public reaction to developments in the use of nuclear power by interest groups, community organizations and state and local government agencies leading to enforcement actions, litigation or other legal proceedings could result in significant delays and cost overruns and could affect our ability to deploy, and our customers' ability to operate, the Xe-100. At this stage, we cannot accurately predict the amount of funding or the time required to successfully manufacture and commercialize nuclear fuel in the future at scale. The actual cost and time required to commercialize our fuel may vary significantly from our initial forecasts depending on, among other things:

- the results of our research and product development efforts;
- the cost of developing or licensing our nuclear fuel and TX-1 and any subsequent fuel fabrication facilities;
- the cost to construct and operate TX-1 and subsequent fuel fabrication facilities, including potential construction delays;
- the ability of customers to efficiently adopt and use TRISO-X fuel in their reactors;
- changes in the focus and direction of our research and product development programs;
- access to test facilities;
- competitive and technological advances;
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- the regulatory approval process;
- the fuel manufacturing process;
- adverse public reaction to the developments in the use of nuclear power resulting in litigation or other legal proceedings;
- staffing and training qualified personnel to operate TX-1 and any other TRISO-X fuel fabrication facilities;
- the scalability of our TRISO-X fuel operations;
- availability and cost of LEU and HALEU;
- availability and cost of other raw materials necessary for fuel manufacturing, including process gases, and chemicals, among others; and
- marketing and other costs associated with commercialization of these technologies.

Because of this uncertainty, even if financing is available to us, we may need significantly more financing than anticipated, which may not be available on terms acceptable to us or at all. As a result, our revenues and other expected benefits from our nuclear fuel technology may be delayed or not realized. Any material change to our assumptions or expectations, or any material overruns or other increases in costs, could have a material adverse effect on our expected revenues, gross margins, and other information included in the illustrative unit economics provided in this prospectus or our ability to develop and market other coated particle fuels.

If our customers are unable to access HALEU, our ability to manufacture TRISO-X fuel will be adversely affected, which could have a material adverse effect on our business, financial condition, operating results, and prospects.

Existing commercial nuclear infrastructure, including enrichment facilities and fuel fabrication facilities, were in most cases designed and are currently licensed to handle uranium in pellet form with enrichment up to 5% of the isotope Uranium 235 (LEU). Our fuel designs are based on a pebble design composed of graphite, pyrolytic carbon and silicon carbide encapsulated uranium that have been enriched up to 15.5% for use in the Xe-100. This higher enriched uranium (greater than LEU but still below 20%) is known as HALEU. Supplying HALEU to our TX-1 and any subsequent fuel fabrication facilities, which will manufacture the fuel for our SMRs, will require certain modifications to NRC licensing of existing commercial uranium enrichment facilities, none of which are owned or operated by us, along with the construction of our TX-1 facility and construction and NRC licensing of subsequent fuel fabrication facilities.

Presently, HALEU enrichment services are available only in limited quantities globally. HALEU can be sourced in limited quantities from the DOE, and a limited domestic supply from the government sector that we do not expect will be a significant source of our HALEU. Despite U.S. government programs designed to ensure initial HALEU quantities, including the establishment of the HALEU Program to ensure access to HALEU for civilian domestic research, development, demonstration, and commercial use, the HALEU program is still in its early stages, and significant progress is needed to achieve reliable and scalable production.

Government funding is subject to the political process, which is inherently unpredictable and highly competitive. The funding of government programs is dependent on budgetary

congressional appropriations and administrative allotment of funds, all of which may be affected in U.S. government policies resulting from various political developments, including recent negotiations to negotiate an increase in, or suspension of, the debt ceiling. If an alternative commercial-scale HALEU outside of Russia or China fails to materialize, including as a result of a lack of political support for the prioritization of the development of nuclear energy or otherwise, it may affect our ability to secure HALEU fuel in the future, which would materially adversely affect our business, financial condition, operating results and future prospects.

Our initial ARDP plant deployment may consequently depend on HALEU feedstocks received from the U.S. government, which may initially be in the form of highly enriched uranium, or “HEU” (enriched uranium).

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20%), that will be down-blended to HALEU levels for use in our TX-1, any subsequent fuel fabrication facilities and, ultimately, in ARDP reactors owned and operated by our partners. HEU can be converted to HALEU by only a limited number of licensed U.S. third parties. These third parties do not currently produce commercial levels of HALEU and may require regulatory approvals and process changes to produce the HALEU we require for the ARDP project. Our customers’ longer term fuel feedstock arrangements and subsequent fuel fabrication, whether related to X-energy’s TRISO-X fuel or other particle fuels we may manufacture for others, are likely to rely on commercial suppliers that do not currently produce and market HALEU, and which will also require NRC regulatory licenses in order to produce HALEU. Therefore, our customers’ ability to obtain adequate long term HALEU supplies for our SMRs on a predictable schedule and at predictable cost may be impaired, and activities like fuel loading and reactor ultimate operation of our SMRs may be delayed, and our customers exposed to cost overruns and cost uncertainty, all of which may negatively affect the competitiveness of our SMRs.

Any HALEU enrichment facility will need to secure NRC licenses to enrich uranium to HEU levels, and our TX-1 and any subsequent fuel fabrication facilities will require an NRC license to convert HEU to HALEU and fabricate HALEU into TRISO-X fuel. In February 2026, we received a Special Nuclear Material License from the NRC that establishes TX-1 as the first-ever Category II nuclear fuel facility licensed in the United States, enabling TRISO-X to commercially manufacture fuel using HALEU at its TX-1 site under a 40-year license. We anticipate constructing TX-2 on the same site, which will allow for the license to be shared between both facilities. There is risk of relevant entities within the nuclear power industry being slow to complete required facility infrastructure modifications or to obtain, maintain or extend required licenses to enable such enrichment of HALEU fuel. Finally, there is a risk associated with possible negative public perception of uranium enrichment to greater than 5% that could potentially delay or hinder regulatory approval of our nuclear fuel designs. Should any of these events occur, our business would be materially and adversely impacted.

Unsatisfactory safety performance or security incidents at our facilities, or any nuclear facility around which we operate, could have a material adverse effect on our business, financial condition, operating results and future prospects.

We design and will facilitate the manufacturing of highly sophisticated SMRs that depend on advanced technology. We also work cooperatively with our suppliers, subcontractors, venture partners and other third parties. Failures, disruptions or compromises to our or our third parties’ systems or facilities may result from natural disasters, accidents, power disruptions, telecommunications failures, acts of terrorism, computer viruses, bugs or vulnerabilities, physical or electronic break-ins, human error, cyberattacks, other intentional conduct, or similar events or incidents. While we have built robust processes to ensure that the design, manufacture, performance and servicing of our SMRs meet or exceed safety standards and performance goals, there can be no assurance that we will not experience operational or process failures or other problems, including through manufacturing or design defects, failure to implement safeguards, mishandling or process failures, natural disasters, cyber attacks, or other intentional

could result in potential safety risks. There can be no assurance that our preparations, or the parties, will be able to prevent or address any such incidents.

Any actual or perceived safety issues at our facilities, those of our customers, or any nuclear facility around the world may result in significant reputational harm to our businesses; in addition, safety issues at our or our customers' facilities could result in enforcement proceedings brought by regulators, tort liability, maintenance costs, increased safety infrastructure requirements and other costs may arise. Such issues with our SMRs, facilities, or customer safety could result in delaying the delivery of SMRs to our customers, increased regulation or other systemic consequences. Our inability to meet our safety standards or address adverse publicity affecting our reputation as a result of safety issues, mechanical failures, damages to customer property or medical complications could have a material adverse effect on our business, financial condition and results of operation.

In the nuclear industry, an accident or incident involving the mishandling of nuclear materials at a nuclear facility in the world can have an impact on other nuclear facilities around the world, public acceptance, political pressures, and regulatory requirements and scrutiny. For example, the March 2011 accident at the Fukushima Daiichi plant in Japan resulted in millions of dollars in costs, regulatory reviews and requirements for U.S. nuclear power plants. As a result of the Fukushima accident, some countries that were considering launching new domestic nuclear power programs delayed or canceled the preparatory

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activities they were planning to undertake as part of such programs. If a safety incident occurred at a nuclear facility in the world, it could delay licensing and/or drive up costs to license or own or operate a reactor, which could negatively impact our business or financial condition.

We rely on a limited number of suppliers for certain materials and supplied components, some of which are specialized and are being designed for first-of-a-kind or sole use in the Xe-100. We and our third-party suppliers may not be able to obtain sufficient materials or supplied components to meet our manufacturing and operating needs, or obtain such materials on favorable terms or at expected costs.

We rely on a limited number of suppliers for certain raw materials and supplied components. If we are not able to obtain sufficient raw materials or supplied components to meet our manufacturing and operating needs, or obtain such materials on favorable terms or at expected costs, which could result in our inability to fulfill our orders in a timely manner or increase our costs of production.

We do not directly manufacture any of the components of our SMRs. Our ability to manufacture SMRs is dependent upon sufficient availability of raw materials and supplied components, including highly technical components that are still under design, are being designed for first-of-a-kind use in the Xe-100 and have not yet been qualified for use, are only produced by a limited number of suppliers. Our supply chain may be particularly susceptible to cost increases, supply chain disruptions or inflationary pressures. Supply chain disruption incurred by our third party suppliers or degradation in the quality and quantity of our manufacturer partners, may result in delays, cost overruns or impairments to the development of our reactors.

Certain materials, such as the graphite used to line our reactor cores and for our TRISO fuel, and helium, which is used as a reactor coolant, are currently produced in limited quantities and are available from a limited number of vendors, which in some cases are predominantly outside the United States (e.g., Germany and Japan). There is also increasing scrutiny on the environmental, social, and governance provenance of certain goods, including critical minerals, which may require us to incur costs that may further limit our ability to source certain goods required for our operations. For example, in December 2021, the European Union announced a ban on the export of Russian coal, which could impact our ability to source coal for our operations.

the U.S. adopted the Uyghur Forced Labor Prevention Act (“UFLPA”) which creates a presumption that any goods, wares, articles, and merchandise mined, produced, or manufactured in part in the Xinjiang Uyghur Administrative Region of China or that are produced by certain entities are prohibited from importation into the United States. These import restrictions came into effect in 2022. While we are not presently aware of any direct impacts of these restrictions on our supply chain, UFLPA may have an adverse effect on global supply chains which could adversely impact our results of operations. Although U.S. graphite suppliers are developing the capability and capacity to meet our needs, our current reliance on foreign suppliers to secure raw materials and supplied components exposes us to volatility in the prices and availability of these materials, and may result in our being susceptible to changes in geopolitical relationships. We may not be able to obtain a sufficient quantity of raw materials or supplied components, on favorable terms or at all, which could result in delays in our inability to, manufacture our TRISO-X fuel and SMRs or result in increased costs.

We are dependent on the ability of our supplier partners to scale their production output to meet the requirements of our forecast. If our supplier partners cannot increase their production capacity to meet demand, it could result in delays in our ability to manufacture our reactors at the rates required.

Certain key components of our SMR are currently manufactured outside the U.S. and we rely on the timely and cost-effective transportation of goods via ocean freight. Disruptions to global shipping, such as port congestion, vessel capacity shortages, labor strikes, extreme weather events, or geopolitical tension, or new environmental regulations could significantly delay shipment of components, increase transportation costs, or impair our ability to meet customer demand. If such disruptions occur, we may be unable to secure alternative transportation in a timely manner, which could adversely affect our supply chain, customer relationships and financial performance.

Moreover, a major shipping accident could have a severe, negative impact on the delivery of components and the cost of our early projects, such as in the case of key components being lost at sea due to severe weather.

Additionally, the imposition of tariffs and impacts of inflation on raw materials or supplied components for our reactors could have a material adverse effect on our operations. Prolonged disruptions of our supply chain could have a material adverse effect on our operations.

any of our key raw materials or components, difficulty qualifying new sources of supply, inability to secure replacement materials or new sources of supply or any volatility in prices could have a material adverse effect on our ability to operate in a cost efficient, timely manner. Such prolonged disruptions could cause us to experience cancellations or delays of scheduled launches, customer cancellations, which could have a material adverse effect on our business, financial condition, operating performance and future prospects.

In particular, recent global trade tensions and policy shifts have created an unpredictable environment for businesses operating across international borders. Changes in trade agreements, sanctions, export controls, and customs regulations may limit our ability to source materials from certain countries, potentially forcing rapid and costly adjustments to our supply chain. Trade policies can change without notice, making long-term planning difficult and increasing operational costs.

While we attempt to mitigate these risks through diversification of our supplier base, supply chain management strategies, and contractual protections, there can be no assurance that these measures will be effective. Any significant disruption to our supply chain resulting from tariffs or trade policy changes could have a material adverse effect on our business, financial condition, and ability to meet projected milestones.

We depend on key executives and management to execute our business plan and conduct our operations. The departure of key personnel could have a material adverse effect on our business.

Our success depends, in significant part, on the continued services of our senior management team. We also depend on our ability to attract, motivate, develop and retain a sufficient number of other highly skilled personnel, including engineers, manufacturing and quality assurance, finance, marketing and sales personnel. Our senior management team has extensive experience in the energy industry, including nuclear power and manufacturing industries, and we believe that their depth of experience is instrumental to our success. The loss of any one or more members of our senior management team, for any reason, including resignation or retirement, could impair our ability to execute our business strategy and have a material adverse effect on our business and financial condition if we are unable to successfully attract and retain qualified and highly skilled replacement personnel.

Our business plan requires us to attract and retain qualified personnel including personnel with high levels of technical expertise. Were we not to be able to successfully recruit and retain experienced and qualified personnel, we could have a material adverse effect on our business.

Our future success depends in part on our ability to contract with, hire, integrate, and retain competent nuclear reactor and fuels focused engineers and scientists, and other qualified professionals. Competition for the limited number of these skilled professionals is intense. If we are unable to anticipate our needs for certain key competencies and implement human resource solutions to address them, or improve these competencies, our business, financial condition, operating results and future prospects could suffer. Additionally, U.S. immigration policy may make it more difficult for qualified foreign professionals to obtain or maintain work visas. These visa limitations may make it more difficult and more expensive for us to hire the skilled professionals we need to execute our growth strategy and may adversely affect our business. If we are unable to recruit and retain highly skilled personnel, especially personnel with specialized technical expertise to develop our reactors and fuel, we may experience delays, increased costs, and reputational harm. additional requirements, which could adversely affect our results of operations.

We must complete nuclear grade material qualifications and obtain regulatory approvals for the use of these materials in our TRISO-X fuel and our reactor designs. This includes long lead time irradiation testing and analysis, which may require redesign or use of alternative suppliers if results are unsatisfactory. Furthermore, nuclear grade materials and components, such as graphite, are only produced in limited quantities and are predominantly outside of the U.S. Cultivating expanded foreign or domestic U.S. supply chain capacity for key materials and components depends on cooperation from government and supply chain partners that may result in shortages and delays if not accomplished within assumed timelines or costs. These materials and components may also be particularly vulnerable to inflationary pressures and cost increases.

The equipment, components, and materials used in a nuclear power plant are subject to a high level of manufacturing and quality assurance scrutiny, in compliance with NRC regulatory requirements, codes and

nuclear industry standards. Moreover, it is critical to demonstrate in facility design and development that the materials used in the reactor facility, which will be exposed to radioactive materials, perform in accordance with necessary design parameters. The heightened manufacturing and quality assurance requirements and regulatory oversight limit the number of potential suppliers from whom we can procure materials, equipment, components, and materials used in our reactors and within TX-1, as well as the types of facilities where we can test certain materials. We may not be able to obtain sufficient materials or components to meet our manufacturing and operating needs, or obtain such materials on favorable terms, which could impair our ability to fulfill our orders in a timely manner or increase our costs of production.

Our suppliers' ability to manufacture components for our fuel fabrication facilities is dependent on the sufficient availability of materials and possibly other supplied components, some of which are specialized and are being designed for first-of-a-kind or sole use in our fuel fabrication facilities. A supply chain disruption incurred by our third-party suppliers or degradation in the quality and performance of our manufacturer partners, may result in delays, cost overruns or impairments to the development of our fuel fabrication facilities.

These suppliers and the key materials and essential components may be particularly vulnerable to price increases, as a result of supply and demand dynamics, inflation, the current administration's trade policies, tariffs, and other price pressures. As a result, supplier delays, unexpected performance testing requirements, delays in the manufacturing process or procuring necessary materials, international procurement needs, regulatory compliance issues, component qualification issues or delays, increases in costs as a result of inflation, and otherwise, and geopolitical considerations can all impact our ability to perform necessary functions for our customer in licensing a reactor, license TX-1, assist customers in constructing and operating a reactor, reactor design, or construct and operate TX-1. Such prolonged disruptions could also cause us to experience cancellations or delays of scheduled projects, customer cancellations or reductions in our margins, any of which could harm our business, financial condition, operating results and future prospects.

Our business is subject to the risks of earthquakes, fire, floods and other natural catastrophic events, global pandemics, and interruptions by man-made problems, such as network security breaches, computer viruses, terrorism. Material disruptions of our business or information system resulting from these events could materially and adversely affect our operating results.

We are vulnerable to damage from catastrophic events, such as natural disasters, global pandemics, power loss, and similar unforeseen events beyond our control. Catastrophic events could disrupt our business operations, reduce or restrict our supply of products and services, incur significant costs to repair our employees and facilities, or result in regional or global economic distress, which may materially and adversely affect our business, financial condition, operating results and future prospects. Threats such as threatened war, terrorist activities, political unrest, civil strife, and other geopolitical uncertainties could have a similar adverse effect on our business, financial condition, and results of operations. Any of these events may adversely affect our operation results, or even for a prolonged period of time, materially and adversely affect our business, financial condition, and results of operations.

We cannot guarantee that we are adequately protected from the effects of earthquakes, typhoons, global pandemics, power loss, telecommunications failures, break-ins, war, riots, network security breaches, computer viruses, terrorist attacks or similar events. Any of the foregoing events could give rise to interruptions, damage to our property, delays in production, breakdowns, system outages, technology platform failures, or internet failures, which could cause the loss or corruption of data, malfunctions of our internet system as well as adversely affect our business, financial condition, and results of operations.

If a natural disaster, power outage or other event occurred that prevented us from using a significant portion of our headquarters, damaged critical infrastructure, or otherwise disrupted our operations, it may be difficult or, in certain cases, impossible for us to continue our business for a substantial period of time. The disaster recovery and business continuity plans we have in place are unlikely to provide adequate protection in the event of a serious disaster or similar event. We may incur substantial expenses in connection with the limited nature of our disaster recovery and business continuity plans, which could have a material adverse effect on our business.

Our use of AI-based technology may present new risks and challenges to our business.

The rapidly evolving use of AI in our industry presents significant risks. If we fail to adopt and govern AI responsibly, our brand, regulatory approvals, operations, and financial results could be adversely affected.

We may explore the usage of third-party AI or machine learning (“ML”) platforms, offerings, including AI chatbots and generative AI products (“AI/ML Technology”), in our internal operations. The development and use of AI/ML Technology present various privacy and security risks that may impact our business. AI/ML technology is subject to privacy and data security laws, as well as increasing regulatory scrutiny. The use of AI/ML technology and third-party open-source AI tools by our employees or consultants could pose risks relating to the protection of data, including the potential exposure of proprietary, confidential or otherwise protected information to unauthorized recipients and the loss of or third-party intellectual property. Use of AI tools may result in allegations or claims against us for violation of third-party intellectual property rights, unauthorized access to or use of proprietary information, and failure to comply with open-source software requirements.

We have developed policies governing the use of AI/ML Technology by our employees and authorized agents that are designed to protect our assets, including intellectual property, confidential information, personal information we may collect or process, and customer information. Any failure by our personnel, contractors or other agents to adhere to our policies could result in a violation of our obligations or applicable laws and regulations (including data privacy laws), jeopardize our intellectual property rights, cause or contribute to unlawful discrimination, result in the misuse of confidential or identifiable information, or introduce greater vulnerabilities to cybersecurity attacks or malware on our systems. We also could be subject to claims from providers of third-party AI/ML Technology for using their products, tools or outputs in a manner that is inconsistent with their terms of use. Such claims may result in costly legal proceedings.

The regulatory framework for AI/ML Technology is rapidly evolving as many federal, state and local government bodies and agencies have introduced or are currently considering additional regulations. Additionally, existing laws and regulations may be interpreted in ways that would restrict the use of AI/ML Technology, or could be rescinded or amended as new administrations take differing views to evolving AI/ML Technology. As a result, we cannot be certain that our policies or adherence to existing laws and regulations will offer us sufficient protection or that the use of the AI/ML Technology will not harm our reputation, financial condition or operating results.

Several jurisdictions around the world, including Europe, the U.S. federal government and various U.S. states, have proposed, enacted or are considering laws and policies governing the development, use and use of AI/ML, such as the EU Artificial Intelligence Act. It is possible that further new regulations will be adopted in the U.S. and in other non-U.S. jurisdictions, or that existing regulations, including competition and antitrust laws, may be interpreted in ways that would restrict our ability to use AI/ML Technology for our business. Additionally, certain privacy laws extending to consumers (such as the right to delete certain personal data) and regulate automated decision making may be incompatible with our use of AI/ML. These obligations may make it harder for us to conduct our business using AI/ML, jeopardize our proprietary information, lead to regulatory fines or penalties that require us to change our business practices, retrain our AI/ML, or prevent or limit our use of AI/ML. If our use of AI/ML or our use is restricted, our business may be less efficient, or we may be at a competitive disadvantage. In addition, uncertainties regarding developing legal and regulatory requirements and standards may require significant resources to modify and maintain business practices to comply with laws in the U.S. and other jurisdictions concerning the use of AI/ML Technology, the nature of which are not yet determined at this time.

The use of third-party AI/ML Technology by our business partners and employees with access to our confidential information, including trade secrets, may continue to increase. This could lead to the disclosure of such information, which could negatively impact us, including our ability to realize the full benefits of our intellectual property. The use of AI/ML Technology by our business partners could also introduce cybersecurity risks, which could have a material adverse effect on our operations and reputation. Finally, the operations of any of our business partners. Finally, the use of AI/ML Technology also presents ethical issues, and

if our use of third-party AI/ML Technology becomes controversial, we may experience brand or reputational harm or legal liability or we may be at a competitive disadvantage.

Further, constraints or shocks in AI/ML Technology supply chains could increase costs of our projects as we continue to rely on them for faster design and delivery of our solutions. Availability and pricing of advanced computing, specialized chips, and cloud capacity used for model development and simulation fluctuate with industry demand. Supply constraints could slow our digital engineering and analytics roadmaps or increase spend.

Any of these risks could be difficult to eliminate or manage and, if not addressed, could have a material adverse effect on our business, financial condition, operating results and future prospects.

Risks Related to the Industry, Competition and General Economic Conditions

The market for our products and services is still in the early stages of growth and if it does not continue to grow more slowly than we expect, or fails to grow as large as we expect, our business, financial condition, operating results and future prospects may be adversely affected.

Our success depends substantially on the willingness of consumers to widely adopt our products and services and to pay associated prices for our products and services sufficient to meet our margin targets in terms of percentage and gross receipts. To be successful, we will have to educate consumers about our products and services through significant investment and provide quality products and services that are superior to the products provided by our competitors. It is difficult to predict the future growth rate, timing, and size of our market. We cannot assure you that our market will develop or that our products and services will be widely adopted. If our market does not develop, develops more slowly than expected, becomes saturated with competitors, or if our products and services do not achieve market acceptance, our business, financial condition, operating results and future prospects could be adversely affected.

The market for Gen IV advanced nuclear reactor designs generating electric power and high-temperature process heat is not yet established and may not achieve the growth potential we expect or may grow more slowly than we expect.

The market for Gen IV advanced nuclear reactor designs such as the Xe-100 has not yet been established. SMRs utilizing advanced nuclear technologies have limited operational history and have not been proven at scale. Estimates for the total addressable market and our expectations, including updates, with regard to certain illustrative unit economics provided in this prospectus are based on a combination of internal and third-party estimates, including our potential contracted revenue, the number of customers who have expressed interest in our SMRs, assumed prices and production and regulatory requirements for our SMRs, our ability to leverage our current logistical and operational processes, assumptions about our technology and general market conditions. However, our assumptions and the data underlying our estimates may not be correct and the conditions supporting our assumptions or estimates may change over time, reducing the predictive accuracy of these underlying factors. As a result, our expected production and revenue indicated by the illustrative unit economics provided in this prospectus, our estimates of the total addressable market and serviceable addressable market for our services, as well as the expected production and revenue for the total addressable market and serviceable addressable market for our services, may be incorrect.

In addition, our reactors, financial models and the illustrative unit economics included in this prospectus assume an anticipated plant life of 60 years. However, the licenses we receive from the NRC and other government authorities are limited to an initial term of 40 years, which is the maximum licensing term currently permitted. There is no assurance that we will be able to obtain license extensions, and if we are unable to do so, our reactors may be required to cease operations before the end of their

their useful lives, which could reduce the economic return of our projects, adversely affect demand, and negatively impact our business, financial condition, operating results and future prospects.

Our cost estimates are highly sensitive to broader economic factors, and our ability to control or manage costs may be limited.

Capital and operating costs for the deployment of a first-of-a-kind reactor such as the Xe-100 and TRISO-X fuel may be difficult to project, inherently variable and are subject to significant change based on a variety of factors, including site location, regulatory requirements, and local labor rates.

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specific factors, customer offtake requirements, regulatory oversight, operating agreements, availability, local labor rates, inflation, detailed design, engineering and other factors. Opportunities for cost reductions with subsequent deployments are similarly uncertain. To the extent cost reductions are achieved within the expected timeframe or magnitude, the Xe-100 and TRISO-X fuel may be competitive with alternative technologies, which could materially and adversely affect our revenues, gross margins and the information included in the illustrative unit economics presented in this prospectus.

In addition, our cost estimates and project economics could be materially impacted by trade restrictions imposed on materials, components or equipment required for the construction and operation of our reactors and fuel facilities. While we do not expect to purchase certain key components for several years, changes in tariff regimes or the imposition of new tariffs prior to our procurement period could significantly increase our costs or limit our ability to source needed materials. The potential for future tariffs or trade actions creates additional uncertainty in our cost estimates. Such actions could adversely affect our competitiveness, margins and financial results.

The illustrative unit economics presented in this prospectus are estimates only, reflecting management's current expectations and based on numerous assumptions. The illustrative unit economics presented in this prospectus should not be realized, and actual results could differ materially from the illustrative estimates presented.

Given our early stage of development, it is difficult to predict what results we might achieve. While we present “illustrative unit economics” in this prospectus, these figures are not forecasts, targets or guidance for our future operating results but what we believe to be potential revenues based on customer fees, and associated costs, based on information available as of the date of the prospectus. Our ability to actually achieve these unit economics and our business performance will depend on numerous factors and are based on numerous assumptions. In particular, the illustrative unit economics described in this prospectus are based on management’s current expectations and assumptions regarding customer fees, costs, and margins for a single plant. Actual results will depend on numerous factors outside our control, including customer negotiations and fee structures, market conditions, supply chain costs and dynamics, manufacturing efficiency and utilization, financing terms and conditions, customer adoption and actual operational performance, and are subject to the risks described elsewhere in this “Risk Factors” section. Any variation in these assumptions could cause our actual results to differ materially from the illustrative estimates presented.

Technology Fees. The illustrative unit economics reflect management’s current expectations regarding the amount and timing of technology fees we would expect to charge for customers’ use of our technology. These expectations are based on management’s internal benchmarking of IP and technology fees across various similar industries and sectors, together with the expectations of our sales team and ongoing customer discussions. However, there is no directly comparable technology fee to benchmark the fee, and there can be no assurance that we would be able to collect from customers fees that are typical in other industries or sectors. While we have had preliminary discussions

customers regarding a proposed fee estimate within the middle of the range reflected in the illustrative unit economics, we have not yet entered into any technology, intellectual property or related agreements. We can provide no assurance that customers will accept our anticipated fee structure or pricing because the technology fees relates to the use of our intellectual property, the associated costs are not assumed to be zero.

The final fee will be negotiated on a case-by-case basis with each customer, based on numerous factors specific to each customer, and our actual revenues and customer fees from our technology fees may differ materially from the estimates in the illustrative unit economics, including as a result of the following factors, among others:

- customer negotiations that result in materially lower fees or delayed payment schedules;
- variations in project scope, timing, and customer profile that affect negotiated pricing;
- the potential emergence of competitors or alternative reactor technologies that exert competitive pressure on fees;
- evolving market conditions, regulatory developments, and cost of capital for nuclear power;
- customer reluctance to pay milestone-based technology fees prior to commercial operation.

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- customers not being willing to pay higher fees for later stage plants, even after successful validation; and
- litigation, claims or other events that could result in an associated costs to us related to our intellectual property.

If we are not able to receive our anticipated technology fee structure and pricing, it will have a negative impact on the amount of revenues, the timing of revenues and the percentage of revenues we expect to be generating pre-COD. For example, we expect approximately 15 – 35% of revenue in our reactor business to be realized prior to (and including) COD, and this includes both the technology fees and services fees described below. If the actual technology fee we receive is lower, or customers are not willing to pay the assumed higher fees for later stage plants or on the milestone-based payment structure, our illustrative pre-COD revenues and our illustrative percentage of pre-COD revenue will be lower than expected, which could have a material and adverse impact on our business, results of operations and financial position, including our need to obtain additional financing or to generate sufficient liquidity or revenue to sustain our business pre-COD.

Services. The illustrative unit economics reflect management’s current expectations regarding the amount and timing of services fees we would expect to charge customers for providing them with reactor project development and operating services, including site selection, permitting, engineering, construction, procurement support and, following commercial operation date (“COD”), ongoing maintenance and operator training. Our assumptions regarding services pricing, timing, and margins are based on our own experience to date from the provision of initial strategic business and other services we have provided to customers and potential customers, as well as industry benchmarking and our team’s collective experience. However, there can be no assurance that we would be able to collect from our customers the amounts we expect to be typical in other industries or sectors. To date, we have entered into only limited services or other agreements with potential customers, and we can provide no assurance that customers will accept our anticipated fee structure or pricing. The illustrative unit economics also assume that our customers will continue to use us as a service provider throughout the operational life of a plant, and there is no assurance

continue to do so. In addition, our services business depends on assumptions about customer pricing, and long-term relationships that may not materialize.

- Our actual services revenues, customer fees, costs and margins may differ materially from our estimates in the illustrative unit economics, including as a result of the following factors, among others: our estimates are based on our limited history providing services at commercial rates; actual costs to provide these services may be higher, and we may not be able to pass through the costs to our customers at our estimated fees and our estimated margin;
- we have only entered into a small number of services agreements with potential customers, which may not be representative of future revenues, fees, costs or margins that we are able to achieve;
- third-party service providers could enter the market, introducing price competition and reducing revenue and fees, increasing costs and reducing margins;
- customers may elect to self-perform or contract with alternative vendors for services such as maintenance, or operator training; and
- our ability to maintain efficient staffing, supply-chain arrangements, and project execution may directly affect costs and profitability.

If our services assumptions prove inaccurate, our expected revenues, fees and margins may be materially reduced, and we may experience higher costs than currently anticipated. Because we cannot construct, own, or operate plants, following COD, we depend entirely on customers' continued use of our services to generate recurring revenue from our reactor line of business, with approximately 80% of revenue from our reactor line expected to be from the provision of long-term services offered post-COD. In addition, if the actual pre-COD services fees we receive are lower, then our illustrative pre-COD revenues and our illustrative percentage of pre-COD revenue will be lower, which could have a material adverse impact on our business, results of operations and financial prospects, including our need for additional financing or to generate sufficient liquidity or revenues to operate our business post-COD. Similarly, if we do not generate the anticipated revenues post-COD, or if our costs are higher or lower than anticipated, it could

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have a material and adverse impact on our business, results of operations and financial prospects, including our need to obtain additional financing or to generate sufficient liquidity or revenues to operate our business post-COD.

Fuel Pricing. The illustrative unit economics reflect management's current expectations regarding the amount of TRISO-X fuel we will sell to customers, the price at which we will do so, and our operating and capital costs. Our expectations assume that we will supply the initial core fuel for our customers, and, following COD, provide recurring fuel supply for refueling over the life of each plant. We assume that we will be the initial sole supplier of proprietary TRISO-X fuel. We have not yet entered into definitive fuel supply agreements, and there can be no assurance that customers will agree to the fee structure, pricing, or volume commitments. Customers may seek lower-cost alternative fuel sources, volume discounts, or purchase from other suppliers once alternative sources of TRISO-X fuel are available.

Our actual revenues, fees, costs and margins from selling TRISO-X fuel to our customers may differ materially from the estimates in the illustrative unit economics, including as a result of the following factors, among others:

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the emergence of third-party suppliers or alternative fuel technologies that compete performance;

- changes in uranium, feedstock, or manufacturing costs that adversely affect our margins;
- delays or cost overruns in developing and commissioning our fuel fabrication facilities, costs of labor, materials, and overhead;
- customer decisions to source fuel from other vendors once alternative supply chains and
- the need to finance future fuel facilities at assumed utilization levels that may not be achieved.

If actual utilization of our manufacturing or fuel-fabrication facilities is lower than expected, an increase due to inflation, regulatory delays, or operational inefficiencies, our margins could be reduced. In addition, the assumptions underlying the illustrative unit economics also exclude risk related to uranium procurement or inventory, feedstock procurement, or spent fuel management costs, which are borne by our customers. If we are required to bear any of these costs, or if fuel costs increase due to inflation, supply-chain constraints, or regulatory changes, our expected margins could be reduced. If our fuel pricing assumptions prove inaccurate, expected revenues, fees and margins could be materially reduced, and we may experience higher costs than currently anticipated. In addition, if our customers' costs borne by our customers exceed their expectations, our reputation with our customers may be impacted. Any of the foregoing could have a material and adverse impact on our business operations and financial prospects, including our need to obtain additional financing or generate sufficient liquidity or revenues to operate our business.

Our economics are based on an anticipated 60-year operating life of a reactor. While the Atomic Energy Act of 1954 limits initial licensing to 40 years, the current operating fleet was designed to operate beyond that limit. Similarly, the Xe-100 is designed with substantial margins to support operations beyond the initial 40-year licensing term. The NRC established a license renewal program that permits reactors to seek approval for subsequent 20-year operating periods. In license renewal, the NRC focuses on whether aging management programs adequately address the effects of aging on structures, systems, and components important to safety. For much of the current operating fleet, license renewal requirements were not in place at the time of initial licensing, and comprehensive aging management programs were not developed and reviewed as part of the license renewal process. Almost every U.S. commercial nuclear power plant operates under a renewed license, valid for 40–60 years of potential total operation. The NRC is now reviewing applications for “subsequent license renewal,” valid for 60–80 years of potential operations. The Xe-100 design contemplates long-term operations, and while X-energy is incorporating aging management considerations into the reactor design from the outset to be well positioned for future license renewal applications, NRC approval cannot be assured.

While the NRC license renewal process is well established and the majority of applications are approved, the NRC has, in limited circumstances, denied or reversed license renewal applications that failed to satisfy applicable safety, environmental, or regulatory requirements. According to

X-energy’s proactive approach to aging management and regulatory compliance may not eliminate license renewal risk. If the NRC does not approve our license renewal application, or if the anticipated operating life of a reactor is lower than anticipated, it could have a material and adverse impact on our business operations and financial prospects.

Competition from existing or new companies could cause us to experience downward pressure on customer orders, reduced margins, the inability to take advantage of new business opportunities, and market share.

We operate in highly competitive markets and are subject to competition based upon price, performance, project execution, constructability, pricing, quality, and services, from competitors, suppliers as well as from alternative means of producing electricity and/or heat. There are advanced reactor designs, and some advanced reactor projects, under development in the U.S. These designs are involved in preapplication review with the NRC. Our advanced design, project design performance, engineering expertise, and quality control have been important factors in our success. Nonetheless other companies providing competing technologies could capture customers or market share from us, which could have a material adverse effect on our business or financial condition.

In addition, some markets experience very low power prices due to a combination of renewable energy, renewables and low-cost fuel sources, and we may not be able to compete in these markets. The benefits of the Xe-100 and our TRISO-X fuel are sufficiently valued in the market.

For sales and/or deployments outside of jurisdictions with highly developed nuclear regulatory frameworks, some of our foreign competitors currently benefit from, and others may benefit from, permissive regulatory and licensing regimes and/or from protective measures by their home countries where governments are providing financial support, including significant investments in the development of new technologies. Competitors based in Russia and China, such as China National Nuclear Corporation, currently operate commercial or near-commercial SMRs in their home countries, while competitors in South Korea are pursuing SMR development but do not currently operate commercial SMRs. Although our designs have not been approved or licensed for use by the NRC or any other nuclear regulatory jurisdiction outside of their native countries, those competitors may have a competitive advantage if they are able to obtain approvals, or if they can demonstrate to potential customers the value and benefits of SMRs, particularly in jurisdictions that have less stringent nuclear regulatory requirements. Our competitors may have access to greater sources of funding to develop and commercialize their technologies than we do, whether as a result of potential competitive advantages or from supportive national governments. This market environment may result in increased pressures on our pricing and other competitive factors.

We believe our ability to compete successfully in designing, engineering and manufacturing nuclear products and services at significantly reduced costs to customers does and will depend on many factors which may change in the future due to increased competition, our ability to meet customer needs and the frequency and availability of our offerings. If we are unable to compete successfully in this business, financial condition, operating results and future prospects would be adversely affected.

Changes in the availability and cost of electricity, natural gas, oil and other forms of energy are subject to volatile market conditions that could adversely affect our business.

The prices for and availability of electricity, natural gas, oil and other energy resources are subject to volatile market conditions. We do not control these market conditions, which are, moreover, influenced by political and economic factors beyond our control. Decreases in energy prices, or changes in energy costs relative to other forms of energy, may adversely affect our business. To the extent that such uncertainties cause suppliers and customers to be more cost sensitive or to adjust their business operations, decreased energy prices may have an adverse effect on our results of operations and financial condition.

The cost of electricity generated from nuclear sources may not be cost competitive with other electricity sources in some markets, which could materially and adversely affect our business.

U.S. electricity markets are heavily regulated at the federal, state and local levels. Regulation of electricity markets price electric energy, capacity, and/or ancillary services on a competitive basis may impact market prices.

subject to substantial fluctuations. Other markets require that power purchase decisions by electricity generators be subject to various competitiveness or prudence tests. As a result of competitive pricing in electricity markets experience low marginal energy prices at certain times due to a combination of subsidized generating resources, competitors with low cost or no cost fuel sources, or market design features that create incentives for certain attributes or deliver revenue in unpredictable ways. Our Xe-100 may not be able to compete in these markets unless the benefits of the low carbon, reliable, and resilient energy generation provided by the Xe-100 are sufficiently valued. Even in electricity markets that pay a price for reliable capacity on a long term basis, there is no guarantee that our customers' Xe-100 will be of sufficiently low cost so as to clear auction style capacity markets, and clearing a capacity auction in one year is no guarantee of clearing in successive years.

Given the relatively lower electricity prices and higher availability of power in the U.S. compared to many international markets, the risk associated with lower cost alternatives may be higher with respect to our business and operations in the U.S. Regardless of jurisdiction, however, our ability to build SMRs to provide competitively priced electric energy, capacity and/or ancillary services could be delayed and adversely affect our business.

We and our customers operate in a politically sensitive environment, and the public perception of nuclear energy can affect our customers and us.

Successful execution of our business model is dependent upon public support for nuclear energy in the U.S. and other countries. The risks associated with uses of radioactive materials, both in our proposed fuel fabrication facility, and subsequent fuel fabrication facilities and by our customers in the deployment of our SMR designs, and the public perception of those risks, can affect our business. Opposition by third parties can delay or prevent the licensing and construction of new nuclear facilities and in some cases can limit the operation of nuclear reactors. Adverse public opinion or developments in the use of nuclear power could directly affect our customers and indirectly affect our business. Adverse public reaction, increased regulatory scrutiny and related litigation could result in extended licensing and construction periods for new nuclear reactors, delay construction schedules, or shut down operations at already-constructed reactors. There have only been two new nuclear reactors licensed in the U.S. over the past 30 years, so there is not a good calibration today of the level of public opinion or adverse reactions to nuclear power. Public perception of nuclear energy, as with most energy technologies, tends to be locally generated depending on the politics of the impacted communities rather than being nationally driven. As with other very visible technologies, a potential accident by any company or organization in the nuclear community can negatively impact the public perception of a particular project, such as one of our customer's projects.

From a political perspective, while the current U.S. federal administration and select members of Congress are currently quite favorable towards nuclear energy, we cannot guarantee that such support will last or that opponents of nuclear energy will not be successful in increasing the cost, complexity, or timelines associated with nuclear energy, whether in our operations or those of the broader industry value chain. Any such delays may also impact the marketability of our technology internationally and our competitive position relative to others. For example, China already has a 500 MWt High Temperature Gas Cooled Reactor (HTGR) plant in operation, has a TRISO-X fuel facility, and is already marketing its technology internationally. Although Russian and Chinese competitors have SMR designs that have been approved by the NRC or in any jurisdiction outside of their native countries, those companies may have a competitive advantage if they are able to obtain approval comparable to the NRC's Standard Design Approval, or if they can otherwise demonstrate to potential customers the value and benefits of their technology, particularly in jurisdictions that have less stringent regulatory requirements. In addition, these companies may have access to greater government or other funding to develop and commercialize their SMRs than we can do.

The direct and indirect impact on us and our customers from severe weather and other effects of climate change and the economic impacts of the transition to noncarbon based energy, could adversely affect our business condition, operating results, and cash flows.

There are inherent climate-related risks wherever business is conducted. Our operations and those of our customers, may in the future be adversely impacted by flooding, wildfires, drought and other effects of severe weather conditions. Climate change is expected to increase the frequency and

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severity of such events, as well as contribute to chronic changes (such as changes to meteorological and hydrological patterns) that may result in similar risks. These events can force us or our customers to suspend operations at impacted properties and may result in significant damage to such properties. Even if these events do not directly impact us or our customers they may indirectly impact us and our customers through increased insurance, energy or other costs. The ongoing transition to non-carbon based energy sources presents certain risks, including macroeconomic risks related to higher energy costs and energy supply, among other things, which may also impact us directly or indirectly, such as through our supply chain. The speed and direction of the energy transition are also uncertain and subject to various competitive forces, including as a result of political, market, and other forces. If policymakers or the market favor alternative energy technologies — such as renewables with battery storage, hydrogen, geothermal, and synthetic fuels with carbon capture — such trend may adversely impact our ability to capture benefits as a result of the energy transition. These direct and indirect impacts from climate change could adversely impact our financial condition, operating results, supply chain and cash flows.

We are subject to a series of risks related to sustainability matters.

There is ongoing scrutiny from investors, customers, policymakers and other stakeholders regarding our companies' consideration and management of climate change, human capital, and environmental and social matters. We from time to time engage in various initiatives (including, but not limited to, addressing such matters and related stakeholder expectations; however, such initiatives entail costs and may not have the desired effect. Methodologies, standards, and data associated with sustainability disclosure are often complex and continuing to evolve. As with other companies, our approach to such matters is expected to evolve, and we cannot guarantee that our approach will align with the expectations or preferences of any particular stakeholder. For example, in some instances, companies have been subject to accusations of greenwashing due to alleged deficiencies in disclosure, methodology, or data. Additionally, in some instances, such stakeholders have different, or even conflicting, expectations, which can increase the cost and complexity of response. Failure to successfully navigate such challenges (including any regulatory requirements) may result in reputational harm, loss of customers, loss of engagement from regulators or capital providers, or other adverse impacts to our business. Our suppliers, customers, and other stakeholders are also subject to similar expectations, which may result in new or greater risks, including risks that may not currently be known to us.

Risks Related to X-energy's Use of Technology

Technological changes could render our technology and products uncompetitive or obsolete, which could prevent us from achieving market share and sales.

Our failure to refine or advance our SMR technologies could cause our reactor technologies to become uncompetitive or obsolete, which could prevent us from achieving market share and sales. We may invest significant financial resources in research and product development to keep pace with technological advances in the industry and to compete in the future; we may be unable to secure such financing if competing alternative technologies may be in development by other companies that could result in lower manufacturing or operating costs and/or higher performance than those expected for our technologies.

development efforts may be rendered obsolete by the technological advances of other technologies may prove more advantageous for commercialization.

We and our third-party providers are subject to information technology and cyber security risks which may have material adverse effects to our business, financial condition, operating results and future prospects, and may result in damage to our reputation, material financial penalties, and legal liability.

We are increasingly dependent upon information technology systems, infrastructure and data for our business (collectively “IT Systems”). We own and manage some of these IT Systems but rely on third parties for a range of IT Systems and related products and services. In the ordinary course of our business, we collect, store and transmit confidential information (including but not limited to intellectual property, proprietary business information and personal information) (collectively, “Confidential Information”) through our IT Systems. It is critical that we do so in a secure manner to maintain the confidentiality and integrity of

such Confidential Information. We also have outsourced elements of our operations to third parties. As a result we manage a number of third-party contractors who have access to our Confidential Information.

The IT Systems and those of our contractors and consultants face numerous and evolving risks that threaten the confidentiality, integrity and availability of our IT Systems and Confidential Information, including breakdown or other damage or disruption from service interruption, hardware malfunction, natural disasters, terrorism, war and telecommunication and electrical failures, security breaches from inadvertent or intentional actions by our employees, contractors, business partners, and/or other third parties, or from cyberattacks by malicious third parties (including the deployment of harmful malware, ransomware, denial-of-service attacks, social engineering and other attacks) that may affect service reliability and threaten the confidentiality, integrity and availability of information.

Cyberattacks are expected to accelerate on a global basis in frequency and magnitude as threat actors are becoming increasingly sophisticated in using techniques and tools, including AI, to bypass security controls, evade detection and remove forensic evidence. As a result, we may be unable to investigate, remediate or recover from future attacks or incidents, or to avoid a material adverse effect on our IT Systems, Confidential Information or business. There can also be no assurance that our risk management program and processes, including our policies, controls or procedures, are fully implemented, complied with or effective in protecting our IT Systems and Confidential Information. Furthermore, given the nature of complex systems, software and services like ours, and the scale of the IT systems that we deploy across our networks and products, we regularly identify and track security vulnerabilities that include potential risk level and severity. We are unable to comprehensively apply patches or other security measures are in place to mitigate all such vulnerabilities, or that patches will be applied in a timely manner or vulnerabilities are exploited by a threat actor.

While we have not experienced any material cyberattacks or other incidents to date, we cannot assure you that such incidents will not occur in the future, which could have a material adverse effect on our reputation, business, financial condition and results of operations. For example, we maintain a database comprised of our Xe-100 nuclear design technical engineering information and operations data, which have been and will continue to be used to design the Xe-100 reactors and will be utilized in the “twin” construction and operations environments to allow for highly efficient construction and operation of these designs. If this database were to be lost or compromised, our ability to efficiently deploy and operate our reactors could be significantly impaired.

Furthermore, any adverse impact to the availability, integrity, or confidentiality of our Confidential Information could result in financial, legal, business, and reputational harm to us. If any such event that leads to unauthorized access, use, or disclosure of Confidential Information or personal information related to our employees, could harm our reputation directly, compel us to comply with federal and/or state breach notification laws and foreign law equivalents, subject us to corrective action, and otherwise subject us to liability under laws and regulations that protect the privacy and security of personal information, which could result in significant legal and financial damages and reputational damages that could potentially have an adverse effect on our business. Finally, we do not guarantee that any costs and liabilities incurred in relation to an attack or incident will be covered by our existing insurance policies or that applicable insurance will be available to us in the future on reasonable terms or at all.

If we fail to continuously offer innovative products and services and keep pace with changes in technology and our ability to obtain new engagements will be adversely affected and our revenues and profitability may decline.

Our business depends, in part, on our ability to anticipate industry, business and technological changes and help businesses to become more competitive and profitable. To remain competitive, we will continue to develop new services and products for our clients that distinguish us from our competitors. We also will need to identify new and evolving technologies that will gain acceptance in the market and develop relationships with new leaders in technology as they emerge. If we fail to accomplish our business objectives, we may be unable to compete effectively for new engagements, which may cause our revenues and profitability to decline. We may also invest resources in new services that are not competitive and may not gain wide acceptance from our clients, which may cause our profitability to decline. While we offer innovative services that distinguish us from our competitors, we expect to be able to obtain higher fees. Over time, however,

if our competitors begin to offer services which we may have developed or pioneered, we may experience decreases in the amount of fees we can charge for these services.

Risks Related to Ongoing and Future Changes to Our Reactor and Fuel Design

Our reactor and fuel designs have evolved over time and are expected to continue to evolve as we progress through development, regulatory review, and commercialization. The process of optimizing our technology is ongoing and may result in material modifications to the design, cost, or performance characteristics of our products. These changes may be driven by regulatory requirements, advances in technology, supply chain constraints, customer requirements, or lessons learned from prototyping, testing, and early deployment.

Any significant changes to our design may require additional engineering, testing, and regulatory review, which could result in increased costs, delays in our development timeline, or the need for certain qualification or licensing activities. Design changes may also impact our ability to achieve standardization, scale manufacturing, or meet previously communicated performance, safety, or other requirements. Furthermore, customers or partners may delay purchase decisions or require renegotiation of terms in response to design changes, which could adversely affect our business prospects and financial performance.

There is also a risk that changes made to address one set of requirements may introduce new challenges or unforeseen issues, potentially impacting the commercial viability or regulatory approval of our products. If we are unable to effectively manage ongoing design changes, or if such changes result in negative perceptions among customers, regulators, or investors, our business, financial performance, and results of operations could be materially and adversely affected.

If we are unable to scale our fuel fabrication facilities, our ability to manufacture pebble fuel will be affected, which could have a material adverse effect on our business prospects, financial condition, operations and cash flows.

The Xe-100 requires pebble fuel in order to function, the manufacturing of which is not yet commercial. If we may not be able to manufacture a sufficient amount of pebble fuel in order to support our business, our ability to become profitable in the future will not only depend on our ability to successfully manufacture Xe-100 and TRISO-X fuel, but also to ensure we have a sufficient supply of fuel pebbles to support our business. If we are unable to efficiently design, manufacture, market, sell, distribute and service our pebble fuel, our margins, profitability and prospects would be materially and adversely affected.

Our ability to introduce new features, integrations, capabilities and enhancements is dependent on our research and development resources.

To remain competitive, we must maintain adequate research and development resources, including appropriate personnel and development technology, to meet the demands of the market. If we are unable to develop features, integrations, capabilities and enhancements internally due to certain constraints, such as employee turnover, a lack of management ability or a lack of other research and development resources, our business may be harmed. Moreover, research and development projects can be technically challenging and expensive. The nature of these research and development cycles may cause us to experience delays between the time we incur expenses associated with research and development and the time we are able to introduce compelling features, integrations, capabilities and enhancements and generate revenue, if any, from that investment. If we expend a significant amount of resources on research and development and it does not lead to the successful introduction or competitive improvement of features, integrations, capabilities and enhancements, it could harm our business, financial condition, operating results and future prospects. In addition, our failure to maintain adequate research and development resources or to compete effectively with the research and development programs of our competitors may harm our business, financial condition, operating results and future prospects.

Operating a nuclear reactor in an unusual environment, whether due to unusual siting or an industrial application, has additional risks and costs compared to conventional electric power applications.

We expect our initial deployment of the Xe-100 under the ARDP to include industrial heat applications. This deployment is expected to be the first instance of using SMR technology for such applications.

deployment will require additional overhead associated with the licensing process, configuration of the plant, chemistry control, Tritium containment, minimum operating staff, training, security infrastructure, radiation protection, government reporting, and nuclear insurance, all of which may be costly. We may require separate operating agreements to provide necessary nuclear overhead without disrupting our manufacturing processes. While we have attempted to address such increases in our updated cost estimates, additional costs or increased regulatory requirements may cause delays under our project timelines and could have an adverse impact on our ARDP partner.

Additionally, our technologies are also designed to be able to serve customers in locations that are atypical for nuclear energy, whether due to being far away from urban infrastructure, in closer proximity to population centers, and/or areas that may experience more extreme planetary conditions. Such deployments may come with additional risks and costs, including beyond any projections we may have made, which could adversely impact our business.

Risks Related to X-energy's Proprietary and Intellectual Property Rights

We rely heavily on our intellectual property portfolio. Our ability to maintain, protect or enforce our other intellectual property rights may be challenged and is not guaranteed. If we are unable to maintain, protect or enforce our intellectual property rights, our business and competitive position may be harmed.

Our success depends in part on our ability to maintain, protect and enforce our intellectual property rights, including our patents in connection with our SMRs. We rely on a combination of the property protections afforded by patent, trademarks/service mark, copyright and trade secret laws in the U.S. and other jurisdictions, as well as contractual restrictions in our agreements such as confidentiality agreements, assignment agreements, and license agreements with our employees, consultants, and other third parties with whom we have a business relationship, to protect, establish, maintain and enforce our intellectual property rights, including rights associated with our SMRs and related technologies. However, the steps we take to protect our intellectual property and other confidential information may not adequately secure our intellectual property rights, and we may not be able to prevent the unauthorized disclosure or use of confidential information. Furthermore, if any of our employees, consultants, suppliers, or other third parties with whom we have a business relationship breach the terms of any of these agreements, we may not have adequate remedies for any such breach. It is also possible that our confidential or other proprietary information could be obtained by third parties as a result of breaches of our physical or electronic security systems. Even where remedies are available, enforcing a claim that a party illegally disclosed or misappropriated our confidential or other proprietary information is expensive and time consuming, and the outcome is unpredictable. In addition, we cannot be certain that our competitors will not independently develop same or similar technology, obtain the same or similar intellectual property rights, or design around intellectual property rights of ours.

For example, on August 13, 2024, X-energy, through counsel, notified Ultra Safe Nuclear Corporation (“Ultra Safe”) of our belief that Ultra Safe could be infringing certain of our patents relating to the fabrication. Ultra Safe subsequently filed for bankruptcy and we have filed an objection and motion to set aside rights to Ultra Safe’s debtors’ bidding procedures and sale before the Delaware Bankruptcy Court in connection to the potential sale of the certain assets believed by us to be infringing on our patents. While we are seeking to resolve the matter, an adverse outcome or prolonged litigation could materially impact our ability to operate or commercialize our technology.

Our success depends in large part on our ability to obtain and enforce patent protection for our SMR technology. We either own or have license rights to certain intellectual property applicable to our SMR technology, including issued patents and patent applications. However, merely holding patents and patent applications on our nuclear power reactors and fuel technologies is not a guarantee of protection. Furthermore, we cannot be certain that our patent applications will result in patents being issued. Even if existing or future issued patents will afford protection against competitors with similar technology, during the patent prosecution process, a patent office may require us or our licensors to narrow the claims of our or our licensors’ patent applications. This may limit the scope of patent protection and our licensors’ ability to assert patent infringement if the patent is subsequently issued. In some cases, a patent may not be issued if we or our licensors are unable to overcome rejections from a patent office, which may have a material

adverse effect on our ability to prevent others from commercially exploiting products similar to ours. If we are pursuing patent rights by filing a patent application, we or our licensors may lose trade secret protection if we have otherwise been protected had a patent not been sought, and third parties may be able to use the published information in our patent application. Additionally, even if we obtain a patent in one jurisdiction (e.g., the U.S.), we cannot guarantee that we will obtain a corresponding patent in another jurisdiction (e.g., China) as patent laws differ from jurisdiction to jurisdiction. Additionally, maintaining and enforcing patent rights can involve complex legal and factual questions, and may be subject to litigation.

cases. For example, third parties may challenge the validity or enforceability of our or our licensors' patents at a tribunal such as the Patent Trial and Appeal Board at the U.S. Patent and Trademark Office or a federal court. Third parties may prevail in invalidating a patent or preventing a patent application from being issued as a patent. Furthermore, issued patents or patent applications owned by third parties exist in the field in which we have developed, are developing or will develop in the future our technology. In addition, the risk of infringing those patents, those patents may also be used as a basis to invalidate our patents or prevent our patent applications from issuing as patents.

Even if our patent applications succeed and we are issued patents, it is still uncertain whether our patents will be contested, circumvented, infringed upon, invalidated or limited in scope in the future. The rights granted under any issued patents may not provide us with meaningful protection of our technology advantages, and some foreign countries provide significantly less effective patent enforcement than the U.S. In addition, the claims of our issued patents may not be broad enough to prevent others from developing technologies that are similar or that achieve results similar to ours. The intellectual property rights of others could also bar us from licensing and exploiting our issued patents. In addition, patents issued to us may be infringed or designed around by others and others may obtain patents that we can't license or design around, either of which would increase costs and may adversely affect our operating financial condition, operating results and future prospects.

We currently enjoy only limited geographical protection with respect to certain issued patents and may not be able to protect our intellectual property rights throughout the world.

We do not have worldwide patent rights for our SMRs and related proprietary technologies. Intellectual property rights are territorial. Accordingly, we may not be able to protect our intellectual property rights in certain jurisdictions. Filing, prosecuting and defending patents on our SMRs in multiple jurisdictions can pose several challenges. First, procuring patent rights in multiple jurisdictions can be costly and prohibitive because individual patent offices in different jurisdictions examine each patent application separately. Once a patent is issued, we or our licensors will also have the continued obligation to pay maintenance fees periodically to avoid patents from becoming abandoned or lapsed. Second, the scope of claims in patents may vary from jurisdiction to jurisdiction. For instance, certain patent offices may require narrower claims, resulting in patent rights that are less extensive. Further, we may not be able to obtain patents in some jurisdictions even if we obtain patents in other jurisdictions. Accordingly, our technology may operate in countries where we do not have patent protection and can freely use our technology and discoveries in such countries to the extent such technologies and discoveries are publicly available or disclosed in countries where we have issued patents or patent applications.

Further, many countries have compulsory licensing laws under which a patent owner may be required to grant licenses to third parties. Many countries also limit the enforceability of patents against certain government agencies or government contractors. In these countries, the patent owner may have limited remedies and this could materially diminish the value of such patent. If we or any of our licensors are forced to grant licenses to third parties with respect to any patents relevant to our business, our competitive position may be impaired, and our business and financial condition may be adversely affected.

We may be subject to intellectual property rights claims by third parties, which are extremely costly to defend and could require us to pay significant damages and could limit our ability to use certain technologies.

We cannot be certain the conduct of our business does not or will not infringe, misappropriate or otherwise violate the intellectual property rights of a third party. Third parties, including our current and future competitors, may currently hold or obtain in the future patents, trademarks/service marks or other intellectual property rights that cover aspects of our proprietary technology or other intellectual property that we use in our business.

would prevent, limit or interfere with our ability to develop our intellectual property and develop, import, offer to sell or sell our SMRs and related technology, which may have a material effect on our business and financial condition. From time to time, we have received and may receive communications from holders of patents or other intellectual property rights inquiring whether we are infringing their intellectual property rights, seeking court declarations that we do not infringe their intellectual property rights or requesting us to obtain a license from them. If we are determined to have infringed a third party's intellectual property rights, we may be required to do one or more of the following among other things: (i) cease selling, incorporating or using SMRs that are found to be infringing; (ii) pay substantial damages; (iii) pay for and obtain a license from the holder of the infringed intellectual property right, which may not be available on reasonable terms or at all; or (iv) redesign part or all of our technology. In the event of a successful claim of infringement against us and our failure or inability to obtain a license to the infringed technology, our business, financial condition, operating results and future prospects may be materially adversely affected. In addition, any litigation or claims, whether or not valid, could result in substantial costs and diversion of resources and management's focus and attention.

We also license patents and other intellectual property from third parties, and we may face claims that the use of this intellectual property infringes the rights of other third parties. In such cases, we may be required to obtain indemnification from the licensors under our license contracts with those licensors, or otherwise to settle the claims. However, our rights to indemnification or damages may be unavailable or insufficient to cover our losses, depending on our use of the technology, whether we choose to retain control over our technology, litigation, and other factors.

We review and assess third-party patents and other intellectual property rights relevant to our SMR and fuel fabrication processes. However, we may not identify all relevant third-party patents or may incorrectly interpret their scope, relevance, or expiration. Any such oversight could result in third parties bringing infringement claims, require us to modify our technology, or limit our ability to develop, manufacture, sell or market our SMRs and related products. This could result in costly litigation, delays, or the need to obtain licenses on unfavorable terms, any of which could adversely affect our business, financial condition and results of operations.

In addition, there are several circumstances under which a patent application may not be published or accessible to us or our licensors. For example, patent applications in the U.S. and many foreign countries are typically not published until 18 months after filing, but some patent applications in the U.S. are maintained in secrecy until the patents are issued. Publications in the scientific literature are often published behind actual discoveries. Therefore, we cannot be certain that others have not filed patent applications covering technology covered by our issued patents or patent applications, or that we were the first to develop the technology or to file a patent application covering the technology. Our competitors may have filed or in the future file, patent applications covering our SMRs or technology similar to ours without our knowledge. Any such patent application may have priority over our patent applications or patents, which could require us to procure rights to issued patents covering such technologies in order to avoid infringement.

We may be subject to claims of ownership and other rights to our patents and other intellectual property from third parties.

While we require our employees, consultants, contractors and other third parties to assign to us intellectual property conceived by them to us in the event that the intellectual property is not otherwise owned by us, those agreements may not be effective or honored, and our attempts to enforce or assign intellectual property may be challenged or breached. Moreover, there may be some circumstances where we are unable to negotiate for such ownership rights, or where others misappropriate those rights.

We may be subject to claims that former or current employees, consultants, contractors or other third parties have an interest in our patents or other intellectual property. Litigation may be necessary to resolve these claims challenging inventorship and ownership. Furthermore, we may enter into agreements that limit the scope of our rights in such intellectual property. If we fail in defending any such claims, we may be required to pay monetary damages, we may lose exclusive ownership of, or the right to use or license, our intellectual property. Such an outcome could have a material adverse effect on our business. E

successful in defending against such claims, litigation could result in substantial costs and be to management and other employees.

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Risks Relating to Compliance with Law, Government Regulation and Litigation

Our business and the businesses of our customers are subject to the policies, priorities, regulations, funding levels of multiple governmental entities and may be negatively impacted by any change there

We and our customers are subject to a wide variety of complex laws and regulations relating to various aspects of our business, including with respect to the use, possession and manufacturing of nuclear materials; design, manufacture, operations, marketing and export of nuclear technologies; employment; labor; tax; data security of the operational and information technology we use; the U.S. Foreign Corrupt Practices Act and other applicable anti-bribery laws; health and safety; zoning and environmental regulations. Our facilities and our customers' projects are subject to various regulations regarding matters relating to human health and safety, including, among others, wastewater, stormwater, air emissions, investigation and cleanup of contaminated sites, and storage of hazardous materials, including petroleum. We must comply with the Occupational Safety and Health Act (OSHA). Laws and regulations at the federal, state and local levels frequently change and are often interpreted in different ways, particularly in relation to new and emerging industries, and we cannot reasonably predict the impact from, or the cost of compliance with, current or future law or regulatory or administrative changes. We cannot guarantee that our measures to monitor these developments and the time and resources we spend to comply with laws, regulations, and guidelines will be satisfactory to regulators or other third parties, and our customers, who are also subject to extensive government regulation. Our efforts to comply with changing laws and regulations may result in increased general and administrative expenses and a diversion of management time and attention. Our insurance and compliance costs may increase as a result of changes in environmental, health and safety laws and regulations or changes in enforcement. Moreover, changes in law or the interpretation of existing laws, the imposition of new or additional regulations, or the enactment of any new or more stringent legislation that impacts our business could require us to change the way we operate and could have a material adverse effect on our sales, profitability, cash flows, financial condition, and lead to regulatory delays that could impact our ability to obtain licenses, certificates, and permits, approvals, and/or certifications from regulatory agencies (collectively "Regulatory Approvals").

Failure to comply with laws, regulations or Regulatory Approvals may result in civil or criminal penalties, administrative enforcement actions and penalties, the imposition of remedial obligations, including the installation of additional pollution control equipment, the issuance of orders enjoining our operations, private civil liability for property damage and personal injuries, or the suspension or revocation of those Regulatory Approvals, any of which would have a negative impact on our or our customers' business and could prevent us from operating our business. With respect to our TX-1, we require regulatory approval from the NRC to construct and operate the facility under a nuclear materials license, in addition to local and state regulatory requirements. While we have already obtained NRC approval, we must comply with the regulatory requirements to maintain this approval and may need to extend the license in the future. Any of these Regulatory Approvals may be subject to denial, revocation or modification under various circumstances, including:

- failure to provide adequate financial assurance for closure;
- failure to comply with environmental, health and safety laws and regulations or permit conditions;
- local community, political or other opposition;
- executive action; and
- legislative action.

Failure on our or our customers' part to comply with these laws and regulations, obtain Regulatory Approvals, or receive exemptions from such regulations when available could result in penalties, or the inability to operate our business. Any delays in regulatory actions also could delay the manufacture of our TRISO-X fuel and thereby affect our financial performance.

Our SMRs are subject to regulations in all jurisdictions related to nuclear safety, environmental and safety and financial qualification. Regulatory Approvals, such as construction permits and licenses issued by the NRC, are necessary for our customers to construct and operate our SMRs. Our customers to deploy SMRs rely on timely receipt of such Regulatory Approvals in the jurisdictions in which they do

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business. Such regulatory approval processes may be subject to change, can be technically complex to address, may result in the imposition of conditions that impact the financial viability of our SMRs, and may also provide opportunities for third parties to lodge objections or seek more stringent requirements for our products that, in each case, could hinder or prevent developments of our or our customers' products.

Over the past several years, Congress has enacted laws that aim to put nuclear energy on a level playing field with respect to government incentives, tax credits, and other financial instruments to make nuclear energy more economically competitive with other energy sources. These incentives were signed into law through the Infrastructure Investment and Jobs Act (IIJA) and the OBBBA. To the extent these government financial tools are incorporated into our business model and that of our customers, our benefit from these subsidies are subject to cancellations and sunsets or changes to sunset provisions by Congress and the current administration. The impact of changes to these financial benefits could significantly impact the demand for our products.

The reduction or elimination of Section 48C, the Qualifying Advanced Energy Project Credit under the Reduction Act of 2022 ("48C"), investment tax credits ("48E") and production tax credits ("45Y") received through the U.S. government and Clean Technology Investment Tax Credits ("CT ITCs") received through the U.S. government related to clean energy solutions could adversely affect our business and reduce the demand for our Xe-100 and other technologies.

Our business model and financial performance are significantly dependent on the availability of 48C, 48E, CT ITCs and 45Y. The reduction, denial, delay, recapture, or adverse modification of federal and local tax incentives for clean energy — including 48C, 48E, 45Y, and CT ITCs — could adversely affect our business and reduce demand for our Xe-100 and other technologies. Our business model and the timing of customer projects, supplier expansions, and financing structures are significantly dependent on the availability, amount, timing, and monetization of these incentives.

Many of our counterparties rely on the receipt and monetization of 48C, 48E, CT ITCs to finance or price projects involving our technologies. If these parties do not receive, cannot receive, or later lose expected credits — due to changes in law, allocation caps, eligibility requirements, regulatory failures, or market conditions — they may delay, renegotiate, or terminate contracts, reduce or default on obligations to us. In addition, changes in transferability, direct pay rules, or market conditions for these credits could further diminish project returns and impair demand.

Any reduction or unavailability of these incentives, or counterparties' inability to obtain them as anticipated, would likely reduce the demand for the Xe-100, SMRs and nuclear energy in general, which would have a material, adverse impact on our business, financial condition, operations, and future prospects.

The U.S. government's budget deficit and the national debt, as well as any inability of the U.S. to complete its budget or appropriations process for any government fiscal year could have an adverse impact on our business, financial condition, results of operations and cash flows.

The U.S. government's budget deficit and the national debt, along with any negotiated increase or suspend the so-called debt ceiling, as well as any inability of the U.S. government to complete its budget process for any government fiscal year and consequently having to shut down or reduce funding levels equivalent to its prior fiscal year pursuant to a "continuing resolution," could have an adverse impact on our business, financial condition, results of operations and cash flows.

Uncertainty will continue to exist regarding how future budget and program decisions, including the energy spending priorities of the U.S. government, what challenges budget reauthorization present for the energy industry and whether annual appropriations bills for all agencies will be enacted for U.S. government fiscal 2026 and thereafter. Some of the changes in the political environment, including the change to the leadership within the current administration, and any resulting uncertainty of future policy or priorities and resultant funding. There can be no assurance that increases in funding for our programs currently experience will continue, and any plateau or reduction in funding for our programs could adversely affect our ability to execute our strategy, meet milestones, and achieve projected financial goals. The U.S. government's

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budget deficit and the national debt could have an adverse impact on our business, financial condition, results of operations and cash flows in a number of ways, including the following:

- the U.S. government could reduce or delay its spending on, reprioritize its spending or fail to appropriate funding to decline to provide funding for the government programs in which we participate, or fail to provide the level of funding as anticipated;
- U.S. government spending could be impacted by arrangements similar in effect to the current arrangement, which increases the uncertainty as to U.S. government spending priorities and levels; and
- we may experience declines in revenue, profitability and cash flows as a result of reduced orders or payments or other factors caused by economic difficulties of our customers and our customers, including U.S. federal, state and local governments.

Other contributing factors that could impact the company's financial situation are rising interest rates, as more U.S. government spending must be appropriated to servicing the national debt or the impact of tariffs on our supply chain as we begin to long lead procurements on certain nuclear power systems. Budget and program decisions made in this environment would have long-term implications for us and the entire nuclear energy industry.

We are required to comply with numerous laws and regulations, some of which are highly complex, and our failure to comply could result in fines or civil or criminal penalties or suspension or debarment by the U.S. government, which could result in our inability to continue to work on or receive U.S. government contracts, which could adversely affect our results of operations.

We must comply with laws and regulations relating to the formation, administration, and performance of U.S. government contracts, which affect how we do business. Such laws and regulations may impose added costs on our business and our failure to comply with those laws and regulations could result in civil or criminal penalties, including termination of our U.S. government contracts, including our participation in the Agreement, and/or suspension or debarment from contracting with federal agencies. Some significant laws and regulations that affect us include:

- the Federal Acquisition Regulation (the “FAR”), and agency regulations supplemental to the FAR, which regulate the formation, administration, and performance of U.S. government contracts; for example, the FAR 52.203-13 requires contractors to establish a Code of Business Ethics, implement a comprehensive internal control system, and report to the government if there is credible evidence that a principal, employee, agent, or subcontractor, in connection with a government contract, has violated certain federal criminal laws, violated the civil False Claims Act, or has received a significant overpayment;
- the False Claims Act, which imposes civil and criminal liability for violations, including monetary penalties, for, among other things, presenting false or fraudulent claims for payment or approval;
- the False Statements Act, which imposes civil and criminal liability for making false statements to the U.S. government;
- the Truthful Cost or Pricing Data Statute (formerly known as the Truth in Negotiations Act), which requires certification and disclosure of cost and pricing data in connection with the performance of certain contracts, modifications, or task orders;
- the Procurement Integrity Act, which regulates access to competitor bid and proposal information and certain internal government procurement sensitive information, and our ability to receive compensation to certain former government procurement officials;
- laws and regulations restricting the ability of a contractor to provide gifts or gratuities to officials of the U.S. government;
- post-government employment laws and regulations, which restrict the ability of a contractor to recruit and hire current employees of the U.S. government and deploy former employees of the U.S. government;

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- laws, regulations, and executive orders restricting the handling, use and dissemination of information that is classified for national security purposes or determined to be “controlled unclassified information,” “for official use only” and the export of certain products, services, and technical data; and requirements regarding any applicable licensing of our employees involved in such work;
- laws, regulations, and executive orders regulating the handling, use, and dissemination of information that is identifiable information in the course of performing a U.S. government contract;
- international trade compliance laws, regulations and executive orders that prohibit trade with certain sanctioned entities and require authorization for certain exports or imports in order to protect national security and global stability;
- laws, regulations, and executive orders governing organizational conflicts of interest that may restrict our ability to compete for certain U.S. government contracts because of the nature of the work we currently perform for the U.S. government or may require that we take measures such as terminating or taking off certain employees or restricting their future work activities due to the current work we are performing under a U.S. government contract;
- laws, regulations and executive orders that impose requirements on us to ensure compliance with certain requirements and protect the government from risks related to our supply chain;
-

laws, regulations and mandatory contract provisions providing protections to subcontractors seeking to report alleged fraud, waste, and abuse related to a government contract and

- the FAR Cost Accounting Standards and Cost Principles, which impose accounting and requirements that govern our right to reimbursement under certain cost-based U.S. government contracts and require consistency of accounting practices over time.

In addition, the U.S. government adopts new laws, rules, and regulations from time to time that may have a material impact on our results of operations. Adverse developments in legal proceedings on matters relating to, among other things, cost accounting practices and compliance interpretations and statute of limitations, could also result in materially adverse judgments, withheld payments, penalties, or other unfavorable outcomes.

Our performance under our U.S. government contracts and our compliance with the terms of our contracts and applicable laws and regulations are subject to periodic audit, review, and investigation by various agencies of the U.S. government, and the current environment has led to increased scrutiny and sanctions for non-compliance by such agencies generally. In addition, from time to time we may report potential or actual violations of applicable laws and regulations to the relevant government authority. Any such report of a potential or actual violation of applicable laws or regulations may result in an audit, review, or investigation by the relevant agencies of the U.S. government. If such an audit, review, or investigation uncovers a violation of a law or regulation, or improper or illegal activities related to our U.S. government contracts, we may be subject to civil or criminal penalties or administrative sanctions, including the termination of contracts, forfeiture of profits, the triggering of price reductions, withholding or suspension of payments, fines and suspension, or debarment from contracting with government agencies. Such penalties and sanctions are not uncommon in the industry, and there is uncertainty as to the outcome of any particular audit, review, or investigation. If we incur a major civil or administrative sanction or otherwise suffer harm to our reputation, our profitability, cash flow, and future prospects could be materially and adversely affected.

Further, if the U.S. government were to initiate suspension or debarment proceedings against us, or if we are indicted for or convicted of illegal activities relating to our U.S. government contracts, an audit, review, or investigation, we may lose our ability to be awarded contracts in the future, the renewals of existing contracts for a period of time which could materially and adversely affect our results of operations or financial condition. We could also suffer harm to our reputation if allegations of wrongdoing were made against us, which would impair our ability to win awards of contracts in the future and the renewals of existing contracts.

Uncertain global macro-economic and political conditions could materially adversely affect our results of operations and financial condition.

Our results of operations are materially affected by economic and political conditions internationally, including inflation, deflation, interest rates, availability of capital, energy and commodity prices, trade laws and the effects of governmental initiatives to manage economic conditions. Our potential customers may delay or decrease spending on our products and services as their budgets are impacted by economic conditions. The inability of current and potential customers to pay for our products and services may adversely affect our earnings and cash flows.

We are subject to stringent U.S. export and import control laws and regulations. Unfavorable changes to these laws and regulations or U.S. government licensing policies, our failure to secure timely U.S. government approvals under these laws and regulations, or our failure to comply with these laws and regulations could have a material adverse effect on our business, financial condition, and results of operations.

The inability to secure and maintain required export licenses or authorizations could negatively impact our ability to compete successfully or market our SMR technology and TRISO-X fuel for certain applications outside the U.S. For example, if we were unable to obtain or maintain our licenses to export certain nuclear hardware, software, source code, or technical assistance, we would be effectively prevented from exporting our SMR and TRISO-X fuel technology to non-U.S. locations, limiting our customer base outside the U.S. and providing a competitive advantage to international SMR suppliers. In such cases, export restrictions could also require us to implement design changes to our SMRs to address issues with our domestic supply chain, which may increase costs or result in delays in delivery of new SMRs and subsequent additional SMRs when ordered. Certain key components of our SMRs are currently manufactured outside the U.S., and the imposition of sanctions, tariffs, or material changes in export requirements on a nation-by-nation basis, on materials or supplied components for our SMRs, could have a material adverse effect on our operations. Additionally, we may require U.S. approval to import certain materials and components that may be predominantly produced outside of the U.S. Failure to comply with import and export requirements and regulations could expose us to civil and criminal penalties, fines, investigations, more onerous compliance requirements, loss of export privileges, and loss from government contracts or limitations on our ability to enter into contracts with the U.S. Any changes in export control regulations or U.S. government licensing policy, such as that resulting from the implementation of U.S. government commitments to multilateral control regimes, may restrict our operations, cause delays, conditions or unexpected requirements may increase costs for us or our customers and create uncertainty regarding the ability to deploy our technology in a predictable way, which may adversely impact our competitiveness.

We are part of the highly regulated nuclear power industry. Our fuel designs differ from fuels currently used by commercial nuclear power plants, and our SMR designs similarly differ from reactor designs currently in operation, including with respect to potential industrial applications. As a result, the regulatory approval process for our nuclear power plants and for others that operate with our nuclear fuels will be more extensive and prolonged than for those of other nuclear power plants of this kind and therefore may experience delays and increased costs as part of the regulatory review process. The regulatory acceptance of our nuclear fuels may also be hindered.

All entities that operate nuclear power facilities, fabricate nuclear fuel, or transport nuclear materials within the U.S. are subject to the jurisdiction of the NRC (except for those facilities and applications that are regulated by the DOE or the U.S. Department of Defense (the “DOD”)), and entities performing nuclear activities in other countries are subject to regulation by the NRC’s counterparts around the world. Our fuel designs also differ significantly from the reactors used today by commercial nuclear power facilities. These differences may result in a more extensive and prolonged review by the NRC and its counterparts around the world, potentially causing delays in fuel and reactor development program and in commercial operation.

TRISO-X fuel cannot be used in existing nuclear reactors, and no currently available commercial fuel will work in our SMRs. Our fuel development timeline depends on the relevant nuclear regulatory requirements and approving technical information and documentation that is generated during the fuel development program. There is a risk that regulators may require additional information regarding the fuel development program performance that could necessitate unplanned analytical or experimental work, resulting in schedule delays and increased research and development costs. Similarly, our reactor development timeline depends on the relevant nuclear regulatory requirements and approving technical information and documentation that is generated during the reactor development program.

regulator accepting and approving technical information and documentation about our reactor in the course of any design-specific licensing, certification, approval or similar process, or failure to obtain such licensing. There is a risk that regulators may require additional information regarding the reactor design or performance that could cause delays and additional costs. They may impose operational restrictions such as lower fuel burnup limits, power levels, or more frequent maintenance and testing of components. These could adversely affect the commercial viability of our products.

We must obtain governmental licenses to possess and use radioactive materials, including isotopes used in our TX-1 operations. Failure to obtain or maintain, or delays in renewing, such licenses could impact our ability to fabricate TRISO-X fuel for our customers, who will initially be reliant on us for TRISO-X fuel, and have an adverse effect on our business, financial condition and results of operation.

In February 2026, TRISO-X received an initial 40-year Special Nuclear Material License under Part 70 from the NRC enabling TRISO-X to commercially manufacture X-energy's TRISO-X fuel at a facility intended to manufacture the fuel for our reactor designs and for other advanced reactors that utilize coated particle fuel. We expect this license to also cover our TX-2 facility if built and planned on the same site. Failure to maintain compliance with the NRC license during commercial operation of TX-1 or to appropriately extend such license could interfere with or prevent production of TRISO-X fuel to initial and subsequent Xe-100 reactors, inhibit interest in our reactors by future customers, impair our ability to produce and market other coated particle fuels, or impact our ability to build and operate TX-2, our second facility. We also face the risk that the NRC could impose conditions in future extensions of the license that are not acceptable to us, which could materially and adversely affect our business and commercial viability of our nuclear fuel products.

Given the nature and length of the permitting process, significant turnover and organizational changes within the NRC's workforce, including at the Commission, project management, and technical staff, could cause a reduced and transitioning workforce and could lengthen review timelines for future extensions of the license — particularly in light of the increasing number of applications being submitted to the NRC by other companies. The combination of staffing adjustments and a heavier workload could cause delays in the review and approval of any future license extension applications, which would impact our business and ability to develop our and our customers' projects.

The operations of our planned TX-1 facility in Tennessee, and any future fuel fabrication facilities, will be regulated by U.S. federal and state level governmental authorities, including the NRC as well as the State of Tennessee and any other state jurisdictions in which we may establish operations. Our operations could be significantly impacted by changes in government policies and priorities.

The NRC has the authority to issue notices of violation for violations of the Atomic Energy Act of 1954, as amended (the "Atomic Energy Act"), NRC's regulations and conditions of licenses, orders, and compliance, and orders. The NRC has authority to impose civil penalties (the maximum amount of which is adjusted annually to account for inflation) or additional requirements and to order cessation of activities for violations of these requirements. Penalties under the NRC regulations and applicable agency orders could include substantial fines, imposition of additional requirements, or withdrawal or suspension of licenses or certificates. Any penalties imposed on us could adversely affect our results of operations and liquidity. The NRC also has the authority to issue new regulatory requirements or to change existing requirements. Changes to the regulatory requirements also could adversely affect our results of operations and financial condition.

Our TX-1 in Oak Ridge, Tennessee will also be regulated by the State of Tennessee pursuant to the Atomic Energy Act authority transferred under the NRC's Agreement State Program and applicable state regulations. Any future fuel fabrication facilities will be highly regulated in the location in which they are located.

Additionally, certain aspects of our current fuels R&D activity take place within DOE's ORNL National Laboratory (separate and apart from our TX-1 within the City of Oak Ridge). These activities are subject to DOE regulation and contractual requirements. Changes to those requirements could materially affect our operations.

Changes in federal, state or local government laws, regulations, policies and priorities can impact our operations. These could include changes in interpretations of regulatory requirements, increased enforcement activities, changes in budgetary priorities, changes in tax laws and regulations and other actions or inaction. Any of these local, state and federal agencies may have the authority to impose criminal penalties and our designs, facilities and projects must be approved by various regulatory bodies that have not been approved. The public has the ability to intervene in licensing proceedings before the construction of a fabrication facility or a reactor.

The Xe-100 design has not yet been approved or licensed for use at any site by the NRC, the national regulatory body, and approval or licensing of these designs is not guaranteed. Under the Energy Act and the implementing NRC regulations, members of the public, including state, provincial and federal governments, have the right to request a public hearing or file a petition to intervene in connection with the NRC's review of a license or permit application. Such requests may oppose the issuance of a license or permit in whole, or challenge specific aspects of the application or the NRC's review. These proceedings can be complex and time-consuming, requiring the devotion of significant management attention and incurring substantial legal and technical expenses to prepare responses, participate in hearings and address contentions. These proceedings may also delay or prevent the issuance of required Regulatory Approvals for any of our or our customer's projects, including TX-1 or a customer's Xe-100. The outcome of these proceedings is inherently uncertain and could result in the imposition of additional licensing conditions or, in some cases, the denial of a Regulatory Approval. Any such delays or adverse outcomes could materially affect the timing, cost and feasibility of the projects and operations. Additionally, these hearings and proceedings may also have adverse impacts on the local population's perception of the safety of our facilities, potentially impacting their timely deployment and efficient operations.

For example, as of August 12, 2025, one such public entity has filed a petition to intervene in connection with a hearing in connection with the pending application for a construction permit for one of our projects which will utilize our SMR technology under NRC Docket No. 50-614-CP. The petition challenges certain safety and environmental findings contained in the construction permit application. As of August 22, 2026, the NRC's Atomic Safety and Licensing Board granted Waterkeeper's intervention in the proceedings, and litigation two contentions concerning financial qualifications, and rejected the remaining contentions. As of the date of this prospectus, the adjudicatory proceeding is ongoing. The Board also directed the applicant to proceed with mandatory disclosures and scheduling, and indicated the proceeding will be completed by October 10 C.F.R. Part 2, Subparts C and L, which may extend timelines and increase costs. In addition, the applicant has received admissible contentions from Waterkeepers, an advocacy group, related to their concerns about our fuel fabrication facility plans. Should such a petition result in delay or denial of regulatory approvals, our business would be materially adversely impacted. We are also subject to Canadian regulatory review processes for a portion of our business seeking approval in Canada. We must work through the Canadian review processes and engage with the NRC to address technical, policy, and other matters ahead of formal application reviews for new projects.

To date, the Xe-100 has not yet been licensed, certified or approved by the U.S. NRC, U.S. Nuclear Regulation ("ONR") or the Canadian Nuclear Safety Commission ("CNSC"), and no operating NRC or CNSC-regulated reactor uses high temperature gas-cooled reactor technology or TRISO-X fuel.

If the NRC, ONR or CNSC disagrees with our, or our customers', licensing approach or the scientific bases supporting the nuclear safety and environmental impact evaluations, the construction permit application or license application processes could take longer than currently expected, or a license may not be issued at all, which would materially and adversely affect our business. Further, the NRC or CNSC could impose conditions in a license that are not favorable or acceptable to us or our customers, which could

and adversely affect our business. Any delays, conditions or unexpected requirements may increase costs for us or our customers and may result in uncertainty regarding the ability to deploy our technology in a predictable way, which may adversely impact our competitiveness.

Since taking office in January 2025, President Trump has announced, revised, paused, and rescinded various executive orders and tariffs which could have wide-reaching economic and regulatory impacts, including but not limited to causing the NRC to change its regulatory standards and through other means, unexpected changes to

the NRC's requirements may result in substantial delay or increased costs for us or our customers, which may adversely impact our competitiveness.

Even if the Xe-100 is licensed in the U.S., U.K. and Canada, we must still obtain approvals on a country-by-country basis to deploy this reactor technology, which approvals may be delayed or denied or which may require modification to our design.

Even if the Xe-100 is licensed, certified and/or approved in the U.S., U.K. and Canada, we may not be able to deploy our technology in other countries, we must first obtain regulatory approvals for our technology in those countries. The regulatory framework to obtain approvals is complex, varies from country to country, and may involve authorities on a subnational or local level. Timelines are likely to be long for the deployments of our technology in any jurisdiction, as regulatory agencies may not be familiar with our technology and how it differs from the technology used in legacy nuclear power facilities. Moreover, other countries' approval processes may differ markedly from the NRC process or the CNSC process, and may require that we alter aspects of our design before providing approval. Design modifications to meet specific requirements could adversely impact our ability to achieve standardization and therefore increase our design, construction, and/or operational costs. Denial or delay in approvals abroad could materially adversely affect our business outside of the U.S. and Canada.

Our operations involve the use, transportation and disposal of toxic, hazardous and/or radioactive materials, which could result in liability without regard to fault or negligence.

Our operations involve the use, transportation, and disposal of toxic, hazardous and/or radioactive materials. A release of these materials could pose a health risk to humans, plants and animals, and the environment. If an accident were to occur, its severity would depend on the volume and location of the release and the speed of corrective action taken by emergency response personnel, as well as factors beyond our control, such as weather and wind conditions.

Under federal, state and local laws and regulations, a current or former owner or operator of property may be liable for costs to remediate contamination resulting from the presence of hazardous substances, wastes or petroleum products. These laws and regulations continue to evolve and have generally become more stringent over time. The costs associated with remediating contamination can be substantial, and liability under such laws is strict, joint and several and may attach whether or not the owner or operator knew of or caused such contamination. Moreover, the presence of contamination can expose us to third-party claims for property damage or bodily injury, subject our properties to eminent domain of the government for damages and cleanup costs, impose restrictions on the manner in which we use our properties, and materially adversely affect our ability to sell, lease, insure or develop our properties. We may be liable for costs of remediating third-party disposal sites to which we arranged for the treatment of hazardous substances without regard to whether such disposal occurred in compliance with environmental laws. These matters could have an adverse effect on our financial condition.

Additionally, we may be responsible for decontamination or decommissioning of facilities, conduct, or previously conducted, operations. Activities of our contractors, suppliers or other parties similarly may involve toxic, hazardous, and radioactive materials, and we may be liable under applicable law, to contribute to remedy damages or other costs arising from such activities or the decontamination or decommission of third-party facilities.

In the U.S., the nuclear liability law codified at 42 U.S.C. 2210 (along with subsequent amendments, the “Price Anderson Act”) and applicable NRC regulations and corresponding insurance requirements channel liability to the nuclear operator of a nuclear power plant for third-party offsite damages arising from a nuclear incident or a precautionary evacuation due to a possible or actual nuclear incident. Unlike global nuclear liability regimes wherein operators are subject to nuclear liability channeling and financial protection regimes, such as required insurance policies or indemnification, to cover the operator’s financial risk in the event of a nuclear incident that results in third-party offsite liability. If, however, an incident or precautionary evacuation is not covered by the nuclear liability limiting regulatory framework we could be financially liable for damages arising from such incident or evacuation, which could have an adverse effect on our results of operations and financial condition.

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The Price Anderson Act does not, however, cover onsite loss or damage to property due to a nuclear incident. Rather, the NRC, like many nuclear regulators around the world, requires nuclear operators to maintain onsite property damage insurance for loss or damage to property due to a nuclear incident. If an incident resulting in onsite property damage is not otherwise covered by the mandatory insurance maintained at the facility or the damages exceed policy coverage limits, then we could be potentially liable for damages arising from such incident, which could have an adverse effect on our results of operations and financial condition. We cannot provide assurance that any insurance we obtain will be adequate to cover our losses or liabilities or that such coverage will continue to be available, or available on acceptable rates.

There is no assurance that our contractual limitations on liability will be effective in all jurisdictions. The costs of defending against a claim arising out of a nuclear incident or precautionary evacuation not otherwise covered by insurance, and any damages awarded as a result of such claim, could adversely affect our results of operations and financial condition.

Unresolved spent nuclear fuel storage and disposal issues and associated costs could have a significant impact on X-energy’s business operations if potential Xe-100 customers view the risks associated with storage and costs as unacceptably high. Additionally, U.S. policy related to storage and disposal of used fuel from the TRISO plant and/or negative customer perception of risks relating to these policies could have a significant impact on our business prospects, financial condition, results of operations and cash flows.

During the licensing process, a nuclear power plant operator must indicate how it will manage its power plant and must have a “standard agreement” with the DOE related to the storage of fuel created during its operating life. Therefore, the requirement for our customers’ facilities to manage disposal of fuel may create challenges related to timeline and optimal use of our TRISO technology. The Nuclear Waste Policy Act (“NWPA”) of 1982 requires the DOE to take title to, and to provide for permanent geologic disposal of, spent nuclear fuel (“SNF”) and associated high-level nuclear waste (“HLW”) generated by domestic nuclear reactors. In 1987, Congress amended the NWPA to designate Yucca Mountain, in Nevada, as the only site that the DOE could consider for a permanent repository. The DOE has suspended the Yucca Mountain licensing process due to lack of congressional approval, and the site remains designated in federal law. Under the NWPA and the Standard Contracts DOE entered into with nuclear utilities, DOE remains obligated to provide for permanent disposal of all SNF

Interim storage of SNF and HLW is authorized under the NWPA and NRC regulations and construction and maintenance of NRC licensed SNF/ HLW storage facilities. While the costs of and maintaining these interim storage facilities can have a significant effect on the costs associated with waste storage and disposal for nuclear reactors, including X-energy’s reactors, these costs could be impacted by the timing of the opening of a disposal facility, as well as any possible future cost of interim storage or transportation requirements for SNF and other forms of HLW, and the extent to which operators are able to continue to successfully recover costs from DOE through breach-of-contract claims for its continued failure to provide for permanent disposal.

There are currently two consolidated interim storage (“CIS”) facilities under development for the interim storage of SNF/HLW. Both facilities — Holtec International’s HI-STORE in Mexico and Interim Storage Partners’ facility in Texas — have been issued licenses by the NRC for construction and operation. These licenses were challenged in federal court, and in June 2021 the Supreme Court upheld the NRC’s authority to issue the licenses; however, further engagement with the states and DOE is necessary to construct the projects. It is possible that SNF/HLW generated at an X-energy reactor could be stored at one of these CIS facilities; however, it is also possible that these CIS facilities are never built or become operational, or are unable to store such waste from an X-energy reactor. In either case, the waste would need to be stored onsite or at another interim SNF storage facility until a permanent disposal option became available, such as a U.S. government-determined permanent national repository or other government storage facility.

The establishment of a national repository for the storage and/or permanent disposal of SNF/HLW, the one previously considered at Yucca Mountain, Nevada, the timing of such a facility’s opening, the ability of such a facility to accept waste from an X-energy reactor, and any related regulatory action could impact the costs associated with our Xe-100 customers’ storage and/or disposal of SNF/HLW. In the absence of the establishment of a CIS for the storage of SNF/HLW, the timing of such a facility’s opening and

accept waste from an X-energy reactor, and any related regulatory action, could impact our costs associated with storage of SNF/HLW. These waste storage issues, and changes to the current waste management practices or changes to reactor operators’ ability to recover storage costs from DOE through litigation, could be material to X-energy’s operations if potential customers view waste disposal as problematic or a negative factor when considering an investment in an X-energy reactor.

Our SMRs may not qualify as low-emissions or emissions-free pursuant to regulatory or incentive programs that consider emissions on a lifecycle basis or that otherwise account for fuel cycle emissions or energy consumed

While our SMRs directly generate virtually no air emissions, including greenhouse gases, during operations, our SMRs may nonetheless not qualify as providers of emissions-free, carbon-free, or carbon or similar generating resources under emissions-limitation schemes that assess emissions on a lifecycle basis or that otherwise consider emissions from energy consumed in our fuel cycle because of the use of carbon in our TRISO-X fuel production and because our TRISO-X fuel and its feedstocks require substantial amounts of energy to produce. Our fuel fabrication facilities and our suppliers’ operations may rely on the local electric grid and its mix of generating sources for electricity or may rely on a power supply that does not provide a choice among electric generation sources. We cannot control the timing and electricity purchasing decisions of local electric utilities and their suppliers or of electric utilities or our third-party partners. Certain regimes, such as various sustainability “taxonomies,” may restrict eligibility based on a wider array of environmental objectives, which we cannot guarantee our customers will be able to meet. The failure of our SMRs to qualify for inclusion in emissions-based or climate change related emissions control schemes, or emissions-based incentive programs may result in higher costs or lower revenues for us or our customers, as well as eligibility for financing

providers focused on certain sustainability criteria, and may adversely impact the demand for our products from our customers, which could materially and adversely affect our business, financial condition, results and future prospects.

Changes in effective tax rates or adverse outcomes resulting from examination of our income or other tax returns could adversely affect our results of operations and financial condition.

We are subject to taxation by U.S. federal, state, and local tax authorities. Our future effective tax rate could be subject to volatility or adversely affected by a number of factors, including:

- allocation of expenses to and among different jurisdictions;
- changes to our assessment about our ability to realize, or in the valuation of, our deferred tax assets that are based on estimates of our future results, the prudence and feasibility of possible tax strategies, and the economic and political environments in which we do business;
- expected timing and amount of the release of any tax valuation allowances;
- tax effects of stock-based compensation;
- costs related to intercompany restructurings;
- changes in tax laws, regulations, or interpretations thereof;
- the outcome of current and future tax audits, examinations, or administrative appeals;
- lower than anticipated future earnings in jurisdictions where we have lower statutory rates compared to higher than anticipated future earnings in jurisdictions where we have higher statutory rates;
- limitations or adverse findings regarding our ability to do business in some jurisdictions.

Any changes in U.S. taxation may increase our effective tax rate and harm our business, financial condition, and results of operations. In particular, new income or other tax laws or regulations enacted at any time, which could adversely affect our business operations and financial condition. Further, existing tax laws and regulations could be interpreted, modified, or applied adversely to us.

The unavailability, reduction or elimination of government and economic incentives and credits could have a material adverse effect on our business, prospects, financial condition, results of operations, and cash flows.

Any unavailability, reduction, or elimination of government and economic incentives and credits, because of policy changes, or the reduced need for such incentives and credits due to the perceived safety of nuclear

energy or other reasons, may result in the diminished competitiveness of the alternative fuel industry generally or our reactors, software and services in particular. Any of the foregoing could materially and adversely affect the growth of the alternative fuel automobile markets and our business, prospects, financial condition, results of operations, and cash flows.

While certain tax credits and other incentives for alternative energy production, alternative nuclear energy have been available in the past, there is no guarantee these programs will be available in the future. Some of these tax credits and incentives require interpretations from government bodies, and changes could impact the applicability of these tax credits and incentives. Incentives provided by state authorities may have predetermined expiration dates, may conclude once allocated funds are exhausted, or could be reduced or discontinued due to changes in regulatory or legislative priorities. Con-

availability of tax credits or other government incentives and our ability and that of our competitors to benefit from these credits and incentives remain uncertain at this time.

We may become involved in litigation that may materially adversely affect us.

From time to time, we may become involved in various legal proceedings relating to matters outside the ordinary course of our business, including intellectual property, commercial, product liability, employment, class action, whistleblower, personal injury, property damage, and other litigation and governmental and other regulatory investigations and proceedings. Such matters could be time consuming, divert management's attention and resources from the operation of our business and incur significant expenses or liability or require us to change our business practices. Because of the potential risks, expenses and uncertainties of litigation, from time to time, we may settle disputes where we believe that we have meritorious claims or defenses. Because litigation is inherently unpredictable, we cannot assure you that the results of any of these actions will not have a material adverse effect on our business.

Some of our customers' projects will be regulated by U.S. federal energy regulators and may be materially adversely affected by changes in law, policy or regulation, or a failure to comply with laws or regulations.

Some of our customers' projects in the U.S. will be subject to regulation at the federal level by the Federal Energy Regulatory Commission ("FERC"). Under the Federal Power Act, FERC regulates the production of electric energy at wholesale in interstate commerce and the transmission of electric energy in interstate commerce. FERC rate regulation will subject projects making wholesale sales of energy to FERC rate regulations as well as certain risks, including the possibility that FERC may revoke or otherwise limit their authorizations to make sales if FERC determines that such a company and its affiliates exercise undue horizontal or vertical market power, create barriers to entry or engage in abusive affiliate transactions or market manipulation. Public utilities and their holding companies are subject to FERC regulatory requirements that impose administrative burdens and that can expose a public utility to criminal penalties for failure to comply with such requirements. Our projects will also likely be subject to reliability standards overseen by the North American Electric Reliability Corporation ("NERC"). Failure to comply with NERC reliability standards could lead to sanctions, including substantial monetary penalties.

Some of our customers' U.S. projects will be located within organized bulk power markets administered by electric grid operators and some of our U.S. project companies will sell power and participate in those markets. These electric grid operators each have their own rules set forth in approved tariffs and related documents that govern the functioning of the wholesale electricity markets they oversee, as well as interconnection to the transmission system. These electric grid operators, which act like quasi-regulatory bodies, can impose new rules or adopt new interpretations thereof that could have a material adverse effect on our business. Electric grid operators market rules regularly change and such changes may materially adversely affect our project companies' ability to sell energy, ancillary services at a beneficial price and materially adversely affect our business.

A failure to comply with U.S. energy laws or FERC, NERC or electric grid operator rules or regulations could have a material adverse effect on our customers, which could in turn have a material adverse effect on our business, including any existing or future financing arrangements.

Any actual or perceived failure to comply with new or existing laws, regulations and other requirements relating to the privacy, security and processing of Personal Information could adversely affect our business operations, or financial condition.

In connection with running our business, we receive, store, use and otherwise process information that relates to individuals and/or constitutes “personal data,” “personal information,” “personally identifiable information,” or similar terms under applicable data privacy laws (collectively, “Personal Information”), including from and about actual and prospective customers, as well as our employees and business partners. We therefore may be subject to laws, regulations and other requirements relating to the privacy and handling of Personal Information.

The application and interpretation of such laws, regulations, and other requirements are constantly evolving and are subject to change, creating a complex compliance environment. In some jurisdictions, requirements may be either unclear in their interpretation and application or they may have inconsistent or conflicting requirements with each other. Further, there has been a substantial increase in legislative and regulatory focus on data privacy and security in the U.S. and elsewhere, including in connection with cybersecurity incidents.

It is possible that new laws, regulations and other requirements, or amendments to existing laws, interpretations of existing laws, regulations and other requirements, may require us to incur significant costs, implement new processes, or change our handling of information and business operations. Any failure or perceived failure by us to comply with laws, regulations and other requirements relating to the privacy, security and handling of information could result in legal claims or proceedings, government investigations, or enforcement actions. We could incur significant costs in investigating and defending such claims and, if found liable, pay significant damages or fines or be required to make changes to our information handling practices. These proceedings and any subsequent adverse outcomes may subject us to significant negative publicity and an erosion of trust. If any of these events were to occur, our business, results of operations and financial condition could be materially adversely affected.

Risks Relating to X-energy’s Capital Resources

In order to fulfill our business plan, we will require additional funding. To the extent we require such investor funding in the future, such funding may be dilutive to our investors and no assurances can be given as to terms of any such funding. Any such funding and the associated terms will be highly dependent on market conditions and the progress of our business at the time we seek such funding. The terms of any financing we pursue may be less favorable than previously anticipated and could become even less favorable depending on the amount of funds we may require.

Our business is capital intensive. We expect that significant additional capital will be required in the future to continue our planned operations, including commercialization efforts, expanded development activities, investment in our customers’ projects, risk-sharing arrangements and other activities associated with operating as a public company. To raise capital, we may enter into financing arrangements that may be costly or impose certain restrictive covenants or otherwise restrict our ability to secure additional leverage or financing. We may also seek to sell common stock, convertible securities or other equity securities in one or more transactions at prices and in a manner we determine from time to time. If we sell common stock, convertible securities or other equity securities, investors may be materially diluted in subsequent sales. Such sales may also result in material dilution to our existing stockholders. If our investors could gain rights, preferences and privileges senior to the holders of our common stock, the above events could significantly harm our business, prospects, financial condition and results of operations and cause the price of our common stock to decline.

Our corporate expenditures, including our corporate level outspend, are subject to numerous uncertainties.

Our current and future operating expenses are uncertain and impacted by various factors outside of our control, including rising costs and other impacts of inflation, evolving regulatory requirements, supply chain availability, global conflicts, global supply chain challenges and component manufacturing challenges and other uncertainties, among other factors. Accordingly, it is possible that our overall expenses and results of

could be higher than the levels we currently estimate, and any increases could have a material effect on our business, financial condition, operating results and future prospects.

We may experience a disproportionately higher impact from inflation and rising costs.

Inflation has resulted in, and may continue to result in, higher interest rates and capital costs, higher shipping costs, higher material costs, supply shortages, increased costs of labor and other significant costs. Although the impact of material cost, labor, or other inflationary or economically driven factors on the entire nuclear and energy transition industry (including renewable sources of electricity, solar, wind), the relative impact may not be the same across the industry, and the particular effect on our industry will depend on a number of factors, including material use, design, structure of supply chain, project management and others, which could result in significant changes to the competitive landscape, our technology and our ability to sell Xe-100 reactors and our TRISO-X fuel, which could have an adverse effect on our business, financial condition, operating results and future prospects.

We have a history of losses and may not achieve profitability in the future. We will need substantial additional capital to fund our operations. If we fail to obtain significant additional capital in connection with the consummation of this offering, we will be unable to sustain operations unless we are able to raise additional capital following this offering from additional funding sources.

We expect to continue to incur operating losses for the foreseeable future as we continue to develop, and we will need additional capital from external sources. If we are unable to raise additional capital, we will have to significantly delay, scale back or discontinue one or more of our development programs. We may be required to cease operations or seek partners for our products at an earlier stage than otherwise would be desirable and on terms that are less favorable than otherwise be available. In the absence of additional capital, we may also be required to relinquish or otherwise dispose of rights to technologies, product candidates or products that we would otherwise develop or commercialize on terms that are less favorable than might otherwise be available. If we are unable to secure additional capital, we may be required to take additional measures to reduce our operating expenses, to conserve our cash in amounts sufficient to sustain operations and meet our obligations. These measures may significantly alter our business plan and could cause significant delays in the development of our product candidates.

Future indebtedness could expose us to risks that could adversely affect our business, financial condition, operating results and future prospects.

In the future, we may incur indebtedness. Future indebtedness could have significant consequences for our security holders, business, results of operations and financial condition, including the following other things:

- increasing our vulnerability to adverse economic and industry conditions;
- limiting our ability to obtain additional financing;
- requiring the dedication of a substantial portion of our cash flow from operations to service our indebtedness, which will reduce the amount of cash available for other purposes;
- limiting our flexibility to plan for, or react to, changes in our business; and
- placing us at a possible competitive disadvantage with competitors that are less leveraged and have better access to capital.

Our business may not generate sufficient funds, and we may otherwise be unable to maintain cash reserves, to pay any additional indebtedness that we may incur. Any future indebtedness we incur may contain financial and other restrictive covenants that will limit our ability to operate, raise capital or make payments under our indebtedness. If we fail to comply with such covenants, we may be required to make payments under any of our indebtedness when due, then we would be in default under that indebtedness, which could, in turn, result in that indebtedness becoming immediately payable in full and cross-acceleration under our other indebtedness and other liabilities.

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Our actual operating results may differ significantly from our guidance.

From time to time, we may release guidance in our quarterly earnings releases, quarterly conference calls, or otherwise once we are a public company, regarding our future performance. This guidance represents our management's estimates as of the date of release. This guidance, which includes forward-looking statements, will be based on projections prepared by our management. These projections are prepared with a view toward compliance with published guidelines of the American Institute of Certified Public Accountants, and neither our registered public accountants nor any other independent outside party compiles or examines the projections. Accordingly, no such person expresses an opinion on, or any other form of assurance with respect to the projections.

Projections are based upon a number of assumptions and estimates that, while presented with a degree of specificity, are inherently subject to significant business, economic, and competitive uncertainties and contingencies, many of which are beyond our control, such as the effects of inflation on our costs and expectations, and are based upon specific assumptions with respect to future business developments, some of which will change. Any material change to the assumptions or estimates underlying the projections that management prepares, or any material overruns or other unexpected increase in costs, could have an adverse effect on the projections and the guidance on which it is based. The rapidly evolving nature of the market in which we operate may make it difficult to evaluate our current business and our future prospects, which may affect our ability to plan for and model future growth. We intend to state possible outcomes as high as reasonably achievable, which are intended to provide a sensitivity analysis as variables are changed. However, actual results may vary from our guidance and the variations may be material. The principal reason that we release this guidance is to provide a basis for our management to discuss our business outlook as of the date of release with analysts and investors. We do not accept any responsibility for any projections or reports published by third parties. Investors are urged not to rely upon our guidance in making an investment decision in our common stock.

Any failure to successfully implement our operating strategy or the occurrence of any of the circumstances set forth in this "Risk Factors" section could result in our actual operating results being different from our guidance, and the differences may be adverse and material.

Our financial results may vary significantly from quarter to quarter.

We expect our revenue and operating results to vary from quarter to quarter. We may incur significant operating expenses during the startup and early stages of large contracts and may not be able to generate the corresponding revenue in that same quarter. We may also incur additional expenses when contracts are terminated or expire and are not renewed.

Payments due to us from our customers may be delayed due to billing cycles or as a result of delays in government budgets to gain congressional and administration approval in a timely manner. The government's fiscal year ends September 30. If a federal budget for the next federal fiscal year is not approved by that date in each year, our customers may have to suspend engagements that we are performing.

until a budget has been approved. Any such suspensions may reduce our revenue in the fourth federal fiscal year or the first quarter of the subsequent federal fiscal year.

Additional factors that may cause our financial results to fluctuate from quarter to quarter addressed elsewhere in this “*Risk Factors*” section and the following factors, among others:

- the terms of customer contracts that affect the timing of revenue recognition;
- variability in demand for our services and solutions;
- commencement, completion or termination of contracts during any particular quarter;
- timing of shipments and product deliveries;
- timing of award or performance incentive fee notices;
- timing of significant bid and proposal costs;
- the costs of remediating unknown defects, errors or performance problems of our products.

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- variable purchasing patterns under blanket purchase agreements and other indefinite quantity contracts;
- restrictions on and delays related to the export of nuclear articles and services;
- costs related to government inquiries;
- strategic decisions by us or our competitors, such as acquisitions, divestitures, spin-offs and joint ventures;
- strategic investments or changes in business strategy;
- changes in the extent to which we use subcontractors;
- seasonal fluctuations in our staff utilization rates;
- changes in our effective tax rate, including changes in our judgment as to the need for a valuation allowance recorded against our deferred tax assets; and
- the length of sales cycles.

Changes in our accounting estimates and assumptions could negatively affect our financial position and operations.

We prepare our consolidated financial statements in accordance with U.S. GAAP. These principles require us to make estimates and assumptions that affect the reported amounts of assets, liabilities and the disclosure of contingent assets and liabilities at the date of our financial statements. We are also required to make certain judgments that affect the reported amounts of revenues and expenses during each reporting period. We periodically evaluate our estimates and assumptions including, but not limited to, those relating to amortization and depreciation, estimates of costs to complete contracts and the associated revenues, the valuation of profits interests, unit-based compensation, preferred units, warrants, exchanged debt, and liabilities measured at fair value. We base our estimates on historical experience and various assumptions that we believe to be reasonable based on current circumstances. These assumptions and estimates involve the exercise of judgment and discretion and may evolve over time in light of operational experience, regulatory direction, developments in the industry, principles and other factors. Actual results could differ from these estimates as a result of

circumstances, assumptions, policies or developments in the business, which could materially affect our consolidated financial statements.

Risks Related to our Capital Structure

Our principal asset after the completion of this offering will be our interest in XERC, and, as a result, we will depend on distributions from XERC to pay our taxes and expenses (including payments under the Tax Receivable Agreement) and pay dividends. XERC's ability to make such distributions may be subject to various limitations and restrictions.

Upon the consummation of this offering and the Transactions, we will be a holding company and will have no material assets other than our ownership of Common Units. As such, we will have no other means of generating revenue or cash flow, and our ability to pay our taxes and operating expenses and pay dividends in the future, if any, will be dependent upon the financial results and cash flow of XERC and distributions we receive from XERC. There can be no assurance XERC will generate sufficient cash flow to distribute funds to us or that applicable state law and contractual restrictions, including restrictive covenants in any applicable debt instruments, will permit such distributions. XERC is currently subject to debt instruments or other agreements that restrict its ability to make distributions to us, which could affect XERC's ability to pay distributions to us and thereby adversely affect our cash flows.

XERC will continue to be treated as a partnership for U.S. federal income tax purposes and will generally not be subject to any entity-level U.S. federal income tax. Instead, any taxable income of XERC will be allocated to holders of Common Units, including us. Accordingly, we will incur taxes on our allocable share of any net taxable income of XERC. Under the terms of the XERC LLA, XERC will be obligated, subject to various limitations and restrictions, including with respect to debt agreements, to make tax distributions to holders of Common Units, including us. In addition to our expenses, we will also

incur expenses related to our operations, including payments under the Tax Receivable Agreement, which we expect will be significant. See “Certain Relationships and Related Party Transactions — Tax Receivable Agreement.” We intend, as its managing member, to cause XERC to make cash distributions to holders of Common Units in an amount sufficient to (i) fund all or part of their tax obligations in respect of their share of income allocated to them and (ii) cover our operating expenses, including payments under the Tax Receivable Agreement. However, XERC's ability to make such distributions may be subject to various limitations and restrictions, such as restrictions on distributions that would either violate an applicable agreement to which XERC is then a party, including debt agreements, or any applicable law, which could have the effect of rendering XERC insolvent. If we do not have sufficient funds to pay our operating expenses, liabilities, or to fund our operations (including, if applicable, because of an acceleration of our obligations under the Tax Receivable Agreement), we may have to borrow funds, which could materially and adversely affect our liquidity and financial condition, and subject us to various restrictions imposed by any lender in connection with such funds. To the extent we are unable to make timely payments under the Tax Receivable Agreement for any reason, such payments generally will be deferred and will accrue interest until paid; provided, however, that nonpayment for a specified period may constitute a Material Breach under the Tax Receivable Agreement resulting in the future payments under the Tax Receivable Agreement for each tax year in which any such Material Breach being calculated utilizing certain deemed exchanges and valuation adjustments. See “Certain Relationships and Related Party Transactions — Tax Receivable Agreement.” If XERC does not have sufficient funds to make distributions, our ability to declare and pay cash dividends will also be restricted or impaired, although we do not anticipate declaring or paying any cash

our Class A common stock in the foreseeable future. See “Risk Factors — Risks Related to the Ownership of our Class A common stock” and “Dividend Policy.”

In addition, the tax distributions that XERC may be required to make may be substantial. The amount of any additional tax distributions XERC is required to make likely will exceed the amount that would be owed by a corporate taxpayer similarly situated to XERC. As a result of differences in the amount of net taxable income allocable to us and to the other holders of Common Units, (ii) the lower tax rate applicable to corporations as opposed to individuals, and (iii) certain tax benefits covered by, and payments under, the Tax Receivable Agreement, these tax distributions may be substantial that exceed our tax liabilities. Our board of directors will determine the appropriate uses for any such cash, not so accumulated, which may include, among other uses, the payment of obligations under the Tax Receivable Agreement and the payment of other expenses. We will have no obligation to distribute such cash (or any available cash) to our shareholders. No adjustments to the exchange ratio for Common Units to corresponding shares of Class A common stock will be made as a result of any cash dividend or distribution by us or any retention of cash by us. As a result, the holders of Common Units (other than us) will not receive from value, if any, attributable to such cash balances if they acquire shares of Class A common stock in exchange for their Common Units, notwithstanding that such holders may have participated in distributions to holders of Common Units in distributions that resulted in such excess cash balances to us. See “Description of Capital Stock.” To the extent we do not distribute such excess cash as dividends on our Class A common stock we may take other actions with respect to such excess cash, for example, holding such cash, lending or contributing it (or a portion thereof) to XERC, which may result in shares of our Class A common stock increasing in value relative to the value of Common Units, or repurchasing shares of our Class A common stock. Following a contribution of such excess cash to XERC, we are not required to, make an adjustment to the outstanding number of Common Units held by holders of Common Units (other than us).

Funds used by XERC to satisfy its obligation to make tax distributions will not be reinvested in our business, except to the extent we or certain other Continuing Equity Owners may elect to reinvest excess cash received to reinvest in XERC for additional Common Units.

Moreover, because cash available for additional tax distributions will be determined by the amount of cash available to XERC, account the ability of XERC and its subsidiaries to take on additional borrowing, XERC may be required to increase its indebtedness in order to fund additional tax distributions. Such additional borrowing may adversely affect our results of operations, cash flows and financial position by, without limitation, increasing our ability to borrow in the future for other purposes, such as capital expenditures, and increasing our interest expense and leverage ratios.

The Tax Receivable Agreement with XERC and the TRA Holders requires us to make cash payments to TRA Holders in respect of certain tax benefits to which we may become entitled, and we expect that such payments will be substantial.

In connection with the consummation of this offering, we will enter into a Tax Receivable Agreement with XERC and the TRA Holders. Under the Tax Receivable Agreement, we will be required to make cash payments to the TRA Holders equal to 85% of the cash tax savings, if any, that we actually realize in certain circumstances are deemed to realize, as a result of (i) Basis Adjustments, (ii) Existing Tax Attributes, (iii) Tax-Deferred Attributes and (iv) Interest Deductions. We will be required to make such payments to the TRA Holders even if all of the TRA Holders were to exchange or redeem their remaining Common Units.

The payment obligations under the Tax Receivable Agreement are an obligation of the company, not of XERC. We expect that the amount of the cash payments we will be required to make under the Tax Receivable Agreement will be substantial. Any payments made by us to the TRA Holders under the Tax Receivable Agreement will not be available for reinvestment in our business and will generally reduce the amount of overall cash flow that might have otherwise been available to us. To the extent that we are unable to make timely payments under the Tax Receivable Agreement for any reason, the unpaid amounts will be deferred and will accrue interest until paid by us; provided, however, that nonpayment for a specified period may constitute a Material Breach under the Tax Receivable Agreement. In the case of a Material Breach, the event of certain changes of control (as defined under the Tax Receivable Agreement), including certain mergers, asset sales and other forms of business combinations, or upon X-Energy, Inc. electing to early terminate the Tax Receivable Agreement (and X-Energy, Inc. does not make an election to make a single, lump sum payment) the Tax Receivable Agreement will not terminate nor will a single, lump sum payment be due; however, the calculation of certain future payments made under the Tax Receivable Agreement will utilize certain valuation assumptions, including that (i) any Common Units that have not been exchanged are deemed exchanged for the market value of the shares of our Class A common stock at the time of the change of control and (ii) X-Energy, Inc. will have sufficient taxable income to utilize the tax attributes covered by the Tax Receivable Agreement. See “Certain Relationships and Related Party Transactions — Tax Receivable Agreement.” These payment obligations could (i) make the company an attractive target for an acquisition, particularly in the case of an acquirer that cannot use some of the tax benefits that are the subject of the Tax Receivable Agreement and (ii) result in holders of our Class A common stock receiving substantially less consideration in connection with a change of control than they would receive in the absence of such obligation. Accordingly, the TRA Holders’ interests may conflict with those of the holders of our Class A common stock. Furthermore, if we exercise our right to terminate the Tax Receivable Agreement, we would be obligated to make an immediate payment. Such payment may be significantly in advance of, and may materially exceed, the actual realization of, the future tax benefits to which the payment relates.

Assuming no material changes in the relevant tax laws and that we earn sufficient taxable income to realize all tax benefits that are subject to the Tax Receivable Agreement, we expect that the cash payments associated with the purchase of Common Units in connection with this offering, together with the cash payments from redemptions or exchanges of all remaining Common Units owned by the TRA Holders pursuant to the XERC LLC Agreement as described above, would aggregate to approximately \$ _____ million over the next _____ years from the date of this offering based on the assumed initial public offering price of \$ _____ per share of our Class A common stock, which is the midpoint of the range set forth on the cover of this prospectus, and assuming all redemptions or exchanges would occur immediately after the completion of this offering for the remaining ownership of XERC not acquired by X-Energy, Inc., which is assumed to be 100% for purposes of the pro forma information presented herein and elsewhere in this prospectus. Under such scenario, assuming future payments are made on the date each relevant tax liability is due, without extensions, we would be required to pay approximately 85% of such amount, or a total of approximately \$ _____ million, over the _____-year period from the date of this offering, to the TRA Holders. The amount of Basis Adjustments, Existing Basis, Blocker Tax Attributes and Interest Deductions and the amount of any resulting tax benefits, as well as the amount and timing of any payments under the Tax Receivable Agreement, will vary depending upon a number of factors including: the timing of redemptions or exchanges by TRA Holders; the price of shares of our Class A common stock at the time of the exchange; the extent to which such exchanges are taxable; the amount of gain recognized by such TRA Holders; the amount of taxable income allocated to us or otherwise

generated by us in the future; the portion of our payments under the Tax Receivable Agreement not subject to the federal and state tax rates then applicable; and the federal and state tax rates then applicable.

Our organizational structure, including the Tax Receivable Agreement, confers certain benefits on the Continuing Equity Owners that will not benefit holders of our Class A common stock to the same extent that it will benefit the Continuing Equity Owners.

Our organizational structure, including the Tax Receivable Agreement, confers certain benefits on the Continuing Equity Owners that will not benefit the holders of our Class A common stock to the same extent that it will benefit the Continuing Equity Owners. We will enter into the Tax Receivable Agreement with XERC and the TRA Holders in connection with the completion of this offering and the terms of the TRA, which will provide for the payment by us to the TRA Holders of 85% of the amount of cash taxes, if any, that we actually realize, or in some circumstances are deemed to realize, as a result of certain tax Adjustments, (ii) Existing Basis, (iii) Blocker Tax Attributes and (iv) Interest Deductions. The TRA is referred to as the “Tax Receivable Agreement—Tax Receivable Agreement.” Although we will not be able to quantify the amount of such tax benefits, this and other aspects of our organizational structure may impact the future trading market for our Class A common stock.

In certain cases, payments under the Tax Receivable Agreement to the TRA Holders may be significantly in excess of the actual benefits we realize in respect of the tax attributes subject to the Tax Receivable Agreement.

The Tax Receivable Agreement will generally apply to each of our taxable years, beginning with our first taxable year ending after the consummation of the Transactions. There is no maximum term of the Tax Receivable Agreement. Importantly, upon a change of control, the Tax Receivable Agreement will not terminate nor will the lump sum payment be due. However, the Tax Receivable Agreement will terminate if we (i) commit a Material Breach under the Tax Receivable Agreement or (ii) undergo a change of control, which includes certain mergers, asset sales, other forms of business combinations or other transactions resulting in a change of control occurring after the consummation of this offering, then our obligations, or our obligations, under the Tax Receivable Agreement to make payments will be calculated utilizing certain valuation assumptions, including that (x) any Common Units that have not been exchanged for shares of our Class A common stock at the time of the change of control will be valued at the market value of the shares of our Class A common stock at the time of the change of control and (y) X-Energy, Inc. will have sufficient taxable income to fully utilize the tax attributes subject to the Tax Receivable Agreement. Furthermore, the Tax Receivable Agreement will provide that, in the event of an early termination of the Tax Receivable Agreement, then, at our sole election, either (i) our obligations will be paid out over time and determined in the same manner as described above in the event of a change of control and Material Breach or (ii) the payment obligations will be accelerated and we will make a single, lump sum payment to the TRA Holders based on certain assumptions, including the assumption that we will have sufficient taxable income to fully utilize all potential future tax attributes subject to the Tax Receivable Agreement.

As a result of the foregoing, we would be required to make an immediate cash payment to the TRA Holders of the present value of the anticipated future tax benefits that are the subject of the Tax Receivable Agreement, based on certain assumptions, which payment may be made significantly in advance of the realization, if any, of such future tax benefits. Such cash payment to the TRA Holders could be a significant percentage of any actual benefits we ultimately realize in respect of the tax attributes subject to the Tax Receivable Agreement. In these situations, our obligations under the Tax Receivable Agreement could have a substantial negative impact on our liquidity and could have the effect of deferring, or preventing certain mergers, asset sales, other forms of business combinations or other transactions resulting in a change of control. For example, should we elect to terminate the Tax Receivable Agreement immediately after this offering, assuming no material changes in the relevant tax laws or tax rates and that we do not have sufficient taxable income to realize all potential tax benefits that are subject to the Tax Receivable Agreement, we estimate that the aggregate of termination payments would be approximately \$ 100 million (assuming a discount rate of SOFR plus 100 basis points) based on the assumed initial public offering price of \$ 100 of our Class A common stock, which is the midpoint of the range set forth on the cover of this prospectus, and assuming SOFR (as defined in the Tax Receivable Agreement) were to be 100%. There can be no assurance that we will be able to fund or finance our obligations under the Tax Receivable Agreement. We may need to incur

debt to finance payments under the Tax Receivable Agreement to the extent our cash is insufficient to meet our obligations under the Tax Receivable Agreement as a result of timing or otherwise.

We will not be reimbursed for any payments made to the TRA Holders under the Tax Receivable Agreement in the event that any tax benefits are disallowed.

Payments under the Tax Receivable Agreement will be based on the tax reporting position we determine, and the Internal Revenue Service (“IRS”), or another tax authority, may challenge the Basis Adjustments, Existing Basis, Blocker Tax Attributes, Interest Deductions or other tax claims, as well as other related tax positions we take, and a court could sustain such challenge. If the outcome of any such challenge would reasonably be expected to materially and adversely affect the rights and obligations of TRA Holders under the Tax Receivable Agreement, then we will not be required to settle such challenge without the consent (not to be unreasonably withheld or delayed) of the TRA Holders. The interests of the TRA Holders in any such challenge may differ from or conflict with our interests, and the TRA Holders may exercise their consent rights relating to any such challenge in a manner adverse to our interests and your interests. We will not be reimbursed for any cash payments previously made to the TRA Holders under the Tax Receivable Agreement in the event that any tax benefits initially claimed by us and for which payment has been made to a TRA Holder are subsequently disallowed by a taxing authority and are ultimately disallowed. Instead, any excess cash payments made by us to a TRA Holder will be netted against future cash payments, if any, that we might otherwise be required to make to such TRA Holder, under the terms of the Tax Receivable Agreement. However, we might not be able to determine whether we have effectively made an excess cash payment to a TRA Holder for a number of years after the initial time of such payment. Moreover, the excess cash payments we made previously under the Tax Receivable Agreement could be greater than the amount of future cash payments against which they would otherwise be permitted to net such excess. The applicable U.S. federal income tax rules for determining the amount of applicable tax benefits we may claim are complex and factual in nature, and there can be no assurance that the IRS or a court will agree with our tax reporting positions. As a result, payments could be made under the Tax Receivable Agreement significantly in excess of any actual cash tax savings that we realize from the tax attributes with respect to a TRA Holder that are the subject of the Tax Receivable Agreement.

The acceleration of payments under the Tax Receivable Agreement in the case of an early termination, the application of certain valuation assumptions under the Tax Receivable Agreement in the case of an early termination, certain changes of control or a Material Breach may negatively impact the value received by our Class A common stock.

The Tax Receivable Agreement will provide that if we elect an early termination of the Tax Receivable Agreement, at X-Energy, Inc.’s election, either (i) X-Energy, Inc.’s (or its successor’s) payment obligations with respect to the Tax Receivable Agreement would be accelerated and a single, lump sum payment would be due, which payment is based on certain assumptions, including that we (or our successor) will have sufficient taxable income to fully utilize the benefits arising from the increased tax deductions and other benefits covered by the Tax Receivable Agreement or (ii) the payment obligations will be made over time and determined in the same manner as described below for a change of control or a Material Breach. In the event of certain changes of control, a Material Breach of the Tax Receivable Agreement, X-Energy, Inc., an insolvency event, or X-Energy, Inc.’s election with respect to an early termination, the calculation of certain future payments made under the tax receivable agreement will utilize certain valuation assumptions, including that (i) any Common Units that have not been exchanged are deemed to be worth the market value of the shares of our Class A common stock at the time of the change of control.

Energy, Inc. will have sufficient taxable income to fully utilize the tax attributes covered by the Tax Receivable Agreement. Such payments may significantly exceed the actual benefits X-Energy, Inc. may realize in respect of the tax attributes subject to the Tax Receivable Agreement. Consequently, it is possible that, in these circumstances, that the actual cash tax savings realized by us may be significantly less than the corresponding tax benefit payments under the Tax Receivable Agreement. We expect that the cash tax savings we may make under the Tax Receivable Agreement following a change of control will be substantially less than we may be in excess of 85% of X-Energy, Inc.'s actual cash tax benefits. X-Energy Inc.'s accelerated tax obligations and/or assumptions adopted under the Tax Receivable Agreement may negatively impact the value received by owners of our Class A common stock in a change of control transaction.

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The Continuing Equity Owners, including the X-energy Founder and certain of their affiliates, may have conflicting interests with holders of shares of our Class A common stock.

Immediately following this offering and application of the net proceeds therefrom, the Continuing Equity Owners, including the X-energy Founder and certain of their affiliates will beneficially own approximately 10% of the common units of the Company, which will entitle them to exercise approximately 10% of the voting power of our Class A common stock or Class B common stock (or approximately 10% if the Continuing Equity Owners exercise in full their option to purchase additional shares of Class A common stock). Each share of Class A common stock entitles the holder to one vote per share and each share of Class B common stock entitles the holder to one vote per share on all matters on which shareholders are entitled to vote generally. For a description of our multi-class structure, see “Description of Capital Stock.”

In addition, immediately following this offering and application of the net proceeds therefrom, the Continuing Equity Owners, including the X-energy Founder and certain of their affiliates, will own approximately 10% of the Common Units (or approximately 10% if the underwriters exercise in full their option to purchase additional shares of Class A common stock). Because they hold this interest in our business directly in XERC, rather than through the Company, the Continuing Equity Owners, including the X-energy Founder and certain of their affiliates, may have conflicting interests with holders of shares of our Class A common stock. For example, if XERC makes distributions to the Company, the managing members of XERC will also be entitled to receive such distributions pro rata in accordance with their ownership of Common Units and their preferences as to the timing and amount of such distributions may differ from those of our public shareholders. The Continuing Equity Owners, including the X-energy Founder and certain of their affiliates, may also have different tax positions from the TRA Holders, which may influence their decisions regarding whether and when to dispose of assets, especially in connection with the existence of the Tax Receivable Agreement that we entered into in connection with this offering. The Continuing Equity Owners, including the X-energy Founder and certain of their affiliates, may also have different tax positions from the TRA Holders, which may influence their decisions regarding whether and when to incur new or refinance existing indebtedness and when the Company should terminate the Tax Receivable Agreement and accelerate its payments thereunder. In addition, the structuring of future transactions may take into consideration our tax positions, the Continuing Equity Owners tax or other considerations even where no similar benefit would accrue to us. See “Description of Capital Stock,” “Relationships and Related Party Transactions — Tax Receivable Agreement.”

Our shares of Class B common stock will not have economic rights. Immediately following the consummation of this offering, all of our Class B common stock will be held by certain Continuing Equity Owners, and certain of their affiliates.

Changes in tax laws and unanticipated tax liabilities could adversely affect our effective income tax rate and profitability.

We are subject to income taxes in the U.S. Our effective income tax rate and profitability may be adversely affected in the future by several factors, including changes in tax laws, regulations, and guidance or interpretations at the federal, state, or international level and changes in the valuation of deferred tax assets and liabilities.

On July 4, 2025, the President signed into law the OBBBA, a sweeping tax and spending bill that makes permanent many provisions of the 2017 Tax Cuts and Jobs Act (the “TCJA”), while introducing new tax policies and restructuring others. While certain provisions may reduce our tax liability, others may introduce new complexity and audit risk. We will continue to monitor the potential impact of the OBBBA. Because tax laws are dynamic and often retroactive or uncertain in interpretation, our tax liabilities may differ significantly from eventual obligations. The net impact remains uncertain. A misapplication of the new rules could lead to materially adverse outcomes.

We regularly assess all of these tax-related matters to determine the adequacy of its tax provision. If our current tax strategies are ineffective or not in compliance with domestic and international tax laws, our financial position, operating results, and cash flows could be adversely affected.

Risks Related to the Offering and Ownership of our Class A common stock

There has been no prior public market for our Class A common stock, the stock price of our Class A common stock may be volatile or may decline regardless of our operating performance, and you may not be able to sell your shares at or above the initial public offering price.

There has been no public market for our Class A common stock prior to this offering. The offering price for our Class A common stock was determined through negotiations between us and the underwriters and may vary from the market price of our Class A common stock following this offering. The market prices of the securities of newly public companies such as us, particularly in the technology sector, have historically been highly volatile. The market price of our Class A common stock may fluctuate significantly in response to numerous factors, many of which are beyond our control, including

- overall performance of the equity markets and the performance of technology companies in particular;
- variations in our operating results, cash flows, and other financial metrics and non-financial factors and how those results compare to analyst expectations;
- changes in the financial projections we may provide to the public or our failure to meet those projections;
- failure of securities analysts to initiate or maintain coverage of us, changes in financial estimates by any securities analysts who follow our company, or our failure to meet these estimates and the expectations of investors;
- recruitment or departure of key personnel;
- the economy as a whole and market conditions in our industry;
- negative publicity related to problems in our manufacturing or the real or perceived quality of our products, as well as the failure to timely launch new products or services that attract market acceptance;
-

rumors and market speculation involving us or other companies in our industry;

- announcements by us or our competitors of new products, services, features and content, technical innovations, acquisitions, strategic partnerships, joint ventures, or capital commitments;
- new laws or regulations or new interpretations of existing laws or regulations applicable to our business;
- lawsuits threatened or filed against us, litigation involving our industry, or both;
- developments or disputes concerning our or other parties' products, services, or intellectual property rights;
- other events or factors, including those resulting from war, incidents of terrorism, or natural disasters, and the effects of these events;
- the expiration of contractual lock-up or market standoff agreements; and
- sales of shares of our Class A common stock by us or our stockholders.

In addition, the stock markets have experienced extreme price and volume fluctuations that have affected and continue to affect the market prices of equity securities of many companies. Stock prices of many companies have fluctuated in a manner unrelated or disproportionate to the operating performance of those companies. In the past, stockholders have instituted securities class action litigation following periods of market volatility. If we were to become involved in securities litigation, it could subject us to substantial costs, divert resources and the attention of management from our business, and adversely affect our business.

Sales, directly or indirectly, of a substantial amount of our Class A common stock in the public market by existing security holders may cause the price of our Class A common stock to decline.

Sales of a substantial number of shares of our Class A common stock into the public market, particularly sales by our directors, executive officers, and principal stockholders, or the perception that these sales might occur, could cause the market price of our Class A common stock to decline and could impair our ability to

raise capital through the sale of additional equity securities. Many of our existing security holders may have substantial unrecognized gains on the value of the equity they hold, and may take, or attempt to take, action to sell, directly or indirectly, their shares or otherwise secure, or limit the risk to, the value of their unrecognized gains on those shares.

All of the shares of Class A common stock sold in this offering will be freely tradable without restrictions or further registration under the Securities Act except that any shares held by our directors, executive officers, and principal stockholders, as defined in Rule 144 under the Securities Act, would only be able to be sold in compliance with Rule 144 and any applicable lock-up agreements described below.

In connection with this offering, subject to certain customary exceptions, we, all of our directors, executive officers, and substantially all of the holders of the Common Units, have entered into lock-up agreements with us or lock-up agreements with the underwriters that prohibit them from selling, contracting to sell, granting any option for the sale of, transferring, or otherwise disposing of our Class A common stock, stock options, or any security or instrument related to common stock or convertible preferred stock without the permission of J.P. Morgan Securities LLC on behalf of the underwriters for a period of 90 days from the date of this prospectus, subject to early termination as described below. See “Underwriting” for more information.

Holders of our outstanding shares of Class A common stock and securities convertible into or exercisable or exchangeable for shares of our Class A common stock are subject to restrictions on their ability to sell or transfer their equity either prior to the pricing of this offering or from the pricing of this offering through the date that is 180 days after the date of this prospectus. We refer to such restrictions as a “lock-up period”. Pursuant to the lock-up agreements with the underwriters, if (1) at least 180 days have elapsed since the date of this prospectus, (2) we have publicly released our earnings results for the period during which this offering occurred, and (3) such lock-up period is scheduled to end during the five trading days prior to a broadly applicable period during which trading in our securities is permitted under our insider trading policy, or a blackout period, such lock-up period will end five trading days prior to the commencement of such blackout period. We and the underwriters may restrict our stockholders from the market standoff agreements or lock-up agreements prior to the end of the lock-up period. Record holders of our securities are typically the parties to the lock-up agreements with the underwriters and to the market standoff agreements with us referred to above, while holders of beneficial interests in our shares who are also not holders in respect of such shares are not typically subject to such agreements or other similar restrictions. Accordingly, we believe that holders of beneficial interests who are not holders and are not bound by market standoff or lock-up agreements could engage in transactions with respect to those beneficial interests that negatively impact our stock price. In addition, an equity holder who is neither subject to a market standoff agreement with us nor a lock-up agreement with the underwriters may be able to sell, short sell, transfer, hedge, pledge, or otherwise dispose of their equity interests at any time, which could cause our market price to decline. Any such transaction described above involving shares of our Class A common stock, or any perception by the market that such transaction may occur, could cause our market price to decline.

When the applicable lock-up and market standoff periods described above expire, we and our stockholders subject to a lock-up agreement or market standoff agreement will be able to sell our shares in the public market. In addition, J.P. Morgan Securities LLC on behalf of the underwriters, may, at its discretion, release all or some portion of the shares subject to lock-up agreements prior to the expiration of the lock-up period. Sales of a substantial number of such shares upon expiration of the lock-up period, the market standoff agreements, or the perception that such sales may occur, or early release of these agreements, could cause our market price to fall or make it more difficult for you to sell your Class A common stock at the price and price that you deem appropriate.

We may also issue our shares of common stock or securities convertible into shares of our Class A common stock from time to time in connection with a financing, acquisition, investment, or otherwise. Such future issuance could result in substantial dilution to our existing stockholders and cause the market price of our Class A common stock to decline.

If securities or industry analysts do not publish research, or publish inaccurate or unfavorable research about our business, the price of our Class A common stock and trading volume could decline.

The trading market for our Class A common stock will depend in part on the research and reports that securities or industry analysts publish about us or our business, our market, and our competitors.

have any control over these analysts. If few securities analysts commence coverage of us, or if all of the analysts cease coverage of us, the trading price for our Class A common stock would be negatively affected. If one or more of the analysts who cover us downgrade our Class A common stock or publish unfavorable research about our business, our Class A common stock price would likely decline. If more of these analysts cease coverage of us or fail to publish reports on us regularly, demand for our Class A common stock could decrease, which might cause our Class A common stock price and trading volume to decline.

We have identified a material weakness in our internal control over financial reporting. If our remediation of this material weakness is not effective, or if X-energy experiences additional material weaknesses or other deficiencies, we may not be able to design and maintain effective internal control over financial reporting in the future, X-energy's ability to accurately report its financial condition and results of operations or comply with applicable laws and regulations could be impaired, which may adversely affect investor confidence in X-energy and, as a result, the trading price of X-energy Class A common stock.

In connection with the preparation of our consolidated financial statements for the years ended December 31, 2024 and 2023, we identified a material weakness in our internal control over financial reporting. A material weakness is a deficiency, or combination of deficiencies, in internal control over financial reporting such that there is a reasonable possibility that a material misstatement of our financial statements or interim consolidated financial statements will not be prevented or detected on a timely basis.

For the years ended December 31, 2024 and 2023, the material weakness primarily resulted from the following matter that is relevant to the preparation of our consolidated financial statements:

- we lacked a sufficient complement of accounting and financial reporting personnel to interpret complex technical agreements and related valuations and ensure we recorded transactions appropriately.

The material weakness described above was not remediated during the year ended December 31, 2024 and if not remediated in the future, could result in a misstatement of one or more accounts or disclosures that would result in a material misstatement to the annual or interim consolidated financial statements that would not be prevented or detected.

In order to remediate this material weakness, we have taken and plan to take the following actions:

- continuing to hire additional, qualified accounting and finance personnel; and
- designing and implementing additional controls and processes that operate at an appropriate level of precision to ensure adequate review of highly complex technical agreements and valuations.

We cannot assure you that the measures we have taken to date, and that we are continuing to implement, will be sufficient to remediate the material weakness we have identified or prevent the identification of additional material weaknesses in the future. If the steps we take do not remediate the material weakness in a timely manner, there could continue to be a reasonable possibility that a deficiency or others could result in a material misstatement of our annual or interim consolidated financial statements that would not be prevented or detected on a timely basis.

The process of designing and implementing internal control over financial reporting required to comply with the disclosure and attestation requirements of Section 404 of the Sarbanes-Oxley Act of 2002 is a consuming and costly. If, during the evaluation and testing process we identify additional material weaknesses in our internal control over financial reporting or determine that the existing material weaknesses have not been remediated, our management will be unable to assert that our internal control over financial reporting is effective and our independent registered public accounting firm may conclude that we have material weaknesses with respect to our internal control over financial reporting. If we are unable to assert that our internal control over financial reporting is effective, or when required in the future, our independent registered public accounting firm is unable to express an unqualified opinion on the effectiveness of our internal control over financial reporting, investors may lose confidence in the accuracy and completeness of our financial reports, the market price of our common stock could be adversely affected and we could become subject to litigation or investigations by the stock exchange, the SEC, or other regulatory authorities, which could require additional management resources.

An active, liquid trading market for X-energy's securities may not develop, which may limit your ability to sell your securities.

Although we intend to apply to list the Class A common stock on the Nasdaq under the XE, an active trading market for the Class A common stock may never develop or be sustained through this offering. The initial valuation of \$_____ per share may not be indicative of the market value of the Class A common stock that will prevail in the open market after this offering. A public trading market having the desirable characteristics of depth, liquidity and orderliness depends upon the existence of sufficient buyers and sellers at any given time, such existence being dependent upon the individual decisions of buyers and sellers over which neither we nor any market maker has control. The failure of a liquid trading market to develop and continue would likely have a material adverse effect on the value of the Class A common stock, and you may not be able to sell your Class A common stock at an attractive price, if at all. An inactive market may also impair our ability to raise capital to continue to fund our operations and issuing additional shares of Class A common stock.

We will have broad discretion in the use of the net proceeds we receive in this offering and may not use them effectively.

We will have broad discretion in the application of the net proceeds we receive in this offering, including for any of the purposes described in the section titled "Use of Proceeds," and you will not have the opportunity as part of your investment decision to assess whether we are using the net proceeds appropriately. Because of the number and variability of factors that will determine our ultimate use of the net proceeds from this offering, their ultimate use may vary substantially from their currently intended use. If we do not use the net proceeds that we receive in this offering effectively, our business, financial condition, operating results, and prospects could be harmed, and the market price for our Class A common stock could decline. Pending their use, we may invest the net proceeds from this offering in short-term, investment-grade interest-bearing securities such as money market accounts, certificates of deposit, commercial paper, and guaranteed obligations of the U.S. government that may not generate a high yield to our investors. These investments may not yield a favorable return to our investors.

We do not intend to pay dividends on the common stock of X-energy for the foreseeable future.

We have never declared or paid any cash dividends on the common stock of X-energy and do not intend to pay any cash dividends in the foreseeable future. Additionally, our ability to pay dividends on our common stock is limited by the restrictions under the terms of our loan and security agreement. We do not anticipate that for the foreseeable future we will retain all of our future earnings for use in the operations of our business and for general corporate purposes. Any determination to pay dividends in the future will be at the discretion of our board of directors. Holders of our Class B common stock do not have any dividend rights or any right to receive dividends, or to receive a distribution upon a liquidation, dissolution, or winding up of X-Energy, Inc., with respect to their Class B common stock. Accordingly, investors purchasing shares of our Class A common stock after price appreciation, which may never occur, as the result of a sale, will not realize any future gains on their investments.

X-energy will incur significant transaction and transition costs in connection with the registration of its securities.

X-energy will have incurred and expects to continue to incur significant, nonrecurring costs in connection with consummating the initial public offering and operating as a public company. As a public company, we will be subject to the reporting requirements of the Exchange Act and will be required to comply with the applicable requirements of the Sarbanes-Oxley Act and the Dodd-Frank Wall Street Reform and Consumer Protection Act, as well as rules and regulations subsequently implemented by the SEC. The costs of the establishment and maintenance of effective disclosure and financial controls, changes in corporate governance practices and required filing of annual, quarterly and current reports with respect to our financial condition and operating results. These requirements will increase our legal and financial compliance costs and make some activities more time-consuming and costly. In addition, our management and other personnel will need to divert attention from operational and other business matters to devote substantial time to public company requirements. We will also need to hire additional accounting and financial personnel.

appropriate public company experience and technical accounting knowledge and will need to build an internal audit function. We also expect that

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operating as a public company will make it more expensive for us to obtain director and officer liability insurance, and we may be required to accept reduced coverage or incur substantially higher costs to obtain coverage. This could also make it more difficult for us to attract and retain qualified people to join our board of directors, our board committees or as executive officers.

In addition, after we no longer qualify as an “emerging growth company,” as defined under the Securities Act, we expect to incur additional management time and cost to comply with the more stringent disclosure requirements applicable to companies that are deemed accelerated filers or large accelerated filers, including complying with the auditor attestation requirements of Section 404 of the Sarbanes-Oxley Act. We are just beginning the process of compiling the system and processing documentation needed to comply with such requirements. We may not be able to complete our evaluation, testing and remediation in a timely fashion. In that regard, we currently do not have an internal audit function, and we will need to hire or contract for additional accounting and financial staff with appropriate public company experience and technical accounting knowledge.

We cannot predict or estimate the amount of additional costs we may incur as a result of becoming a public company or the timing of such costs.

We are eligible to be treated as an emerging growth company, as defined in the Securities Act, and we believe that certain if the reduced disclosure requirements applicable to emerging growth companies will make our common stock less attractive to investors because we may rely on these reduced disclosure requirements.

We are eligible to be treated as an emerging growth company, as defined in Section 101 of the Securities Act, as modified by the JOBS Act. Under the JOBS Act, emerging growth companies can delay adopting new or revised financial accounting standards until such time as those standards apply to all public companies. We intend to take advantage of this extended transition period under the JOBS Act to adopt new or revised financial accounting standards.

For as long as we continue to be an emerging growth company, we may also take advantage of certain exemptions from various reporting requirements that are applicable to other public companies that are not emerging growth companies, including presenting only limited selected financial data and not being required to comply with the auditor attestation requirements of Section 404 of the Sarbanes-Oxley Act. As a result, our shareholders may not have access to certain information that they may deem important. We can remain an emerging growth company for up to five years, although circumstances could cause us to cease to be an emerging growth company earlier, including if our total annual revenue exceeds \$1.235 billion, if we issue more than \$1 billion in nonconvertible debt securities during any three-year period, or if before that time we are deemed an “accelerated filer” under U.S. securities laws. We cannot predict if investors will find our Class A common stock less attractive because we may rely on these exemptions. If some investors find our Class A common stock less attractive, as a result, there may be a less active trading market for our Class A common stock and our share price may be more volatile.

We will incur increased costs as a result of operating as a public company, and our management will need to devote substantial time to new compliance initiatives and corporate governance practices.

As a public company, and particularly after we are no longer an emerging growth company, we may incur significant legal, accounting and other expenses that we did not incur as a private company. In addition, the Sarbanes-Oxley Act, the Dodd-Frank Wall Street Reform, the Consumer Protection Act and other applicable securities requirements of and other applicable securities rules and regulations impose various requirements

companies, including establishment and maintenance of effective disclosure and financial corporate governance practices. Our management and other personnel will need to devote a significant amount of time to these compliance initiatives. Moreover, these rules and regulations will increase our legal and financial compliance costs and will make some activities more time-consuming and costly. For example, these rules and regulations may make it more difficult and more expensive for us to obtain and maintain director and officer liability insurance and make it more difficult for us to attract and retain qualified members on our board of directors.

We are evaluating these rules and regulations and cannot predict or estimate the amount of costs we may incur or the timing of such costs. These rules and regulations are often subject to varying interpretations, in many cases due to their lack of specificity, and, as a result, their application may

evolve over time as new guidance is provided by regulatory and governing bodies. This could result in continuing uncertainty regarding compliance matters and higher costs necessitated by ongoing changes to disclosure and governance practices.

Key members of our management team have limited experience managing a public company.

Many members of our management team have limited experience managing a publicly traded company, interacting with public company investors and complying with the increasingly complex laws and regulations applicable to public companies. Our management team may not successfully or efficiently manage our transition to a public company subject to significant regulatory oversight and reporting obligations under federal securities laws and the continuous scrutiny of securities analysts and investors. These new obligations and constituents will require significant attention from our senior management and could divert time and resources away from the day-to-day management of our business, which could adversely affect our business performance, financial condition, operating results and future prospects.

CAUTIONARY NOTE REGARDING FORWARD-LOOKING STATEMENTS

This prospectus contains forward-looking statements. These forward-looking statements include, without limitation, statements relating to expectations for future financial performance, business operations or expectations for X-energy's business. These statements are based on the beliefs and assumptions of the management of X-energy. Although X-energy believes that their plans, intentions and expectations are reasonable, in or suggested by these forward-looking statements are reasonable, X-energy cannot assure you that they will achieve or realize these plans, intentions or expectations. These statements constitute projections and forward-looking statements, and are not guarantees of performance. Such statements can be identified by the fact that they do not relate strictly to historical or current facts. When used in this prospectus, words such as "anticipate," "believe," "can," "continue," "could," "estimate," "expect," "forecast," "may," "might," "plan," "possible," "potential," "predict," "project," "seek," "should," "strive," "will," "would" and similar expressions may identify forward-looking statements, but the absence of such words does not mean that a statement is not forward-looking.

Forward-looking statements in this prospectus and in any document incorporated by reference into this prospectus may include, for example, statements about X-energy, including:

- the ability to obtain and maintain the listing of the Class A common stock on the Nasdaq Global Market following this offering;
- the ability to raise financing in the future and to comply with restrictive covenants relating to our indebtedness;
- the future financial performance of X-energy;
- anticipated trends, growth rates, and challenges in our business and in the markets in which we operate;
- the expectations and estimates presented regarding certain illustrative unit economic metrics and other expectations with respect to costs, revenue and sources of revenue, and gross margins;
- the anticipated timeline for the completion of the design and target delivery estimates of the ARDP under the ARDP;
- X-energy's ability to maintain, protect, and enhance its intellectual property;
- X-energy's ability to retain or recruit, or to effect changes required in, its officers, key management personnel and directors following this offering;
- X-energy's ability to comply with laws and regulations applicable to its business; and
- expansion plans and opportunities.

These forward-looking statements are based on information available as of the date of this prospectus and X-energy's management teams' current expectations, forecasts and assumptions. They involve a number of judgments, known and unknown risks and uncertainties and other factors, many of which are beyond the control of X-energy and its directors, officers and affiliates. Accordingly, forward-looking statements should not be relied upon as representing X-energy's management teams' views as of any subsequent date. X-energy does not undertake any obligation to update, add to or otherwise correct any forward-looking statements contained in this prospectus to reflect events or circumstances after the date they were made, whether as a result of new information, future events, inaccuracies that become apparent after the date of this prospectus or otherwise, except as may be required under applicable securities laws.

You should not place undue reliance on these forward-looking statements in deciding whether to purchase or should be cast or in voting your shares on the proposals set out in this prospectus. Should one or more of the number of known and unknown risks and uncertainties materialize, or should any of our assumptions prove incorrect, our actual results or performance may be materially different from those expressed in or implied by these forward-looking statements. Some factors that could cause actual results to differ include, but are not limited to:

- changes in applicable laws or regulations;
- changes to applicable government policies, priorities, regulations, mandates and funding levels relating to X-energy's business with government entities;

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- the impact to X-energy and its potential customers from changes in interest rates or rising costs, including commodity and labor costs;
- changes to the appropriations or funding available under the ARDP, including any failure by the government to appropriate additional funding to the ARDP, either to complete the existing projects or in light of increased project costs and inflationary pressures;
- construction delays affecting the initial deployment of the Xe-100 under the ARDP;
- the ability to raise sufficient capital to fund X-energy's business plan, including limited amount of capital raised in the initial public offering;
- the impact and potential extended duration of the current supply/demand imbalance in the market for high-assay low-enriched uranium;
- X-energy's business with various governmental entities being subject to the policies, regulations, mandates and funding levels of such governmental entities and being negatively impacted by any change to such policies, priorities, regulations, mandates and funding levels;
- X-energy's and its commercial partners' ability to obtain regulatory approvals necessary to build and operate small modular reactors in the U.S. and abroad in a timely way, or at all;
- X-energy's and its commercial partners' ability to deliver cost-competitive electricity from X-energy's small modular reactors and to deliver cost-competitive fuel for such reactors;
- the inclusion or exclusion of advanced nuclear technologies in regulatory schemes relating to carbon capture and/or reductions in carbon emissions;
- costs related to the initial public offering;
- the ability to accurately assess costs associated with developing the Xe-100 and other facilities; and
- other risks and uncertainties indicated in this prospectus, including those set forth under the heading "Risk Factors" of this prospectus entitled "*Risk Factors*" beginning on page [24](#).

USE OF PROCEEDS

We estimate that we will receive net proceeds of approximately \$ million (or approximately \$ million, if the underwriters exercise in full their over-allotment option) from the sale of shares of Class A common stock in this offering, assuming an initial public offering price of \$ per share, at the midpoint of the estimated price range set forth on the cover page of this prospectus, and after deducting estimated underwriting discounts and commissions and estimated offering expenses payable by us. If the number of shares of common stock to be sold in this offering is increased (decreased) by 1,000,000 shares from the expected number of shares of common stock to be sold in this offering, assuming no change in the assumed initial offering price per share, would increase (decrease) the net proceeds from this offering by \$ million. A \$1.00 increase (decrease) in the assumed initial public offering price would increase (decrease) the net proceeds to us from this offering by \$ million, assuming the number of shares of common stock offered by us, as set forth on the cover page of this prospectus, remains the same, and after deducting the underwriting discounts and commissions and estimated offering expenses payable by us. We intend to use the proceeds from this offering to purchase newly issued Common Units from XERC. In the event the underwriters exercise their option to purchase additional shares of Class A common stock, we intend to use any proceeds from such exercise to purchase newly issued Common Units from XERC. The foregoing purchases of Common Units will be made at a price per unit equal to the initial public offering price per share of Class A common stock in this offering, after deducting estimated underwriting discounts and commissions. XERC currently intends to use the net proceeds it receives from this offering for working capital and other general corporate purposes, which may include research and development and sales and marketing activities, general and administrative matters, and other expenditures, including spending necessary for supply chain and procurement activities. We may also use a portion of the proceeds for future growth projects.

DIVIDEND POLICY

We currently expect to retain all future earnings for use in the operation and expansion of our business and have no current plans to pay dividends on our common stock. Holders of our Class B common stock are not entitled to participate in any dividends declared by our Board. The declaration, amount and timing of any future dividends will be at the sole discretion of our board of directors, and will depend on a number of other things, general and economic conditions, our results of operations and financial condition, the amount of available cash and current and anticipated cash needs, capital requirements, contractual, legal and regulatory restrictions and implications on the payment of dividends by us to our stockholders, and our subsidiaries to us, including restrictions under our credit agreements and other indebtedness, and such other factors as our board of directors may deem relevant. If we elect to pay such dividends in the future, we may reduce or discontinue entirely the payment of such dividends at any time. X-Eno is a holding company and its operations are conducted through its wholly owned subsidiaries. In order for us to pay a dividend, we do pay a dividend, we intend to cause our operating subsidiaries to make distributions to us in an amount sufficient to cover such dividend. Any additional financing arrangement we enter into in the future may include restrictive covenants that limit our subsidiaries' ability to pay dividends to us. In addition, applicable law may impose requirements that may restrict our ability to pay dividends to holders of our common stock.

Immediately following this offering, we will be a holding company, and our principal asset will be our Common Units we purchase from XERC. If we decide to pay a dividend in the future, we will intend to cause XERC to make distributions to us in an amount sufficient to cover such dividend. If we make such distributions to us, the other holders of Common Units will be entitled to receive pro rata distributions. See "Risk Factors — Risks Related to Our Capital Structure — Our principal asset after the offering will be our interest in XERC, and, as a result, we will depend on distributions from XERC to pay our taxes and expenses (including payments under the Tax Receivable Agreement) and pay dividends. XERC's ability to make such distributions may be subject to various limitations and restrictions."

CAPITALIZATION

The following table sets forth the cash and cash equivalents and capitalization as of December 31, 2019, of X-Energy Reactor Company, LLC on an actual basis and of X-Energy, Inc. on a pro forma basis to reflect:

- the Transactions; and
- the sale of shares of our common stock offered by us in this offering at an assumed offering price of \$ 15.00 per share, which is the midpoint of the estimated price range set forth on the cover page of this prospectus, after deducting the underwriting discounts and commissions, estimated offering expenses payable by us, and the application of the net proceeds to us as described under “Use of Proceeds.”

You should read this table in conjunction with the information contained in “Use of Proceeds,” “Summary Historical and Pro Forma Financial Information of X-energy,” and “Management’s Discussion and Analysis of Financial Condition and Results of Operations of X-energy” as well as our consolidated financial statements and related notes, each included elsewhere in this prospectus.

	As of December 31, 2019	
	X-Energy Reactor Company, LLC	X-Energy, Inc.
	Actual	Pro Forma
Cash and cash equivalents	\$ 458,932	\$ —
Mezzanine Equity:		
Class A Common Units	1,800	—
Class B Common Units	93,353	—
Series A Redeemable Convertible Preferred Units	218,408	—
Series A-1 Redeemable Convertible Preferred Units	21,477	—
Series B Redeemable Convertible Preferred Units	101,382	—
Series C Redeemable Convertible Preferred Units	265,797	—
Series C-1 Redeemable Convertible Preferred Units	686,715	—
Series D Redeemable Convertible Preferred Units	677,623	—
Total mezzanine equity	2,066,555	—
Members’ Deficit:		
Class A common stock	—	—
Class B common stock	—	—
Additional paid-in-capital	12,167	—
Accumulated and other comprehensive income	(117)	—
Accumulated deficit	(1,236,345)	—
Total members’ deficit / stockholders’ equity (deficit) attributable to X-Energy, Inc.	(1,224,295)	—
Non-controlling interest	—	—

Total capitalization

1,211,271

A \$1.00 increase (decrease) in the assumed initial public offering price of \$ per share of Class A common stock, which is the midpoint of the estimated price range set forth on the cover page of this prospectus, would increase (decrease) the net proceeds that we receive in this offering and the additional paid-in capital, total stockholders' equity and total capitalization by approximately \$1,000,000, assuming the number of shares of Class A common stock offered by us remains the same as set forth on the cover page of this prospectus and after deducting the underwriting discounts and commissions and estimated offering expenses payable by us. An increase (decrease) of 1,000,000 shares in the number of shares of Class A common stock to be sold by us in this offering, assuming no change in the assumed initial public offering

price, would increase (decrease) our net proceeds from this offering and each of additional paid-in capital, total stockholders' equity and total capitalization by approximately \$1,000,000. The pro forma condensed balance sheet reflects the issuance of shares of common stock of X-Energy, Inc. in connection with the contribution of the X-Energy Reactor Company, LLC by its unitholders and related reorganization transactions and related cancellation of all outstanding members' equity at X-Energy Reactor Company, LLC in connection with the offering.

The table above does not include:

- shares of Class A common stock reserved for issuance upon redemption or exchange of Units that will be held by the Continuing Equity Owners on a one-for-one basis;
- shares of Class A common stock reserved for future issuance under the 2026 Plan that will become effective upon the effectiveness of the registration statement of which this prospectus is a part (which number includes shares of our Class A common stock subject to stock unit awards and stock options that will be granted to certain of our employees pursuant to our 2026 Plan in connection with the consummation of this offering, based on the assumed initial public offering price of \$ per share (the midpoint of the estimated price range set forth on the cover page of this prospectus)); and
- shares of Class A common stock reserved for future issuance under the ESPP, which will become effective upon the effectiveness of the registration statement of which this prospectus is a part.

The Continuing Equity Owners will own Common Units after the Transactions. Continuing Equity Owners will not have any right to receive distributions from X-Energy, Inc. to their Common Units, we have presented dilution in pro forma net tangible book value per share before and after this offering assuming that all of the holders of Common Units (other than X-Energy, Inc.) had their Common Units redeemed or exchanged for newly issued shares of Class A common stock on a one-for-one basis (rather than for cash) and the transfer to the Company and cancellation of the shares in consideration of all of their shares of Class B common stock common stock (which are not entitled to receive distributions or dividends, whether cash or stock from X-Energy, Inc.) in order to more meaningfully present the dilutive impact on the investors in this offering. We refer to the redemption or exchange of all Common Units for shares of Class A common stock as described in the previous sentence as the Assumed Redemption.

Dilution is the amount by which the offering price paid by the investors of the Class A common stock in this offering exceeds the pro forma net tangible book value per share of Class A common stock as of the offering. XERC's pro forma net tangible book value (deficit) as of [redacted] prior to this offering, after giving effect to the other Transactions, and the Assumed Redemption was \$ [redacted] million. The pro forma net tangible book value per share prior to this offering is determined by subtracting our total liabilities from the total book value of our tangible assets and dividing the difference by the number of shares of Class A common stock deemed to be outstanding after giving effect to the Assumed Redemption.

If you invest in our Class A common stock in this offering, your ownership interest in us will be diluted to the extent of the difference between the initial public offering price per share of our Class A common stock and the pro forma net tangible book value (deficit) per share of our Class A common stock as of the offering. Dilution results from the fact that the per share offering price of the Class A common stock is substantially in excess of the pro forma net tangible book value per share attributable to the owners.

Our pro forma net tangible book value (deficit) as of [redacted] was approximately \$ [redacted] per share of our Class A common stock. We calculate pro forma net tangible book value per share by taking the amount of our total tangible assets, reduced by the amount of our total liabilities, and then dividing that amount by the total number of shares of Class A common stock outstanding as of the pro forma basis giving effect to the Reorganization Transactions.

After giving effect to (i) our sale of shares of Class A common stock in this offering at the initial public offering price of \$ [redacted] per share, which is the midpoint of the estimated price range set forth on the cover page of this prospectus, and (ii) the use of proceeds therefrom, after deducting the offering discounts and commissions and estimated offering expenses payable by us, our pro forma net tangible book value (deficit) as of [redacted] would have been \$ [redacted] million, or \$ [redacted] per share of our Class A common stock. This amount represents an immediate increase in pro forma net tangible book value per share of \$ [redacted] per share of Class A common stock to our existing owners and an immediate and substantial increase in pro forma net tangible book value (deficit) of \$ [redacted] per share of Class A common stock to the investors purchasing shares in this offering.

The following table illustrates this dilution on a per share of common stock basis assuming that the underwriters do not exercise their option to purchase additional shares of Class A common stock.

Assumed initial public offering price per share of Class A common stock
Pro forma net tangible book value (deficit) per share of Class A common stock as of [redacted]
Increase in pro forma net tangible book value per share of Class A common stock attributable to investors in this offering
Pro forma net tangible book value (deficit) per share of Class A common stock after giving effect to this offering
Dilution in pro forma net tangible book value per share of Class A common stock to investors in this offering

Dilution is determined by subtracting pro forma net tangible book value (deficit) per share of Class A common stock after the offering from the initial public offering price per share of Class A common stock.

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Each \$1.00 increase or decrease in the assumed initial public offering price per share of common stock would increase or decrease, as applicable, the pro forma net tangible book value per share and the dilution to new investors in the offering by \$ per share, assuming the number of shares offered in this offering, as set forth on the cover page of this prospectus, remains the same. The pro forma information discussed above is for illustrative purposes only. Our net book value (deficit) following the completion of the offering is subject to adjustment based on the actual price of our common stock and other terms of this offering determined at pricing.

The following table summarizes, on the same pro forma basis as of , the total number of shares of Class A common stock purchased from us, the total cash consideration paid to us and the average price per share of Class A common stock paid by our existing owners and by new investors purchasing shares of Class A common stock in this offering.

	Shares Purchased		Total Consideration	
	Number	Percent	Amount	Percent
			(in thousands)	
Existing owners		%	\$	%
New investors in this offering				
Total		%	\$	%

If the underwriters were to exercise in full their option to purchase additional shares of common stock from us, the percentage of shares of our Class A common stock held by existing owners would be %, the percentage of Class A common stock held by the Continuing Employees would decrease to approximately % of the total number of shares of our Class A common stock outstanding after this offering and the percentage of shares of our common stock held by new investors in this offering would be %.

UNAUDITED PRO FORMA CONDENSED CONSOLIDATED FINANCIAL INFORMATION

Defined terms included below will have the same meaning as terms defined and included in this prospectus.

The unaudited pro forma condensed consolidated financial information has been prepared in accordance with Article 11 of Regulation S-X under the Securities Act, as amended. The pro forma condensed consolidated financial information is presented to provide relevant information necessary for an understanding of X-energy upon consummation of the Company's initial public offering after giving pro forma effect to the following (collectively, the "pro forma events"):

- the amendment and cashless exercise of the 2024 Warrants in exchange for Series C Units, as described in the subsequent event footnote to the Company's historical financial statements as of and for the year ended December 31, 2025, which is presented below within Other Adjustments;
- the Reorganization Transactions;
- the impacts associated with the Tax Receivable Agreement, as described under "Relationships and Related Party Transactions — Tax Receivable Agreement;" and
- this offering and the application of the estimated net proceeds from this offering as described in "Use of Proceeds."

The unaudited pro forma condensed consolidated balance sheet as of December 31, 2025 gives effect to the pro forma events as if they had been consummated on December 31, 2025.

The unaudited pro forma condensed consolidated statement of operations for the year ended December 31, 2025 gives effect to the pro forma events as if they had been consummated on December 31, 2025.

We have derived the unaudited pro forma condensed consolidated balance sheet and the unaudited pro forma condensed consolidated statement of operations from our consolidated financial statements as of and for the year ended December 31, 2025, which are included elsewhere in this prospectus. The historical financial statements have been prepared in accordance with GAAP.

Except as otherwise indicated, the unaudited pro forma condensed consolidated financial information presented assumes no exercise by the underwriters of their over-allotment option to purchase additional shares.

As a public company, we will be implementing additional procedures and processes for the purpose of addressing the standards and requirements applicable to public companies. We expect to incur additional annual expenses related to these additional procedures and processes and, among other things, directors' and officers' liability insurance, director fees, additional expenses associated with complying with the reporting requirements of the SEC, transfer agent fees, costs relating to additional accounting and administrative personnel, increased auditing, tax, and legal fees, stock exchange listing fees and other public company expenses. We have not included any pro forma adjustments relating to these expenses in the information below.

The pro forma adjustments are based on available information and upon assumptions that management believes are reasonable in order to reflect, on a pro forma basis, the effect of the pro forma events on the historical financial information. The adjustments are described in the notes to the unaudited pro forma condensed consolidated financial information.

condensed consolidated balance sheet and the unaudited pro forma condensed consolidated operations. The unaudited pro forma condensed consolidated financial information is presented for illustrative purposes only and do not purport to represent the results of operations or the financial position that would actually have occurred had the pro forma events been consummated on the dates and periods presented. We do not project the Company's results of operations or financial position for any future date or period.

The unaudited pro forma condensed consolidated financial information should also be read in conjunction with "Basis of Presentation — Organizational Structure," "Capitalization," "Management's Discussion and Analysis of Financial Condition and Results of Operations of X-energy," and our financial statements and accompanying notes included elsewhere in this prospectus.

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Offering

The Company is offering _____ shares of common stock in this offering at an assumed offering price of \$ _____ per share, which is the midpoint of the estimated offering price range set forth on the cover page of this prospectus.

Reorganization Transactions

Reorganization Transactions refer to the organizational transactions immediately before the consummation of the offering, which (i) the eighth amendment and restatement of the XERC LLC Agreement to, among other things, effect a recapitalization in which all existing ownership interests in XERC are converted into Common Units; (ii) the amendment and restatement of the X-Energy, Inc. certificate of formation to, among other things, authorize two classes of common stock; (iii) X-Energy, Inc.'s designation as manager of XERC, (iv) X-Energy, Inc.'s acquisition of Common Units held by the Blocker Companies in connection with the Blocker Mergers, (v) X-Energy, Inc.'s acquisition of all of the Common Units held by the Former Owners (except for Management LLC, who is addressed in clauses (vi) and (vii), below) and the Common Units held by the Continuing Equity Owners, in each case, in exchange for an equal number of shares of Class A common stock, (vi) the second amendment and restatement of the Management LLC Agreement to, among other things, effect a recapitalization in which all existing ownership interests in Management LLC are converted into one class of Common Units, (vii) Management LLC's conversion of all of its Common Units of XERC to X-Energy, Inc. in exchange for an equal number of shares of Class A common stock, which shares shall remain subject to the same vesting conditions applicable to the corresponding Common Units immediately prior to such contribution and (viii) X-Energy, Inc.'s acquisition of the Continuing Equity Owners of a number of shares of Class B common stock (equal to the number of Common Units held by the Continuing Equity Owners) in exchange for a nominal cash contribution. Following the consummation of the offering, the combined company organized in an Upstream Merger will hold substantially all of the assets and the business of the company will be held by X-Energy Reactor LLC, as more fully described elsewhere in this prospectus.

Following the consummation of this offering, Continuing Equity Owners may redeem their Common Units for cash or Class A stock on a one-for-one basis.

Tax Receivable Agreement

Upon closing, we will enter into a Tax Receivable Agreement with XERC and the Tax Receivable Trust. Pursuant to the Tax Receivable Agreement, we will generally be required to pay the TRA Holders the amount of the cash tax savings, if any, in U.S. federal, state, and local taxes that are attributable to the tax attributes measured with respect to, net income or profits, and any interest related thereto that we are expected to realize, as a result of the tax attributes subject to the Tax Receivable Agreement, in cash or Class A stock.

- Basis Adjustments;
-

Existing Basis;

- Blocker Tax Attributes; and
- Interest Deductions.

Due to the uncertainty as to the amount and timing of future exchanges of Common Units by TRA Holders and as to the price of our Class A common stock at the time of any such exchanges, our pro forma condensed consolidated financial information does not assume that any Continuing Equity Owners have exchanged Common Units that would create an obligation under the Tax Receivable Agreement. Therefore, no increases in tax basis in the XERC Group's assets or other tax benefits that may be realized under the Tax Receivable Agreement with respect to the Continuing Equity Owners' right, have been reflected in the unaudited pro forma condensed consolidated financial information. Future exchanges will result in incremental tax attributes and potential cash tax savings for us. Depending on our assessment of the realizability of such tax attributes, the arising Tax Receivable Agreement liability will be recorded at the exchange date against equity, or at a later point through income.

However, if the Continuing Equity Owners as TRA Holders were to exchange or sell all of their Common Units, we would recognize an incremental deferred tax asset of approximately \$ 10.0 million, and an incremental liability under the Tax Receivable Agreement of approximately \$ 10.0 million, and

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(i) all exchanges or purchases occurred on the same day; (ii) a price of \$ 10.00 per share; (iii) the corporate tax rate; (iv) that we will have sufficient taxable income to fully utilize the tax attributes; and (v) no material changes in tax law. These amounts are estimates and have been prepared for informational purposes only. The actual amount of deferred tax assets and related liabilities that we will recognize will differ based on, among other things, the timing of the exchanges, the price per share of our Class A common stock at the time of the exchange, and the tax rates then in effect and certain change of control, Breach or early termination events occurring.

The term of the Tax Receivable Agreement will continue until all such tax benefits have been realized or expired unless we exercise our right to terminate the Tax Receivable Agreement, in which case our obligations under the Tax Receivable Agreement will be accelerated and we will be required to make a lump sum payment to the TRA Holders in an amount equal to the present value of future payments under the Tax Receivable Agreement. This payment would be based on certain assumptions, including that we will have sufficient taxable income to fully utilize the benefits arising from the tax attributes subject to the Tax Receivable Agreement. Importantly, upon a change of control, the Tax Receivable Agreement will not terminate nor will a single, accelerated lump sum payment be due. If there is a Material Breach of the Tax Receivable Agreement, we experience a change of control (as defined in the Tax Receivable Agreement) or we elect an early termination of the Tax Receivable Agreement and do not elect a single, lump sum payment, our (or our successor's) future payments under the Tax Receivable Agreement for each taxable year after any such event would be calculated utilizing certain valuation assumptions, including that (i) any Common Units that have not been exchanged are deemed exchanged for their fair market value of the shares of our Class A common stock at the time of the change of control and (ii) XERC will have sufficient taxable income to fully utilize the tax attributes covered by the Tax Receivable Agreement.

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Unaudited Pro Forma Condensed Consolidated Balance Sheet
As of December 31, 2025
(in thousands)

	<u>X-Energy Reactor Company, LLC</u>	<u>Other Material Transactions</u>	<u>Reorganization Transaction Adjustments</u>	<u>Subtotal</u>	<u>Other Transactions</u>
ASSETS					
Current assets					
Cash and cash equivalents	\$ 458,932	\$ —	\$	\$	
Short-term investments	304,908	—			
Accounts receivable	32,940	—			
Unbilled receivables and contract assets	41,529	—			
Prepaid and other current assets	11,491	—			
Due from related parties	4,580	—			
Total current assets	854,380	—			
Long-term investments	261,458			261,458	
Restricted cash	3,698	—			
Property and equipment, net	50,105	—			
Operating lease right-of-use-assets	22,696	—			
Other long-term assets	18,934	—			
Total assets	\$ 1,211,271	\$ —	\$	\$	
LIABILITIES, MEZZANINE EQUITY, AND MEMBERS' DEFICIT					
Current liabilities:					
Accounts payable	\$ 3,363	\$ —	\$	\$	
Accrued liabilities	51,217	—			
Due to related parties	4,225	—			
Total current liabilities	58,805	—			
Long-term deferred revenue	15,153	—		15,153	
Long-term operating lease liabilities	20,887	—			
Warrant liabilities	274,166	(263,390)	(A)	(C)	
Total liabilities	\$ 369,011	\$ (263,390)	\$	\$	
Mezzanine equity					
Class A Common Units	1,800	—		(B)	
Class B Common Units	93,353	—		(B)	
Series A redeemable convertible preferred units	218,408	—		(B)	
Series A-1 redeemable convertible preferred units	21,477	—		(B)	
Series B redeemable convertible preferred units	101,382	—		(B)	
Series C redeemable convertible preferred units	265,797	—		(B)	
Series C-1 redeemable convertible preferred units	686,715	263,390	(A)	(B)	
Series D redeemable convertible preferred units	677,623	—		(B)	
Total mezzanine equity	2,066,555	263,390			
Class A Common Stock	—	—		(B)	
Class B Common Stock	—	—		(B)	

	X-Energy Reactor Company, LLC	Other Material Transactions	Reorganization Transaction Adjustments	Subtotal	Other Transactions
Accumulated deficit	(1,236,345)	—	(C)		
Accumulated other comprehensive income	(117)	—			
Additional paid-in capital	12,167	—	(B)		
			(D)		
Members' deficit / Total shareholders' equity (deficit) attributable to X-energy	(1,224,295)	—			
Noncontrolling Interest	—	—	(D)		
Total shareholders' equity / members' deficit	(1,224,295)	—			
Total liabilities and shareholders' equity / members' deficit	\$ 1,211,271	\$—	\$	\$	\$

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Unaudited Pro Forma Consolidated Statement of Operations
For the Year Ended December 31, 2025
(in thousands, except share and per share amounts)

	XERC	Transaction Accounting Adjustments – Reorganization	Subtotal	Transaction Accounting Adjustments – Offering
Services revenue	\$ 94,260	\$	\$	\$
Grant income	14,838			
Total revenues and grant income	109,098	—		—
Operating expenses				
Direct costs	161,367			
Selling, general and administrative	116,318			
Research and development	1,708			
Total operating expenses	279,393	—		—
Operating loss	(170,295)	—		—
Other income (expense)				
Interest expense	(475)			
Interest income	20,293			
Other income (expense), net	(239,301)			
Total other income (expense), net	(219,483)	—		—
Net loss	(389,778)	—		—
Net loss attributable to noncontrolling interest	—		(AA)	
Net loss attributable to common shareholders	<u><u>\$(389,778)</u></u>	<u><u>\$</u></u>	<u><u>\$</u></u>	<u><u>\$</u></u>
Pro forma net income per share:				
Basic and diluted				
Weighted average shares outstanding				
Basic and diluted				

Please refer to the notes to the unaudited pro forma condensed consolidated financial info

Notes to Unaudited Pro Forma Condensed Consolidated Financial Information

Note 1 Basis of Presentation

The unaudited pro forma condensed consolidated financial information has been prepared in accordance with Article 11 of Regulation S-X and presents the pro forma financial condition and results of operations of the Company based upon the historical financial information after giving effect to the pro forma events and related adjustments set forth in the notes to the unaudited pro forma condensed consolidated financial information.

The unaudited pro forma condensed consolidated financial information does not include management adjustments for expected effects of the pro forma events.

The unaudited pro forma condensed consolidated balance sheet as of December 31, 2025, is based on the balance sheet as of December 31, 2024, adjusted to the pro forma events as if they had occurred on December 31, 2025. The unaudited pro forma consolidated statement of operations for the year ended of December 31, 2025, gives effect to the pro forma events as if they had occurred on January 1, 2025.

The unaudited pro forma condensed consolidated financial information does not give effect to the income tax benefit or expense associated with the pro forma adjustments due to the history of income tax in current and previous periods and full valuation allowance on our deferred tax assets.

Note 2 Adjustments to the Unaudited Pro Forma Condensed Consolidated Balance Sheet

The following adjustments were made related to the unaudited pro forma condensed consolidated balance sheet as of December 31, 2025:

- A. Reflects the cashless exercise of the 2024 Warrants in exchange for Series C-1 Preferred Units, which occurred in March 2026.
- B. Immediately preceding the closing, as part of the Reorganization Transactions, XERC will convert its Series A redeemable convertible preferred units, Series A-1 redeemable convertible preferred units, Series B redeemable convertible preferred units, Series C redeemable convertible preferred units, Series C-1 redeemable convertible preferred units, and Series D redeemable convertible preferred units will be converted into Class A common stock, equal to the number of units of each series of units retained by each, and reclassified as permanent equity.

As a result of the Reorganization Transactions, the operating agreement of XERC will be amended and restated to, among other things, modify XERC's capital structure by reclassifying its outstanding units in XERC into a single class of LLC Units, the Common Units.

This adjustment further reflects the acquisition of a number of shares of Class B common stock from the Continuing Equity Holders, equal to the number of Common Units retained by the Continuing Equity Holders, for nominal consideration.

- C. Represents the elimination of XERC's legacy warrants, which were subject to cashless exercise immediately prior to closing. The issuance of shares related to this instrument is reflected in adjustment (B).
- D. Immediately following closing, the economic interests held by the noncontrolling interest holders (comprising Common Units issued at closing to Continuing Equity Owners) will be a

%. The following table shows the economic interest of XERC immediately offering:

	Units
X-energy	
Other Common Units	
Total Common Units	

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- E. Reflects the net effect on cash of the receipt of offering proceeds to us of \$ million on the sale by the Company of million shares of Class A common stock at an initial public offering price of \$ per share (the midpoint of the price range on the cover of this prospectus), after deducting the estimated underwriting commissions.
- F. Reflects estimated transaction costs of \$ million, including certain legal, accounting and other related costs. Upon completion of the offering, deferred transaction costs of \$ million will be charged against the proceeds from the offering with a corresponding reduction to additional paid-in capital; and transaction costs of \$ million will be expensed as incurred based on the actual costs.

Note 3 Adjustments to the Unaudited Pro Forma Condensed Consolidated Statement of Operations

The following adjustments were made related to the unaudited pro forma condensed consolidated statements of operations for the years ended December 31, 2025:

- AA. Represents net income/(loss) attributable to the noncontrolling interest based on the ownership structure at closing, as noted at adjustment (D).
- BB. Represents the elimination of the income statement impact related to the change in XERC's legacy warrants. The elimination of these warrants, which do not survive the offering, is reflected at adjustment (A) and (C), respectively.
- CC. Represents recurring stock-based compensation expense associated with the IPO Equity Incentive Plan, which are expected to be issued at closing.

Note 4 Pro Forma Income/Loss Per Share

Represents the pro forma income/(loss) per share calculated using the weighted average shares outstanding as a result of the pro forma events, assuming the shares were outstanding since January 1, 2023. As the pro forma events are being reflected as if they had occurred at the beginning of the periods presented, the calculation of weighted average shares outstanding for basic and diluted pro forma income/(loss) per share assumes that the shares issuable relating to the pro forma events have been outstanding for the entire periods presented.

Pro forma basic and diluted income/loss per share**Numerator:**

Pro forma net income/(loss)

Less: Pro forma net income/(loss) attributable to noncontrolling interests

Pro forma net income/(loss) attributable to common stockholders – basic and diluted

Denominator:

Pro forma weighted average shares outstanding of Class A common stock – basic and diluted

Pro forma basic and diluted income/(loss) per share, Class A common stock⁽¹⁾

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- (1) Pro forma basic and diluted income/(loss) per share of Class B common stock is not presented as Class B shares have no economic rights.

The computation of pro forma diluted loss per share does not assume conversion of preferred stock or the contingent issuance of securities as their inclusion in the pro forma loss per share calculation would not have been antidilutive.

MANAGEMENT’S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS OF X-ENERGY

The following discussion and analysis of X-energy’s financial condition and results of operations should be read with X-energy’s audited consolidated financial statements and notes thereto included elsewhere in this prospectus. Certain of the information contained in this discussion and analysis, set forth elsewhere in this prospectus, including information with respect to plans and strategy for the future of the business, includes forward-looking statements that involve risks and uncertainties. As a result, actual results, factors, including those factors set forth in the section “Risk Factors,” X-energy’s actual results may differ materially from the results described in or implied by the forward-looking statements contained in this following discussion and analysis. Refer to the section entitled “Risk Factors” to gain an understanding of the important factors that could cause actual results to differ materially from X-energy’s forward-looking statements. For more information, see the section entitled “Cautionary Note Regarding Forward-Looking Statements.”

Unless otherwise indicated or the context otherwise requires, references in this section to “Company,” “we,” “us,” “X-energy,” or “our” refer to the business of XERC for the period from the initial offering and X-Energy, Inc. and its subsidiaries for all periods after the initial offering.

Overview of Our Business and History

X-energy is a leading designer of advanced nuclear reactor technology (commonly referred to as modular reactors, “SMRs”) and manufacturer of advanced nuclear fuels. We believe these small modular generation technologies help satisfy historically unprecedented electricity demand growth, driven by the development of AI and associated data center infrastructure. Total demand for new electricity is expected to increase globally by 7,626 TWh from 2023 to 2030 and the challenges associated

this demand have led policymakers and industry leaders to recognize nuclear energy, particularly nuclear, as a key component to address this need.

Founded in 2009 by Dr. Kamal “Kam” Ghaffarian to bring clean, safe, secure and advanced nuclear technology to market, X-energy is seeking to redefine the energy industry through its flagship Xe-100, an advanced small modular HTGR, in development for nearly a decade. The Xe-100 is designed to generate 80 megawatts of electric power or 200 megawatts of thermal output, or a combination thereof. This reactor technology builds on more than 50 years of research and development in the global nuclear industry and the operating experience of previous HTGRs including the Peach Bottom in the U.S., and Dragon in the U.K. in the 1960s-1970s, and more recently with Chinese HTGR deployments in the 21st century.

The Xe-100 has several technological attributes that we believe make it advantaged compared to other sources of baseload generation. These include advanced safety features, virtually no direct greenhouse gas emissions during generation, high thermal output, load-following capabilities and modularity, all of which enable X-energy to more specifically meet a customer’s power and/or industrial heat needs. X-energy’s modular Xe-100 design directly translates into simplicity of project delivery through reduced construction complexity and labor intensity during construction, which we believe will lead to lower costs and shorter deployment timelines when compared with conventional nuclear energy sources. X-energy has announced the deployment of its Xe-100 into a four-reactor format that outputs 320 MWe (or 800 MWt). By deploying four independent reactor modules instead of a single unit, this optimized four-reactor design inherently delivers the high levels of reliability and redundancy required for both AI and industrial applications.

X-energy’s reactors use a TRISO coated particle fuel in the form of a spherical ‘pebble’, or TRISO-X fuel. This pebble fuel consists of HALEU fuel kernels individually encapsulated in layers of silicon carbide and pyrolytic carbon, forming miniature containment systems that trap fission products. These particles are then embedded in a graphite matrix to make fuel pebbles that possess exceptional safety margins and compact, enabling operations at very high temperatures. The HALEU fuel used in X-energy’s TRISO-X pebble fuel is enriched to 15.5%, a higher energy density form than the less than 5% LEU used in conventional nuclear reactors. Our TRISO-X fuel will be produced at our fuel fabrication facility in Paducah, Tennessee. The first facility, known as TX-1, began construction in October 2024 and is expected to be completed by the first half of 2028. Upon completion, it is expected to be North America’s first commercial advanced nuclear fuel

fuel fabrication facility. The TX-1 facility will have sufficient production capacity to support the deployment of the first 11 Xe-100 reactors at steady state operations.

In addition to its technology leadership, X-energy has three high-quality customers in DTE Energy, Centrica, and Centrica, who we expect will underpin the deployment of the initial fleets of Xe-100 reactors. Together, assuming each customer exercises its contingent rights in full, these three customers represent a more than 11 gigawatts electric (“GWe”), 144 reactor pipeline across the U.S. and the U.K. X-energy has advanced development efforts already underway on the first Dow project at its Seadrift Operations in Texas and the first Amazon project in connection with Energy Northwest.

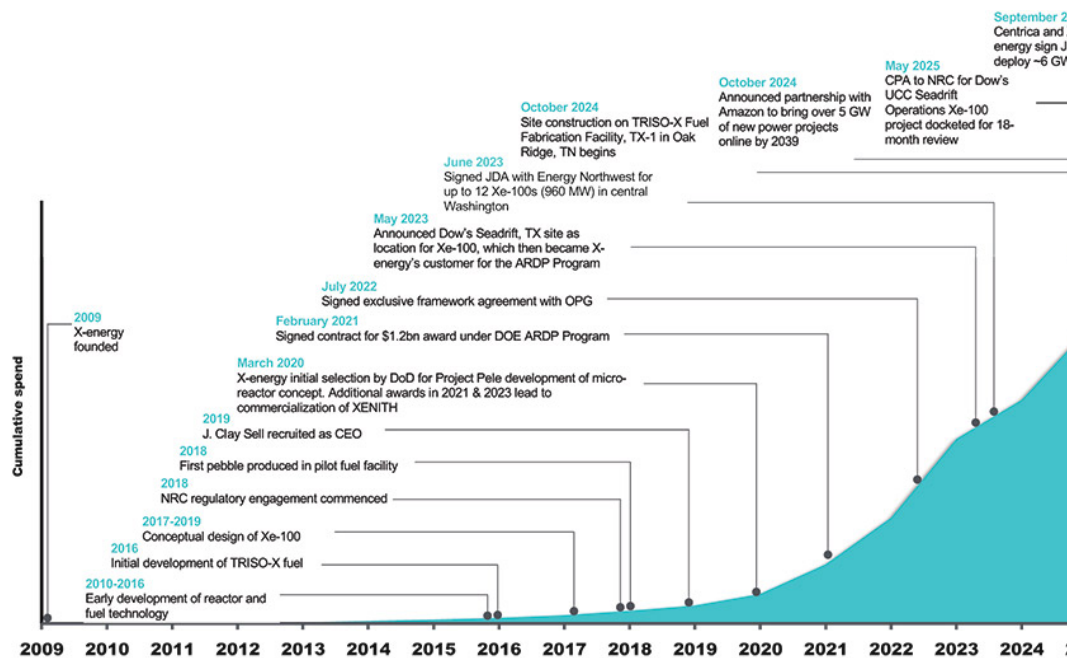
X-energy maintains a strong relationship with the DOE and in December 2020 was awarded a \$1.2 billion as part of its selection as one of two awardees in the ARDP, the most substantial federal commitment to deploying advanced nuclear technology. The cooperative agreement for the project, signed in February 2021, provides 50/50 cost share of \$2.4 billion of eligible costs (\$1.2 billion re-

through 2027, allowing X-energy to continue work toward design, licensing, commercial construction of its first-of-a-kind commercial advanced nuclear plant and commercial TRISO fuel fabrication facility, while benefiting from decades of nuclear experience and knowledge within

As of December 31, 2025, we have been reimbursed approximately \$438 million in funding under the ARDP. We submit our budgets through an ongoing “budget period” basis tied to project milestones under the ARDP Agreement, and our current budget covers a budget period that began in March 2025 and extends through August 2026. We submit a Continuation Application to the DOE to extend the current budget period in subsequent periods. Extensions beyond the current budget period are subject to DOE discretionary approval. Under the terms of the ARDP Agreement that rely on the Office of Management and Budget (OMB) guidance, the total extension of the award may not exceed three years (for a total period of performance of 10 years). Any additional extension would require an approval within DOE at the discretion of the Contracting Officer. If we are unable to obtain extensions and incur eligible costs beyond the currently approved period of performance, we would forgo reimbursement for such costs and would have no obligation of unobligated funds at closeout. There can be no assurance that we will receive additional funding beyond the current budget period or that extensions will be granted.

Our organizational structure following the offering and the Reorganization Transactions is referred to as an umbrella partnership-C corporation (or “Up-C”) structure. Pursuant to the terms of the offering, following this offering we will hold a number of Common Units equal to the number of our outstanding shares of Class A common stock, and holders of Common Units (each, an “Continuing Equity Owner”) (other than us) will hold a number of Common Units equal to the number of our outstanding shares of Class B common stock. The Up-C structure was selected in order to (i) provide Continuing Equity Owners with an option to continue to hold their economic ownership in the business in “pass-through” form for U.S. federal income tax purposes through their ownership of Common Units and (ii) potentially allow our Continuing Equity Owners and us to benefit from certain tax savings that we might realize in the future, as more fully described in the subsection titled “Key Milestones that Transformed Our Business — Tax Receivable Agreement.”

Key Milestones that Transformed Our Business



- **2009:** X-energy founded by Kam Ghaffarian to bring safe, secure, clean and affordable energy to the global marketplace.
- **2010–2011:** Early studies working with a small team of engineers and consultants on various technologies.
- **2012–2014:** Onboarding expertise on HTGRs from South Africa's Pebble-bed Modular program: Dr. Martin van Staden and Dr. Eben Mulder.
- **2015–2016:** Conducted multi-dimensional trade studies on market needs, reactor size, and cost to refine early reactor concepts.
- **2016:** Initial development of TRISO-X pebble fuel at Oak Ridge Pilot Facility.
- **2017–2019:** Conceptual design of the Xe-100 after years of research & development.
- **2018:** NRC regulatory engagement commenced.
- **2018:** First pebble produced on commercial-scale equipment in pilot fuel facility at ORNL National Lab in Tennessee.
- **2019:** J. Clay Sell appointed as CEO.
- **March 2020:** X-energy initial selection by the DOD for Project Pele development of micro-reactor concept. Across three awards (2020: \$14.3 million, 2021: \$28.7 million, 2023: \$17.5 million) X-energy received approximately \$60.0 million from the DOD which eventually led to the commercialization of XENITH microreactor. As of the date of this prospectus, the performance for the XENITH microreactor has expired, but the period of performance for Project Pele remains active.
- **February 2021:** Signed contract for initial \$1.2 billion award won under the DOE's ARDP.
- **July 2022:** Signed exclusive framework agreement with Ontario Power Generation for deployment in Canada. OPG also became an equity investor in this year.
- **May 2023:** Announced Dow's Seadrift, Texas site as location for Xe-100. Dow becomes ARDP from the DOE.
- **June 2023:** Signed JDA with Energy Northwest for up to 12 Xe-100s (960 MW) in Washington.
- **October 2024:** Construction of TX-1 site in Oak Ridge, Tennessee commenced. Vertical deployment began in **September 2025**.

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- **October 2024:** Announced engagement with Amazon for options to bring over 5 GWe projects online by 2039. First Amazon project identified as Energy Northwest site.
- **May 2025:** Construction permit application to NRC for Dow's UCC Seadrift Xe-100 project for 18 month review timeline after March 2025 submission.
- **September 2025:** Centrica and X-energy sign JDA to deploy 6 GWe of new nuclear capacity in the U.K.
- **February 2026:** TRISO-X receives NRC Special Nuclear Material License for a fuel fabrication facility.

Our Business Model

We have an intellectual property-driven business model based on our reactor and fuel. We derive revenues from technology licensing, services and fuel operations that span the development and operation of the reactors.

- **Reactors:** The revenue stream from reactors includes technology fees for the use of our intellectual property of the Xe-100 technology. We will not own and operate the facilities themselves. This model believe significantly reduces the amount of capital needed to operate our business.

We anticipate offering site-specific engineering and site characterization, project planning, construction coordination, construction support, regulatory support, procurement support and long-term operations to customers. Utilizing our knowledge and expertise in licensing, construction, project management and other processes, we plan to provide customers with a full suite of value-added services throughout the development of the nuclear power facilities. At the same time, we expect to generate recurring revenue from services such as the ongoing maintenance and operator training throughout the anticipated 60-year life of a facility.

- **Fuel:** We intend to provide manufacturing services to customers, including producing and assembling a full load of both TRISO-X fuel and an LEU-based TRISO fuel at commissioning of a plant. We intend to generate additional long-term recurring revenue from our own proprietary TRISO-X fuel. We are not required to refuel plants during the anticipated 60-year life of each facility. We expect to manage the inventory risks related to uranium or enriched uranium fuel feedstock. We intend to provide fuel fabrication supply services (e.g., transformation of HALEU into the final TRISO-X fuel) to our customers and assume limited risks associated with holding the uranium or enriched uranium feedstock. Additionally, we will not have any responsibility for spent fuel management. The design of such facilities to adequately handle spent fuel during the life of the plant and its operations, spent fuel remains the responsibility of the plant operator. Thereafter, permanent spent fuel management remains the responsibility of the DOE.

Factors Affecting Our Performance

The growth and future success of our business depends on many factors. While each of our current and present significant opportunities for our business, they also pose important challenges that we must successfully address in order to sustain our growth, improve our results of operations and maintain our long-term profitability.

Our ability to commence and expand commercial operations

Our business model is dependent on our commencing and expanding commercial operations. We currently anticipate initial customer deliveries to achieve mechanical completion in the early 2020s, which we expect to take place 1-2 years ahead of commencement of operations. Commencement of construction for these projects is dependent upon finalizing and achieving design maturity, permitting for customers, and supporting our customers in pursuing necessary permits and licenses from the relevant regulatory team of engineers, scientists, and other staff is highly motivated and committed to accomplishing these challenges. Failure to complete any one of these tasks in a timely manner could result in us being unable to begin production in the anticipated timeframe.

We are developing a global network of potential customers and supply chain partners that we believe will play an integral role in bringing our technology to market. In the near term, TRISO-X and other technologies will

depend on the U.S. government for access to HALEU given the current inability to access global enrichment. The government and commercial enrichers are developing enrichment capabilities for future demand. Additionally, the imposition of tariffs and impacts of inflation on raw materials or supplied components for our reactors could have a material adverse effect on our operations. Management has considered the potential economic impact of these tariffs as they relate to our suppliers and raw materials, and we believe our timeline to expand commercial operations will soften the impact these tariffs may have on the overall cost of our reactors. To the extent the U.S. government restricts our access to HALEU or the U.S. fails to obtain sufficient HALEU for our needs, our ability to commence and expand commercial operations may be significantly impaired.

We have three publicly announced customers in (1) Dow, (2) Amazon, both of whom have made equity investment in X-energy and (3) Centrica. As of the date of this prospectus, we also have a growing pipeline of customer opportunities across both power generation and industrial heat use across multiple geographies. We believe this growing pipeline of announced and potential customers demonstrates the demonstration of the increased market interest in, and provides external validation for, our product.

Widespread acceptance of nuclear power as an emissions-free energy source

Our growth and future success are dependent on public support for nuclear power in the United States and other countries where we intend to market and sell our technology, including Canada, the U.K., France, and other countries in Europe and Asia, among others. Electricity demand is accelerating globally, with global electricity consumption estimated to grow by more than 25% from approximately 30,000 TWh in 2020 to approximately 37,000 TWh by 2030. In the U.S., electricity demand is currently driven by data center buildout from cloud computing providers, industrial growth and reshoring of manufacturing, and electrification (e.g., electric vehicle installed base). Given the importance of nuclear power to decarbonization, center expansion, Amazon, Google and Meta, among other companies, have signed a pledge to achieve global nuclear capacity threefold by 2050. Industrial companies have historically relied on traditional fossil fuels such as natural gas to generate electricity and steam to power their facilities' industrial processes. In order for our business model to succeed, we will depend on energy providers sourcing a larger portion of their energy from nuclear power facilities instead of sourcing energy from fossil fuel facilities. As the market for SMRs has not yet been established, as we are one of the pioneers in the industry. As we continue to invest in the capabilities of our SMRs, we expect a growing number of jurisdictions, including the U.S. and globally to adopt SMRs as an always-on, carbon emissions-free alternative to fossil fuel sources. As nuclear power and SMRs, in particular, gain widespread acceptance, we expect that the revenue generated from sales of our SMRs to increase.

Inflation, supply chain pressures, and rising development costs could increase our operating expenses and affect our margins

We are a development and design stage company that is preparing its flagship product for substantial governmental support and collaboration from a team of commercial partners. As with Xe-100, TRISO-X fuel and other aspects of our business, we have been, and expect to continue to be, adversely affected by price increases from our suppliers and logistics partners as a result of inflation, as well as other factors such as increased development, labor and overhead costs.

The Xe-100 and corresponding TRISO-X fuel are costly, complex and challenging to design. Sources of funding for the estimated cost include U.S. government funding, whether via the ARDP, DOE sources, and additional funding to be provided by X-energy's designated partner under the ARDP. Dow is a sub-awardee and our designated partner under the ARDP. The ARDP grant is included in different components. First, for non-recurring engineering work related to the design of the Xe-100, X-energy is responsible for the funding of such engineering work and is eligible to receive 50% reimbursement for this funding through the ARDP program. Secondly, for TRISO-X, Dow is responsible for the funding and is eligible to receive 50% reimbursement for this funding through the ARDP program. These two ARDP-related programs are not tied to Dow's funding requirements. For the construction of the Xe-100 plant, Dow is responsible for the funding of the Xe-100 plant at the Seadrift site and is eligible to receive 50% reimbursement for this funding through the ARDP program.

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Dow's current funding commitments are representative of a typical energy project development process. At present, Dow's funding is released as project milestones are reached; however, X-energy has a financial obligation to construct the plant without Dow's funding. As we are currently in preliminary design, X-energy is receiving revenues from Dow pursuant to our MPDA for services including design, engineering, and construction services related to the Seadrift site, NRC licensing activities, and other technology use typical of a nuclear power plant. As project milestones are reached, Dow's funding commitments are expected to increase, as Dow will need to fund long-lead procurement and engineering services in advance of commercial operations. X-energy has no obligation to move forward with the project without funding from Dow. If final investment decision is made, Dow is expected to continue to be responsible for the funding of the construction of the Xe-100 plant, which work is eligible under the ARDP grant for a 50% cost share. If Dow does not make a final investment decision with respect to the Seadrift project, X-energy is under no obligation to continue funding to the Dow project or construction on the plant itself. In order to continue our participation in the ARDP program, we would need to identify another funding source within a reasonable amount of time for the demonstration portion of the project, and failure to do so could result in significant delays, increased costs, and loss of revenue.

We continue to work with our commercial partners to seek opportunities for cost reduction and efficiency with ARDP work. Irrespective of ARDP funding, we nonetheless expect sustained and increased costs in the future to directly impact our operating expenses, which could ultimately impact expected margins across our business.

The capital expenditure for our reactors will not be included on our balance sheet.

Our ability to obtain and maintain regulatory approvals at federal, state and local levels

Our capacity for continued growth and ability to achieve and maintain profitability depends in part on our ability to obtain and maintain regulatory approvals across multiple jurisdictions, including international, federal, state and local levels. The federal government, along with each state and jurisdiction in which we operate, maintains distinct regulatory frameworks. These include

regulations that can directly or indirectly affect our operations and those of our customers, including those related to real estate usage, environmental sustainability, employment and labor practices and stakeholder engagement. We believe that we have an experienced licensing team that has developed strong relationships with the NRC and other regulators. Our success will depend on our licensing team's ability to continue to obtain and maintain Regulatory Approvals on commercially reasonable timelines. Our first project's first Construction Permit Application ("CPA") received an 18-month review schedule. Our engagement with the regulator to date has been constructive, but future projects may experience longer review horizons or difficulties related to agency interaction due to staffing changes or shortages. Our CPA was submitted in March 2025, docketed in May 2025, and the review is expected to be complete by the end of the year in 2026, with receipt of the CPA anticipated in the first quarter of 2027; however, any delays or unforeseen challenges to this or other Regulatory Approvals for us or our customers could negatively impact our business, financial condition, or results of operations. In addition, because our current project and future projects represent first-of-a-kind deployments, they may attract heightened public opposition from local communities, non-governmental organizations, or advocacy groups, which could result in additional review, procedural challenges, or delays in obtaining Regulatory Approvals. We have engaged experienced regulatory, environmental, and stakeholder-engagement advisors to anticipate and address potential risks; however, there can be no assurance that these efforts will prevent delays, increased costs, or unfavorable outcomes.

At the same time, certain U.S. federal policy initiatives currently support the advancement of nuclear energy, infrastructure, and domestic manufacturing projects aligned with energy abundance, energy security, and decarbonization objectives, which may support regulatory coordination or resource allocation.

While we operate in an industry that is subject to, and benefits from, safety and environmental regulations, such regulations have generally become more stringent over time, particularly across international markets. While efforts have been underway in recent years to improve regulatory efficiency, new or updated regulations in our target markets include nuclear safety regulations, grid interconnections and cybersecurity requirements, environmental permits and assessments, and export controls. As a company operating in a heavily regulated industry, our margins could be particularly and adversely impacted by increasing regulatory requirements, regulatory developments or regulatory scrutiny. Regulations on nuclear energy, while historically more stringent across the U.S., the U.K., Canada

and the European Union, where most of our production and sales are expected, are subject to regulatory changes and unpredictable change that could impact our ability to meet projected sales or margins. More stringent regulations could impact our customers' ability to obtain Regulatory Approvals and comply with applicable nuclear safety requirements, which may affect our ability to market our technologies and obtain approvals in other countries.

Our dependence on government policy support and funding for nuclear energy development

Our future growth is largely dependent on our ability to continue to capitalize on government policy support and corporate investment in the nuclear energy industry.

Congress has successfully reinvigorated the U.S. nuclear industry with a concentration of legislative priorities: (1) The initiation of the Advanced Reactor Demonstration Program, and associated funding, to mitigate first-of-a-kind reactor cost and schedule risks; (2) Regulatory framework reforms more aligned with the increased safety of advanced reactors through the Nuclear Energy Innovation and Modernization Act (NEIMA) in 2019 and the Accelerating Deployment of Versatile, Advanced Reactors Act (ADVANCE ACT) in 2024; (3) Enacting financial instruments such as Advanced Production Tax Credits, Manufacturing Tax Credits and Production Tax Credits that are on a par with other clean energy tax credits, such as wind and solar, as included in the Inflation Reduction Act of 2022, and (4) Enacting

deploying federal credit support through the DOE’s Loan Programs Office, which provides loan guarantees and other credit assistance to catalyze first-of-a-kind and large-scale nuclear projects, fuel cycle development, life-extension and uprate initiatives, and other qualifying nuclear infrastructure, thereby lowering the cost of capital and accelerating commercial deployment.

X-energy was selected by the DOE as an awardee under the ARDP in 2020 for “demonstration” projects in the United States, and it is particularly critical to our success. The ARDP is structured as a 50/50 cost-share between the DOE and its private sector awardee for eligible commercial projects to reduce first-of-a-kind reactor risks with the goal to attract follow-on customers both domestic and international in the global marketplace. More specifically, through the ARDP, X-energy is eligible to receive funding for approximately 50% of the cost of designing the Xe-100. X-energy is also eligible to receive a maximum of 50% of the cost of TX-1, its first fuel fabrication facility. Finally, X-energy’s first commercial reactor, Dow, is eligible to receive 50% approximately of the cost to build the first Xe-100, which is currently located at its Seadrift site in Texas.

Congress has appropriated funding that was allocated towards X-energy’s award, approximately \$1.1 billion, as well as recent additional appropriations of \$3.1 billion to the ARDP, an incremental portion of which we expect to be allocated to X-energy. DOE’s ability to receive funding from the appropriated portion of the ARDP is subject to the political process, which is inherently unpredictable and highly competitive. As of December 31, 2025, we have been reimbursed over \$438 million in funding from the ARDP. The funding of government programs is dependent on budgetary limitations, annual appropriations and administrative allotment of funds, all of which may be affected by changes in federal government policies resulting from various political developments. If political support for the development of nuclear energy decreases, including due to policy changes by the current administration and future administrations and changing congressional funding priorities, we may not be able to secure continued government funding under the ARDP, which would adversely affect our development timeline, and financial condition.

Our ability to expand our services offerings

We intend to offer customers a diversified suite of services throughout the life of a project, beginning approximately eight years prior to a plant’s commercial operation date. Our envisioned suite of services includes pre- and post-commercial operations date offerings, whereby we intend to provide customers with critical services related to the design, development, licensing, construction, commissioning, operations and maintenance of the Xe-100. We expect that, as we refine our services offerings based on Dow and the early Amazon and Centrica projects, the number of services we offer and the revenue we generate from our services offerings will grow. We anticipate that our services offerings will have high penetration rates across our future clients and will provide consistent, recurring revenue throughout the expected life of each reactor.

Our ability to obtain additional capital

We operate in a capital intensive industry and expect to continue to incur operating losses in the foreseeable future as we continue to expand and develop, and may need to raise additional capital in the future. If we are unable to raise additional capital when needed, we may have to delay, suspend or discontinue one or more of our lines of business. We may be required to cease operations or liquidate some or all of our lines of business at an earlier stage than otherwise would be desirable and on terms less favorable than might otherwise be available. If we are unable to raise additional capital when needed, we may also be required to relinquish, license or otherwise dispose of rights to technologies or lines of business.

that we would otherwise seek to develop or commercialize on terms that are less favorable than otherwise be available. If we are unable to secure additional capital when needed, we may be required to take additional measures to reduce costs in order to conserve our cash in amounts sufficient to maintain our operations and meet our obligations. These measures may significantly alter our business plan and cause significant delays in the development of our product candidates and ultimately our financial performance and ability to operate as a going concern.

Key Components of Results of Operations

Revenues and grant income

At present, our revenues and grant income are generally derived from contract services provided to the U.S. Government and commercial entities. In the future, we expect to generate revenues from technology fees for the use of the design of the Xe-100 technology, project planning, assembly and construction support, regulatory support, procurement support, long-term services to customer supply of fuel and associated services. Our revenues are generally derived from cost-share agreements as the ARDP provided by the U.S. government and research and development, product development and fuel services provided to other government agencies and commercial entities. A majority of our contracts with the U.S. government are generally subject to FAR and are competitively priced based on the costs of providing the contractual goods or services.

Operating expenses

Direct costs

Direct costs include all costs directly attributable to providing services under contracts with the U.S. Government and grants related to income, such as direct labor, direct materials and subcontracting costs. Indirect costs are allocated to direct costs in the same manner as such costs are defined in disclosure statements of the U.S. Government Cost Accounting Standards.

Selling, general and administrative

Selling, general and administrative expenses consist of human capital related expenses for sales and marketing involved in general corporate functions; rent relating to our office space; professional fees; and other general corporate costs.

Research and development

We conduct research and development activities related to the development and improvement of nuclear technologies pertaining to nuclear reactor and fuel design engineering. The costs incurred for our research and development primarily include equipment, material, and labor hours.

Other income (expense)

Interest expense

Interest expense consists of interest paid and accrued on the Live Oak Credit Facility, 2023 New York Credit Facility, 2023 Bridge Loan, 2024 Convertible Note, and the 2024 Bridge Loan discussed below, along with the amortization of deferred financing fees, costs and debt discount.

Interest income

Interest income is related to our investment of excess cash in money market funds and debt

Other income (expense), net

Other income (expense), net consists of miscellaneous income and expenses such as m gains and losses on various instruments detailed in Note 13 — Fair Value Measurements o consolidated financial statements. Other income (expense), net also consists of the gain a conversion of C-1 Notes and C-2 Notes and related reclassification of other comprehensive i on extinguishment of debt, gains and losses on foreign currency transactions, and other r expenses.

Results of Operations

The following tables set forth our results of operations for the periods presented. The per comparison of financial results is not necessarily indicative of future results.

Comparison of Fiscal Years Ended December 31, 2025 and 2024

The following tables set forth our historical results for the periods indicated and the cha periods:

(\$ in thousands)	2025	2024	\$ Change
Services revenue	\$ 94,260	\$ 83,986	\$ 10,274
Grant income	14,838	36,166	(21,328)
Total revenues and grant income	109,098	120,152	(11,054)
Operating expenses			
Direct costs	161,367	130,115	31,252
Selling, general and administrative	116,318	111,887	4,431
Research and development	1,708	1,662	46
Total operating expenses	279,393	243,664	35,729
Operating loss	(170,295)	(123,512)	(46,783)
Other income (expense), net	(239,301)	10,909	(250,210)
Total other income (expense), net	(219,483)	(2,448)	(217,035)
Net loss	<u>\$(389,778)</u>	<u>\$(125,960)</u>	<u>\$(263,818)</u>

Revenues and grant income

Revenues and grant income decreased by \$11.1 million or 9% for the year ended Decem from 2024. The decrease is driven by a \$19.9 million decrease in a service revenue contract to develop a small modular reactor at one of Dow's U.S. Gulf Coast sites as a result of work s after the submission of the application to the NRC in March 2025 and before the start of const \$5.3 million decrease related to a service revenue contract with the DOD. This was offset by a \$12.7 million increase in revenue and grant income from contracts with the DOE attributable Agreement with the DOE and an increase of \$3.4 million related to service revenue contract Northwest, which began during the second quarter of 2024. Total revenues and grant income ended December 31, 2025 and 2024 attributable to the ARDP Agreement with the DOE, which an aggregate of \$1.2 billion in funding to support design, licensing, and commercialization o and construction of the TX-1, were \$89.2 million and \$76.5 million, respectively, of which and \$36.2 million, respectively were recorded within grant income. The increase in reven

income from the ARDP Agreement with the DOE is due to the increase in activities and nature of work performed during the year ended December 31, 2025.

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Operating expenses

Direct costs

Direct costs increased by \$31.3 million or 24% for the year ended December 31, 2025, primarily due to an increase of \$19.9 million in direct labor, which is driven by an increase in headcount and an increase in bonuses, and an increase of \$6.7 million and \$6.6 million in subcontractor costs and direct materials, respectively, which is driven by an increase in activity related to the ARDP Agreement.

Selling, general and administrative

Selling, general and administrative expenses increased by \$4.4 million or 4% for the year ended December 31, 2025 from 2024. This increase was primarily due to a \$26.8 million increase in selling expenses related costs due to an increase in employee headcount and a \$13.5 million increase in compensation due to new grants made during the year ended December 31, 2025. Selling and administrative expenses further increased by \$12.1 million due to contractor costs related to the ARDP projects, a \$5.5 million increase in technology costs, and a \$0.5 million increase in insurance costs.

The increase in selling, general and administrative expenses was partially offset by the \$55.3 million one-time warrant issuance cost incurred in the prior year. In the year ended December 31, 2024, the Company issued the 2024 Warrant for no consideration and without receiving any asset or benefit, requiring the fair value to be expensed immediately upon issuance.

Research and development

Research and development had an immaterial increase for the year ended December 31, 2025 from 2024.

Other income (expense)

Interest expense

Interest expense decreased by \$15.7 million or 97% for the year ended December 31, 2025, due to the settlement, maturity, redemption and conversion of substantially all of the outstanding debt during the year ended December 31, 2024.

Interest income

Interest income increased by \$17.5 million or 616% for the year ended December 31, 2025, due to investment of excess cash into money market funds starting in the fourth quarter of 2024 and investment in held-to-maturity securities starting in the fourth quarter of 2025.

Other income (expense), net

Other income (expense), net, was \$239.3 million of expenses for the year ended December 31, 2025 compared to \$10.9 million of income for the year ended December 31, 2024, representing a

expenses of \$250.2 million, or 2,294%. The increase was primarily due to a mark-to-market loss on warrant liabilities amounting to \$223.5 million for the year ended December 31, 2025 compared to a mark-to-market gain on warrant liabilities amounting to \$7.9 million for the year ended December 31, 2024.

Liquidity and Capital Resources

We assess our liquidity in terms of our ability to generate adequate amounts of cash to meet our future needs. Our expected primary uses of cash on a short-term and long-term basis are for working capital requirements, capital expenditures, and other general corporate services. Our primary working capital requirements are for project execution activities including purchases of materials, subcontractor services, and payroll which fluctuate during the year, driven primarily by the timing and extent of activities on new and existing projects. Management expects that future operating losses and negative operating cash flows may increase from historical levels because of additional costs and expenses related to the development of our

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technology and the development of market and strategic relationships with other businesses. Our continued existence is dependent upon our ability to obtain additional capital to support our operations.

Historically, our primary source of funding to support our operations has been revenue from income from the ARDP Agreement, contributions and loans from members, loans from financial institutions, as well as capital raises. During the year ended December 31, 2025, we successfully raised approximately \$700.0 million from issuance of our Series D Preferred Units and approximately \$53.4 million from issuance of our Series C-1 Preferred Units. During the year ended December 31, 2024, we successfully raised \$626.5 million from issuance of Series C-1 Preferred Units. While we have historically been successful in obtaining the capital necessary to support our operations, there is no assurance that we will be able to secure additional capital or other financing in the future.

We have had, and expect that we will continue to have, an ongoing need to raise additional capital from outside sources to fund our operations and expand our business. If we are unable to raise additional capital when desired, our business, financial condition, operating results and future prospects would be materially adversely affected. We may not be able to construct the TX-1 fuel fabrication facility, support development of the facility, or conduct other research and development or project and fulfill our current business plan, and we may need to delay or abandon these and other projects. A successful transition to attain our business plan depends upon achieving a level of revenue and grant income adequate to support our operations.

We expect that working capital requirements will continue to be funded through a combination of cash on hand, funding awarded under the ARDP, further issuances of securities, and successful capital raises in connection with our business plan, management anticipates additional increases in operating capital expenditures relating to the development and commercialization of the Xe-100 and fuel fabrication facility. We intend to finance these expenses with further issuances of debt or equity securities. If we expect we will need to raise additional capital and generate revenues and grant income to meet our operating requirements. If we raise additional funds through the issuance of equity or convertible securities, the percentage ownership of our equity holders could be significantly diluted, and the terms of any issued securities may have rights, preferences or privileges senior to those of existing equity holders. If we raise additional funds by obtaining loans from third parties, the terms of those financing arrangements may include negative covenants or other restrictions on our business that could impair our operations and also require us to incur interest expense.

Bank of New York Credit Facility

On July 28, 2020, we executed a credit agreement with Pershing LLC, an affiliate of Bank of America, in the form of a revolving credit facility (the “Bank of New York Credit Facility”) subject to the guarantee by Ghaffarian Enterprises, who represented a related party investor. During the year ended December 31, 2024, we entered into Credit Support Fee and Subrogation Agreements (the “Credit Support Fee Agreements”) with GM Enterprises, LLC and Ghaffarian Enterprises, LLC, which are affiliated with an owner and member of our Board of Directors, which increased the availability of the Bank of New York Credit Facility to \$20.0 million and extended the maturity of the credit support facility to March 26, 2025. In conjunction with the agreements, we agreed to pay GM Enterprises, LLC and Ghaffarian Enterprises, LLC, a monthly 12% credit support fee to be paid in-kind. Pursuant to the terms of the Credit Support Fee Agreements, we paid credit support fees and issued 562,483 Class B Common Units to GM Enterprises, LLC and Ghaffarian Enterprises, LLC. The Class B Common Units have the same rights as the Class B Profits Interests. In October 2024, we settled the outstanding principal and interest on the Bank of New York Credit Facility with a payment of \$20.2 million, and the Credit Support Fee Agreements were terminated. The Bank of New York Credit Facility did not have an outstanding balance as of December 31, 2024, and the facility matured with no balance on March 26, 2025.

Apart from the Bank of New York Credit Facility as disclosed above, we had the following debt outstanding during the years ended December 31, 2025 and 2024 which were settled, matured, or converted:

- *Series C-1 Convertible Notes:* During 2022, we issued a series of convertible promissory notes (the “C-1 Notes”), in the aggregate principal amount of \$57.4 million with an annual interest rate of 7.00%.

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Of the \$57.4 million in principal, \$37.4 million was due on March 31, 2024 and \$20.0 million was due on October 7, 2024 to a related party. On December 5, 2023, in connection with the conversion of the Series C Preferred Units, a portion of the C-1 Notes equal to \$37.4 million was automatically converted into 5,957,402 Series C Preferred Units at a discounted price of \$7.94. On March 29, 2024, the remaining \$20.0 million principal outstanding of the C-1 Notes was converted into 3,210,405 Series C Preferred Units.

- *2023 Bridge Loan:* On October 4, 2023, we entered into a credit agreement (the “2023 Bridge Loan”) with Ares Acquisition Holdings, LP, which was amended through various amendments. In connection with the amendments, the maturity date was extended to March 26, 2025. Since the initial October 4, 2023 issuance of \$10.0 million, \$14.2 million and \$25.8 million were drawn during 2024 and 2023, respectively through amendments and delayed draws. On October 11, 2024, in accordance with the 2023 Bridge Loan’s stated terms, we paid the outstanding principal and interest on the 2023 Bridge Loan of \$53.5 million, and the 2023 Bridge Loan was settled.
- *2024 Convertible Note:* On September 26, 2024, we entered into a convertible promissory note with Amazon.com NV Investment Holdings LLC in the principal amount of \$20.0 million (the “2024 Convertible Note”). On October 11, 2024, in conjunction with the Series C-1 Preferred Units financing and in accordance with the 2024 Convertible Note’s stated terms, the outstanding principal and interest of the 2024 Convertible Note of \$20.1 million was converted into 3,097,402 Series C Preferred Units.
- *2024 Bridge Loan:* On September 26, 2024, we entered into a bridge loan with Escalante, an entity affiliated with an owner and member of our Board of Directors (the “2024 Bridge Loan”) in the amount of \$3.8 million. Concurrently with the issuance of this debt, the Company issued 3,097,402 Class B Common Units to Ghaffarian Enterprises. On October 11, 2024, in accordance with the terms of the 2024 Bridge Loan, we repaid the outstanding principal and interest on the 2024 Bridge Loan of \$3.8 million, and the 2024 Bridge Loan was settled.

terms, the 2024 Bridge Loan was automatically redeemed, and we paid the outstanding interest associated with the 2024 Bridge Loan of \$3.8 million.

- **Series C-2 Convertible Notes:** During 2022 and 2023, we issued convertible notes with an aggregate principal amount of \$28.0 million and \$85.0 million (“C-2 Notes”), respectively. \$70.0 million of the C-2 Notes were issued to related parties. The C-2 Notes were mature on September 30, 2025 and accrue 10.0% of payable-in-kind interest annually. On September 11, 2024, a portion of the C-2 Notes with an aggregate principal balance of \$98.0 million was converted into 16,960,021 Series C Preferred Units. On September 30, 2025, the remaining \$18.0 million of outstanding principal and unpaid accrued interest on the C-2 Notes were converted into Series C Preferred Units.
- **Live Oak Credit Facility:** On June 14, 2021, we and one of our subsidiaries entered into an agreement for a revolving credit facility (the “Live Oak Credit Facility”) with Live Oak Bank. The Live Oak Credit Facility was amended various times from the date of entering into the facility to its maturity. In accordance with the Live Oak Credit Facility’s stated terms, we settled the outstanding principal associated with the Live Oak Credit Facility with a payment of \$4.1 million on September 30, 2024. On October 31, 2024, with no outstanding borrowings, the facility matured. On November 1, 2024, we reestablished the facility, with an expiration date of December 1, 2025. There have been no borrowings on the facility during the year ended December 31, 2025. On December 1, 2025, with no outstanding borrowings, the facility matured.

Please refer to Note 7 — Debt of our audited consolidated financial statements included in this prospectus for additional information.

Off-Balance Sheet Arrangements

We have no off-balance sheet arrangements.

Cash Flows for the Years Ended December 31, 2025 and 2024

(\$ in thousands)	2025
Net cash used in operating activities	\$(149,860)
Net cash used in investing activities	(628,344)
Net cash provided by financing activities	726,130

Operating Activities

For the year ended December 31, 2025, our operating activities used \$149.9 million compared to \$96.2 million for the year ended December 31, 2024. The increase in cash used in operating activities is primarily driven by an increase in activity on the ARDP Agreement, an increase in headcount, and an increase in technology and corporate contractors during the year ended December 31, 2025. These increases were partially offset by lower transaction costs compared to the prior year. We also included one-time expenses related to the 2023 SPAC transaction.

Investing Activities

For the year ended December 31, 2025, our investing activities used \$628.3 million compared to \$1.9 million for the year ended December 31, 2024. The increase in net cash used in investing activities was primarily attributable to purchases of investments amounting to \$565.9 million,

million increase in capital expenditures related to the construction of new facilities. This was partially offset by \$52.5 million in reimbursements received during the period for capital expenditures qualifying under government grant programs.

Financing Activities

For the year ended December 31, 2025, financing activities provided \$726.1 million compared to \$598.3 million for the year ended December 31, 2024. The net cash provided by financing activities the year ended December 31, 2025 was primarily due to the November 2025 issuance of Preferred Units of \$700.0 million and January 2025 issuance of Series C-1 Preferred Units of \$100.0 million, offset by \$25.3 million of cash paid for issuance costs. The net cash provided by financing activities the year ended December 31, 2024 was primarily due to the October 2024 issuance of Series C-1 Preferred Units of \$626.5 million and proceeds received from debt instruments of \$139.1 million; these were partially offset by repayment on debt instruments of \$152.6 million, \$10.8 million of cash paid for equity issuance costs and \$3.8 million of cash paid for debt issuance costs.

Contractual Obligations and Commitments

In addition to our contractual obligations and commitments described under “— Liquidity Resources,” we lease real estate for office space. These leases are classified as operating leases with expiration dates through 2037. See Note 8 — Leases of our audited consolidated financial statements included elsewhere in this prospectus for more information regarding our lease commitments.

Critical Accounting Policies and Estimates

We believe that the following accounting policies involve a high degree of judgment and complexity.

Accordingly, these are the policies we believe are the most critical to aid in fully understanding and evaluating our consolidated financial condition and results of our operations. See Note 2 — Significant Accounting Policies and Recent Accounting Pronouncements to our audited consolidated financial statements appearing elsewhere in this prospectus for a description of our other accounting policies.

The preparation of our consolidated financial statements and related disclosures requires the use of estimates and judgements that affect the amounts reported in those financial statements and accompanying notes. Although we believe that the estimates we use are reasonable, due to the inherent uncertainty involved in making those estimates, actual results reported in future periods could differ from our estimates.

Revenue and Cost Recognition

The Company generated all of its services revenue from contracts with customers, a substantial portion of which was generated from contracts with the U.S. Government. A majority of the Company's contracts with the U.S. Government are generally subject to the Federal Acquisition Regulation and are based on estimated costs of providing the contractual services.

The Company accounts for a contract when the parties have approved the contract and are committed to perform on it, the rights of each party and the payment terms are identified, the contract has commercial substance, and collection of substantially all of the consideration is probable.

The Company evaluates if its contracts are partially in the scope of ASC 606, Revenue from Contracts with Customers, and partially in the scope of other guidance. For contracts partially in the scope of other guidance, the Company separates and allocates the arrangement consideration to those contracts in accordance with ASC 606 unless the other guidance provides its own separation and allocation.

At contract inception, the Company determines whether the services to be provided are to be accounted for as a single performance obligation or as multiple performance obligations. This evaluation is based on professional judgment, and it may impact the timing and pattern of revenue recognition.

The Company's contracts may include variable consideration, such as adjustments to price based on performance or other contractual terms. Variable consideration is estimated at contract inception and is updated throughout the contract term as additional information becomes available. The Company includes variable consideration in the transaction price only to the extent that it is probable that a significant reversal of cumulative revenue recognized will not occur when the uncertainty associated with the variable consideration is resolved.

The Company generally recognizes revenue over time throughout the performance obligation as the customer simultaneously receives and consumes the benefits provided on services-based contracts and arrangements. The Company satisfies its performance obligation as services are rendered. An input method is used for cost-based contracts, based on the cost of services which correspond directly with the Company's performance completed to date. For fixed-fee contracts, the Company applies the cost-to-cost method — specifically the cost-to-cost approach — where revenue is recognized in proportion to costs incurred, reflecting progress towards complete satisfaction of the performance obligation.

Contract modifications are reviewed to determine whether they should be accounted for as a modification of the original performance obligation or as a separate contract. When a contract modification changes the scope or price and the additional performance obligations are at their standalone selling price, the original contract is terminated and the Company accounts for the change prospectively when the new services to be transferred are distinct from those already provided. When the contract modification includes services that are not distinct from those already provided, the Company records a cumulative adjustment to revenue based on a remeasurement of progress towards the complete satisfaction of the not yet fully satisfied performance obligation.

The Company utilizes other parties in the performance of some services. Based on the Company's evaluation using a control model, the Company determined that in all of its performance obligations, the Company serves as a principal rather than an agent within its revenue arrangements. Revenue and the related costs and expenses are both reported on a gross basis within the consolidated statements of operations and comprehensive loss.

Financial Instruments and Fair Value Measurements

We estimate fair value based on assumptions that active market participants would use to develop a fair value for an asset or liability in the principal or most advantageous market. When considering market conditions and assumptions in fair value measurements, the following fair value hierarchy distinguishes between observable and unobservable inputs. Fair value measurements are categorized according to the hierarchy below based on the lowest level of input that is significant to the overall fair value measurement of the instrument:

- Level 1 inputs: Quoted prices (unadjusted) in active markets for identical assets or liabilities that the reporting entity can access at the measurement date;
- Level 2 inputs: Inputs other than quoted prices included within Level 1 inputs that are observable for the asset or liability, either directly or indirectly; and
- Level 3 inputs: Unobservable inputs for the asset or liability. These are used to measure fair value to the extent those observable inputs are not available, thereby allowing for situations in which the Company's judgment is minimal, if any, market activity for the asset or liability at the measurement date.

Recent Accounting Pronouncements

Please refer to Note 2 — Summary of Significant Accounting Policies and Recent Pronouncements of our audited consolidated financial statements included elsewhere in this prospectus for additional information.

Quantitative and Qualitative Disclosures about Market Risk

We are exposed to market risks in the ordinary course of our business. Market risk represents the risk of loss that may impact our financial position due to adverse changes in financial market prices. Our market risk exposure is primarily the result of fluctuations in foreign currency exchange rates and interest rates. Information related to quantitative and qualitative disclosure about this market risk is provided below.

Foreign Currency Exchange Risk

Currency exchange rate fluctuations may impact our results of operations and cash flows. Currency translation gains and losses arising primarily from changes in exchange rates on foreign currency transactions and balances are not hedged and are recorded in other expenses, net in the statements of operations and comprehensive loss. We do not trade in financial instruments for speculative purposes. Business is generally transacted in a single currency not requiring meaningful transaction costs. As such, a 10% or greater move in exchange rates versus the U.S. dollar would not have a material impact on our financial results and position.

Interest Rate Risk

The Company had no debt instrument outstanding during the year ended December 31, 2024, with variable interest rates. However, during the year ended December 31, 2024, two debt instruments were outstanding which were subject to a floating interest rate, the Bank of New York Credit Facility and the Oak Credit Facility. Neither debt instrument had an outstanding balance at December 31, 2024. As of December 31, 2025 and 2024, we did not have any interest rate swaps. As no amounts were outstanding as of December 31, 2025 and 2024, a 10% movement in the variable rate on our indebtedness would not have a material impact on our financial results and position.

We had cash and cash equivalents of \$458.9 million and \$514.6 million as of December 31, 2025 and 2024, respectively. Cash and cash equivalents consist of cash deposits, cash held in financial institutions, and short-term investments, including debt securities purchased with an original maturity of three months or less. Our cash and cash equivalents are held for working capital purposes. We also had investments in debt securities, which are classified as held-to-maturity, of \$566.4 million as of December 31, 2025. These securities include corporate bonds, U.S. government treasury bills, commercial paper and certificates of deposit issued by financial institution debt securities. Such interest-earning instruments carry a degree of interest rate risk. The primary objectives of our investment activities are the preservation of capital, the fulfillment of liquidity requirements, and the fiduciary control of cash. We do not enter into investments for trading or speculative purposes. A change in interest rates during any of the periods presented would not have had a material impact on our consolidated financial statements as of December 31, 2025 and 2024.

NUCLEAR INDUSTRY OVERVIEW

This section includes industry and market data, including our general expectations of our position, market opportunity and market size, as well as future growth rates relating to our market opportunity and the industry and markets in which we operate, that is based on industry publications and other published industry sources prepared by third parties. Although we believe the industry data to be reliable as of the date of this prospectus, this information could prove to be inaccurate. Our projections, assumptions and estimates of the future performance of the industry in which we operate, including future growth rates and related estimates, forecasts and projections relating to the industry in which we operate and our market position, market opportunity and market size, are prospective. Any such projections, assumptions and estimates are necessarily subject to a high degree of uncertainty and risk due to a variety of factors, including those described in the sections captioned “Risk Factors” and “Cautionary Note Regarding Forward-Looking Statements” elsewhere in this prospectus. These factors could cause any such projections, assumptions and estimates, to differ materially from those expressed in the projections, assumptions and estimates made by third parties and by us. We caution you not to give undue weight to such projections, assumptions and estimates. See “Industry Data” for further information and important limitations and uncertainties regarding our industry and market data we present.

Evolution of Generation IV nuclear fission technologies

The existing nuclear fleet in the U.S. is largely considered “Generation II”, and consists of Pressurized Water Reactors (PWRs) and Boiling Water Reactors (BWRs), making up about 2/3rd and 1/3rd respectively. Both produce steam directly inside the reactor vessel to power the turbine, with the steam recycled back into the core. Newer “Generation III” systems, all Advanced Boiling Water Reactors and predecessor to Gen III+, were brought online in the 1990s, when few new nuclear reactors were built globally. No Gen III plants are in service in the U.S.

Small modular reactors have become more advanced as close following “Generation III” designs come online, including the AP1000 at Vogtle Units 3 and 4, offering significant improvements in safety with the introduction of passive safety features. Gen III+ includes both large conventional reactors and SMRs that adopt similar new safety features into a smaller, more compact design, generally ranging from about 10 MW to around 300 MW in size.

Advanced nuclear reactor technologies considered Gen IV are differentiated by their coolants and fuels that offer advantages in performance, safety and use cases (e.g., higher heat for industrial processes). Xe-100 is considered a Generation IV reactor, but is based on decades of research that have not yet moved from non BWR/LWR-based design concepts to commercialization. These technologies include:

High Temperature Gas-cooled Reactor: A Generation IV design using helium coolant and a ceramic moderator, operating at 700-950 °C. Fuel uses coated particle technology for high safety margins and can be used for both electricity and industrial process heat. Modular HTGRs emphasize passive safety, low use of materials and energy applications. Currently, only China has commercially deployed HTGR technology. China’s HTR-PM project was formally launched in 2001, with basic design completion in 2008, and commercial operation in 2021 at Shidaowan. It is a 210 MWe demonstration plant consisting of two modular 250 MW HTGR units.

Sodium-Cooled Fast Reactor: A fast-spectrum reactor cooled by liquid sodium, allowing for a high neutron economy and closed fuel cycles. Operates at near-atmospheric pressure with

conductivity. Enables actinide recycling, reducing long-lived waste. Challenges include sodium reactivity with air/water. Demonstration reactors exist globally, with Generation IV designs targeting enhanced efficiency and safety.

Lead-Cooled Fast Reactor: Uses molten lead or lead-bismuth eutectic as coolant in a fast neutron spectrum. Offers excellent passive safety, high boiling point, and corrosion resistance. Capable of long refueling intervals and creating, or breeding, fuel. Challenges include structural material compatibility and coolant handling.

Molten Salt Reactor: Employs liquid fluoride or chloride salt as coolant, with fuel dissolved in the salt in solid form. Operates at low pressure and high temperatures (around 700 °C), enhancing efficiency and safety.

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Compatible with thorium fuel cycles and actinide recycling, which reduces the radioactivity of the nuclear waste. MSR offers inherent safety features but face material and chemistry challenges.

Supercritical Water-Cooled Reactor: A high-efficiency design using water above its critical point (around 374 °C, 22 MPa) as coolant and moderator. Achieves thermal efficiencies greater than conventional Rankine cycles in a simpler, once-through steam cycle. Operates in thermal or fast neutron spectra. Challenges include material durability under extreme conditions.

Gas-cooled Fast Reactor: GFRs use helium as coolant and operate with a fast neutron spectrum. Materials are advanced ceramic composites, tolerating outlet temperatures around 850°C. GFRs offer high efficiency, closed fuel cycles, and hydrogen production. The design faces challenges in fuel fabrication, decay heat removal, and structural material resilience.

See below for a comparison table across Generation IV reactors compiled by the U.S. Nuclear Energy Innovation and Research Office (NIRO) as an objective view of the different advanced reactor technologies.

	HTGR	SFR	LFR	MSR	SCWR	
Timescales for delivery	Very high	Very high	Medium	Medium	Low	
Heat	Very high	Medium	Medium	High	Medium	
Safety	Very high	Medium	Medium	Medium	Low	
Security	Very high	Medium	Medium	Medium	Medium	
Economic Cost	High	Medium	Medium	Medium	Medium	
Deployability	High	High	High	High	Medium	
Adaptability	High	High	High	Very high	Medium	
Waste & Environment	Medium	High	High	High	High	
International	Very high	High	Medium	Medium	Low	

Note: A higher performing system scores a higher number. For example, with economic cost, a low-cost system would score ‘very high’ and a high-cost system would score ‘low.’

Fuel value chain overview

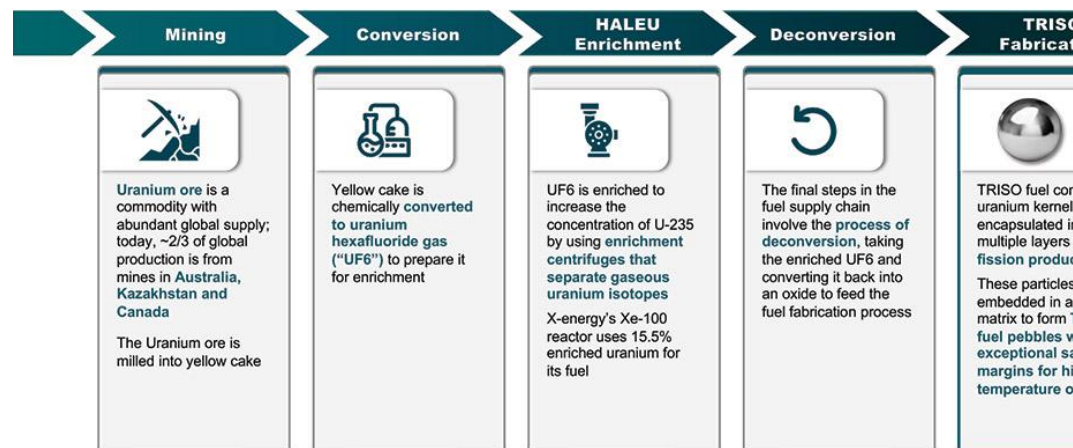
The supply chain for nuclear fuel is dependent on a select few countries that contribute to the supply of uranium and enrichment capabilities. Uranium ore reserves are concentrated in deposits across the world, with two-thirds of the global production of uranium ore coming from mines in Australia, Kazakhstan, and Canada. Once the uranium ore is mined and refined it needs to be enriched. Only nine countries have commercial enrichment capabilities: Russia, the U.K., Germany, Netherlands, the U.S., France, Japan and Brazil. The enrichment process is a critical step in the nuclear fuel cycle that enables energy generation and depends on the following critical steps: (i) Uranium ore is mined from open-pit and underground mines, (ii) Uranium ore is milled into yellow cake, (iii) Yellow cake is converted to uranium hexafluoride gas (UF_6) to prepare it for enrichment, (iv) UF_6 is enriched to increase the concentration of U-235 by using enrichment centrifuges that separate gaseous uranium isotopes. There are three categories of enriched uranium, LEU with enrichment of less than 5%, HALEU with enrichment between 5% – less than 20% and High Enriched Uranium (“HEU”) with enrichment of greater than 20%. X-energy’s Xe-100 reactor uses 15.5% enriched uranium for its fuel. The final steps in the supply chain involve the process of deconversion, fabrication of fuel pellets, kernels and assembly into fuel forms (e.g. TRISO pebbles, compacts) and final fuel configurations which are then used in nuclear energy.

Conventional nuclear reactors based on Generation II and III technology such as PWRs and BWRs use LEU for fuel to generate electricity. There is currently a greater supply of LEU-based fuel due to

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demand, government support, and a simpler enrichment process to achieve a lower (less than 5%) enrichment, and because of this, there is an established global supply chain for this type of fuel. Generation IV technologies including Advanced Small Modular Reactors use HALEU as input for fuel. Generation IV reactors, including the Xe-100, rely on a specific HALEU-based fuel called TRISO fuel (TRISO) to generate electricity.



Differentiation for HTGRs among SMR and Gen IV technologies

Benefits of TRISO fuel

TRISO fuel was developed in the 1960s in the U.K. and Germany for High Temperature Gas-Cooled Reactors and was later adopted in the U.S. for research during the 1970s-1980s. The DOE reviewed

in the 2000s under the Advanced Gas Reactor Fuel Development Program to support research on next generation reactors. Current TRISO fuel production in the U.S. is limited to R&D and production for DOE reactor demonstration programs. This is being expanded through new efforts to secure a U.S.-based uranium enrichment supply chain that will enable additional production. First, the U.S. government plans to bridge the supply of HALEU until private enrichment can operate at scale. Second, private actors such as Centrus and Urenco are being supported through sharing contracts to accelerate commercial operation. DOE’s stated goal for such commercial plants to come online as early as 2027.

TRISO fuel consists of uranium fuel kernels individually encapsulated in multiple layers of carbon and silicon carbide, forming miniature containment systems that trap fission product particles are then embedded in a graphite matrix to make fuel pebbles and compacts that possess wide safety margins and compacts, enabling operations at very high temperatures.

Additional benefits of TRISO fuel include:

- high fuel efficiency that can achieve a 19% maximum burnup rate which is around 4 times current LEU yields and significantly higher than other HALEU-based fuels including uranium metal which have burnup rates of up to 8%, allowing reactors to run longer between refueling and reducing overall downtime;
- longer operational life of reactors due to the fuel’s durable structure and resistance to neutron irradiation, corrosion and oxidation;
- robust structure as TRISO pebbles are manufactured to withstand high temperatures and pressure drop as well as prevent the release of radioactive nuclear materials due to its advanced structure;
- versatile fuel form as it can be used for various reactor technologies including High Temperature Gas-cooled, Molten Salt and Micro reactors;
- can be made from both LEU and HALEU; and

- complex and layered fuel design that makes it more difficult for enriched uranium proliferation

Benefits of Helium as a Coolant

While other gases have been investigated, helium has emerged as the coolant of choice through decades of research on the properties of the gas that make it uniquely suited for high temperature safety requirements. As a noble gas, helium’s chemical inertness, prevents corrosion of components and ensures long-term structural integrity. It aids in efficient heat transfer and does not become radioactive under neutron exposure. As it remains in gaseous form throughout all operating temperatures in the reactor, the use of helium eliminates concerns about phase changes and thermal stress on components. Helium is also transparent and non-toxic, enabling easier visual inspection and safety for the reactor serving lifetime. With a helium outlet temperature of 750°C in the Xe-100 HTGR, the use of helium enables significantly higher steam production temperatures than conventional reactors.

BUSINESS

Unless the context otherwise requires, all references in this section to the “Company,” “we,” “us” or “our” refer to the business of X-Energy, Inc. collectively.

OVERVIEW

Our company

X-energy is a leading designer of advanced nuclear reactor technology (commonly referred to as small modular reactors, “SMRs”) and manufacturer of advanced nuclear fuels. We believe these small modular reactor technologies help satisfy historically unprecedented electricity demand growth, and the development of AI and associated data center infrastructure. Total demand for new electricity is expected to increase globally by 7,626 TWh from 2023 to 2030 and the challenges associated with this demand have led policymakers and industry leaders to recognize nuclear energy, particularly advanced nuclear, as a key component to address this need.

Founded in 2009 by Dr. Kamal “Kam” Ghaffarian to bring clean, safe, secure and advanced nuclear technology to market, X-energy is seeking to redefine the energy industry through its flagship Xe-100, an advanced small modular HTGR, in development for nearly a decade. The Xe-100 is designed to generate 80 megawatts of electric power or 200 megawatts of thermal output, a combination thereof. This reactor technology builds on more than 50 years of research and development in the global nuclear industry and the operating experience of previous HTGRs including the Peach Bottom in the U.S., and Dragon in the U.K. in the 1960s-1970s, and more recently with Chinese HTGR deployments in the 21st century.

The Xe-100 has several technological attributes that we believe make it advantaged compared to other sources of baseload generation. These include advanced safety features, virtually no direct greenhouse gas emissions during generation, high thermal output, load-following capabilities and modularity, all of which enable X-energy to more specifically meet a customer’s power and/or industrial heat needs. X-energy’s Xe-100 design directly translates into simplicity of project delivery through reduced construction complexity and labor intensity during construction, which we believe will lead to lower costs and shorter deployment timelines when compared with conventional nuclear energy sources. X-energy is currently in the deployment of its Xe-100 into a four-reactor format that outputs 320 MWe (or 800 MWt). By utilizing four independent reactor modules instead of a single unit, this optimized four-reactor design inherently delivers the high levels of reliability and redundancy required for both AI and industrial applications.

X-energy’s reactors use a TRISO coated particle fuel in the form of a spherical ‘pebble’, or TRISO-X fuel. This pebble fuel consists of HALEU fuel kernels individually encapsulated in layers of silicon carbide and pyrolytic carbon, forming miniature containment systems that trap fission products. These particles are then embedded in a graphite matrix to make fuel pebbles that possess exceptional safety margins and compact, enabling operations at very high temperatures. The HALEU fuel used in the TRISO-X pebble fuel is enriched to 15.5%, a higher energy density form than the less than 5% LEU fuel used in conventional nuclear reactors. TRISO-X fuel will be produced at our fuel fabrication facility in Tennessee. The first facility, known as TX-1, began construction in October 2024 and is expected to be completed by the first half of 2028. Upon completion, it is expected to be North America’s first built commercial advanced nuclear fuel fabrication facility. The TX-1 facility will have a production capacity to support the fuel needs of the first 11 Xe-100 reactors at steady state operation.

In addition to its technology leadership, X-energy has three high-quality customers in D and Centrica, who we expect will underpin the deployment of the initial fleets of Xe-100 reactors together, assuming each customer exercises its contingent rights in full, these three customers with a more than 11 gigawatts electric (“GWe”), 144 reactor pipeline across the U.S. and the advanced development efforts already underway on the first Dow project at its Seadrift Operations Texas and the first Amazon project in connection with Energy Northwest.

X-energy maintains a strong relationship with the DOE and in December 2020 was awarded \$1.2 billion as part of its selection as one of two awardees in the ARDP, the most substantial federal

commitment to deploying advanced nuclear technology. The cooperative agreement for the project in February 2021 (the “ARDP Agreement”), provides 50/50 cost share of \$2.4 billion of (\$1.2 billion reimbursement) through 2027, allowing X-energy to continue work toward design commercialization and construction of its first-of-a-kind commercial advanced nuclear commercial TRISO-X fuel fabrication facility, while benefiting from decades of nuclear engineering knowledge within the DOE.

As of December 31, 2025, X-energy has been reimbursed approximately \$438 million in funding under the ARDP Agreement. We submit our budgets through an ongoing “budget period” basis to meet milestones under the ARDP Agreement, and our current budget covers a budget period that begins in 2025 and extends through August 2026. We submit a Continuation Application to the DOE for funding into subsequent periods. Extensions beyond the current budget period are subject to DOE review and approval. Under the terms of the ARDP Agreement that rely on the Office of Management and Budget (OMB) guidance, the total extension of the award may not exceed three years (for a total performance of 10 years). Any additional extension would require an approval within DOE at the discretion of the Contracting Officer. If we are unable to obtain extensions and incur eligible costs beyond the currently approved period of performance, we would forgo reimbursement for such costs and have no obligation of unobligated funds at closeout. There can be no assurance that we will receive additional funding beyond the current budget period or that extensions will be granted.

Our Market Opportunity

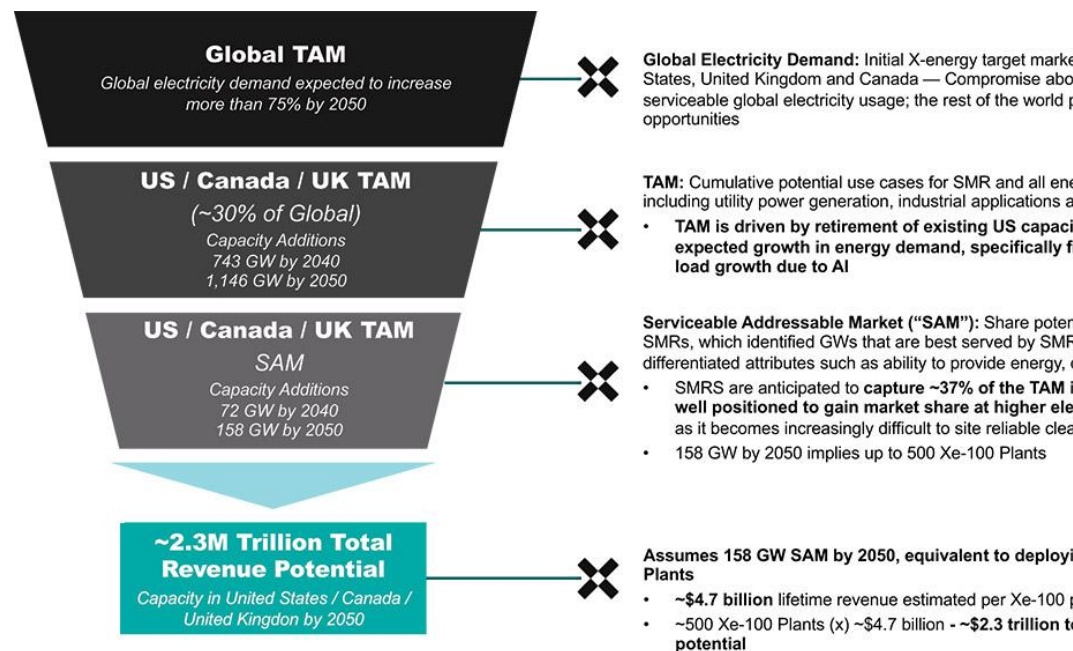
Load growth is accelerating globally with the IEA estimating electricity consumption to grow by more than 25% from approximately 30,000 TWh in 2023 to approximately 37,000 TWh by 2030. Global power demand is being driven by the increasing data center buildout from cloud computing, AI, industrial growth and reshoring of manufacturing, and broader electrification (e.g., electric vehicles, renewable base). Other sources (BNEF New Energy Outlook 2025) also indicate that power demand is being driven by growing load growth demand from AI and industrial use. Policymakers and industry leaders recognize that advanced nuclear, particularly advanced nuclear, will be a key contributor to meeting this load growth and ensuring America’s energy independence, with the stated goal of the current federal administration to increase nuclear capacity to 400 GWe by 2050 (from approximately 97 GWe at present).

We believe the market for SMRs is vast and that X-energy is well positioned to capitalize on this opportunity. According to a 2024 report from PA Consulting, the U.S., the U.K. and Canada are the planned core markets, comprise approximately one-third of potentially serviceable global electricity demand. Across these three geographies, there is an estimated TAM of cumulative capacity additions of 1,146 GWe by 2040 and 2050, respectively. Of this TAM, PA Consulting estimates 72 GWe by 2040 and 158 GWe in 2050 will be best served by SMRs based on needs, siting efficiencies, and deployment

generation. These capacity additions, across a representative set of SMR use cases including process heat, data centers, utility power generation, industrial applications and BTM generation, implies a potential market opportunity of approximately around 1,975 Xe-100 reactors (the equivalent of 494 four-reactor deployments). This represents an estimated \$2.3 trillion market opportunity for X-energy in 2050.

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We believe the following secular movements will continue to support the use cases and potential for SMRs, particularly the Xe-100, can deliver.

- Demand for AI is Prompting Investment in Nuclear Power Generation.** Cloud computing and AI companies are deploying capital across data center infrastructure to keep up with the growing computing demand required by AI. As part of this expansion, electricity demand in data centers is expected to grow from approximately 108 TWh in 2020 to approximately 1,000 TWh by 2030. Since data center facilities must be continuously powered, nuclear generation is a key solution for reliable supply. SMRs are particularly well suited to meet the average 1,000 MW power capacity that data centers require. SMRs have a smaller physical footprint, less siting requirements, and have modular scalability which provides the flexibility to meet specific capacity needs.

- **High Carbon-Intensity Industrials Require Replacement for Industrial Heat Production.** Industrial companies have historically relied on fossil fuel fired boiler electricity and steam for their industrial processes. The current installed base of industrial boilers operates below capacity and is frequently offline for maintenance. The current fleet is approaching end of term replacement cycle due to age. For industrial processes requiring consistent steam, X-energy’s HTGR solution can reliably provide industrial steam in addition to onsite power. With an expected capacity factor of 95%, X-energy’s SMR presents a compelling replacement of aging infrastructure to both decarbonize and achieve greater reliability.
- **Current Alternatives Are Not Well-Suited to Deliver Reliable, Uninterrupted, Clean, and Co-Located Power**
 - *Renewables with or without energy storage* — Solar and wind generation have relatively low capacity factors (i.e., the actual energy output in a given period of time relative to maximum output) of 23% and 33% (EIA) respectively, since they only produce power when the sun is shining or the wind is blowing. Addressing this inherent intermittency issue requires the integration of large-scale battery energy storage systems or supplementary fossil fuel generators for the production capacity of these renewables to even be comparable to the anticipated 95% capacity factor of our reactor. Large-scale deployment of batteries is complicated by global supply chain availability and incremental cost.
 - *Conventional fossil fuel generation with or without carbon capture* — Standalone fossil fuel fired generation can deliver capacity factors similar to those of nuclear power but requires backup generation when maintenance takes key assets offline. Conversely, advanced nuclear can deliver critical redundancy due to its modular nature which results in a reliable power source without requiring

- grid redundancy. Further, many consumers and governments have climate targets, which requires a shift away from carbon-intensive alternatives such as coal, oil, and natural gas generation, which would require pairing them with expensive carbon capture solutions to try to meet climate targets.
- **Traditional Nuclear Power** — Large-scale nuclear, while typically operating at high capacity factors (other than planned outage maintenance and refueling cycles, as compared to advanced nuclear’s online refueling), requires a large footprint and has consistently suffered from historical delays and cost overruns. By contrast, advanced nuclear requires a much smaller footprint, on average about 1/4th to 1/10th that of a traditional nuclear plant. Additional advanced nuclear sites in the U.S. will require a significantly smaller safety zone radius (400 meters vs. 16 kilometers) because of built-in passive safety features. These safety characteristics translate to lower costs and more compact designs due to a reduction in the quantity of concrete and safety infrastructure. The smaller overall footprint of SMRs allows co-location with industrial demand hubs, and the modular nature of the Xe-100 enables scalable power output with additional on-site reactors to match specific needs.
 - **Ongoing Support from the U.S. Government Beyond the ARDP.** Support for nuclear power has been increasing over the past decade with billions of additional dollars being allocated to nuclear through recent energy-related legislation. For instance, the OBBBA, signed in July 2022, provided tax credits that were first introduced as part of the Energy Policy Act of 2005 and later expanded under the Inflation Reduction Act for nuclear projects. Additionally, the U.S. President issued Executive Orders in May 2025, including support for the acceleration of regulatory review for advanced nuclear reactors and promotion of investment in a domestic nuclear supply chain.

believe these broader initiatives, building from U.S. nuclear policy precedent, will provide support for advanced nuclear reactors like the Xe-100.

- **Investing in a Domestic HALEU Supply.** Establishing a domestic HALEU supply for the advanced nuclear reactor market in the U.S. is a top priority for the U.S. Government. The Energy Act of 2020 Congress established the HALEU Availability Program and has directed the DOE to support the availability of HALEU available for civilian use (e.g., initial ARD) until commercial supply chains are in place. The program provides market support through a revolving fund structure that supports offtake commitments and is intended to help with enrichment services and HALEU production, stimulate private investment, and support the development of a commercial-scale U.S. domestic fuel supply chain encompassing uranium conversion, enrichment, fuel fabrication, and transportation. Congress has provided approximately \$3.4 billion in funding authority for these activities.

Key public-private partnerships established under the HALEU Availability Program include enrichment services with companies Centrus Energy, Urenco, Orano and General Motors. These companies place orders through Indefinite Delivery / Indefinite Quantity contracts under a 10-year term to provide enrichment services for the DOE. Centrus is already producing approximately 1 MTU of HALEU per year through its pilot cascade at Piketon, Ohio, and has indicated the capability to add additional cascades over time, enabling scalable increases in HALEU production to meet demand and funding allow. By the first quarter of 2029, Centrus has stated its objective is to increase production to up to 6 MTU of HALEU annually. The DOE has recently announced the allocation of HALEU to select SMR developers, with X-energy receiving the first tranche of 4 MTU. The HALEU Availability Program is expected to provide X-energy a total of around 7.6 MTU, enough for X-energy's first Xe-100 plant with Dow. Consistent with its objectives, in January 2026, the DOE announced an additional \$2.7 billion in awards for domestic uranium enrichment services over a ten-year period, subject to contractual milestones and continued appropriations. These awards are intended to expand U.S. capacity for uranium ("LEU") and to accelerate the development of new HALEU supply chains, distributed through milestone-based task orders. The DOE awarded enrichment task orders of estimated worth of \$900 million each to Centrus Energy's American Centrifuge Operations, General Matter, and Orano Federal Services to support the creation or expansion of domestic HALEU enrichment capacity, as well as additional funding to advance next-generation enrichment technologies. DOE also issued a smaller award of \$28 million Global Laser Enrichment technology to continue advancing next-generation

uranium enrichment technology. GLE is currently undergoing licensing review with the NRC. The enrichment facility to be located in Kentucky, Centrus intends to expand out its American Centrifuge Plant for commercial operations, and General Matter and Orano Federal Services plan to expand their enrichment plants in Kentucky and Tennessee, respectively.

Urenco USA is currently the only commercial enrichment plant operating in the United States. Urenco has not yet expanded its operations to include HALEU production. Urenco USA has indicated it is working with the NRC to amend its license for HALEU enrichment, with enrichment production targeted for the early 2030s. In parallel, Urenco Ltd., headquartered in the United Kingdom, parent company of Urenco USA, is advancing HALEU enrichment capabilities in the United States with support from the UK government. In May 2024, the UK government announced an investment of approximately GBP196 million to support the construction of a HALEU-capable enrichment facility at Urenco's Capenhurst site in northwest England. Urenco has indicated that this facility is expected to have the capacity to produce up to approximately 10 metric tons of HALEU per year.

commercially available by approximately 2031, subject to regulatory approvals, project financing, and continued government support. If successfully developed, these efforts could diversify global HALEU supply and reduce reliance on foreign sources, although timing, and eligibility for specific projects will depend on applicable regulatory, commercial, and policy considerations.

The DOE is also researching advanced fuel cycle and spent fuel recycling technologies for long-term energy security and advanced reactors, although the United States does not currently recycle commercial spent fuel and any future deployment would require significant regulatory, and technological developments.

- **Advanced Reactors Can Provide Supplemental Power to the Grid.** Several utilities have value-add opportunities to use SMRs, like the Xe-100, for power generation to meet electricity demand from their customers. A few examples in North America include the Tennessee Valley Authority and Canadian Ontario Power Generation have each filed regulatory proceedings for SMR projects within their respective jurisdictions, with OPGen receiving a nuclear power reactor construction license from the Canadian Nuclear Commission in April 2025; Dominion Energy is evaluating potential scenarios for SMR development, including publicly discussed concepts involving multiple units beginning construction in the mid-2030s; Constellation Energy is engaging with customers regarding potential SMR deployment at certain existing sites, increasing deployment speed and efficiency by avoiding new transmission connections, security infrastructure, and other incremental requirements for nuclear projects. In late-2025, Duke Energy filed an early site permit (ESP) application with the NRC for a potential SMR project in North Carolina to replace aging coal-fired units at the existing power generation facility.

SMRs like the Xe-100 are attractive to utilities for several reasons including:

- Targeted Incremental Load Growth.** Load growth in many areas of the U.S. does not require the need for very large, centralized capacity additions that traditional nuclear LWRs require; rather, requires more incremental, distributed capacity that SMRs are able to provide. Because of their smaller, configurable capacities, allow for better matching of incremental load needs to more rapidly supplement existing facilities and benefit from existing site infrastructure.
- Safety Case and Relatively Smaller Operational Footprint.** Due to the smaller operational footprint relative to alternatives, SMRs can be sited on existing, soon to be retired, or retired nuclear power plant locations. In doing so, utilities can benefit from existing approved siting and permitting processes, making a compelling option to deliver incremental, continuous baseload power more quickly than greenfield sites.
- Financing Accessibility and Reduced Project Risk.** SMRs, given their size, carry a smaller investment burden than do traditional LWRs, making SMRs more accessible to a wider range of investors, including utilities. The costs to deploy traditional LWRs can reach \$20.0 billion, representing a significant undertaking and financial pressure, even for larger utilities. SMRs, on the other hand, can be financed and built in phases, thereby offering smaller capital requirements that can be spread over several years while delivering power to the grid sooner than a traditional LWR. Phased, repeated deployment

also represents lower risk due to simpler, smaller construction projects with less capex compared to LWR projects, which have traditionally suffered from significant cost overruns.

- iv. **Proximity to Load Centers and Transmission Efficiency.** SMRs' small land requirements and superior safety case enable siting near existing load centers. By locating generation near demand, utilities can reduce or avoid the need for long, new transmission laterals that are required for remote wind or solar resources, lowering interconnection costs, shortening permitting timelines and mitigating right-of-way, permitting, and community opposition risks. Close proximity to load centers curtails line losses and congestion, improving delivered-cost economics and grid reliability, thereby easing pressure on constrained transmission corridors. Repowering existing sites with SMRs will allow utilities to leverage established infrastructure, substation capacity, and prior environmental reviews, further accelerating deployment timelines and reducing execution risk.

Competitive Strengths

As pioneers of next generation nuclear reactor and fuel technologies, we believe that our extensive experience and expertise can allow us to capitalize on our competitive strengths. We believe we will establish a strong market position in the evolving market, including by leveraging the strengths of our early customer relationships, which will give us a differentiated path to success.

Existing Customer Base Provides Initial Momentum and Support. Both Dow and Amazon have signed agreements for our initial projects, and we expect will serve as offtake customers in the future. These initial agreements will enable X-energy's goal of scaling its design to two different offtake customer applications: Industrial Process Heat Applications and Data Center Projects. Dow and Amazon also provide project development in their own operations portfolios. X-energy expects to learn from and engage with these and other customers as it builds the pipeline. This repeatable project schedule, which is also anticipated for our announced Centrica partnership, is expected to enable X-energy to scale from First-of-a-Kind to a multi-unit, First-of-a-Kind model that we believe will reduce our project cost and significantly derisk future projects.

Differentiated Design that Provides Attractive Solution for Customers. The characteristics of X-energy's design provide several advantages over many other energy production alternatives. First, traditional baseload technologies like wind and solar cannot provide reliable baseload power without being paired with longer-term storage technologies. They are also dependent on exogenous factors, which make them incapable of scaling to ramping up to address increases in demand. Second, the Xe-100 has a versatile design that can be deployed in several end markets, including industrial heat and conventional power generation. In contrast to other emerging SMR technologies, many of which are based on conventional LWR technology, X-energy's design is designed to efficiently load-follow or operate at temperatures high enough to provide industrial process heat. X-energy's design is further constrained by the availability of large amounts of water, a constraint that the Xe-100 does not have.

Improved Proven Technology. Unlike some other advanced reactor technologies, X-energy's design is based on HTGR technology that has been previously deployed across the globe, including in the U.S. at the Fort St. Vrain Bottom and in the U.K. at Dorset (Dragon), and building on more than 50 years of development by the global nuclear industry. Both of these reactors served as proof-of-concept for the technology in the U.S. and, along with other HTGRs in Europe and Asia, provide valuable experimental data that X-energy has used to improve its design, leading to a more efficient, commercialized technology.

Intrinsically Safer Based on Physics without Needing Active Safety Systems. Our simplified X-energy design is not dependent upon active safety systems, which are susceptible to failure and can necessitate the redundancy found in a typical LWR reactor design. Our design relies on physics-based passive safety features, such that in the event of a total loss of power to a Xe-100 reactor, the reactor will automatically require no operator or computer actions, grid connections, emergency backup power or additional cooling to cool the reactor. The reactor has a strong negative temperature coefficient, which means that an increase in temperature (such as from the loss of coolant circulation) slows the fission reaction, causing the reactor to shut down. Finally, due to the relatively low power density of the core, the remaining heat load can be dissipated through passive cooling.

Superior Fuel Based on Decades of Research & Development. We expect our TRISO demonstrate technical quality and a streamlined fuel qualification pathway, providing us a competitive advantage in the market.

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the commercial fabrication of TRISO fuel forms. The TRISO-X fuel used in the Xe-100 is a pebble fuel form that is designed to melt, enabling the technological and safety advantages of the pebble fuel form. Based on decades of research, development and testing, including the DOE's Advanced Gas Reactor Development and Qualification Program, the TRISO-X particle fuel is relatively well understood. Historical data has established the parameters for TRISO-X fuel testing and qualification. Our pebble fuel qualification methodology is approved by the NRC, and we have a streamlined pathway to final fuel qualification by the time of our first Xe-100 deployment.

Attractive Intellectual Property-Driven Business Model. X-energy serves as both a reactor technology provider and a fuel fabrication provider, offering customers an integrated solution. In our reactor business line, we expect to receive technology fees for licensing use of our proprietary Xe-100 technology, and we expect to receive fees for coordinating assembly and construction support with customers and anticipated third-party vendors. We also intend to leverage our knowledge and expertise in regulatory affairs, construction, procurement, operations, maintenance and other processes to provide customers with a suite of services from the development of a project, and ultimately for the operating life of the reactor. Since we do not construct the power plants and do not plan to own or operate the power plants once constructed, we, as a result, do not incur capital expenditures relating to constructing, maintaining, owning or operating reactor facilities. However, we intend to remain involved with the EPC process throughout the project lifecycle, providing strategic consulting related to the integration of the reactor technology. We believe this integrated, property-driven business model will position us to generate attractive free cash flow.

Leading TRISO Fuel Provider. From our TRISO-X fuel manufacturing facilities in Tennessee, X-energy plans to provide customers with initial reactor fuel loads of as well as ongoing delivery of TRISO-X fuel required to refuel plants over the lifetime of a plant. This provision of fuel to customers would generate a strong, recurring, revenue stream. X-energy's fuel can also be used in other advanced reactor technologies, making it a key enabler of the broader advancement of the nuclear energy sector. X-energy was selected to receive one of the first allocations of HALEU from the DOE. The DOE is required to begin initial fuel production but X-energy does not, and does not intend to, bear any inventory risk associated with uranium or fuel feedstock. We intend to provide only fabrication and fuel services (e.g., transformation of HALEU into the final TRISO-X fuel form) for customers, thereby limiting risks associated with holding the uranium or enriched uranium fuel feedstock.

Visionary Management Team and Highly Expert Employee Base. We have an experienced and talented team of leaders and innovators who have been directly involved in the development of advanced nuclear technology and who have led large-scale nuclear projects and operations. As of January 1, 2023, we have a highly educated workforce of 889 employees, of whom 312 have master's degrees and 107 have PhDs. Our executive leadership team has experience in nuclear design, nuclear project delivery, nuclear fuel fabrication, operations, government relations and public companies in organizations such as ORNL, the DOE, the Nuclear Regulatory Commission, Constellation Energy, Hunt Consolidated, Westinghouse, Hartree Partners, and Emirates Nuclear Energy Corporation. The management team is led by our CEO, J. Clay Sell, who is one of the foremost leaders in the U.S. energy market. As the former Deputy Secretary of the DOE, he brings the perspective of the U.S. energy industry to X-energy. His experience in renewables development after his time as Deputy Secretary has given him key insights into major project development and valuable insight into the limitations of intermittent renewable energy and in turn the value of nuclear generation.

Strength of Government Relationships. The ARDP significantly derisks the delivery of X-energy of-a-Kind reactor deployments through the Dow Project. Because of the 50/50 cost share program, X-energy has had significant ongoing engagement with the DOE and benefits from the DOE’s collective expertise and support. Further, X-energy has developed high credibility with the Nuclear Regulatory Commission through our ongoing engagement for both our FOAK Fuel Fabrication facility and Dow project. The DOE’s decision by the docketing of the first Construction Permit Application for an 18-month review schedule represents the shortest CPA timelines ever given.

Our Business Model

We have an intellectual property-driven business model based on our reactor and fuel. We derive revenues from technology licensing, services and fuel operations that span the development, construction and operation of the reactors.

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- **Reactors:** The revenue stream from reactors includes technology fees for the use of our intellectual property of the Xe-100 technology. We will not own and operate the facilities themselves. This model believe significantly reduces the amount of capital needed to operate our business.

We anticipate offering site-specific engineering and site characterization, project planning, construction coordination, construction support, regulatory support, procurement support and long-term support to customers. Utilizing our knowledge and expertise in licensing, construction, project management and other processes, we plan to provide customers with a full suite of value-added services throughout the development of the nuclear power facilities. At the same time, we expect to generate recurring revenue from services such as the ongoing maintenance and operator training throughout the anticipated 60-year life of a facility.

- **Fuel:** We intend to provide manufacturing services to customers, including producing and delivering a full load of both TRISO-X fuel and an LEU-based TRISO fuel at commissioning of a plant. We expect to generate additional long-term recurring revenue from our own proprietary TRISO-X fuel throughout the life required to refuel plants during the anticipated 60-year life of each facility. We expect to manage the inventory risks related to uranium or enriched uranium fuel feedstock. We intend to provide fuel fabrication supply services (e.g., transformation of HALEU into the final TRISO-X fuel) to our customers and assume limited risks associated with holding the uranium or enriched uranium feedstock. Additionally, we will not have any responsibility for spent fuel management throughout the design of such facilities to adequately handle spent fuel during the life of the plant. During plant operations, spent fuel remains the responsibility of the plant operator. Thereafter, permanent spent fuel management remains the responsibility of the DOE.

Lifecycle unit economics

We expect each of the four reactor Xe-100 projects (each such optimized four-reactor project referred to as a “plant”) to generate between approximately \$2.4 and \$4.7 billion in aggregate revenues across both our reactor business line and our fuel business lines, beginning with the pre-development phase of the plant and continuing through the plant’s anticipated operational design life of 60 years and beyond of such plant.

Reactor. Revenues from our reactor business line are expected to be comprised of both one-time technology fee and fees from our services offerings. We expect to earn these fees beginning with the development phase of the plant and continuing throughout the operational life of such plant. We expect approximately 15 – 35% of revenue generated in our reactor line of business to be realized during the pre-COD (including) COD. Within that approximately 15-35% of pre-COD revenue, we expect to contribute approximately 10-15% of the total revenue.

technology fee, which we expect to earn across various plant development milestones before reactor development. The remaining approximately 65-85% of revenue from the reactor line of business is expected to be from the provision of long-term services offered post reactor start-up. We expect to charge our services offerings to customers based on time and materials, and based on anticipated pricing of our services fees, we expect to generate approximately 20 – 30% of gross revenue from these services. Our technology fee is expected to be charged to customers for use of our technology property and has no associated cost.

Fuel. Revenues from our fuel pebble fabrication business line are expected to be comprised of revenues generated from selling an initial fuel load at COD and from selling additional pebble fuel throughout the operational life of a plant. We intend to generate fuel revenue from fabrication fees that we charge to our customers throughout the plant’s expected operational life. We expect the initial fuel load at COD to be equivalent to approximately 5% of each plant’s lifetime fuel needs (approximately 3 years of fuel needs). This represents approximately 10% of the total revenue we expect to generate from the fuel business line of each plant. Following COD, we expect continuous online refueling to generate approximately 90% of fuel revenue through the remainder of the plant’s anticipated operational design life of 60 years. Under our vertically integrated business model, we have flexibility in the fabrication fees we charge customers. We anticipate that as our fuel manufacturing capabilities scale and efficiency and utilization increase, our operating costs for fuel fabrication will decrease.

The table below provides an illustration of the expected timing and nature of the cash flows we expect to anticipate being able to generate from a single plant, based on and subject to the assumptions set forth in the table below.

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Dollar figures below are in millions unless otherwise noted. Dollar figures are rounded to the nearest million, percentages are rounded to the nearest 5%:

	Total Plant Life*		
	Customer Fees	Cost to X-energy	Tariffs
Reactor (Technology fee)	\$50 – 300	\$ —	
Reactor (Services)	\$800 – 1,300	\$565 – 930	20%
Fuel	\$1,560 – 3,125	\$1,095 – 1,250	30%
Total	\$2,410 – 4,725	\$1,660 – 2,180	30%

* Total Plant Life includes up to eight years prior to COD, COD, and 60 years following COD.

Key Assumptions Underlying Unit Economics

General

The table above illustrates management’s current expectations regarding the relative contribution of customer fees, our costs, and resulting margins across business lines for a single plant. These figures are illustrative and based on the assumptions summarized below; they are not forecasts, targets, or commitments. Our ability to achieve the illustrative customer fees, cost, revenue and margins are sensitive to a number of factors further described below and under “Risk Factors” elsewhere in this prospectus. Actual results may differ and depend on variables including customer negotiations and fee structures, market conditions, technology costs and dynamics, manufacturing efficiency and utilization, financing assumptions, and customer preferences for optional service packages, any of which could cause material differences from the figures presented.

above. The table above assumes the plant is fully supplied from both TX-1 and TX-2; future requirements for additional fuel facilities may differ in cash flow. This “Lifecycle unit economics” discussion should be read together with “Risk Factors,” “Management’s Discussion and Analysis of Financial Condition and Results of Operations of X-energy,” “Business,” and our consolidated financial statements.

Technology Fee Structure and Pricing Assumptions

We expect to charge a technology fee for customers’ use of our Xe-100 technology and related intellectual property. For purposes of this section, the assumed fee reflects benchmarking of IP licensing fees across various similar industries and sectors, together with the experience of our customers and ongoing customer discussions. The final fee will be negotiated on a case by case basis, reflecting timing, customer profile and scope, and is expected to be paid over multiple development milestones, with the full amount paid by COD. Fees may change over time as plants come online, the technology evolves, and competitors emerge.

The customer fee range reflects variability in the technology fee component, with lower fees for initial plants and higher fees for later plants after scaling and validation. Because the fee relates to use of our intellectual property, the associated cost to us in the table above is assumed to be zero. We have not yet entered into technology, intellectual property or related agreements, and we cannot provide any assurances that customers will accept our anticipated fee structure or pricing.

Services Business Fee Structure and Pricing Assumptions

We plan to provide a full suite of services across all phases of project development, including site selection, permitting, engineering, training, and procurement support, and following COD, ongoing maintenance and operator training through the design life of a plant. These activities are organized into industry-standard front-end loading (FEL 0 – FEL 4) phases. For this section, services pricing is based on a cost-plus basis, comprising direct labor and overhead with a targeted margin informed by our experience generated to date from the provision of initial strategic business and other services we’ve provided to customers and potential customers, as well as from industry benchmarking and our team’s experience. We expect limited pre-COD services revenue, as most services fees are expected to be paid post-COD. Additionally, we assume that post-COD reactor revenue is services-only revenue, as the technology fee is expected to be paid in full by COD. Lastly, for each plant, we assume that customers will continue to rely on us as a service provider throughout

the operational life of such plant. We do not construct, own, or operate plants and therefore do not incur related capital expenditures. As a result, the costs reflected above do not reflect the total cost to build reactors, which cost is primarily borne by our customers and may be substantial.

Reactor costs to us reflect only services costs and consist of estimated direct costs, including labor, overhead, and other costs, which are subject to change with market dynamics and competitive conditions. Our fees include a services component based on the targeted margin applied to these costs. To date, we have entered into limited services agreements and there is no assurance that customers will accept our proposed and anticipated fee structure or pricing or continue to contract solely with us in the future. The presence of third-party service providers could introduce price competition, thereby reducing our margins and increasing costs.

Fuel Model and Assumptions

We intend to provide the initial fuel load at COD and to generate recurring revenue from the sale of our proprietary TRISO-X fuel required for refueling over the anticipated 60-year plant life. Each

an initial core load (covering roughly the first three years of operation), followed by steady replacement. We assume that we are the initial sole supplier of our proprietary fuel and that customers purchase fuel from us. The emergence of third-party TRISO suppliers could introduce price competition, reduce costs, and increase costs.

Fuel customer fees and revenue are based on an assumed price per pebble estimated based on fuel costs, including labor, materials, and overhead. Pricing is intended to recover operating, variable, and fixed costs and to provide a target margin sufficient to support financing of future fuel facilities at a minimum utilization level. We expect to provide fabrication services only and do not anticipate inventory risk for uranium or feedstock. We will not be responsible for spent fuel management or designing facilities to handle spent fuel during plant operations. Accordingly, the illustrative fuel costs reflect inventory or spent fuel management costs.

The range of fuel customer fees primarily reflects the estimated price-per-pebble range, which is affected by efficiency improvements, cost reductions, and external supplier dynamics. The range of costs to us reflects operating, fixed, and variable costs, including labor, overhead, and other costs, which may change with market conditions and competition. We have not yet entered into fuel supply agreements, and there is no assurance that customers will accept the anticipated fee structure or continue to purchase fuel solely from us.

Operating Lifetime Assumptions

Our economics are based on an anticipated 60-year operating life of a reactor, subject to regulatory licensing approvals and renewals. All nuclear power plants in the country have been designed with substantial safety, operational, and component performance margins. While the Atomic Energy Act limits initial licensing to 40 years, the current operating fleet was designed to operate beyond that term. Similarly, the Xe-100 is designed with substantial margins to support operation beyond the initial licensing term, subject to future NRC approval. The NRC established a license renewal process that permits licensees to seek approval for subsequent 20-year operating periods. In license renewal, the NRC focuses on evaluating whether aging management programs adequately address the effects of aging on structures, systems, and components important to safety. For much of the current operating fleet, license renewal requirements were not in place at the time of initial licensing, and comprehensive aging management programs were therefore developed and reviewed as part of the license renewal process. Almost every U.S. commercial nuclear power plant operates under a renewed license, valid for 20 years beyond the 60 of potential total operations, and the NRC is now reviewing applications for “subsequent license renewal,” valid for years 60 – 80 years of potential operations.

For the Xe-100, because the design contemplates long-term operations, X-energy is incorporating aging management considerations into the reactor design from the outset. With the establishment and implementation of these programs from initial commissioning, the Xe-100 is expected to be well positioned to support future license renewal applications, although NRC approval is not assured. It should be noted that the NRC license renewal process is well established, and the NRC has approved thousands of license renewal applications to date; however, approvals are based on the specific facts and circumstances of each application. For these reasons, X-energy believes that its approach to design and aging management will significantly

reduce the risk associated with future license renewal; however, it does not entirely eliminate the risk. While the NRC license renewal process is well established and the majority of applications are approved, the NRC has, in limited circumstances, denied or reversed license renewal applications. Applications that have failed to satisfy applicable safety, environmental, or regulatory requirements. Ac

energy believes that its proactive approach to aging management and regulatory compliance mitigates, but does not eliminate, license renewal risk.

Growth Strategies

We intend to grow our business by leveraging the competitive advantages of our diverse, reliable Xe-100 technology and our strong customer and government relationships to achieve our growth objectives. We believe we have several avenues to achieve our growth objectives:

Continue to Develop our Next Generation Technology. We intend to continue developing our advanced fuel technology with the goal of achieving commercial delivery of our first fleets of reactors in the 2030s. We have substantially advanced detailed design for the Xe-100 and are working with our industrial and construction partners to complete that process. Furthermore, we will work with our industrial customers to complete the site-specific environmental studies required for licensing and progress the submission of the construction permit application, including the preliminary safety analysis report. We have produced TRISO-X pebble fuel in kilogram batch quantities in our fuel fabrication pilot facility and begun construction on North America's first purpose-built commercial advanced nuclear fuel facility in Oak Ridge, Tennessee.

Execute on Attractive Business Development Pipeline. We believe the market for our Xe-100 and TRISO-X fuel technologies is wherever non-intermittent, reliable power is needed. We are initially targeting and deploying our advanced SMRs to both industrial (including large-scale chemical manufacturing) and utility-based service provider customers (e.g. data centers) who have needs for both electric power and the production of high-temperature steam with high reliability needs. We also plan to serve traditional fossil-fueled power plants and IPPs seeking to replace carbon-intensive fossil-fueled power plants in their jurisdictions.

Leverage Repeated Project Execution Learnings to Scale From FOAK to NOAK. Traditional nuclear power has been plagued by cost and schedule overruns. Vogtle units 3 & 4 cost more than \$16.0 billion each and were seven years over schedule in part because they were the first and second AP1000 reactors deployed in the U.S. While the fourth unit was reportedly approximately 20% less expensive than the third, these savings were only achieved when they were applied, achieving expected Nth-of-a-Kind scale with standard costs and schedule timelines only through repeated project delivery over a large order book. The DOE and industry expect that achieving Nth-of-a-Kind delivery can unlock potential savings of more than 30%. Starting with Dow and Centrica, our first customers, and as we deliver more reactors across its first announced sites at Seadrift, Texas, and Richland, Washington, respectively. With substantial potential for further targeted pipeline growth, we expect to be able to achieve reductions in costs and the acceleration of schedule as we deliver more reactors in the future. We believe that NOAK will be achieved through repeated project execution and financial de-risking, including early supply chain engagement, construction management, cost compression, and regulatory streamlining. As we apply lessons learned from each deployment, these steps will significantly reduce overall project execution costs and risks for subsequent projects.

Continue Geographic Expansion. Our initial core markets are the U.S., Canada and the U.K., which have sophisticated regulators that are capable of licensing our technology. Beyond these initial markets, which together represent only one-third of the serviceable energy consumption, we anticipate entering new markets where we already have customer engagement, including countries in Eastern Europe, the Middle East and East Asia.

Drive Technology Advancements. Using our innovative technology platform, we believe we are well-positioned to continue making technology advancements over time. These improvements include optimizing the Xe-100 for industrial heat applications and innovations in the design to support higher operating temperatures greater than 800° C to support more efficient hydrogen production processes.

Develop New Products. We continue to explore the development of innovative new products based on our core technology and varied use cases for our proprietary Xe-100 reactor, TRISO-X fuel and

microreactor technologies TRISO-X is currently performing fuel fabrication work for others who use TRISO-based fuel, including space applications. Similarly, our technologies also use TRISO-based fuel, including space applications. Similarly, our Technologies team is involved in U.S. government-funded studies (such as those from the DOE) for innovative remote powering and lunar applications that may involve our microreactor technologies.

Our Technology

X-energy was founded by Kam Ghaffarian. The technology used today was matured from the Pebble Bed Modular Reactor program, a South African public-private partnership which advanced high-temperature gas-cooled reactor technology. X-energy initially focused on the conceptual design of the Xe-100, discussed below, with an emphasis on top-level requirements designed to meet the broadest use cases. The company has continued to mature the Xe-100 design and has expanded into the development of TRISO fuel. Finally, X-energy has developed the design for the XENITH microreactor and will continue to invest in other research and development efforts.

Xe-100 — Our Advanced Reactor Technology



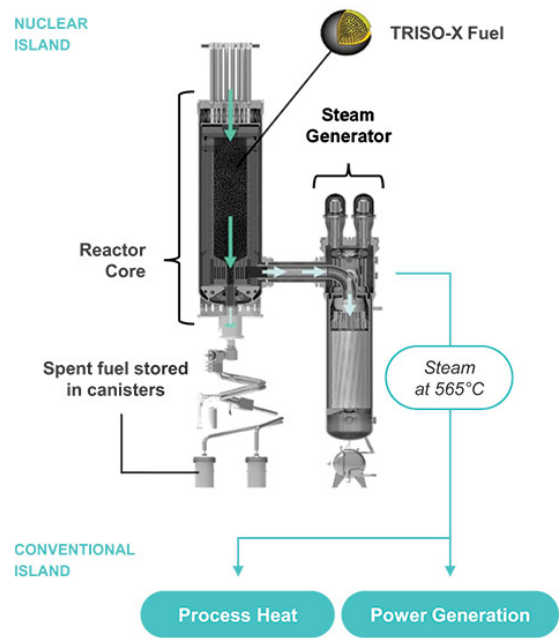
The Xe-100 is an HTGR designed to offer potential advantages in economics, reliability, and safety over conventional LWRs and other advanced SMR designs. The Xe-100 is a Generation IV nuclear technology intended to address certain limitations of LWR designs. The Xe-100 can produce electricity or process heat. When configured for electricity generation, each reactor is designed to support an approximately 80 MW turbine generator. Deployment of these reactors is scaled to be optimized in a four-reactor configuration, forming an approximately 320 MWe power plant.

After several years of development, the Xe-100 design has progressed into what is known as the preliminary design phase, the second of three typical engineering design phases. Design for construction is occurring in parallel with development of initial customer projects to ensure the design reflects customer needs and site-specific requirements. Once the standard reactor design reaches the development and licensing process is expected to be more streamlined for subsequent customers, although project-specific requirements will continue to apply.

The Xe-100 has made substantial progress in the NRC licensing process in the U.S. After several months of pre-application engagement, we supported Dow's subsidiary, Long Mott Energy, LLC, in submitting a construction permit application to NRC for the proposed advanced nuclear project utilizing the Xe-100 reactor design in Dow's Seadrift, Texas site. The pre-application engagement was intended to reduce licensing risk, as regulatory challenges or delays can materially affect a project's timing, cost, and

overall viability. Citing the completeness and quality of the application, and the effective application engagements, the NRC established an 18-month review schedule to complete its technical reviews for the Dow Construction Permit Application.

Xe-100 Schematic



One Xe-100 plant produces 320 MWe¹

Note: Figure is intended to represent an illustrative rendering and is for illustrative purposes only.

The Xe-100 has the following characteristics:

- **Cleaner and more reliable** — Like all nuclear reactors, HTGRs produce virtually zero emissions during energy generation, are “always” on (at 95%+ availability), and provide dispatchable power that is cleaner than most available firm alternatives.
 - Online refueling capabilities designed to enable the Xe-100 to meet a requirement for plant availability, unlike conventional nuclear facilities, which require shutdown every 18 – 24 months.
 - Four individual reactors feeding into a common source allows for enhanced reliability even under downtime for a single reactor, customer electricity and steam requirements continuously met.
 - Nuclear power has no carbon emissions from the power production, offering a viable alternative for major cloud-based service providers and industrial customers to meet decarbonization targets.
 -

Reactor is designed for the 60+ year operational life, longer than traditional nuclear alternatives and providing long-term certainty for customers.

- High burn-up fuel cycle (165,000 megawatt-days per ton of uranium) means the Xe-100 uses its fuel around 4.0x more efficiently than conventional reactors.
- *Designed with Intrinsic Safety Features* — The Xe-100 HTGR is safer by design and by the intrinsic physics of the reactor, requires fewer mechanical safety systems than traditional reactors, as well as fewer personnel for operations.
 - The Xe-100 leverages passive safety features and the inherent physics of its High Temperature Gas-cooled Reactor (HTGR) core, which is designed to shut down safely without operator intervention or external power in the event of an emergency.
 - The Xe-100 requires only about one-sixth the number of active mechanical safety systems typically found in traditional large-scale LWR nuclear plants, such as those used in the U.S. commercial fleet.

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- This reduction is achieved by eliminating the need for systems like emergency backup power, large-scale backup power, and extensive active containment measures, which are required in conventional nuclear designs.
- With only four operator-controlled variables (control rods, helium circulator, feedwater pump, and turbine throttle valve), opportunities for operator error are drastically reduced. Automated operations allow for significantly lower operator workload.
- Because of the low source term, the Xe-100 plant design requires a significantly smaller emergency planning zone of 400 meters compared to 16-kilometer zone required for traditional LWR nuclear plants.
- This compact safety footprint enables the use of commercial-grade structures for the reactor site, rather than the specialized nuclear safety-grade construction required for traditional reactors, and allows for greater flexibility in siting the plant closer to end users and industrial facilities.
- *More Secure* — The TRISO-X pebble fuel is a containment vessel in itself, supporting the inherent nature of the Xe-100.
 - The TRISO-X fuel pebble is a containment vessel in itself, made of layers of graphite, pyrolytic carbon and silicon carbide, and designed not to melt due to its ability to withstand extreme temperatures.
 - TRISO-X fuel retains 99.99% of all radionuclides during operation and severe accidents.
 - Each Xe-100 will use approximately 220,000 of our TRISO-X fuel pebbles, with approximately 18,000 uranium oxycarbide TRISO-X fuel particles in each pebble, which help contain fission products and aids in long term nuclear waste storage.
 - Pebbles remain in the reactor longer and burn off higher-risk byproducts compared to traditional advanced reactor concepts.
 - At full power approximately 175 pebbles are cycled into the reactor daily. Each pebble passes through the fuel handling system core on average 6 times before disposal.
 -

- After a pebble has cycled through the reactor, the spent reactor fuel is itself a vessel, with limited remaining fissile material, substantially limiting proliferation risk.
- Spent fuel can then be transferred directly into dry storage, reducing risk for management. Long-term fuel storage is the responsibility of DOE.
- Scalable — The Xe-100 is expected to be commonly deployed in its optimized configuration of 320 MWe, referred to as the “Xe-100 plant,” with the opportunity to scale additional reactors as needed.
 - We believe the 320 MWe power profile of the four-reactor deployment (and its high power density nature) is uniquely suited to meet data center demand.
 - The Xe-100 is expected to have lower overall construction costs compared to traditional scale nuclear reactors, due to its modular design, use of factory-fabricated components, and simplified safety systems.
 - These features are intended to reduce the complexity and duration of on-site construction, minimize labor requirements, and allow for more predictable project schedules.
 - As a result, historically significant challenges for conventional nuclear projects such as cost overruns and delays is substantially reduced.
 - This cost efficiency not only broadens the addressable market for advanced nuclear power, but also makes the Xe-100 an attractive option for customers seeking firm, clean power at a competitive price point.
- Streamlined — Due to its superior intrinsic safety attributes, the Xe-100 design needs fewer related specialized materials and therefore utilizes substantially more off-the-shelf components than traditional nuclear reactors, allowing for scalability from commercial vendors.

- Our elegant and simple design maximizes the use of off-the-shelf and factory-built components that are expected to be shipped to site using existing road and rail networks.
- A simpler reactor concept also minimizes complexity of the nuclear island development and construction.
- Strategic supply chain partners enable more off-site constructability and integration of components and drastically reduces on-site construction costs, personnel, and time.
- We expect approximately 1/10th the number of people needed to construct a Xe-100 plant compared to traditional nuclear.
- Versatile — In addition to electricity, the Xe-100 plant can be configured to deliver high temperature steam at 565°C, providing a solution for difficult-to-decarbonize industrial heat applications such as oil sands operations, mining, chemical production and petroleum refining and other industrial processes.
 - By using helium as a coolant, the Xe-100 is configured to withstand higher pressures than traditional LWRs, allowing the reactor to deliver heat at higher temperatures (750°C helium outlet temperature vs. 300°C steam from the steam generator) than conventional light water reactors (around 300°C). This is because LWRs are impacted by a phase change (i.e., higher temperatures would require higher pressures).
 -

The Xe-100 provides necessary reliability of industrial heat applications in its expected multiple-reactor deployment configuration, opening the market for replacements and expanding TAM to behind-the-meter applications.

- Load-following — The Xe-100 is designed to ramp down from full power to 40% power and ramp back up in a similar short time span.
 - This capability allows for faster ramp for load-following than existing Generation I conventional large-scale nuclear technologies, providing near-immediate response to energy needs.
 - Allows the Xe-100 to be relevant in both restructured wholesale markets and regional markets, as well as supporting intermittent wind and solar generation.

TRISO-X Pebble Fuel — Our Proprietary TRISO Fuel



Our reactors use TRISO particle fuel, a technology first developed in conjunction with the Dragon reactor in the 1960s. The DOE describes TRISO particles as “the most robust nuclear fuel on Earth.” We manufacture our own TRISO fuel using proprietary methods, through our wholly owned subsidiary, TRISO-X, LLC, to ensure supply and quality control. Further, many of the advanced reactor designs in development are expected to be powered by TRISO fuel or other coated-particle fuels. As part of our business model, we intend to not only fabricate and sell TRISO-X fuel to our customers, we also fabricate and sell TRISO-X fuel and other encapsulated fuels to other advanced nuclear customers, including governmental and private entities.

We do not intend to hold significant inventory of uranium feedstock (i.e., HALEU and UO₂) and expect our customers to procure this feedstock. Our business will be responsible for manufacturing

fuel form using this feedstock and other components such as graphite at our TRISO-X facility. The TRISO-X pebble fuel form, in particular the TRISO-X pebble, will be delivered to customers for use in the reactor.

To be able to use this fuel in operating reactors, the fuel must first be qualified by the NRC. The TRISO-X pebble fuel qualification methodology has already been approved by the NRC. For this methodology, X-energy is now undergoing irradiation testing on the TRISO-X pebbles at Idaho National Lab. This process began in 2025 and the campaign has an expected end date in the fourth quarter of 2026. Data from this campaign is then used in the subsequent testing, evaluation, and ultimate qualification of the fuel form by the NRC.

At commissioning, the initial fuel load for the reactor core of the Xe-100 will include graphite pebbles, LEU-based TRISO pebbles and HALEU-based TRISO-X fuel. This configuration allows the reactor core to transition to an equilibrium condition after core power density is established.

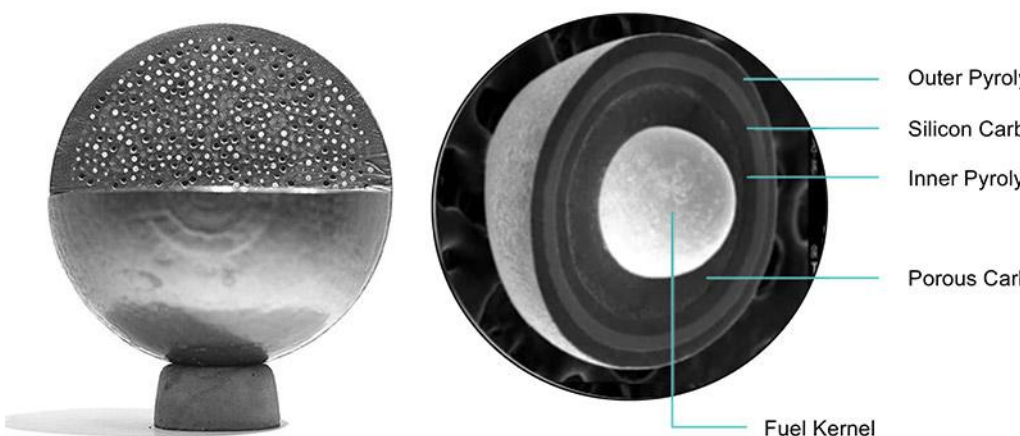
Using the reactor's fuel handling system, the Xe-100 reactor will first cycle out the graphite for the LEU-based pebbles to establish full power and temperature. The LEU-based TRISO-X fuel will then cycle out of the reactor for the HALEU-based TRISO-X fuel pebbles until the reactor reaches a steady state of continuous refueling of TRISO-X fuel. This initial core load is expected to support the first 5% of the plant's life, but the initial core load is expected to generate approximately 10% of the plant's revenues, with ongoing needs for additional refueling being sold through additional contracts.

Safe, established fuel form: While TRISO fuel is already an established fuel after decades of use and development, we expect our proprietary fabrication method to further differentiate our TRISO fuel and deliver a competitive edge in the commercial fabrication of TRISO fuel forms. We anticipate that our patented and novel process will demonstrate superior quality characteristics as well as improved performance compared to historical TRISO and potential future TRISO competitors. The TRISO fuel is a containment vessel in itself, encased in graphite, pyrolytic carbon and silicon carbide, which does not melt due to its ability to withstand extreme temperatures. After HALEU has been encased in the pebble form, security and proliferation risks are substantially reduced. Its robust nature allows for enhanced safety and, we expect a small emergency planning zone for our reactors, providing additional flexibility in site design and access to non-traditional nuclear markets and customers.

Robust integrity: Our TRISO-X pebble fuel is designed not to melt due to its robust construction. The structure minimizes the requirement for the extensive use of expensive and large concrete containment structures typically required in conventional reactors. TRISO-X fuel's unique characteristics include:

- At the heart of the TRISO-X fuel particle is the uranium oxycarbide kernel, which utilizes enriched uranium for higher energy content than the low-enriched uranium (LEU) used in conventional LWRs.
- Each fuel kernel is surrounded by four barrier layers that act in concert to support the fuel kernel, contain the fuel, moderate nuclear reactions, ensure passive safety and contain fission products.
- These particles are designed to retain their integrity during most foreseeable adverse conditions.
- The set of boundary layers act as functional containment for the radioactive fission products within the fuel kernel.
- The robust graphite pebble, about the size of a billiard ball, contains approximately 18,000 TRISO particles.

TRISO-X Fuel ***X-energy's Proprietary TRISO Fuel***



Integrated fuel fabrication business: We currently manufacture TRISO-X fuel in our TRISO-X Fuel Facility in Oak Ridge, Tennessee, which has operated since 2016. In February 2026, we received our Nuclear Material License from the NRC that establishes TX-1 as the first-ever Category II nuclear facility licensed in the United States. TX-1 will be a state-of-the-art Category II nuclear facility specifically for handling and processing HALEU feedstock for the fabrication of TRISO-X pebbles. We anticipate constructing TX-2 on the same site, which will allow for the license to extend to beyond TX-1. We do not expect to hold significant inventory of the fuel feedstock, which we expect to be procured from our customers. Our business will be fabricating the pebble fuels into their final fuel form for sale to our customers for purchase for operation of the Xe-100 reactor and its refueling needs. There are currently no large-scale facilities in the U.S. capable of fabricating HALEU fuel, allowing us the opportunity to fill a gap in the advanced nuclear fuel supply chain through TX-1. Planning for TX-2 is already underway, and we will further expand our TRISO-X fuel production.

Path to qualification due to extensive R&D: The ability to leverage decades of prior TRISO fuel development and development, particularly the DOE's Advanced Gas Reactor (AGR) Fuel Development Program, provides X-energy with a well-defined pathway for fuel qualification. X-energy's TRISO-X Fuel Qualification Methodology has been approved by the NRC and references the AGR test results for TRISO fuel qualification. This established framework provides X-energy a streamlined path towards obtaining NRC approval for TRISO-X fuel by the time of its first Xe-100 deployment. Based on the approved fuel qualification methodology, we are currently in the irradiation test phase and in the final stages of qualifying our TRISO-X pebble fuel.

Long-term revenues derisked: We plan to provide our Xe-100 customers with their initial fuel at no cost while generating additional long-term and recurring revenue streams by refueling Xe-100 reactors throughout their anticipated 60-plus year lives. We further assume that we will be the initial supplier of our proprietary TRISO-X fuel. Our business model is structured so that customers retain ownership and responsibility for procuring uranium and fuel feedstock, which is intended to insulate X-energy from commodity price volatility and supply chain disruptions. Further, we have no responsibility for the management of spent fuel as that function is the responsibility of plant owner and operators, which roles we do not intend to participate in. We believe that our ability to achieve economies of scale through dedicated fuel fabrication facilities and our vertically integrated business model will enable us to be a low-cost provider of TRISO-X fuel, which we believe will be a key driver of customer retention and market competitiveness.

Research and Development

In addition to our core Xe-100 reactor, we continue to innovate and develop new products through our Emerging Technologies team. We are expanding our portfolio to address a broader range of applications and use cases beyond utility-scale power generation.

Our design team brings decades of experience from HTGR pebble-bed design programs in the U.S. and South Africa, combined with veterans of the DOE National Laboratories.

XENITH Microreactor



We have developed the XENITH microreactor, a compact solution designed to deliver reliable power in locations where traditional power infrastructure is unavailable, impractical or vulnerable. It is designed to be deployed in months and operate continuously for 20 years, delivering 3-10 MWe with minimal maintenance requirements.

XENITH is not yet operational but is currently exploring commercialization opportunities. The development of the microreactor began in the Pele program through the U.S. Department of Energy, which X-energy was awarded approximately \$60 million in various contracts. Conceptual design progressed during Pele Phase 1 and 2, and design work has continued into what is now the XENITH microreactor. The microreactor is likely to be deployed in a 5 MWe format, or as dual reactors in an electric format.

Like the Xe-100, XENITH is also a High Temperature Gas-cooled reactor that runs on TRISO fuel. The required fuel is expected to be in the form of compacts rather than pebbles. The expected operating temperature is 750 degrees Celsius. The value proposition is expected to be as follows:

- Scalable Power. Factory-built, transportable nuclear energy solution that prioritizes deployment readiness and compatibility with existing infrastructure to support deployment in a variety of locations.
- Resilient & Reliable. Provides reliable, long-term power with no required refueling over its operational lifetime, eliminating vulnerability to supply chain disruptions while providing 24/7 reliable power.
- Tailored to the Target Market. Military, defense, and remote applications, including black-starts, and remote communities that require secure, independent power.

Space Applications

Our innovative HTGR design and high-temperature-resistant fuel make a unique and powerful combination for nuclear electric propulsion, nuclear thermal propulsion and lunar surface power.

Our use of robust TRISO fuel in combination with innovative reactor designs and advanced materials, all work together to increase the efficiency of the fission reaction while minimizing weight, keeping the reactor light enough for spaceflight. These characteristics have allowed for viable concepts for a Fission Surface Power (FSP) reactor. This concept was studied through a

award from the DOE and the U.S. National Aeronautics and Space Administration (NASA) in advance the design of a FSP solution.

OUR PROJECT DEVELOPMENT APPROACH

X-energy works closely with our customers throughout the project lifecycle to help avoid cost overruns in the future. In some cases, our experienced team has been able to identify design solutions to help reduce costs. We also expect to use learnings on the initial projects to reduce future costs.

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We believe the ability to identify cost-reduction solutions, particularly compared to traditional nuclear, gives us an advantage over other competitors who may face cost overruns or other project delays.

Traditional nuclear has consistently suffered from historical project delays and cost overruns. For example, Vogtle units 3 & 4 were \$16.0 billion+ and seven years over schedule in part because they were the second AP1000 reactors deployed in the U.S. While numerous factors contributed to the delays, for Vogtle units 3 & 4, core causes included an incomplete design and a fixed price agreement that required a change in contractor after the original EPC contractor, Westinghouse, filed for bankruptcy after construction had already begun. The project subsequently was taken over by Southern Company who acted as project manager and hired Bechtel as the prime construction contractor to bring Vogtle units 3 & 4 to completion.

We are currently focused on delivering our first two projects with Dow and Energy North Carolina. We believe achieving Nth-of-a-kind scale with standard costs and schedule timelines is possible through repeated project delivery over a large orderbook. As part of our commercialization strategy, to deliver at scale, X-energy intends to support delivery of its own projects. We believe our success is dependent on leveraging economies of learning and replication, and overseeing the full development of projects enables customer-centric support.

X-energy intends to partner with customers and EPC contractors to finance, build, and operate projects. We expect to oversee projects from origination to delivery and coordinate project development. As part of our business model, in addition to fuel fabrication services, we intend to offer customers a suite of added services during development of the reactor given our expertise on regulatory, engineering, procurement and other processes. Additionally, we may deploy development capital to fund early stage future projects and to invest alongside future customers to drive an accelerated path to commercial deployments.

Similar to the “design one, build many” approach that has been utilized in the oil & gas industry to deliver large-scale projects like offshore oil platforms and liquefied natural gas terminals, X-energy will leverage its repeated order book learnings to accelerate construction timelines and reduce costs across successive deployments. Along with our project development platform, our supply chain and manufacturing are crucial to our strategy. We intend to optimize our supply chain around the modular nature of our reactors to drive construction timeline acceleration. Additionally, we intend to partner with experienced engineering partners to ensure repeatable project delivery, bringing together the policy, finance and commercial expertise to help deliver our projects.

Early in the construction process, we will work with our customers to obtain the permits required to build and operate the projects. In determining what permits are required, we intend to engage regulatory experts to help us understand the requirements of state and local authorities applicable to each project. We intend to commence the permitting or licensing process during the early stages of development to reduce the length of time it typically takes to complete. For the Dow project, the CPA was docketed with the NRC in May 2025 with an 18 month review schedule. X-energy maintains a strong relationship with the NRC.

works closely with customers on submitting the necessary permits. Getting an early start on permitting and licensing is important in helping to de-risk our projects, because permitting and licensing delays can have significant impacts on a project's timeline or design, or in some cases its viability. As part of the permitting and licensing process, we will also seek to promote long-term community support for our projects.

As part of the permitting and licensing processes, we not only intend to support our customers in securing the permits and licenses necessary to construct and operate projects, but we also plan to engage our and our customers' EPC and O&M contractors and partners to ensure they have plans in place to monitor permit compliance during construction and operational phases.

OUR CUSTOMERS

Dow

Dow is X-energy's first customer to receive a reactor and is a global leader in the specialty chemicals industry, involved in the manufacturing and distribution of a wide range of chemical and plastic products. In the course of its annual operations, Dow produces over 12 million MWh of electricity and is committed to the ongoing decarbonization of its operations.

X-energy has partnered through an MPDA and CCA with Dow to provide our services in support of the first-of-its-kind (FOAK) deployment of four Xe-100 reactors to provide power with virtually zero direct GHG emissions and industrial steam at Dow's UCC Seadrift site in Texas. The CCA establishes governance and operational frameworks to advance the ARDP-supported deployment of Xe-100 reactors at Dow's Seadrift site and guide future cooperation, including a joint steering committee with core principles of mutual benefit and shared FOAK risk and benefit. With the support and assistance of X-energy, Long Mott Energy Services, a wholly owned subsidiary of Dow, filed a CPA with the NRC in March 2025 which was approved in May 2025 for an 18-month review period with an expected review completion by late 2026 and the CPA expected in the first quarter of 2027. Initial construction can commence after receipt of the NRC's final approval.

Dow provides expertise in plant design and industrial heat as well as in the development of the Xe-100's first plant, which will be a first-of-its-kind co-generation facility with unique heat capacity not available in other technologies. Under the CCA, as between the parties, X-energy will own the intellectual property for the reactor systems, controls, software, and related nuclear plant technologies, as well as fuel cycle related IP, created by the parties for this project. As between the parties, Dow will own all of the intellectual property for the non-nuclear systems that turn reactor steam into electricity and run steam and water for the plant, as well as IP for equipment and processes that sit outside both the nuclear and conventional plant boundaries, created by the parties for this project.

Amazon

After an extensive evaluation of potential carbon free generation solutions for its extensive data center footprint, Amazon made an equity investment in X-energy in 2024 and announced options to build more than 5 GWe of new Xe-100 projects online across the U.S. by 2039, which assuming full execution of the options, will represent the largest commercial deployment target of SMRs to date as of August 2024.

The first deployment under this 5 GWe total potential target is a project with Energy Northwest in central Washington. Amazon and Energy Northwest entered into a Carbon Free Development Agreement for an initial deployment of four reactors representing 320 MWe, with the potential to expand to a power capacity to 960 MWe.

X-energy has entered into commercial arrangements with Amazon, which provide for, things, a deployment and financing model that X-energy plans to pursue more broadly with infrastructure and utility partners. The arrangements also grant Amazon options and allocations intended to facilitate project development. For more information regarding our agreements with Amazon, see “Risk Factors — Our sales and profitability may be impacted by, and we may incur liabilities from, terms to certain customers, our failure to meet performance guarantees under customer contracts, and our customer safety standards.”

Centrica

Centrica is a major provider of energy services across the U.K. and owns a 20% stake in the country of operating nuclear reactors in the country. In September 2025, X-energy and Centrica signed a joint development agreement dedicated to building and operating Xe-100 reactors in the U.K. The JDA identifies initial areas of collaboration including U.K.-specific planning model translation, and execution of de-risking work packages in the areas of reactor technology, and commercial domains, largely on a non-binding basis. Under the JDA, Centrica and X-energy Company have the opportunity to consider options to collaborate on other Xe-100 deployment projects following further diligence and negotiation. The announcement followed a pledge from President Donald Trump and Prime Minister Keir Starmer to work together on nuclear power and constitutes a new strategic partnership alliance to accelerate the deployment of SMRs, bringing together a leading player in delivering clean energy future and the developer of the world’s most advanced nuclear technology.

The Xe-100 was selected after a significant review period by Centrica of advanced nuclear technologies. Centrica prioritized partnership with X-energy because of the reactor’s safety and ability to provide heat for industrial heat that Centrica sees as key to expansion in the U.K. market. The U.K. government is currently pursuing a “three-legged” approach to nuclear development, designed to ensure a resilient nuclear energy sector. Under this strategy, the first “leg” focuses on large-scale nuclear power plants such as those being developed in partnership with EDF, to provide substantial baseload power. The second “leg” supports the

deployment of Gen III+ light water SMRs, with Rolls Royce’s SMR technology selected as the preferred solution in this category. The third “leg” is dedicated to other small and advanced modular reactors, including HTGRs like Xe-100, which are prioritized for their potential to deliver both electricity and high-temperature grade industrial heat, supporting decarbonization across a range of sectors.

X-energy and Centrica have identified Hartlepool as the preferred site for the first of a fleet of approximately six GWe (representative of 76 reactors likely deployed as 19 80 MWe configurations). A project at Hartlepool will be comprised of up to twelve 80 MWe reactors, with the capability to provide high temperature steam for industrial decarbonization. Subject to securing all necessary permissions and licenses, the first electricity generation is expected to be in the mid-2030s.

OTHER KEY PARTNERSHIPS

Department of Energy

Through the ARDP, we are partnering with DOE and others to build the world’s first commercial advanced nuclear reactor. In December 2020, X-energy, initially in collaboration with Energy Transfer, was selected for an award, initially in the amount of \$1.2 billion to deliver a first-of-a-kind advanced nuclear plant and TRISO-X fuel fabrication facility. This project is now being developed by X-energy and Dow with continued financial support from the DOE on a 50/50 cost share basis. As one of our

selected out of many applicants for the ARDP, X-energy was recognized as having an advanced technology of choice.

The cooperative agreement for the program, signed in February 2021 (the “ARDP Agreement”) provides 50/50 cost share of \$2.4 billion of eligible costs (\$1.2 billion reimbursement) to X-energy, allowing X-energy to continue work toward design, licensing, commercialization and construction of a first-of-a-kind commercial advanced nuclear plant and commercial TRISO-X fuel fabrication facility, benefiting from decades of nuclear experience and knowledge within the DOE.

The Company is constructing and will own the fuel facility subject to the government license and property set forth in 2 CFR § 200.311 and 2 CFR § 910.360, and insure it in accordance with 2 CFR § 200.310-16. The DOE will obtain rights to the intellectual property (“IP”) developed under the ARDP Agreement consistent with typical government retained IP rights under the Bayh-Dole Act, Uniform Government Patent Business Patent Waiver, and DOE regulations. The federal government retains a nontransferable, irrevocable, paid-up license for its own benefit. The DOE also retains management rights under the ARDP Agreement which allow the government, in specified circumstances, to grant a license to a contractor or successors in title to the patent to grant a nonexclusive, partially exclusive, or exclusive license to a responsible applicant or applicants. If the patent owner refuses to do so, the government may grant the license itself. The DOE will fund a portion of the direct and indirect costs incurred for research and development costs to develop the IP and the costs of developing and constructing both the fuel facility and Xe-100 demonstrator reactor. The funding is subject to future government action which may not occur, and the government is able to cancel the contract at any time without substantive penalty.

As of December 31, 2025, we have been reimbursed approximately \$438 million in funding under the ARDP. We submit our budgets through an ongoing “budget period” basis tied to project milestones under the ARDP Agreement, and our current budget covers a budget period that began in March 2021 and extends through August 2026. We submit a Continuation Application to the DOE to extend the award for subsequent periods. Extensions beyond the current budget period are subject to DOE discretion and approval. Under the terms of the ARDP Agreement that rely on the Office of Management and Budget (OMB) guidance, the total extension of the award may not exceed three years (for a total project performance of 10 years). Any additional extension would require an approval within DOE at the discretion of the Contracting Officer. Based on the original February 2021 award date, the maximum amount of funding under the ARDP Agreement, assuming all extensions are granted, is expected to be available through 2030. If we are unable to obtain extensions and incur eligible costs beyond the currently approved performance period, we would forgo reimbursement for such costs and may face de-obligation of funds at closeout. There can be no assurance that we will receive additional ARDP funding beyond the current budget period or that extensions of the award will be granted. Dow, as our sub-awardee under the ARDP, is similarly subject to the budget period and extension

requirements, but does not need to separately seek or be included in any DOE-approved extension. We will continue to seek extensions for us to continue receiving reimbursement from the DOE for eligible costs.

If our award is not extended, we would expect a lapse in our ability to receive ARDP funding beyond the current approved extension, and any unobligated or unneeded DOE funds at closeout would not be available. Extensions under the ARDP Agreement are at the DOE’s discretion and are subject to criteria including: (1) availability of appropriations; (2) availability of future-year budget authority; (3) substantial progress toward meeting project objectives; (4) submittal of required reports; and (5) compliance with the terms and conditions of the award. We continue to engage with the DOE to extend funding with additional

have been allocated to X-energy by the Office of Nuclear Energy, including through additional budget periods to cover subsequent design activities as part of the overall ARDP project. X-energy has been named a sub-awardee under the ARDP in connection with their involvement with the design and construction of the demonstrator reactor. The Dow project's construction permit is anticipated to be received in the first quarter of 2027, after which initial construction can commence, with commercial operations date in the early 2030s. The Dow project timeline extends beyond our current reporting period and may extend beyond the maximum period of performance under the ARDP Agreement, depending on whether extensions are granted and the pace of project development. If the ARDP funding period is not extended before the Dow project reaches completion, Dow would then be requested to cover remaining project costs without the benefit of ARDP reimbursement, which could affect Dow's ability to proceed with the project and our ability to complete the ARDP demonstrator reactor.

Ontario Power Generation

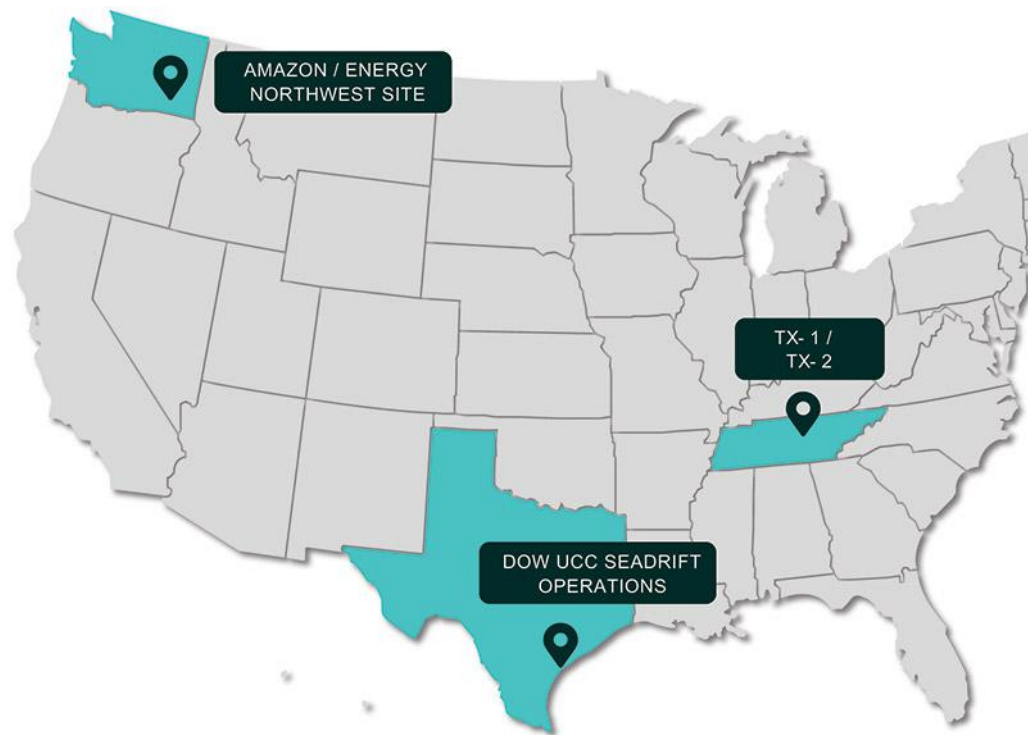
In 2022, X-energy and OPG entered into a framework agreement to work exclusively with OPG with respect to advanced nuclear power industrial applications in Ontario, Canada, and to co-develop and advance the Xe-100 as the nuclear technology of choice for industrial applications throughout Canada. X-energy will be the operator of any Xe-100 facilities that are deployed under this agreement. OPG is a significant equity investor in X-energy.

Department of Defense

We have an agreement with the U.S. DOD Defense Innovation Unit (“DIU”) and the U.S. Air Force to advance the development of X-energy's commercial microreactor in alignment with the President's executive order to deploy advanced nuclear technologies at DOD installations to enhance national security. The agreement supports continued design and development for the X-energy microreactor under the Advanced Nuclear Power for Installations program, an initiative led by the Air Force in partnership with the Department of the Air Force. X-energy was also part of the 2019 initial Project Pele to develop a prototype mobile nuclear reactor. Through these contracts, X-energy has been awarded over \$60.0 million on microreactor activities including engagement with the NRC. While the period of performance for the XENITH microreactor has expired, the DOD has announced initiatives that maintain interest in the microreactor program.

Under the Project Pele Agreement, the Company leads the design, analysis, testing and engineering needed to develop a prototype mobile nuclear reactor and works with the DOD to achieve its objectives under Project Pele. The Company is paid a fixed amount for each milestone accomplished in accordance with the key objectives and milestones set out in the agreement, plus reimbursement of certain costs incurred. The DOD has certain rights in patents and data produced under the Project Pele Agreement, and may terminate the Project Pele Agreement if such termination is in the Company's best interest.

Below is a geographic overview of our three current projects, which is followed by a description of each project.



Reactor Projects

Dow's UCC Seadrift

Dow and X-energy have entered into an MPDA to deliver the inaugural X-energy advanced nuclear reactor at Dow's UCC Seadrift manufacturing site ("Seadrift") in Texas. The project will be a large-scale advanced nuclear reactor deployment for an industrial site in North America and will help decarbonize the operation by providing both process heat and electric power. Seadrift site is a large industrial complex that includes consumer products used every day in homes, schools, offices, hospitals, vehicles, and many other industries, producing more than 4 billion pounds of materials per year across a wide variety of industries, including food packaging and preservation, footwear, wire and cable insulation, solar cell manufacturing, and packaging for medical and pharmaceutical products.

The plant will be comprised of four reactors (320 MWe) and supply all of the site's electricity and steam demand. Dow's UCC Seadrift site spans around 4,700 acres and is Dow's second largest site in Texas. This project supports Dow's corporate sustainability goals and is expected to reduce greenhouse gas emissions by 440,000 MT CO₂e/year.

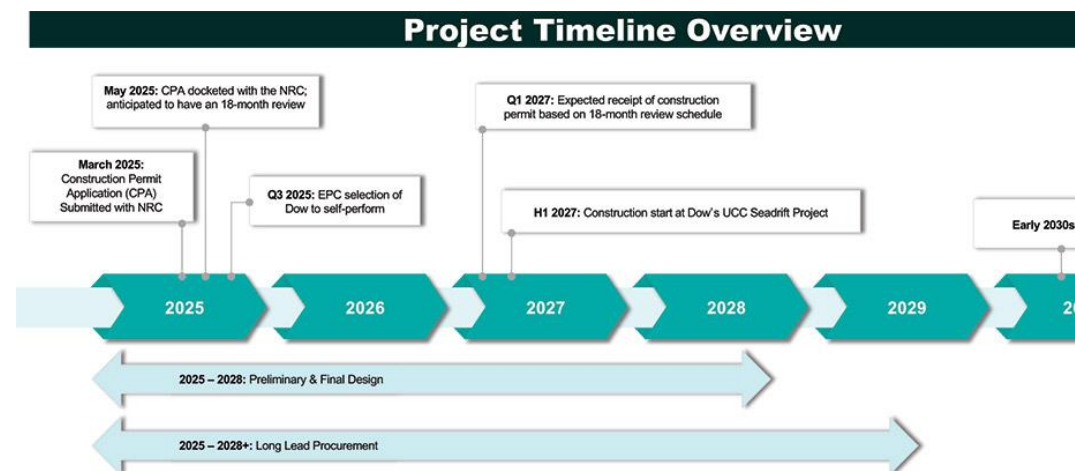


Note: Figure is intended to represent a proposed rendering and is for illustrative purposes only.

The project benefits from the ARDP cooperative agreement with the DOE. This includes a 50% cost share in the project. The project also includes substantial production and investment tax incentives under the Inflation Reduction Act. Through the ARDP, X-energy receives from the DOE 50% of the cost of the Xe-100. X-energy also receives 50% of the cost of TX-1, its first fuel fabrication facility. As X-energy's first customer to construct a reactor, Dow, is eligible to receive 50% of the cost to build a four-pack of Xe-100s, at Seadrift, Texas, which significantly mitigates the risks of building a first-of-its-kind advanced reactor project.

We continuously update the DOE throughout the year regarding our spending against our ARDP and are currently funded through at least August 2026 through a Continuation Application. As part of the Continuation Application process, we submit updated budgets to the DOE for each budget period, including cost estimates and supporting justification. These updated estimates are subject to DOE review and may differ materially from prior estimates.

Key Timeline Highlights of our Dow Project



Development of project milestones:

- Once Dow became our customer, our engineering design process was done in conjunction with them. The Xe-100 is in the second of three engineering design phases. Ongoing engineering design work includes adapting to the commercial expectations of the customer, including specific seismic requirements and customer use cases. In the case of Dow, we expect the output of the four Xe-100 reactors to be for industrial steam for Dow's chemical production.
- We expect to enter into the final design phase (phase 3) according to Dow's schedule in 2027. The final design phase would culminate in a final investment decision, which would lead to the beginning of major nuclear construction, though preliminary construction is expected to begin before this time. These stage gates are designed to unlock key future milestones.

— Licensing:

- March and May 2025: With the support and assistance of X-energy, Long Mott Energy Services, a wholly owned subsidiary of Dow submitted a CPA to the Nuclear Regulatory Commission ("NRC") in March 2025. The NRC docketed the application in May 2025 and established a 12-month review schedule, with completion of the review currently expected by late 2026. Issuance of the CPA anticipated in the first quarter of 2027. The NRC's review process includes environmental, geological, and geotechnical work intended to evaluate site characteristics to help ensure that the Xe-100 plant is designed and constructed based on the site's characteristics. Subject to receipt of the CPA and other required approvals, construction is expected to begin in 2027, with a target commercial operations date in the early 2030s.
- At the appropriate time we expect our customers would submit an operating license application (OLA) to the NRC to be able to operate the reactor.

— Construction & readiness:

- Site construction is expected to begin in the first half of 2027 after receipt of the construction permit application from the NRC.
- Dow and X-energy have been coordinating on the financing and strategy for the project, including procurement items, beginning in 2025, that are required to support the construction, commissioning and commencement of operations of the reactor expected in the early 2030s.

The agreement supports Dow's sustainability targets. The project is set to create thousands of jobs during peak construction and hundreds of full-time, family-wage jobs during operation.

The Dow project's Construction Permit Application is anticipated to be received in the first half of 2027, following which construction can commence, and its commercial operations date is expected in the early 2030s. This timeline extends beyond our current ARDP budget period, which runs through 2026, and beyond the maximum period of performance under the ARDP Agreement, even if extensions are obtained. Based on the original February 2021 award date, the outside date for ARDP funding is expected to be in or around 2030 (assuming the maximum three-year extension). Accordingly, construction activities scheduled to occur after 2030 would not be eligible for ARDP reimbursement without a modification or change in the ARDP award, which would require approval from the DOE at a level above the Contracting Officer.

For activities occurring beyond the ARDP funding period, Dow would be requested to fund the share of remaining project costs without the benefit of 50% ARDP reimbursement if they continue the project. The precise funding requirement that would need to be borne by Dow absent ARDP reimbursement depends on the stage of project completion at the time ARDP funding expires, the scope of construction activities, and market conditions at that time.

As of December 31, 2025, we have been reimbursed approximately \$438 million in fund ARDP. We submit our budgets through an ongoing “budget period” basis under the ARDP A, our current budget covers a budget period that began in March of 2025 and extends through A, extensions are not granted or if the project timeline extends further than currently an incremental funding requirement that Dow would need to bear could increase materially. Any could affect Dow’s final investment decision, cause delays or modifications to the project scop or lead Dow to

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terminate its participation in the project. If Dow does not proceed with the project, X-energy obligation to continue funding to the Dow project or construction on the plant itself; however need to identify an alternative partner under the ARDP Agreement. See “Risk Factors — The timeline extends beyond the ARDP funding period, which could result in significant incremental requirements for Dow and adversely affect the project.”

Amazon / Energy Northwest — Cascade Advanced Energy Facility

Energy Northwest is a Washington state public power joint operating agency and a premier carbon-free electricity through its existing ownership of the Columbia Generating station nuclear power plant in Richland, WA. In 2024 Energy Northwest and Amazon entered into the Development and Funding Agreement for an initial deployment of four reactors representing the Columbia site, with the potential to upsize the power capacity to 960 MWe enough equivalent of 770,000+ U.S. homes.



Note: Figure is intended to represent a proposed rendering and is for illustrative purposes only

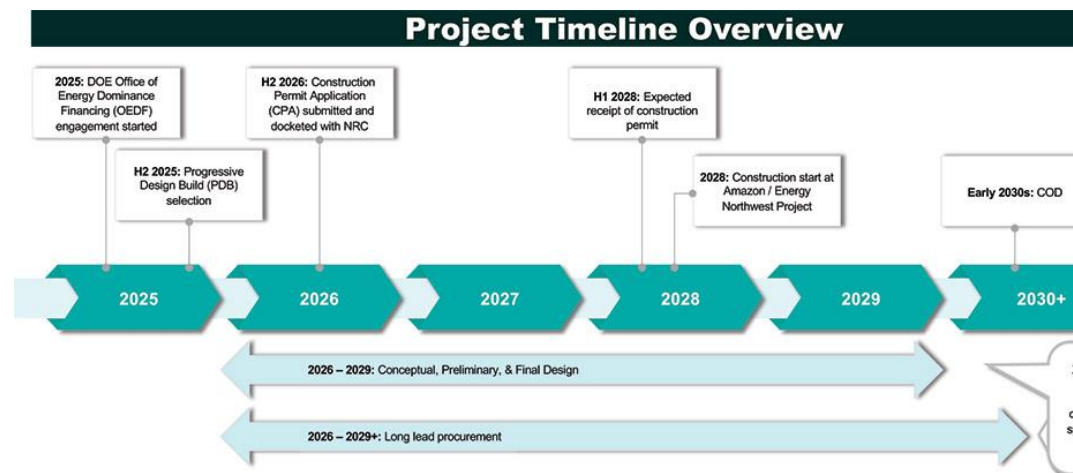
As part of the agreement Amazon also made a direct investment in the project and the power will be available to Amazon and northwest utilities to power homes and businesses. Public power utilities have the opportunity to purchase up to 50% of the power from the full 960 MW project. Initial site work is underway. CPA is targeted for submission to the NRC by the end of 2026.

The agreement supports Amazon's Climate Pledge commitment to reach net-zero carbon emissions by operation by 2040. The project is set to create 1,000-2,000 jobs during peak construction and 1,000 full-time, family-wage jobs during operation.

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Key Timeline Highlights of our Amazon Project



Development of project milestones:

— Conceptual, Preliminary & Final Design:

- As we finalize our design as part of the Dow project, we will integrate the site-specific requirements at the Energy Northwest project, which is expected to generate electricity for four reactors. As of the date of this prospectus, we are in the Conceptual Site Design phase.
- The Energy Northwest project utilizes a progressive design build process, so the project is brought on board in 2025.

— Licensing:

- The Amazon / Energy Northwest CPA is targeted for submission to the NRC by the end of 2026. The construction permit is anticipated to be received in 2028, with construction starting shortly thereafter.

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By the start of construction, engagement with the Office of Energy Dominance is expected to be underway, and engineering work is expected to have commenced. Targeting Commercial Operations in the early 2030s.

- At the appropriate time we expect our customer (Energy Northwest) would submit a license agreement (OLA) to the NRC to be able to operate the reactor.

— Construction & readiness:

- Site construction is expected to begin in the first half of 2028 after receipt of the permit application from the NRC.
- Energy Northwest, Amazon, and X-energy have been coordinating on the financing for long lead procurement items, beginning in 2026, that are required to start construction, commissioning and commencement of operations of the reactor expected in the early 2030s.

Fuel Projects

Fuel Fabrication Pilot Facility

X-energy has a TRISO-X fuel fabrication pilot facility at Oak Ridge National Laboratory. The plant has been operational since 2016 and is currently producing kilogram batch quantities of TRISO fuel for development purposes. The Pilot Facility is planned to be temporarily shut down and relocated to a new site at Oak Ridge National Laboratory to property in Oak Ridge, TN for “TX-L”, which will be adjacent to the existing site. The proposed TX-L will then serve as the ‘laboratory’ for future TRISO-X fuel development.



TX-1

TX-1, located in Oak Ridge, Tennessee, will be the first-in-the-nation commercial advanced nuclear fuel fabrication facility. The facility will manufacture X-energy’s TRISO fuel using proprietary technology.

Once complete, TX-1 will be the first Category II Fuel Fabrication Facility licensed by the NRC with an estimated throughput of 5 metric tons of uranium (“MTU”) per year or approximately 11 TRISO-X pebbles per year, enough fuel for up to 11 Xe-100 reactors.

Part of X-energy’s participation in the DOE ARDP award encompasses the Phase 2A facility construction, including the completion of the core and shell of the 214 thousand square feet. In parallel, X-energy has received approval from the DOE to authorize an additional \$30.0 million for early procurement of critical long-lead equipment and materials to support the delivery of subsequent construction phases and ensure adherence to the overall project schedule. TX-1 was granted an approximately \$150.0 million tax credit by the DOE in recognition of its status as a U.S. FOAK advanced nuclear fuel fabrication facility, subject to the project successfully reaching key milestones.

X-energy has completed an internal 60% Design Review and is planning for internal Design Reviews to begin in May 2026. In February 2026, we received a Special Nuclear Material License from the NRC that establishes TX-1 as the first-ever Category II nuclear fuel facility licensed in the United States, enabling TRISO-X to commercially manufacture fuel using HALEU at its TX-1 site under an industrial license. We anticipate constructing TX-2 on the same site, which will allow for the license to be used for multiple facilities. This NRC license approval formally establishes TX-1 and TX-2 as the first new Category II fuel facility licensed by the NRC in approximately 50 years, with TX-1 set to become the first-ever Category II fuel facility licensed in the United States. X-energy is using Geiger Brothers for the site development of the TX-1 site. Site preparation is substantially complete. Clark Construction Group has begun to conduct the vertical construction phase (2A), which began in September 2025. Facility construction Phase 2B construction is set to commence mid-2026 with expected completion in late 2027 for target operations beginning in the first half of 2028.



Note: Figure is intended to represent a proposed rendering and is for illustrative purposes only.

TX-2

Planning is also underway for a subsequent advanced nuclear fuel fabrication facility, which would occupy the available space on the same site at TX-1 (as seen in the graphic above as “TX-2”). Fuel fabrication facility modelled after TX-2 would be expected to have larger production output than TX-1 and support up to 44 Xe-100 reactors annually. The expected capacity would be 25 MTU with an anticipated annual throughput. This equates to about 2,800,000 TRISO-X pebbles per year. An NRC license would be required for these facilities. X-energy anticipates building TX-2 adjacent to and on the same property as TX-1, in which case a new license for TX-2 would not be required. If TX-2 is built on a different site, as is currently contemplated, a separate NRC operating license would be required; in that case, X-energy expects that elements of the TX-1 licensing basis, programs and regulatory approvals would be leveraged or referenced, as appropriate, to support the license application for the TX-2 facility.



Note: Placement of proposed TX-1 and TX-2 sites are not to scale and should be considered illustrative.

Other

Existing Frederick Operator Training Facility

Funded partially through the ARDP, our existing Operator Training Facility in Frederick opened in March 2023. This facility encompasses 10,447 square feet and supports several functions related to the Xe-100 reactor program. One of the primary functions at this site is the development of a full-scale ANSI 3.5 Xe-100 simulator, which is intended to serve as a true representation of the reactor. The simulator will incorporate all process models of Xe-100 reactors, instrumentation and control systems, and other essential elements to create an environment suitable for training and licensing Xe-100 operators prior to the initial fuel loading of the first Xe-100 reactor. This facility supports designing the reactor control room and constructing a replica control room where operators will receive their training. The Xe-100 control room is expected to be capable of managing up to twelve Xe-100 reactors.

In addition to simulator development, the Operator Training Facility is responsible for developing operational procedures required to control the Xe-100 plant. These procedures will be displayed on a computerized procedure system that we are developing, and will serve as the primary training materials used by Xe-100 operators. Operator training itself is conducted in the replica control room environment, where our customers will be able to utilize both the simulator and our training materials to gain practical experience.

The Operator Training Facility supports elements of human factors engineering, including workload analysis consistent with NRC guidance such as NUREG-0711, which is used to inform the design of the Xe-100 control room and simulator. This analysis also helps determine the workload on Xe-100 operators, which is necessary for calculating the appropriate number of operators required in the control room.

Expanded Frederick Testing and Training Facility — XTAC

For its long-term ambitions, X-energy is developing a state-of-the-art 90,000 square foot test and training facility in Frederick, MD. This facility will ensure key components of the Xe-100 are tested safely and reliably under real-world conditions before full-scale manufacturing and deployment. The building is anticipated to accommodate a helium test facility designed for full-scale integration testing within a

pressurized helium environment. Additionally, it will feature an experimental test facility designed for testing and prototyping of essential reactor components and materials.

X-energy had received conditional approval from the DOE to proceed with building the test facility, which has now been completed, with a 50% cost share arrangement with the DOE through the HALEU program. X-energy also has the potential to receive up to \$6.15 million in state and local support through various programs, including tax credits, conditional loans, and grants. X-energy has secured a DOE National Environmental Policy Act (NEPA) Categorical Exclusion for the building, meaning no Environmental Impact Statement or EIS is required.

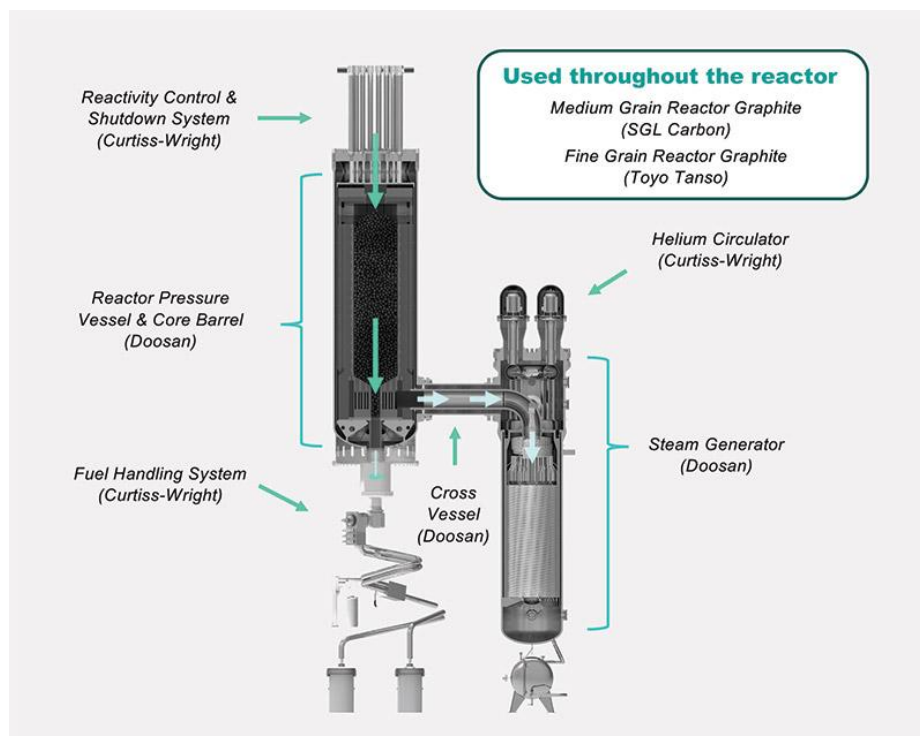
For a description of our other facilities and properties, see “Facilities Overview” included elsewhere in this prospectus.

SUPPLY CHAIN

X-energy has established an extensive, global supply chain ecosystem for the Xe-100, including strategic and commercial partnerships that allow X-energy to leverage established players in the nuclear industry to manufacture key components, including the reactor pressure vessel & core barrel, steam generator, system, burnup measurement block, helium circulator, steam generator, fine grain reactor graph, fine grain reactor graphite, and reactivity control & shutdown system. X-energy has signed Preferred Supplier Agreements (“PSSAs”) with key suppliers for critical Xe-100 components and systems. X-energy has built partnerships with key component providers. Throughout the process, X-energy has been working closely with suppliers in all phases of design, equipment supply, fabrication, and construction.

X-energy is differentiated in its vertically integrated fuel supply model with in-house fuel cycle capabilities. The U.S. Government, through the DOE and HALEU Availability Program, is supporting pathways toward a U.S.-based commercial enrichment supply chain with capabilities to produce HALEU domestically. X-energy received a granting of initial allocation of HALEU in April 2025. X-energy is actively working to provide surety of offtake to establish a domestic commercial supply chain for HALEU uranium. Through partnerships with companies like Centrus Energy, which has already produced approximately 1 MT of HALEU to date for the DOE through its pilot cascade, the DOE is creating domestic avenues for HALEU procurement.

Key components of the reactor with planned suppliers



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Note: Figure is intended to represent an illustrative rendering and is for illustrative purposes only.

Our Key Supply Chain Partners

We have a network of partners that enable our growth and success. These partners include:

- Curtiss-Wright.** Curtiss-Wright is a leading U.S. designer and supplier of critical nuclear power plant systems, equipment, services, and spare parts to the U.S. domestic and global nuclear power industry. Following several competitive bid processes, X-energy selected Curtiss-Wright as a successful bidder for multiple Xe-100 systems. Curtiss-Wright was selected as a preferred supplier to X-energy for the ARDP and subsequent Xe-100 projects in the U.S. Curtiss-Wright is an equity investor in X-energy.
- Doosan Enerbility.** Doosan is a preferred strategic supplier to X-energy. Doosan is a global manufacturer and supplier of core components of nuclear power plants, such as reactor pressure vessels, steam generators, and steam turbines. Doosan has a vertically integrated manufacturing facility in Changwon, Korea, which is capable of raw material production to final assembly of nuclear components. Doosan has manufactured and supplied 34 reactor pressure vessels and 20 steam generators globally. Doosan is an equity investor in X-energy. In December 2023, Doosan signed a Reservation Agreement with X-energy, committing to the construction of a manufacturing and fabrication facility to support the execution of X-energy's more than 11 GWe commercial nuclear program. Under the Reservation Agreement, Doosan will reserve forging capacity for 16 sets of reactor boundary components for the Xe-100 program in exchange for a non-refundable reservation fee payable in two \$6 million installments and creditable against future purchases.

with a \$1.5 million commitment fee per set if purchase orders are not placed on the agreed schedule. Doosan will reserve forging slots upon receipt of the reservation fee, and such fees may be applied as down payments if orders are placed within specified windows, in connection with the proposed SMR fabrication facility. The Reservation Agreement provides for termination for convenience or cause, with customary notice and cure mechanics.

- *DL E&C.* DL E&C will work with X-energy to identify opportunities around the world to implement best practices to support the development and deployment of Xe-100 plants on a global scale. Founded in 1939, DL E&C has the longest business history among construction companies in Korea and has maintained a top ten Korean engineering and construction company ranking for over 60 years. DL E&C is the flagship company of DL Group, and has a broad range of experience in the mid/downstream energy sector engineering, procurement and construction, providing engineering and solutions in more than 35 nations, focusing on a more sustainable and better future. DL E&C is an equity investor in X-energy.
- *Korea Hydro & Nuclear Power.* KHNP is the Korean entity that operates its nuclear power plants in the hydroelectric sector as a subsidiary of KEPCO, the majority state-owned utility. Since 1978, KHNP has successfully constructed and operated 30 nuclear power plants still in operation, 22 in South Korea and four in the UAE. KHNP provides significant expertise in the construction and operation of reactors. In August 2025, KHNP signed a joint compact with X-energy, Amazon, and Microsoft, outlining the intention to collaborate on the deployment of the Amazon order book in the United States.
- *SGL Carbon LLC.* SGL is a developer and manufacturer of advanced carbon materials. SGL and X-energy have collaborated since 2015 on the qualification of NBG-18 graphite for use in Xe-100, leveraging SGL's experience manufacturing graphite for high-temperature gas-cooled reactors. In January 2026, SGL announced a 10-year graphite supply agreement for the components of X-energy's Xe-100 SMRs, and has commenced production for the Dow Chemical Company site as part of an initial three-year award valued at over \$100 million. The graphite supply agreement contains standard terms including a ten-year initial term with automatic renewals, purchase order based releases with firm-fixed pricing and milestone payments secured by a corporate guarantee, delivery, packaging and shipping requirements with liquidated damages for delay, and compliant quality assurance and audit rights. The graphite supply agreement provides for termination for convenience or cause, with customary notice and cure mechanics. It also includes intellectual property protections customary for such agreements.

nuclear-grade materials supply arrangements, including an intellectual property escrow agreement and a first refusal for the Company related to the NBG-18 intellectual property.

COMPETITION

Our competitors include other power generation technologies, including traditional baseload power producers, other advanced nuclear technologies, renewables with or without storage, advanced nuclear storage, fossil fuels with carbon capture, and combinations of these technologies.

- *Traditional baseload* — Traditional baseload power includes natural gas, coal, oil and nuclear.
- *Fossil fuels with carbon capture* — Fossil fuel sources provide firm, baseload power using carbon capture technology to provide clean power. Carbon capture has not yet been demonstrated to be scalable to meet the market need.
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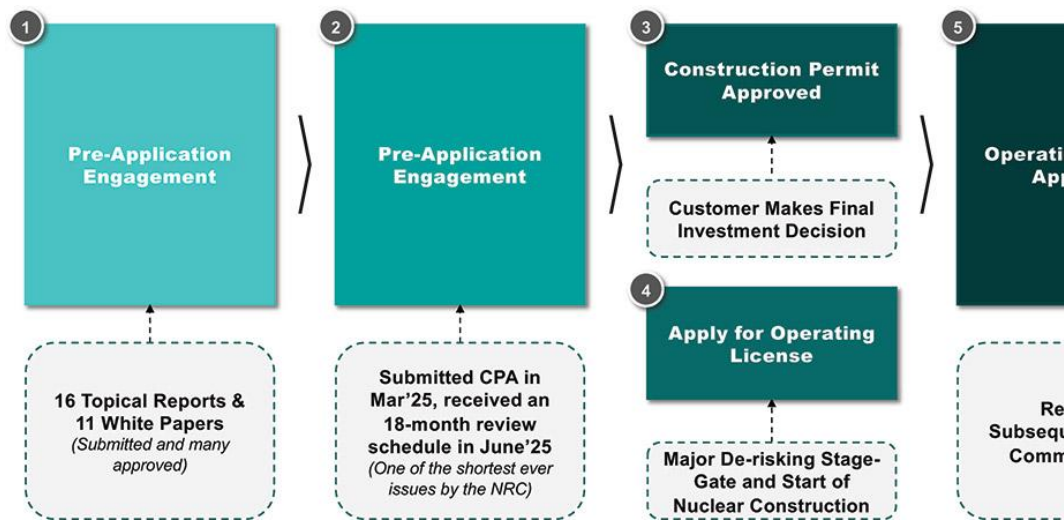
Renewables with or without energy storage — Wind and solar provide clean energy, but they cannot provide firm, baseload energy due to their intermittency. When paired with energy storage, they can provide a more dispatchable energy supply, but battery storage technology has not yet been demonstrated to be scalable or cost-effective to meet the growing market need, while hydrogen production is not yet reliable, particularly in energy-intensive applications.

- *Other advanced nuclear reactors* — There are several advanced reactor and SMR technologies at various stages of development, including high temperature gas-cooled reactors, molten salt reactors, fusion technologies, and advanced light water reactor designs.

REGULATORY OVERVIEW

Licensing

The regulatory pathway with the NRC for the FOAK reactor project is as follows:



X-energy has pursued a Part 50 two-step licensing process timeline, whereby the company first seeks a Construction Permit (up to 18 months) and then seeks an Operating License (up to 18 months). The two-step process gives the company flexibility to modify reactor design if warranted to achieve better performance without needing to restart the licensing process. The alternative, Part 52 combined licensing process, provides a streamlined combined Construction and Operating License Application. However, recent submissions under Part 52 have resulted in application re-submissions caused by design changes or NRC findings or insufficient initial applications.

X-energy expects the regulatory review timeline to shrink materially as the NRC’s familiarity with the Xe-100 design increases. Additionally, the current U.S. government is focused on legislation to

permitting, and the current Administration has issued several Executive Orders to streamline licensing. The DOE is also working to streamline the NRC, accelerate advanced reactor deployment at the DOE and DOD sites for national security and grid resilience, center needs, overhaul the domestic nuclear fuel cycle, and strengthen the U.S. nuclear industry. Further, once X-energy achieves NOAK status and is no longer concerned with design changes, the regulatory design, a COLA process would further abbreviate the regulatory approval timeline.

is developing novel strategies that are anticipated to enable the NRC regulatory review process short as 6 months.

Potential Tax Credits per Standard Configuration of Xe-100 Reactors for Customers

The One Big Beautiful Bill Act (OBBBA) was signed into law in July 2025, maintaining the Electricity Production Tax Credit (§45Y) and Clean Electricity Investment Tax Credit (§48E). These credits will remain available to certain nuclear facilities that begin construction prior to 2033 before being phased out over four years. The OBBBA also expanded the 10% Production Tax Credit (§45Y) added in the FY26 appropriations bill, the “Commerce, Justice, Science; Energy and Water Development and Environment Appropriations Act, 2026,” signed into law on January 23, 2026, in addition to the ongoing budget for nuclear energy, includes additional funding of \$3.1 billion to repurpose appropriated funds towards the DOE’s advanced nuclear programs including the Advanced Reactor Demonstration programs, some portion of which we expect will be allocated to X-energy. This support demonstrates the U.S. federal government’s focus on accelerating the nuclear, particularly advanced nuclear, sector and will fundamentally enhance economics and solidify advanced nuclear reactors as a viable power alternative.

Clean Electricity Production Tax Credit (§ 45Y)^{(1), (2), (3)}

Up to around \$83.0 million of potential tax credits to a customer per four 80 MWe reactors (that can produce electricity) per year for 10 years, adjusting each year for inflation



How it works:

Assuming certain labor requirements are met or an exception applies, a tax credit of 1.5¢/kWh annually for inflation (3.00¢/kWh in 2025) of electricity produced and sold for up to 10 years after a facility is placed in service.

- + 10% additional credit if the facility is located in either a designated “nuclear energy corridor” or “energy community” (e.g., certain metropolitan statistical areas with historic fossil fuel production, certain census tracts where a coal mine has closed (or adjoining tracts), or brownfield sites)
- + 10% additional credit if the facility is constructed solely with domestic iron and steel with a minimum of 40% (minimums increase over time) domestic manufactured products

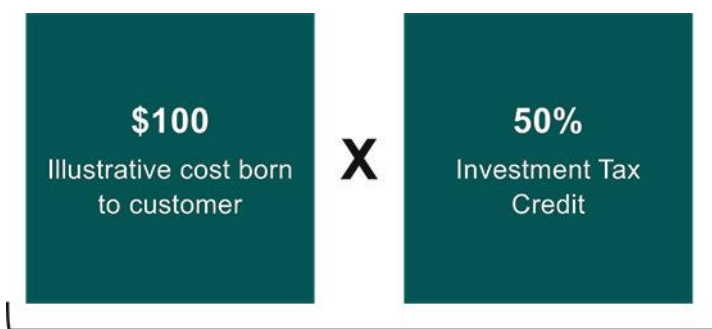
= up to around 3.60¢/kWh total credit

(1) Eligibility requirements: Only certain facilities used for the generation of electricity placed in service after December 31, 2024. Also, facilities whose “greenhouse gas emission rate” do not exceed 100 lbs/mmBtu whose electricity is produced in the U.S. and sold to “unrelated persons;” and that meet “apprenticeship” requirements are eligible to claim these tax credits.

- a. The amount of the tax credit determined with respect to a project is subject to a four percent cap for projects that begin construction in 2033 and later.
 - b. Projects that begin construction after December 31, 2025, may not receive material from certain prohibited foreign entities in constructing such facilities.
 - c. A taxpayer claiming a tax credit in any tax year beginning after December 31, 2025, may not be a prohibited foreign entity and may not make certain payments to specified foreign entities that would give the specified foreign entity effective control over the project.
- (2) Hours in a year multiplied by 1,000.
 - (3) Rates as of 2026. Tax credit is adjusted annually based on inflation.

U.S. Clean Electricity Investment Tax Credit (§ 48E)^{(1), (2)}

Up to around \$50 of potential illustrative tax credits to a customer for every \$100 of cost of 80 MWe reactors purchased, claimed in the year when the plant is put in service



How it works:

Assuming certain labor requirements are met or an exception applies, a tax credit of 30% of the capital cost in a facility with respect to certain eligible costs,

- + 10% additional credit if the facility is in an “energy community” (e.g., certain census tracts with historic fossil fuel employment, certain census tracts where a closed (or adjoining tracts), or brownfield sites)
- + 10% additional credit if the facility is constructed solely with domestic iron and steel, or a minimum of 40% domestic manufactured product

= up to 50% tax credit on certain upfront capital costs

- (1) Eligibility requirements: The credit is available only to certain facilities used for the production of electricity placed in service after December 31, 2024. The facility’s “greenhouse gas emissions” cannot exceed zero and must meet “prevailing wage” and “apprenticeship” requirements and may be subject to recapture if eligibility requirements are not maintained or if the facility (or the taxpayer’s interest therein) is sold.
 - a. The amount of the tax credit determined with respect to a project is subject to a four percent cap for projects that begin construction in 2033 and later.
 - b. Projects that begin construction after December 31, 2025, may not receive material from certain prohibited foreign entities in constructing such facilities.

- c. A taxpayer claiming a tax credit in any tax year beginning after December 31, 2025, prohibited foreign entity, and may not make certain payments to specified foreign entity that would give the specified foreign entity effective control over the project.

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- (2) Subject to diligence and IRS rulemaking process on expense eligibility for investment tax credit; energy assumption not explicitly included in IRA; similar provisions have been included in other renewables tax credit programs.

Potential Tax Credits Related to Construction of Fuel Fabrication Facility TX-1

Qualifying Advanced Energy Project Credit (§ 48C)

The Qualifying Advanced Energy Project Credit (§ 48C) was initially created under the American Recovery and Reinvestment Act (2009) and later expanded by the Inflation Reduction Act (“IRA”). The program creates a qualifying investment tax credit for facilities that manufacture equipment, nuclear fuel or related supply chain technologies. In March of 2024, TRISO-X received around \$150.0 million investment tax credit for the construction of its first purpose-built commercial fuel fabrication facility, TX-1, in Oak Ridge, TN. Upon meeting certification and ‘placement’ qualifications, the credit will be applied against the company’s federal income tax liability or be transferred to an unrelated 3rd party per IRC § 6418 that equals 30% of the eligible capital costs, up to allocated award, in construction and equipping of the TX-1 facility.

GOVERNMENT AND REGULATION

Policymakers and industry leaders increasingly recognize that nuclear, and particularly advanced nuclear, will be a key component of the clean energy transition. Growing energy demand and enduring considerations around managing longer-term carbon risk, creates a clear need for what advanced nuclear offers: carbon-free, consistent baseload energy generation. According to the IEA, fossil fuels supply around 60% of global electricity generation, and the IEA projects that, under the net-zero scenario, global electricity demand could increase by more than 50% by 2050. As such, nuclear is expected to play a significant role in virtually every credible pathway to achieving net-zero.

While renewables have made and continue to make a meaningful contribution to this transition, they lack several key advantages that advanced nuclear offers; specifically, the ability to deliver consistent and readily dispatchable energy. Further, advanced nuclear offers expanded use cases, such as industrial heat, while having lower capital costs, geographic flexibility, simplified operations and substantial safety improvements over legacy nuclear plants. We believe these factors uniquely position advanced nuclear as a central component of the global energy system moving forward.

Nuclear energy has received significant federal- and state-level support in the U.S. in recent years. Currently, about half of U.S. states have, or are considering, policies to support advanced nuclear. Many have also adopted binding deadlines — via legislation, executive orders, or administrative actions — to mandate decarbonization of energy purchased, sold or generated within their borders.

At the U.S. federal level, support for advanced nuclear power has been steady since the enactment of the Nuclear Energy Innovation Capabilities Act in 2018. Since that time, billions of dollars have been authorized and appropriated to support nuclear energy in recent significant energy-related legislation, including in the Nuclear Energy Innovation and Modernization Act of 2019, the Infrastructure

and Jobs Act of 2021, the Inflation Reduction Act in 2022, and the Accelerating Deployment of Advanced Nuclear for Clean Energy Act in 2024, all passed with strong bipartisan and bicameral support.

Similarly, U.K. and Canada policymakers have also expanded support of advanced nuclear. In December 2021, the U.K. government designated HTGRs as the technology focus for its Advanced Reactor (“AMR”) demonstration program, citing their potential to contribute to net-zero goals, produce high-temperature industrial heat, relatively high technical maturity, and the proven safety of the U.K.’s second-generation Advanced Gas-cooled Reactors.

United States

- ***Nuclear Energy Innovation Capabilities Act (“NEICA”).*** Signed into law in September 2021, NEICA directs the DOE to prioritize research, development, and demonstration of advanced nuclear technologies, authorizes the creation of a National Reactor Innovation Center to facilitate

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sector access to DOE facilities for testing and demonstration, and promotes public-private partnerships to accelerate the commercialization of innovative nuclear energy solutions.

- ***Nuclear Energy Innovation and Modernization Act (“NEIMA”).*** Signed into law in January 2022, NEIMA directs the NRC to modernize its licensing processes for advanced nuclear reactors, establishes a new fee structure to ensure more predictable and transparent costs for applicants, and requires the NRC to develop a framework for licensing advanced reactors.
- ***Infrastructure Investment and Jobs Act (“IIJA”).*** Signed into law in November 2021, the IIJA provides \$65.0 billion for power and grid investments, including grid reliability, resilient energy technologies such as carbon capture, hydrogen and advanced nuclear. The IIJA also provides \$2.47 billion funding for the two ARDP demonstration projects (of which X-energy is a partner) following its creation in the Energy Act of 2020.
- ***Inflation Reduction Act.*** Signed into law in August 2022, the IRA appropriated \$700 million to support the transportation and availability of HALEU fuel and increased DOE’s loan guarantee authority to \$250.0 billion, both measures that will support the acceleration of reactor deployment and improve projects economics.
- ***Accelerating Deployment of Versatile, Advanced Nuclear for Clean Energy Act (ADVANCE Act).*** Signed into law in 2024, this historic nuclear energy package brings the U.S. closer to its goal of energy independence and international market leadership. The Act includes NRC licensing reform, a new door to foreign investment, and directs the NRC to develop a program to enhance reactor safety, preparedness and coordination in qualifying and licensing advanced nuclear fuels.
- ***Recent Executive Orders.*** The current Administration unveiled four Executive Orders in 2025 directing federal agencies to streamline licensing at the NRC, accelerate advanced nuclear deployment at the DOE and DOD sites for national security and AI/data-center needs, streamline the domestic nuclear fuel cycle, and strengthen the U.S. nuclear industrial base.
 - ***“Reforming the Nuclear Regulatory Commission”*** which directed a reorganization of the Nuclear Regulatory Commission structure to promote faster processing of license applications and mandated an 18-month review period for a decision on an application to construct and operate new reactors (including advanced nuclear reactors).
 - ***“Reinvigorating the Nuclear Industrial Base”*** seeks to reduce U.S. dependence on foreign sources of nuclear fuel by funding a buildout of domestic supply chains to support

nuclear fleet and an expansion of production by four times by 2050 with an emphasis on establishing a domestic nuclear fuel cycle — including uranium enrichment, restarting, uprates, or construction of reactors; and expanding the nuclear energy workforce.

- “*Deploying Advanced Nuclear Reactor Technologies for National Security*” describes a national security objective and the order sets a priority for the DOE and the DOD to encourage private industry to accelerate deployments of advanced nuclear technology to power. The goals include designating DOE sites appropriate for advanced nuclear reactor construction within 30 months, deploying an advanced nuclear reactor within three years at a domestic site, completing installation and releasing 20 MTU of HALEU for advanced reactor design use at DOE sites, and completing construction and operation of test reactors for research purposes and that do not produce commercial power to fall under the purview of the DOE and directs DOE to expedite environmental reviews and expedite qualified test reactors for operation at DOE sites and controlled facilities within two years of an application.
- **One Big Beautiful Bill Act.** Signed into law in July 2025, the OBBBA is a comprehensive spending package that makes permanent many provisions of the 2017 Tax Cuts and Jobs Act, introducing new tax policies and restructuring others. The OBBBA generally maintains the Electricity Production Tax Credit and Clean Electricity Investment Tax Credit for nuclear projects that were added by the IRA but subjects it to a phase-down schedule that begins for projects that commence construction after 2033. Projects beginning construction in 2034 and 2035 will qualify for tax credits but with a 75% and 50% credit value, respectively. These projects will be treated the traditional

continuity test, which requires placement in service by the end of the fourth year after the first year in which construction starts, or else demonstrate that construction was continuous throughout the project. The OBBBA also provides an expanded 10% tax credit adder definition for nuclear facilities located in designated “energy communities” and authorizes additional subsidies and loan guarantee authority to support the deployment of advanced nuclear technologies.

- Additionally, through the OBBBA, the U.S. Congress reinforced earlier legislation that accelerates the acceleration of environmental review requirements under NEPA for project projects that pay certain fees. Further, DOE amended its NEPA guidelines further to set page limits to encourage greater collaboration between the agencies, and expand the scope of cases that can be reviewed through a shorter and streamlined review process because the projects have a limited environmental impact (such as siting new small-scale projects on existing brownfield or DOE sites). On February 2, 2026, DOE announced a new categorical exclusion that covers the siting, construction, operation, reauthorization and decommissioning of advanced nuclear reactors, including SMRs. The categorical exclusion is effective immediately and is intended to reduce environmental review barriers for qualifying projects. NRC has also begun updating its NEPA guidelines, which are anticipated to similarly streamline project review.
- **Other Legislative Support Provisions under Consideration in the U.S. Congress.** Additional funding in the appropriations bill, the “Commerce, Justice, Science; Energy and Water Development and Environment Appropriations Act, 2026,” signed into law on January 23, 2026, in the ongoing budget for nuclear energy, includes additional funding of \$3.1 billion in additional funding previously appropriated funds towards the DOE’s advanced nuclear programs including the Advanced Reactor Demonstration program, some portion of which we expect will be allocated to

energy. This financial support demonstrates the U.S. federal government’s focus on advanced nuclear, particularly SMR, sector and will fundamentally enhance economics and solidify nuclear reactors as an attractive power alternative.

- ***Several States Showing Strong Commitment to SMR Development.*** With increasing electricity demand and a desire to attract AI & data centers, there has been significant movement at the state level to incentivize the deployment of nuclear across U.S. states. Many have shown strong interest towards developing advanced nuclear technologies, including SMRs, and putting forth the necessary legislation to facilitate nuclear power build-out in their states. For example:
 - Texas: Enacted House Bill 14 to create the Texas Advanced Nuclear Deployment Fund, which established the Texas Advanced Nuclear Development Fund, which provides \$350.0 million to support both development and construction of advanced nuclear reactors.
 - Tennessee: Created the Tennessee Nuclear Energy Fund to expand nuclear development and manufacturing ecosystem for the state of Tennessee, which included an initial \$10.0 million which has increased to \$70.0 million by additional \$10.0 million installments through FY26.
 - Washington: Enacted a clean electricity standard in 2019 which would require 100% clean energy generation by 2025 and mandate 100% clean energy by 2045, including nuclear. The state allocated \$25.0 million to Energy Northwest in support of the utility’s application to the Washington State Energy Programs Office.
 - Maryland: Established greenhouse gas emissions reduction targets and formally recognized the critical role that nuclear energy plays in the state’s clean energy generation profile. In 2025, the state passed the Next Generation Energy Act which establishes a framework for nuclear energy that includes incentives aimed at advancing nuclear energy generation across the state.
 - Indiana: Signed a new law in 2025 that supported the development of small modular reactors in Indiana, including a 20% Investment Tax Credit for SMR manufacturing and allowing utilities to seek preapproval and timely recovery for project development costs related to advanced nuclear energy.
 - Virginia: Established the Virginia Nuclear Energy Consortium (VNEC) to sustain and enhance leadership in nuclear energy through interdisciplinary business development, workforce training related to nuclear energy; passed legislation allowing investor-owned electric utilities to accelerate cost recovery for project development of SMRs.

- Louisiana: Established the Louisiana Advanced Nuclear Competitive Edge (LANCE) Framework, authorized by the Public Service Commission, to position the state as a leader in advanced nuclear reactors, particularly SMRs.
- West Virginia: Repealed its ban on construction of nuclear power plants, removing regulatory constraints that previously blocked new nuclear capacity, including SMRs.
- Illinois: Lifted or partially eased its moratorium on new nuclear plants to allow construction (specifically reactors up to around 300 MW) beginning in 2026, while putting in place new regulatory and safety frameworks.
- New York: Launched a Master Plan for Responsible Advanced Nuclear Development guided by a blueprint from NYSERDA. The plan includes efforts to deploy advanced nuclear

technology as well as development of early site permits for advanced reactors at Nine Mile Point.

Canada

- **Clean Technology Investment Tax Credit.** The clean technology investment tax credit (“30% refundable investment tax credit applicable to the capital cost of clean technology in Canada including certain small modular reactor equipment. The CT ITC is applicable to the cost of eligible property that was acquired and became available for use on or after January 1, 2023. The CT ITC rate is reduced to 15% for eligible property that becomes available for use on or after January 1, 2034, and the CT ITC is not available for property that becomes available for use on or after January 1, 2035. If certain labor conditions are not satisfied then the tax credit rate is reduced to 20%. The CT ITC is proposed to be amended pursuant to the Budget Implementation Act, No. 1 tabled in the House of Commons on November 4, 2025, is available in respect of certain small nuclear energy properties that use a fixed location system that is used all or substantially all to generate electrical or heat energy (or a combination of electrical energy and heat energy) from nuclear fission as determined by the Minister on a case-by-case basis and that is located at a nuclear facility where, at the time the property becomes available for use, the total combined gross-rated thermal generating capacity of all planned and existing nuclear fission reactors at the facility is reasonably expected not to exceed 1,400 MWt.

United Kingdom

- In 2025, the U.K. government announced its intent to create a new ‘Golden Age of Nuclear’ in the U.K. and encourage private sector investment in next-generation nuclear technology. The U.K. government implemented significant reforms to accelerate the deployment of small modular reactors (SMRs) and Generation IV advanced modular reactors (AMRs) including High Temperature Gas-cooled Reactors (HTGRs). Planning reforms were launched to expand eligible sites for SMRs beyond the current eight locations to cover all of England and Wales, supporting growing energy demand in residential centers and heavy industry. The Government declared it would offer value-for-money projects and debt provision from the National Wealth Fund, and financial risk mitigation and revenue sharing mechanisms, negotiated with the DOE Security and Net Zero. A regulatory task force was established to shorten nuclear licensing and consenting timelines, and the U.S. and U.K. Government signed the Atlantic Partnership for Advanced Nuclear Energy, including a regulatory co-operation agreement designed to achieve efficiencies through collaboration on assessment data.

Nuclear Safety Regulation

The commercial nuclear industry is heavily regulated worldwide. In general, regulators require licensing for the design, construction and operation of every nuclear plant, as well as the facilities that supply them. Regulations are designed to ensure protection of people and the environment. Regulations typically address: (1) design safety and robustness against internal hazards (e.g., component failure) and external hazards (e.g., earthquakes and extreme weather), (2) environmental impacts of construction and operations (e.g., water use, preservation of historical sites, animal and plant species), and (3) safety and insurance. Although regulations are country-specific, regulators often collaborate when projects are deployed in multiple jurisdictions.

X-energy’s licensing strategy has two goals: (1) obtain approval of a reactor project for X-energy’s technology within the shortest possible time, through early regulatory engagement and pre-application discussions; and (2) maintain a common Xe-100 design across markets.

After several years of pre-application engagement, in March 2025, Long Mott Energy, L, owned subsidiary of Dow, submitted a construction permit application to the NRC for a proposed nuclear project deploying four Xe-100 reactors at a Dow plant in Seadrift, Texas. The application was docketed by the NRC in May 2025. Citing the completeness and quality of the application and the effectiveness of pre-application engagements, the NRC published an 18-month review timeline for the project and began a concurrent environmental assessment. All current pre-licensing work is completed and submitted by X-energy in order to de-risk future construction permit and operating license applications to the NRC, whether ARDP funded or not, that will be submitted by our customers, the plant owner.

Environmental review under the National Environmental Policy Act is required for facilities, including approval of the construction permit application. During the environmental review process, the NRC consults with collaborating federal agencies, as well as state, local, and tribal authorities, and the relevant parties are aware of and acknowledge the licensing actions being considered. The NRC provides public notice and granted opportunities for comment. In certain cases, individual states may have additional environmental review requirements related to the construction and operation of a nuclear power plant. The environmental review requires the NRC to consider any state, local and tribal stakeholders. In addition to NRC review, other federal and state government permits and authorizations will be required to construct and operate the Long Mott Generating Station facility, including those discussed below.

Environmental, Occupational Health and Safety, and Other Regulations

X-energy is also subject to regulations regarding nuclear material safeguards, non-proliferation, restrictions, liability insurance regimes, and various other matters. Both we, for our products, and customers purchasing our reactors must obtain a variety of permits, licenses, and insurance in the jurisdiction where the facility will be located. We and customers purchasing our reactors are subject to stringent and complex federal, state, local and international laws and regulations governing the handling, use, and disposal of nuclear materials into the environment or otherwise relating to protection of worker health, safety, and the environment. Compliance with all of these laws and regulations may expose us to significant financial liabilities and cause us to incur significant capital expenditures in our operations. Any failure to comply with these laws and regulations may result in the assessment of administrative, civil and criminal penalties, imposition of remedial obligations, and the issuance of injunctions delaying or prohibiting operations. Private parties may also have the right to pursue legal actions to enforce compliance as well as seek damages for non-compliance with environmental laws and regulations or for personal injury or property damage. In addition, the trend in environmental regulation is to place more restrictions on activities that may affect the environment, and thus, any changes in, or more stringent enforcement of, these laws and regulations that result in more stringent and costly pollution control equipment, the occurrence of delays in the permitting or performance of projects, or waste handling, storage, transport, disposal or other requirements could have a material adverse effect on our operations and financial position.

Our business and the business of customers purchasing our SMRs must obtain required permits, licenses and insurance for the jurisdiction where the facility will be located. In the U.S., customers are subject to NRC regulation and licensing as well as other federal and state regulations regarding nuclear materials licensing. For example, we recently received a Special Nuclear Material License from the NRC for the possession and use of enriched uranium and will be required to receive a Specific Nuclear Material License and a Radioactive Material License from the State of Tennessee for our TX-1 facility.

Development and operation of our TX-1 facility, the Long Mott Generating Station, our other facilities, and our customers' other facilities are also subject to federal, state and local regulation, including the Clean Water Act, Clean Air Act and Occupational Health and Safety Act, and required to obtain permits to maintain permits such as National Pollutant Discharge Elimination System (NPDES) Permits, and other permits for Water Discharges from Construction Activities and local construction permits. Furthermore, environmental laws impose substantial penalties for noncompliance, and others, such as the Comprehensive Environmental Response, Compensation, and Liability Act ("CERCLA"), the Resource Conservation and Recovery Act ("RCRA"), or similar state laws, may impose strict, retroactive and several liabilities

for the remediation of releases of hazardous substances. Stated differently, liability under CERCLA or similar state and local laws, may be imposed as a result of conduct that was lawful at the time or for the conduct of, or conditions caused by, prior operators or other third parties. Failure to handle, transport, store or dispose of hazardous materials or otherwise conduct our operations in compliance with RCRA and other EHS laws or similar state and local laws could expose us to liability for penalties, cleanup costs and civil or criminal liability associated with releases of such materials into the environment, damages to property, natural resources and other damages, as well as potentially impairing our customers' ability to conduct our operations.

Finally, siting, cost recovery, interconnection and other state and federal issues for power generation type are also often subject to occasionally lengthy approval processes by state utility commissions, independent transmission system operators, and FERC approvals.

Our compliance with these and other laws, regulations, and contractual commitments may increase and could, individually or in the aggregate, increase our cost of doing business, impact the viability of our business model, limit our ability to pursue certain business practices or products and services, cause us to change our business models and operations, affect our competitive position relative to our peers, and/or otherwise harm our business, reputation, financial condition, and operating results.

We also cannot be assured that future events, such as changes in existing laws or enforcement, or the promulgation of new laws or regulations or the development or discovery of new facts or events, adverse to our operations will not cause us to incur significant costs. While we are currently in compliance with current environmental regulations, we acknowledge the potential for policy changes that could impact our operations. On January 20, 2025, President Trump issued a series of executive orders rescinding approximately 80 Biden-era executive orders related to public health, the environment, climate change, and climate-related financial risks. President Trump also declared a "national energy emergency" and directed agencies to expedite conventional and nuclear energy projects. While the extent of the Administration's changes to the environmental regulatory landscape in the U.S. is unknown at this time, it is possible that additional changes in the future could impact our results of operation and our relationships with customers.

Additional information regarding certain risks related to government regulations is included in our *Factors — Risks Relating to Compliance with Law, Government Regulation and Litigation.*"

Export Controls

Our business is subject to and must comply with stringent U.S. import and export controls, including DOE regulations under 10 CFR Part 810 (governing the export of nuclear technology and technical assistance by U.S. persons); NRC regulations under 10 CFR Part 110 (governing the export of nuclear hardware, equipment, and materials); and Export Administration Regulations administered by the Bureau of Industry and Security within the U.S. Department of Commerce (covering "dual-use" items and associated foreign assistance).

Export control regulations safeguard U.S. national security, support foreign policy objectives, and uphold nonproliferation commitments. Governmental authorizations (e.g., DOE Part 810 approvals) are required before we export technology, equipment, materials, or services, or collaborate with foreign entities. X-energy maintains an extensive export control program to ensure compliance.

The U.S. government agencies responsible for administering the nuclear export controls have a degree of discretion in interpreting and enforcing these rules, and in approving, denying, or conditioning authorizations. Their decisions are also shaped by multilateral export control agreements and geopolitical developments.

FACILITIES OVERVIEW

We own or lease facilities and property in the following geographic locations:

- Rockville, Maryland — The following facilities are located in Rockville, Maryland:

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- X-energy’s corporate headquarters consists of 39,000 square feet and houses engineering teams performing engineering, design, operations, testing, code quality assurance, licensing and project management. In addition, the Rockville facility houses the executive management team as well as accounting, marketing, human resources, procurement and legal staffs.
- X-energy also holds a lease with a third-party for offices consisting of 49,000 square feet in Rockville, Maryland that is expected to house the teams named above beginning in 2025.
- Frederick, Maryland — The following facilities are located in Frederick, Maryland:
 - *The Operator Training Facility* — This facility became operational in October 2023 and consists of 10,447 square feet and houses the following activities:
 - *Xe-100 Simulator Development* — This activity includes the development of a full-scale ANSI 3.5 Xe-100 simulator that is a true representation of the Xe-100 plant. The simulator model will include all process models of Xe-100 reactors, instrumentation, sensors and other elements to provide an environment to train and license Xe-100 operators before the first Xe-100 reactor’s fuel is loaded. This activity also includes the development of the Xe-100 control room and the construction of a replica control room where operators will be trained. It is expected that a single Xe-100 control room will be sufficient to operate up to twelve Xe-100 reactors.
 - *Xe-100 Operator Procedure Development* — This activity includes the development of the operational procedures needed to control the Xe-100 plant. The procedures will be developed then displayed on a “computerized procedure system” that X-energy is currently developing, and will be used to create training material for Xe-100 operators.
 - *Xe-100 Operator Training* — This activity includes training Xe-100 operators in a classroom environment. Students will use the simulator and X-energy training materials.
 - *Human Factors Engineering Analysis* — This activity includes the task analysis and human factors analyses needed for the Xe-100 control room and simulator design, testing and analyses. This activity is also necessary to determine the workload on Xe-100 operators and calculate the necessary number of operators in the control room.
 - *The Frederick Testing Facility (to be known as “XTAC” or “X-energy Technology Assessment Center”)* — This facility was purchased in July 2025 and consists of 90,000 square feet. It will be used for office space, major component testing, machining and manufacturing and training collateral, including the following:
 - *eXperimental Test Facility (“XTF”)* — The XTF is expected to begin testing in 2026 and will allow for bench-scale and fractional scale testing of our first-of-a-kind core technologies, allowing for rapid refinement of designs. Specifically, the XTF will test the steam generator interfaces, the reactor cavity cooling system, component testing, fuel handling system and mechanical properties of the fuel pebbles, among other systems.
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Machine Shop — The machine shop is expected to begin testing in 2026. It will enable rapid prototyping of critical components in the reactor. Specifically, the machine shop will be manufacturing fuel handling system components and main system components, including machining graphite blocks, among other systems.

- *Helium Test Facility (“HTF”)* — This testing facility is expected to begin construction in mid-2027. It will allow for testing of our first-of-a-kind technologies in a simulated high-pressure, high temperature helium environment. The HTF will include a fuel handling system, the helium circulator, and the RODS, among other systems.
- *Training Facility* — The facility will house training facilities for operators, engineers, chemistry, security and other reactor support staff. The location will allow for curriculum design. The facility will begin construction in 2026 and be built out by 2028 for Operational Readiness.

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- *Office space* — The facility will feature 22,000 square feet of office space, including meeting and conference rooms for facility staff and visiting employees and guests.
- Oak Ridge, Tennessee — The following facilities and sites are located in Oak Ridge, Tennessee:
 - *Oak Ridge National Lab (“ORNL”)* — This space located within ORNL is called the Pilot Facility, which houses a single commercial scale TRISO-X fuel fabrication facility to produce HALEU-bearing fuel. It has been used to test new fabrication processes, has issued U.S. patents, and five provisional U.S. patent applications. We anticipate full-scale operations at this space through first half of 2026.
 - *Lafayette Road Facility* — The facility began operations in January 2023. It is approximately 42,000 square feet and houses TRISO-X staff, performing process development, licensing engagement, support for the deployment of the TF3 and process equipment.
 - *Commercial Fuel Fabrication Facility (TX-1)* — Our first fuel facility began construction in September 2025. Once operational, the facility is expected to be America’s first purpose-built commercial advanced nuclear fuel fabrication facility, covering 110 acres, and is part of a former DOE site known as Horizon Center Industrial Park.
 - *Apollo Emory Valley Site* — Leased for storage space that we anticipate moving in 2026.
 - *Planned TX-L site* — Known as ‘Area 7’, this site is on adjacent land to our pilot facility location.
- Toronto, Canada — X-energy’s Toronto office space houses staff and is the headquarters of the wholly owned subsidiary, X-Energy Canada, Inc.
- Warrington, England — X-energy’s Warrington office space houses staff and is the headquarters of the wholly owned subsidiary, X-Energy UK.

The company believes the locations of its facilities provide it with unique access to the DOE, government partners and customers, and the DOE’s extensive ORNL lab facilities and personnel. These locations are being attractive locations within some of the most well-educated metropolitan corridors in North America.

The existing facilities accommodate current operating needs, and the company will expand its facilities as needed to accommodate a growing workforce as it moves closer to deploying the reactor and associated services.

INTELLECTUAL PROPERTY

We rely on a combination of patent and trade secret protection laws, as well as contractual relationships, to protect and enhance the proprietary technology, inventions and improvements commercially important to our business. We also rely on trademark laws to protect our brands.

Our policy is to protect our competitive position by, among other methods, filing patent applications in the U.S. and other jurisdictions, related to our proprietary technology, inventions, improvements and products that are material to our business. We also rely on our trade secrets and know-how to protect our proprietary technology, inventions, improvements and products as well as certain innovative licensing opportunities to develop, strengthen and maintain our competitive position in the field. Such rights can be difficult to protect and provide us with limited protection. We take steps to preserve our trade secrets and know-how by employing various methods, including confidentiality, non-disclosure and non-compete agreements with certain of our employees and including our vendors, contractors and potential business partners, to protect our intellectual property and proprietary information.

As of the date of this prospectus, we own approximately 13 issued U.S. patents, 14 pending patent applications, two U.S. trademark registrations and two U.S. trademark applications. The issued patents have expiration dates between 2036 and 2045. Of those approximately 13 issued U.S. patents, 10 cover certain aspects of our reactor and fuel technology. We also own 15 issued foreign patents in Canada, Japan and South Korea, and 48 pending patent applications. The issued foreign patents have expected expiration dates between 2038 and 2045.

EMPLOYEES AND HUMAN CAPITAL

As of March 17, 2026, we have a highly educated workforce of 916 employees, of which 104 have master's degrees and 104 have Ph.Ds. Our employees are located across 40 different states and countries. We are dedicated to our corporate goal of changing the world through innovative implementable energy solutions.

Due to the highly specialized nature of our business, we need to hire and train skilled personnel to design, build and operate our state-of-the-art equipment, and to perform a broad range of services to support our company and our customers. Our success as a company depends on our ability to attract, develop and retain people with highly technical and specialized expertise who are consistently performing quality work and, for some of our positions, can obtain a security clearance. We are dedicated to promoting the health, welfare and safety of our employees. We strive to treat all employees with dignity and respect and provide them with fair, market-based, competitive and equitable compensation. We aim to recognize and reward the performance of our employees in line with our pay-for-performance philosophy and provide a comprehensive suite of benefit options that enables our employees and dependents to live healthy and productive lives. Moreover, we aim to foster an environment where all individuals feel a sense of belonging and can achieve their highest potential, regardless of background. We are an equal opportunity employer and work to create an inclusive environment. We have a foundation of compliance with applicable civil rights laws.

Safety in our workplaces is paramount. We take measures to prevent workplace hazards, enforce safety behaviors and enforce a culture of continuous improvement to foster processes that help prevent incidents and injuries and comply with applicable health and safety laws.

LEGAL PROCEEDINGS

From time to time, we may be subject to various claims, lawsuits and other legal and administrative proceedings that may arise in the ordinary course of business. Some of these claims, lawsuits or proceedings may range in complexity and result in substantial uncertainty; it is possible that they could result in damages, fines, penalties, non-monetary sanctions or relief. We currently do not have any pending lawsuits or proceedings against X-energy that, individually or in the aggregate, would be material to our business or likely to result in a material adverse effect on our future operating performance or financial condition or cash flows.

BOARD OF DIRECTORS AND MANAGEMENT OF X-ENERGY

The table below sets forth information about the executive officers and directors that are expected to be in place upon consummation of this offering. With respect to our directors, each biography includes information regarding the person's service as a director, business experience, director position held currently or at any time during the past five years, information regarding involvement in legal or administrative proceedings and the experience, qualifications, attributes or skills that caused our board of directors to determine that the person should serve as a director of the Company.

The initial Class I directors shall serve for a term expiring at the first annual meeting of stockholders. The initial Class II directors shall serve for a term expiring at the second annual meeting of stockholders and the initial Class III directors shall serve for a term expiring at the third annual meeting of stockholders. At each annual meeting of stockholders beginning with the first annual meeting of stockholders, the successors of the class of directors whose term expires at that meeting shall be elected to hold office for the term expiring at the third annual meeting of stockholders to be held following their election. Each director in each such class shall hold office until such director's successor is duly elected and qualified to serve, or such director's earlier death, resignation or removal.

Name	Age	Position
J. Clay Sell	58	Chief Executive Officer and Director
Daniel Gross	55	Executive Vice President and Chief Financial Officer
Dragan Popovic	53	Executive Vice President and Chief of Global Operations
Kamal Ghaffarian	67	Chairman
Edward Sonnenschein	71	Director
Michael J. Wallace ⁽³⁾	78	Director
Kathleen W. Hyle ⁽¹⁾⁽²⁾	67	Director
Christopher F. Ginther ⁽³⁾	64	Director
Gregory J. Goff ⁽¹⁾⁽³⁾	69	Director
David B. Kaplan	58	Director
Allyson Satin ⁽¹⁾⁽²⁾	40	Director

(1) Member of the audit and risk committee

(2) Member of the compensation and culture committee

(3) Member of the nominating and corporate governance committee

The officers and members of the Board of Directors of the Company (the “Board”) are well-qualified leaders. In their prior positions, they have gained experience in core management skills, such as strategic planning, financial planning, financial reporting, compliance, risk management, and leadership. Several of Company’s officers and directors also have experience serving on boards of directors and committees of other public companies and private companies, and have an understanding of corporate governance practices and trends, which provides an understanding of different business challenges, and strategies. Further, certain officers and directors have other experience that is valuable, such as prior experience in mergers and acquisitions, in financial services, managing large portfolios of assets.

We believe that the above-mentioned attributes, along with the leadership skills and other qualifications of the officers and board members described below, will provide the Company with a diverse set of perspectives and judgment necessary to facilitate our goals and be good stewards of capital.

Executive Officers and Directors

J. Clay Sell. Mr. Sell will serve as Chief Executive Officer (CEO) of the Company and as a member of the Board. Mr. Sell has served as Chief Executive Officer of the Company since January 2019 and as a member of the Board since July 2021. Over the last seven years, Mr. Sell has led the company through a dramatic transformation and growth in product development investment, U.S. government funding, and private equity investment.

capital investments, customer commitments, regulatory achievements and executive talent. From 2008 – 2018, Mr. Sell was the president of Hunt Energy Horizons, LLC, the renewable energy subsidiary of Hunt Consolidated, Inc., a multinational energy, real estate, and investment company controlled by the Ray L. Hunt family of Dallas. Previously, he held positions for 14 years in the U.S. government, as Deputy Secretary of Energy in the George W. Bush Administration from 2001 to 2003, Special Assistant to the President from 2003 – 2005, and in senior staff positions in the U.S. Department of Energy.

House of Representatives. Mr. Sell holds a J.D. from the University of Texas School of Law and a Master's degree in Accounting from Texas Tech University. We believe Mr. Sell is qualified to serve on the Board because of his experience and leadership in the nuclear energy industry.

Daniel Gross. Mr. Gross will serve as Chief Financial Officer of the Company. Mr. Gross has served as Chief Financial Officer of the Company since December 2025 and was an Amazon designee as Chief Financial Officer from October 2024 to December 2025. From August 2021 to December 2025, Mr. Gross was a Senior Director of Corporate Development at Amazon with a principal focus on the Climate Pledge Fund, a corporate capital fund dedicated to low-carbon and other sustainability technologies. Prior to Amazon, Mr. Gross served as Chief Investment Officer at Climate Real Impact Solutions from July 2020 to July 2021. Mr. Gross also served as Managing Director at Oaktree Capital Management from 2013 to 2015, a Partner at Lazard Energy from 2007 to 2013 and a Vice President at Goldman Sachs from 2005 to 2007. Mr. Gross holds masters degrees in business administration and environmental management, as well as a Bachelor's degree from Yale University, where he has also served as a renewable energy finance lecturer since 2010.

Dragan Popovic. Mr. Popovic will serve as Chief of Global Operations of X-energy. Mr. Popovic previously was at Ontario Power Generation ("OPG"), where he served as President of SMR Project Execution from February 2002 to October 2025. During his tenure, Mr. Popovic successfully guided integrated project teams through critical milestones including site preparation, construction approval, and final investment decisions for the development of first small modular reactor project in G7 countries. Prior to this role, Popovic held several leadership positions at OPG at the Darlington Nuclear Stations, with experience spanning the entire nuclear power lifecycle including financing, licensing, construction, operations, maintenance, and decommissioning of nuclear reactors, including key leadership roles at several major projects such as successful refurbishment of the Darlington Nuclear Station. Mr. Popovic holds a degree in electrical engineering from Conestoga College.

Kamal Ghaffarian. Dr. Ghaffarian will serve as a member of the Board. Throughout his 20-year career, Dr. Ghaffarian has created multiple successful companies and has gained extensive experience working at the intersection of government contracting and technological innovation.

Dr. Ghaffarian started his entrepreneurial career in 1994 by founding Stinger Ghaffarian Technologies, Inc. (acquired by KBR, Inc. (NYSE: KBR)), a government services company focusing on IT, engineering, and science applications. Dr. Ghaffarian also held numerous technical and management positions at Lockheed Martin, Ford Aerospace and Loral. Dr. Ghaffarian holds a B.S. in Computer Engineering from Catholic University of America, a B.S. in Electronics Engineering from California State University, and an M.S. in Science in Information Management from George Washington University, a Master's in Management Information System from Kennedy Western University and a Ph.D. in Technology from Capital Technology University.

Dr. Ghaffarian is the co-founder and chairman of several companies including Intuitive Machines (NASDAQ: LUNR), a space services and technologies company, since December 2012, Axion Space, a private space infrastructure company, since August 2016, Quantum Space, LLC, a space infrastructure company, since May 2021, and IBX, LLC, an innovation and investment firm, since October 2021. We believe Dr. Ghaffarian is qualified to serve on the Board because of his role as the executive chairman of Intuitive Machines, extensive experience in the field and deep understanding of company leadership.

Edward Sonnenschein. Mr. Sonnenschein will serve as a member of the Board. Mr. Sonnenschein has served on the Board since December 2021 and on the board of directors of Quantum Space, LLC, a private space infrastructure company, since September 2022, and on the board of directors of Axion Space, a private space infrastructure company, since March 2026. He serves as Vice Chairman of IBX, LLC, where he provides leadership on governance and major transactions, as well as strategic guidance on M&A, and new initiatives, and leads the legal function. Previously, Mr. Sonnenschein was an attorney at a leading global law firm Latham & Watkins, LLP from November 1979 to June 2021, where he

respected deal maker and counselor for numerous public and private companies, private sovereign wealth funds and family offices in hundreds of cutting-edge matters. Mr. Sonnenschein has held numerous leadership positions at Latham, including as the firm's Global Chair of the Corporate Governance Committee, Chair of the Strategic Initiatives Committee, Chair of the Business Development and Finance Committees and member of the Executive Committee. He was also a member of Coastview Capital, a biotechnology venture capital firm, from July 2001 to October 2003. Mr. Sonnenschein holds a Bachelor's degree in Social Studies from Harvard College and a Juris Doctorate from Harvard Law School. We believe Mr. Sonnenschein is qualified to serve on the Board because of his experience and leadership in strategic transactions, financing and venture capital fields.

Michael J. Wallace. Mr. Wallace will serve as a member of the Board. Mr. Wallace has served on the Board since December 2021. In addition, he served as a member of the board of directors of Constellation Energy Nuclear Energy Corporation from February 2013 to November 2025, and the boards of NextEra Energy Company, and Barakah One Holding Company from October 2017 to , and the U.S. National Infrastructure Foundation. Since October 2017, Mr. Wallace has also served as a Senior Advisor at the National Electric Reliability Council, where he advised stakeholders on matters related to assuring the reliability and security of the electricity grid. He was a member of the National Infrastructure Advisory Board from October 2017 to April 2022, where he advised the President of the United States on matters related to the security of the country's infrastructure. Mr. Wallace served as the senior-most nuclear industry executive at both Exelon (then known as ComEd) and Constellation Energy, where he also served as President and Chairman of the Company. Mr. Wallace holds a B.S. in Electrical Engineering from Marquette University and an M.B.A. from the University of Chicago. We believe Mr. Wallace is qualified to serve on the Board because of his experience and leadership in the nuclear energy industry.

Kathleen W. Hyle. Ms. Hyle will serve as a member of the Board. Ms. Hyle was appointed to the Board in January 2023. She served as Senior Vice President of Constellation Energy Corporation (CEG) and Chief Operating Officer of Constellation Energy Resources from November 2008 to November 2022. From June 2007 to November 2008, she served as Chief Financial Officer for Constellation Energy Group and for UniStar Nuclear Energy, LLC. Prior to joining Constellation Energy in 2008, Ms. Hyle served as the Chief Financial Officer of ANC Rental Corporation, Vice President and Treasurer of AutoNation, Inc. (NYSE: AN), and Vice President and Treasurer of Stanley Black & Decker, Inc. (NYSE: SWK). Ms. Hyle previously served on the board of Cencora, Inc. (NYSE: COR), formerly AmerisourceBergen Corporation (NYSE: ABC) from May 2010 to March 2025. From July 2023 to January 2025, Ms. Hyle served on the board of Bunge Ltd (NYSE: BG), during which she held the position of Chair from 2018 to 2023. Ms. Hyle holds a B.A. in Accounting from Loyola College. She is also a Certified Public Accountant. We believe Ms. Hyle is qualified to serve on the Board because of her experience and leadership in the energy industry.

Christopher F. Ginther. Mr. Ginther will serve as a member of the Board. Mr. Ginther was appointed to the Board since January 2023. He also served as Executive Vice President Enterprise Strategy, Business Development and Commercial Management at OPG, a globally recognized leader in the development and production of clean energy and nuclear energy projects, from July 2015 to his retirement in May 2024, where he oversaw the company's business strategy, development and structuring, negotiations and commercial positioning in support of OPG's growth strategy. Prior to OPG, Mr. Ginther served as Vice President and General Counsel at Bell Canada Enterprises Inc. (NYSE: BCE) and as Vice President, General Counsel and Corporate Secretary at Ontario Lottery and Gaming Corporation. Mr. Ginther began his legal career at the law firm Torys LLP in Toronto. Mr. Ginther holds a B.A. in Law and Political Science from the University of Western Ontario and an L.L.B and L.L.M from the University of Western Ontario Law School. We believe Mr. Ginther is qualified to serve on the Board because of his experience and leadership in the energy industry.

Gregory J. Goff. Mr. Goff will serve as a member of the Board. Mr. Goff has served on the Board since July 2023. He currently serves on the board of directors of Avient Corporation (NYSE: AVN) and is a member of the boards of directors of Enbridge Inc (NYSE: ENB) from February 2020 to July 2025, and Exxon Mobil Corporation (NYSE: XOM) from June 2021 to October 2024. Mr. Goff is the Chief Financial Officer of Claire Technologies, Inc., a technology company developing solutions to de-carbonize the energy industry.

modes of transportation using hydrogen, and the founder of GJG Energy, LLC, a company with opportunities in the refining, marketing and logistics business. Mr. Goff was the Vice Chairman of Petroleum

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Corporation (NYSE: MPC), an integrated downstream energy company, from October 2019 to December 2019. He also served on the board of directors of MPLX L.P. (NYSE: MPLX) from October 2019 to December 2019. Prior to Marathon's acquisition of Andeavor Corp. ("Andeavor") (NYSE: MFC), Mr. Goff served as Chairman, President and Chief Executive Officer of Andeavor, a leading oil refining and marketing company, from May 2010 to September 2018. He also served as Chairman and Chief Executive Officer of Andeavor Logistics, LP (NYSE: ANDX), an NYSE-listed master limited partnership that owned, operated and developed crude oil and refined products and logistics assets, from September 2018 to September 2018. Prior to that, Mr. Goff worked for ConocoPhillips Corporation, an integrated oil and gas company, where he held a number of senior leadership positions, most recently Senior Vice President, Commercial from 2008 to 2010. Mr. Goff holds a B.S. in Management and an M.B.A. from the University of Utah. We believe Mr. Goff is qualified to serve on the Board because of his experience and knowledge in the energy industry.

David B. Kaplan. Mr. Kaplan will serve as a member of the Board. Mr. Kaplan has served on the Board since December 2023. Mr. Kaplan is a Co-Founder, Director and Partner of Ares Management Corporation. Mr. Kaplan serves on several Ares Investment Committees including, among others, the Ares Private Equity Group's Corporate Opportunities, Energy Opportunities and Extended Value Opportunities Committees. Additionally, Mr. Kaplan served as Chief Executive Officer and Co-Chairman of the board of directors of Ares Acquisition Corporation II from March 2021 to September 2025. Mr. Kaplan joined Ares in 2003 from Shelter Capital Partners, LLC, where he was a Senior Principal from June 2000 to June 2003. From 1991 through 2000, Mr. Kaplan was a Senior Partner of Apollo Management, L.P. and Apollo Management. Prior to Apollo, Mr. Kaplan was a member of the Investment Banking Department at Donaldson & Jenrette Securities Corp. Mr. Kaplan currently serves on the supervisory board of LuxExperience B.V., formerly known as MYT Netherlands Parent B.V. Mr. Kaplan also serves as Chairman of the board of directors of the parent entity of Cooper's Hawk Winery & Spirits, LLC. Mr. Kaplan's previous public company board experience includes Floor & Decor Holdings, Inc., Maidenform Brands, Inc., where he served as the company's Chairman, GNC Holdings, Inc., GNC Supermarkets, Inc., Stream Global Services, Inc., Orchard Supply Hardware Stores Corporation, Final Stores, Inc. and Allied Waste Industries Inc. Mr. Kaplan also currently serves as Chairman of the Board of Directors of Cedars-Sinai Medical Center and is on the Board of Trustees at the Los Angeles County Museum of Art (LACMA). Mr. Kaplan sits on the President's Advisory Group of the University of Michigan, where he graduated with High Distinction, Beta Gamma Sigma, with a Bachelor of Science in Business Administration degree, concentrating in Finance. We believe Mr. Kaplan is well qualified to serve on the Board due to his knowledge of and extensive experience with leveraged finance, acquisition, and equity investments, in addition to his service as a director of other public and private companies.

Allyson Satin. Ms. Satin will serve as a member of the Board. Ms. Satin has served on the Board since December 2025 and previously served on the Board from December 2023 to July 2025. Ms. Satin is a Senior Partner in the Ares Corporate Strategy Group of Ares Management Corporation, where she focuses on the firm's SPAC Business. Ms. Satin previously served as the Chief Operating Officer of Ares Management Corporation I from January 2021 through November 2023 and Chief Operating Officer of Ares Management Corporation II from March 2023 through September 2025, and has served as a member of the board of directors of its successor, Kodiak AI, Inc. (NASDAQ: KDK) since September 2025. From 2015 to 2025, Ms. Satin was an investment professional in the Ares Private Equity Group where she participated in various leveraged buyouts, growth equity and distressed debt transactions. Prior to joining Ares, Ms. Satin was an investment banking analyst in the Global Financial Sponsors Group at Barclays.

(formerly Lehman Brothers). Ms. Satin holds a B.S. from the University of California, B School of Business in Business Administration. We believe Ms. Satin is well qualified to serve due to her knowledge of and extensive experience with leveraged finance, acquisitions and investments, in addition to her service as a director of other companies.

Corporate Governance

Composition of the Board

Our business and affairs will be managed under the direction of the Board. The Board will consist of Clay Sell, Kamal Ghaffarian, Edward Sonnenschein, Michael J. Wallace, Kathleen W. Hyle, Christopher F. Ginther, and Allyson Satin.

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Gregory J. Goff, David B. Kaplan and Allyson Satin as members. The Board is expected to determine whether Michael J. Wallace, Kathleen W. Hyle, Christopher F. Ginther, Gregory J. Goff, David B. Kaplan and Allyson Satin qualify as independent in accordance with applicable Nasdaq rules. Subject to the approval of the Amended and Restated Certificate of Incorporation and Amended and Restated Bylaws, the number of directors will be fixed by the Board.

When considering whether directors and director nominees have the experience, qualifications, attributes and skills, taken as a whole, to enable the Board to satisfy its oversight responsibilities in light of its business and structure, the Board expects to focus primarily on each person's background and experience as reflected in the information discussed in each of the directors' individual biographies above in order to provide an appropriate mix of experience and skills relevant to the size and complexity of the business.

Director Independence

Nasdaq listing standards require that a majority of the board of directors be independent. "Independent director" is defined generally as a person other than an officer or employee of the Company or its subsidiaries or any other individual having a relationship which in the opinion of the board of directors would interfere with the director's exercise of independent judgment in carrying out the responsibilities of a director.

The Board will undertake a review of its composition, the composition of its committees and the independence of its directors and consider whether any director has a material relationship with the Company that could compromise his or her ability to exercise independent judgment in carrying out her responsibilities.

We anticipate that the Company will have a majority of "independent directors" as defined in the Nasdaq listing standards and applicable SEC rules.

Classified Board of Directors

Pursuant to the Certificate of Incorporation, the Company's directors will be divided into three classes, with each class serving staggered three-year terms. The Board will initially consist of nine directors. The Board will be divided among the three classes as follows:

- The Class I directors are _____, _____ and _____ ;
- The Class II directors are _____, _____ and _____ ; and

- The Class III directors are _____, _____ and _____.

Committees of the Board

The Board will direct the management of its business and affairs, as provided by Delaware law, to conduct its business through meetings of the Board and standing committees. The Board will have standing audit and risk committee, compensation and culture committee and nominating and governance committee, each of which will operate under a written charter that will be posted on the Investors Relations page of the Company's website at <https://x-energy.com/investors> as may be required by applicable Nasdaq rules.

From time to time, special committees may be established under the direction of the Board if the Board deems it necessary or advisable to address specific issues.

Audit and Risk Committee

The Board's audit and risk committee will be responsible for, among other things:

- appointing, compensating, retaining and overseeing the work of our independent registered public accounting firm and any other registered public accounting firm engaged for the purpose of preparing or issuing an audit report or related work or performing other audit, review or assurance services for us;
- assessing the independence of our independent registered public accounting firm;

- pre-approving all audit and non-audit services provided to us by our independent registered public accounting firm (other than those provided pursuant to appropriate preapproval policies established by the audit committee or exempt from such requirement under the rules of the SEC);
- discussing with our independent registered public accounting firm any audit problems and management's response;
- reviewing and discussing our annual and quarterly financial statements with management and our independent registered public accounting firm;
- reviewing, discussing and evaluating our policies with respect to risk assessment and risk management;
- reviewing and approving or ratifying any related person transactions;
- reviewing and discussing our internal control over financial reporting and code of business conduct and ethics;
- setting clear hiring policies for employees or former employees of our independent registered public accounting firm;
- establishing procedures for the receipt, retention and treatment of complaints received by or on behalf of the company regarding accounting, internal accounting controls or auditing matters, and for the confidential, anonymous submission by our employees of concerns regarding questionable accounting or auditing matters; and
- preparing the audit committee report for inclusion in our annual proxy statements.

Our audit and risk committee is expected to consist of Kathleen W. Hyle, Gregory J. Goff and Jeffrey A. Satin with Kathleen W. Hyle serving as chair. All members of our audit and risk committee are expected to satisfy the requirements for financial literacy under the applicable Nasdaq rules and regulations. The Board expects to affirmatively determine that each member of the audit and risk committee qualifies as “independent” under the Nasdaq’s additional standards applicable to audit committee members and Rule 10A-3 of the Securities Exchange Act applicable to audit committee members. The Board expects to determine that Kathleen W. Hyle qualifies as an “audit committee financial expert,” as such term is defined in Item 407 of Regulation S-K.

Compensation and Culture Committee

The Board’s compensation and culture committee will be responsible for, among other things:

- reviewing and approving corporate goals and objectives with respect to the compensation of our Chief Executive Officer, evaluating our Chief Executive Officer’s performance in light of such goals and objectives and setting our Chief Executive Officer’s compensation;
- reviewing and setting or making recommendations to the Board regarding the compensation of our other executive officers;
- reviewing and making recommendations to the Board regarding director compensation;
- reviewing and approving or making recommendations to the Board regarding our compensation and equity-based plans, policies and programs and administering the plans and programs;
- reviewing and approving all employment and severance arrangements for our executive officers;
- reviewing and discussing annually with management our “*Compensation Discussion and Analysis*” to the extent it is required to be included in our annual report on Form 10-K or our proxy statement;
- periodically reviewing our compensation policies and practices and assessing the risk associated with such policies and practices;
- administering and overseeing our compliance with our clawback policies;
- overseeing and reviewing with management our strategies, policies and practices regarding executive capital management and talent development;
- preparing the annual compensation committee report required by SEC rules, to the extent applicable, and

- reviewing and discussing the results of our stockholder advisory votes on executive compensation and reviewing and recommending to the Board the frequency of such votes.

Our compensation and culture committee is expected to consist of Kathleen W. Hyle, Allyson J. Smith and Jeffrey A. Satin with Kathleen W. Hyle serving as chair. The Board expects to determine that Kathleen W. Hyle, Allyson J. Smith and Jeffrey A. Satin qualify as “independent” under the Nasdaq’s additional standards applicable to compensation committee members.

Nominating and Corporate Governance Committee

The Board’s nominating and corporate governance committee will be responsible for, among other things:

- identifying individuals qualified to become members of the Board and ensuring that the requisite expertise and consists of persons with sufficiently diverse and backgrounds;
- recommending to the Board the persons to be nominated for election as directors committee of the Board;
- developing and recommending to the Board corporate governance guidelines, and reassessing the adequacy of such corporate governance guidelines and recommending changes to the Board for approval from time to time; and
- overseeing the annual self-evaluations of the Board and its committees.

Our nominating and corporate governance committee is expected to consist of Gregory J. Goff, Christopher F. Ginther and Michael J. Wallace, with Gregory J. Goff serving as chair. The Board has determined that Gregory J. Goff and Michael J. Wallace qualify as “independent” under the standards applicable to nominating and corporate governance committee members.

Code of Ethics

The Company will adopt a Code of Business Conduct and Ethics that applies to all of our officers, directors and employees, including our principal executive officer, principal financial officer, principal accounting officer or persons performing similar functions, which will be available on our website. Our Code of Business Conduct and Ethics will be a “code of ethics,” as defined in Item 405 of Regulation S-K.

We intend to make any legally required disclosures regarding amendments to, or waivers from, our code of ethics on our website.

Compensation Committee Interlocks and Insider Participation

No anticipated member of the Board’s compensation committee was at any time during fiscal year 2024 and 2025, or at any other time, one of our officers or employees. None of our executive officers served as a director or member of a compensation committee (or other committee serving an equivalent function) of any entity, one of whose executive officers is anticipated to serve as a member of the Board’s compensation committee.

This section discusses the material components of the executive compensation program for the named executive officers who are named in the “2025 Summary Compensation Table” below. With the year ended December 31, 2025, our “named executive officers” or “NEOs” were as follows:

- J. Clay Sell, Chief Executive Officer;
- Daniel Gross, Executive Vice President and Chief Financial Officer; and
- Dragan Popovic, Executive Vice President and Global Chief Operating Officer.

This discussion may contain forward-looking statements that are based on our current considerations, expectations and determinations regarding future compensation programs. Compensation programs that we adopt may differ materially from the currently planned programs summarized in this discussion.

2025 Summary Compensation Table

The following table sets forth information concerning the compensation of our named executive officers for the year ended December 31, 2025.

Name and Principal Position	Salary (\$) ⁽¹⁾	Bonus (\$)	Stock Awards (\$) ⁽²⁾	All Other Compensation (\$) ⁽³⁾
J. Clay Sell Chief Executive Officer	612,000	583,848	9,188,142	113,692
Daniel Gross ⁽⁴⁾ Executive Vice President and Chief Financial Officer	36,264		6,339,673	1,058
Dragan Popovic ⁽⁵⁾⁽⁶⁾ Executive Vice President and Global Chief Operating Officer	83,673	134,375	4,437,768	

- (1) Amounts reflect base salary earned during 2025.
- (2) Amounts reflect the aggregate grant date fair value of profits interests in Management Incentive Plan during the year ended December 31, 2025 computed in accordance with FASB ASC Compensation — Stock Compensation and does not correspond to the actual economic value that will be received by the named executive officers from these awards. We provide information regarding the assumptions used to calculate the value of all profits interests made to named executive officers in Note 12 — Unit-Based Compensation Expense to the financial statements included in this report.
- (3) Amounts reflect (i) \$19,701 and \$1,058 in employer matching contributions made to the company's defined contribution plan, for Messrs. Sell and Gross, respectively; and (ii) \$93,991 in reimbursement to Mr. Sell for expenses incurred in connection with travel to and from his primary residence.
- (4) Mr. Gross commenced employment as our Executive Vice President and Chief Financial Officer in November 2025.
- (5) Mr. Popovic commenced employment as our Executive Vice President and Global Chief Operating Officer in October 2025.
- (6) During his period of employment prior to December 22, 2025, Mr. Popovic was employed by a subsidiary, X-Energy Canada, Inc., and was paid in local currency (CAD). The amounts of compensation Mr. Popovic earned in local currency have been converted to USD using an exchange rate for the year ended 2025 of 1 USD to 1.37 CAD.

Narrative to Summary Compensation Table

2025 Salaries

Each of our named executive officers receives an annual base salary to compensate the services rendered to us. The base salary payable to each named executive officer is intended to be a fixed component of compensation reflecting the executive's duties and authorities, contributions, experience and sustained performance.

The 2025 annual base salaries for our named executive officers were \$612,000 for Mr. Sell, \$500,000 for Mr. Gross, and \$450,000 for Mr. Popovic.

The "Salary" column of the 2025 Summary Compensation Table above shows the actual salary earned by each named executive officer in 2025.

Cash Incentive Compensation

Annual cash bonuses are determined by our Compensation and Culture Committee and are intended to be commensurate with the executive's duties and authorities, contributions, prior experience and sustained performance. The discretionary cash bonus earned by Mr. Sell for fiscal year 2025 was \$134,375. Mr. Gross was not eligible for an annual bonus for fiscal year 2025, and Mr. Popovic received an annual cash bonus of \$134,375 for fiscal year 2025, as described in the section entitled "Compensation Arrangements" below.

Equity Compensation

2025 Equity Awards

Prior to the closing, Management LLC typically granted Class B-2 Common Units of Management LLC Profits Interest Plan (the "Profits Interest Plan"). We refer to these Management LLC Class B-2 Common Units as "Incentive Units." We granted equity awards in the form of Class B Units of Management LLC intended to qualify as profits interests within the meaning of applicable SEC guidance. The grant of Class B Units is implemented through an "upstairs-downstairs" structure. Our employees, including our named executive officers, receive Class B Units of Management LLC that mirror the economics (including vesting schedules) of corresponding units that X-energy receives from Management LLC.

In June 2025, we granted 2,576,000 Incentive Units to Mr. Sell and, in December 2025, we granted the following Incentive Units to our named executive officers: 1,147,800 Incentive Units to Mr. Sell, 1,436,171 Incentive Units to Mr. Gross and 1,436,171 Incentive Units to Mr. Popovic.

The Incentive Unit awards each vest in substantially equal annual installments on each second, third and fourth anniversaries of the applicable vesting commencement date, subject to the executive officer's continuous employment through the applicable vesting date. The vesting commencement date for the Incentive Unit awards granted to Mr. Sell in June 2025 is January 1, 2024, and the vesting commencement date for the Incentive Unit awards granted to Messrs. Sell, Gross and Popovic in December 2025 is December 23, 2025.

In connection with this initial public offering, both XERC and Management LLC will complete a recapitalization, pursuant to which all outstanding units in XERC and Management LLC will be recapitalized into common units in accordance with the applicable operating agreement (on a "pro rata" value" basis based on the fair market value of the equity interests, including the Incentive Units) at the time of the offering and the common stock price in the offering). Following this recapitalization, the common units will continue to be subject to any applicable vesting conditions of the offering. The Incentive Units before the applicable recapitalization. Thereafter, Management LLC will contribute

Common Units of XERC to X-energy in exchange for an equal number of shares of X-energy common stock, which Class A common stock shall be subject to the same vesting conditions as the corresponding Common Units of XERC before the exchange. In addition, in connection with this offering, we expect the Profits Interests Plan will terminate and we will not make further awards under the Profits Interest Plan.

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2026 Equity Incentive Plan

In connection with this offering, we intend to adopt the 2026 Equity Incentive Plan, or the 2026 Plan, in order to facilitate the grant of cash and equity incentives to directors, employees (including executive officers) and consultants of our company and employees and consultants or consultants of our affiliates and to enable our company and certain of our affiliates to obtain and retain services of certain individuals, which is essential to our long-term success. For additional information about the 2026 Plan, please see the section titled “2026 Equity Incentive Plan” below.

IPO Equity Awards

Our Board intends to approve the grant of restricted stock unit awards and stock options pursuant to the 2026 Plan to certain of our employees, including Messrs. Sell, Gross and Popovic, and our non-executive directors, which grants will become effective in connection with the consummation of this offering (“IPO Equity Awards”), subject to continued service through the grant date. Stock options will be granted to our employees and non-employee directors who hold Incentive Units.

The IPO Equity Awards that will be granted to each of Messrs. Sell, Gross and Popovic will be comprised of stock options and are expected to have a cumulative value of \$1,000,000, \$1,000,000, and \$1,000,000, respectively, as of the closing of this offering. In addition, certain of our directors who hold Incentive Units will receive stock options, which are expected to have a cumulative value of \$1,000,000 in aggregate.

Each stock option granted to our named executive officers and directors will vest in equal installments and be exercisable in substantially equal annual installments pursuant to the same vesting schedule as the Incentive Units.

The IPO Equity Awards that will be granted to our non-employee directors will be comprised of restricted stock unit awards, which are further described under the section titled “Executive Compensation — Director Compensation — Non-Employee Director Compensation Program” below.

The aggregate value of restricted stock unit awards to be granted to our employees and non-employee directors in connection with the offering will equal approximately \$1,000,000. The aggregate value of stock options and restricted stock unit awards to be granted to our employees in connection with the offering will equal approximately \$1,000,000.

Other Elements of Compensation and Compensation Policies

We provide benefits to our named executive officers on the same basis as provided to our other employees, including health, dental and vision insurance; critical illness insurance; life insurance; hospital indemnity insurance; short-and long-term disability insurance; and a Section 401(k) plan for which we make safe harbor contributions on behalf of our employees. We have also provided Mr. Sell with compensation to cover expenses incurred by him in connection with his travel and from his primary residence (including commuting, lodging and dining expenses).

In addition to the health and welfare benefits described above, prior to his relocation to the United States in December 2025, Mr. Popovic was eligible to receive similar health and welfare benefits in connection with his employment in Canada.

We believe the employee benefits and perquisites described above are necessary and appropriate to provide a competitive compensation package to our named executive officers.

No Tax Gross-Ups

We do not make gross-up payments to cover our named executive officers' personal income taxes that may pertain to any of the compensation or perquisites we pay or provide.

Clawback Policy

In connection with this offering, our Board intends to adopt a compensation recovery policy that complies with the listing standards of Nasdaq, as required by the Dodd-Frank Act.

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Outstanding Equity Awards at Fiscal Year-End

The following table summarizes the number of outstanding unvested Incentive Units held by each named executive officer as of December 31, 2025.

Name	Grant Date	Number of Incentive Units That Have Not Vested (#)	Market Value of Incentive Units That Have Not Vested
J. Clay Sell	June 13, 2025	1,932,000 ⁽²⁾	16,920,000
	December 23, 2025	1,147,800 ⁽³⁾	3,171,960
Daniel Gross	December 23, 2025	2,051,674 ⁽³⁾	6,155,020
Dragan Popovic	December 23, 2025	1,436,171 ⁽³⁾	4,308,513

- (1) The Incentive Units are not publicly traded and, therefore, there was no ascertainable market value for the Incentive Units as of December 31, 2025. The value of the Incentive Units is based on the closing price of our common stock as of December 31, 2025 based on the number of outstanding unvested Incentive Units as of December 31, 2025 taking into account applicable distribution thresholds, and was estimated to be \$3.09 (for Incentive Units granted in December 2025) and \$8.36 (for Incentive Units granted in June 2025).
- (2) Represents unvested Incentive Units that will vest in substantially equal annual installments on the first, second, third and fourth anniversaries of January 1, 2024, subject to Mr. Sell's continued service through the applicable vesting date. The distribution threshold is \$1,729,728,630.
- (3) Represents unvested Incentive Units that will vest in substantially equal annual installments on the first, second, third and fourth anniversaries of the grant date, subject to the named executive officer's continuous service through the applicable vesting date. The distribution threshold is \$5,000,000,000.

Executive Compensation Arrangements

Each of Messrs. Sell, Gross and Popovic is party to an offer letter with X Energy, LLC that provides for at-will employment that will continue until terminated at any time by either party. Pursuant to his offer letter, Mr. Gross is eligible for an annual cash incentive bonus (targeted at 75% of his base salary) commencing in the 2026 performance year, and is entitled to receive a monthly stipend covering expenses related to executive housing, travel and other expenses associated with commuting to the company's Maryland headquarters, and reimbursement of up to \$175,000 in relocation expense reimbursements. Pursuant to the terms of his offer letter, Mr. Popovic is eligible for an annual cash incentive bonus (targeted at 75% of his base salary), with a guaranteed bonus of \$134,375 for 2025, and is entitled to receive reimbursement of travel expenses for commuting, lodging, and house-hunting trips, and an additional \$135,000 in relocation expense reimbursements. As of the end of 2025, Mr. Gross had not received any commuting expense reimbursements, and neither Mr. Gross nor Mr. Popovic had received any relocation expense reimbursements. Pursuant to their offer letters, each of our named executive officers is eligible to participate in the benefit plan and programs maintained by us for the benefit of our employees. None of the offer letters provide for severance payments on termination of employment.

Each of Messrs. Sell, Gross and Popovic also entered into the Company's standard employment agreement, disclosure and non-solicitation agreement, which includes a company information non-disclosure covenant that lasts during employment and for five years thereafter, a perpetual government data non-disclosure covenant, and an employee and client non-solicitation covenant that lasts during employment and for two years thereafter. In addition, as a condition to their receipt of Incentive Units, they each entered into a restrictive covenant letter, which includes non-competition and employee and business relationship non-solicitation covenants that apply during employment and for one year following termination of employment. Each of Messrs. Sell, Gross and Popovic also entered into a company information non-disclosure covenant and non-disparagement covenant that each applies during the period they provide services to X-energy and thereafter, and an assignment of intellectual property covenant.

2026 Equity Incentive Plan

In connection with this offering, we intend to adopt the 2026 Plan, under which we may make equity and cash incentive awards in order to attract, motivate and retain the talent for which we compete. The 2026 Plan will require stockholder approval. The material terms of the 2026 Plan are summarized below.

Eligibility and Administration. Our employees, consultants and directors, and executive officers and consultants of our affiliates, will be eligible to receive awards under the 2026 Plan. Following the completion of this offering, the 2026 Plan will be administered by our board of directors with respect to awards to non-employee directors and by our compensation committee with respect to other eligible persons. Each of which may delegate its duties and responsibilities to committees of our directors and officers (referred to collectively as the plan administrator), subject to the limitations imposed under the Securities Act, Section 16 of the Exchange Act, stock exchange rules and other applicable laws. The plan administrator will have the authority to take all actions and make all determinations under the 2026 Plan, to interpret the 2026 Plan and award agreements and to adopt, amend and repeal rules for the administration of the 2026 Plan as it deems advisable. The plan administrator will also have the authority to determine which eligible persons receive awards, grant awards and set the terms and conditions of all awards under the 2026 Plan, including any vesting and vesting acceleration provisions, subject to the conditions and limitations of the 2026 Plan.

Limitation on Awards and Shares Available. The initial aggregate number of shares of our common stock that will be available for issuance under the 2026 Plan will be equal to 10% of the number of shares of our Class A common stock and Class B common stock outstanding (assuming all shares of Class A common stock and Class B common stock are outstanding).

stock reserved under the 2026 Plan have been issued) as of immediately following the completion of the offering (which is expected to be _____ shares, assuming an initial public offering price of \$ _____ per share, which is the midpoint of the estimated price range set forth on the cover of the prospectus). In addition, the number of shares of our common stock available for issuance under the 2026 Plan will be subject to an annual increase on the first day of each calendar year beginning on January 1, 2027 and ending on and including January 1, 2036, equal to the lesser of (A) 5% of the number of shares of our Class A common stock and Class B common stock outstanding on the last day of the immediately preceding calendar year and (B) such smaller number of shares as is determined by the board of directors. The maximum number of shares that may be issued pursuant to the exercise of stock options, or ISOs, granted under the 2026 Plan, will be 100,000,000. Any shares issued pursuant to the 2026 Plan will be Class A common stock and may consist, in whole or in part, of authorized common stock, treasury common stock or common stock purchased on the open market.

If an award under the 2026 Plan expires, lapses or is terminated, exchanged for or settled in cash, the shares subject to such award (or portion thereof) may, to the extent of such expiration, lapse, termination or cash settlement, be used again for new grants under the 2026 Plan. Shares tendered or withheld to satisfy the exercise price or tax withholding obligation for any award will not reduce the shares available for grant under the 2026 Plan. Further, the payment of dividend equivalents in cash in conjunction with the exercise of an award under the 2026 Plan will not reduce the shares available for grant under the 2026 Plan. The following shares may not be used again for grant under the 2026 Plan: (i) shares subject to restricted stock, restricted stock units, or SARs, that are not issued in connection with the stock settlement of an award, and (ii) shares purchased on the open market with the cash proceeds from the exercise of an award.

Awards granted under the 2026 Plan upon the assumption of, or in substitution for, awards granted or outstanding under a qualifying equity plan maintained by an entity with which we enter into a similar corporate transaction will not reduce the shares available for grant under the 2026 Plan and will not count against the maximum number of shares that may be issued upon the exercise of ISOs.

The 2026 Plan provides that, commencing with calendar year 2027, the sum of any cash bonus and the aggregate grant date fair value (determined as of the date of the grant under Financial Accounting Standards Board Accounting Standards Codification Topic 718, or any successor thereto) of awards granted to a non-employee director as compensation for services as a non-employee director in any fiscal year, or director limit, may not exceed an amount equal to \$750,000 (increased to \$1,000,000 in the calendar year of a non-employee director's initial service as a non-employee director or any successor director). During which a non-employee director serves as chair of our Board or lead independent director, the director limits shall not apply to the compensation for any non-employee director who serves in any such capacity in addition to that of a

non-employee director for which he or she receives additional compensation or any compensation in excess of the director limit in the calendar year following the calendar year in which the 2026 Plan becomes effective.

Awards. The 2026 Plan provides for the grant of stock options, including ISOs and nonqualified stock options, or NSOs, SARs, restricted stock, dividend equivalents, restricted stock units, or RSUs, and cash-based awards. Certain awards under the 2026 Plan may constitute or provide for "nonqualified deferred compensation" under Section 409A of the Code, which may impose certain requirements on the terms and conditions of such awards. All awards under the 2026 Plan will be governed by award agreements, which will detail the terms and conditions of the awards, including the vesting and payment terms and post-termination exercise limitations. Awards other than restricted stock, restricted stock units, or SARs will generally be settled in shares of our common stock, but the applicable award agreement may provide for cash settlement of any award. A brief description of each award type follows.

- *Stock Options and SARs.* Stock options provide for the purchase of shares of our common stock in the future at an exercise price set on the grant date. ISOs, in contrast to NSOs, may have a deferral beyond exercise and favorable capital gains tax treatment to their holders if certain conditions, period and other requirements of the Code are satisfied. SARs entitle their holder, upon exercise, to receive from us an amount equal to the appreciation of the shares subject to the award from the grant date and the exercise date. Unless otherwise determined by our board, the exercise price of a stock option or SAR may not be less than 100% of the fair market value of the underlying shares on the grant date (or 110% in the case of ISOs granted to certain significant stockholders). In the case of a stock option or SAR that is a substitute award granted in connection with a corporate transaction, the exercise price of a stock option or SAR may not be longer than ten years (or five years in the case of ISOs granted to certain significant stockholders). Conditions applicable to stock options and/or SARs may include, but are not limited to, continuing service, the attainment of performance goals and/or such other conditions as the plan administrator may determine.
- *Restricted Stock and RSUs.* Restricted stock is an award of nontransferable shares of our common stock that are subject to certain vesting conditions and other restrictions. RSUs are awards that promise to deliver shares of our common stock in the future, which may also remain subject to certain restrictions unless and until specified conditions are met and may be accompanied by the right to receive dividend equivalent value of dividends paid on shares of our common stock prior to the delivery of the underlying shares (i.e., dividend equivalent rights). The plan administrator may determine whether the delivery of the shares underlying RSUs will be deferred on a mandatory basis or at the election of the participant. The terms and conditions applicable to RSUs will be determined by the plan administrator, subject to the conditions and limitations contained in the 2026 Plan. Conditions applicable to restricted stock and RSUs may be based on continuing service, the attainment of performance goals and/or such other conditions as the plan administrator may determine.
- *Other Stock or Cash-Based Awards.* Other stock or cash-based awards are awards of shares of our common stock or vested shares of our common stock and other awards valued wholly or partially by reference to the value of, or otherwise based on, shares of our common stock. Other stock or cash-based awards may be awarded to participants and may also be available as a payment form in the settlement of other awards, as standalone payments and as payment in lieu of compensation to which a participant may be entitled.
- *Dividend Equivalents.* Dividend equivalents represent the right to receive the equivalent of dividends paid on shares of our common stock and may be granted alone or in tandem with other awards other than stock options or SARs. Dividend equivalents are credited as of the dividend payment date during the period between the date an award is granted and the date such award vests, is exercised, is distributed or expires, as determined by the plan administrator. Dividend equivalents may be paid in respect to an award prior to the vesting of such award instead will be paid out to the participant upon the date to the extent that the vesting conditions are subsequently satisfied and the award vests.

Certain Transactions. The plan administrator has broad discretion to take action under the 2026 Plan, as well as make adjustments to the terms and conditions of existing and future awards, to prevent the dilution or enlargement of intended benefits and facilitate necessary or desirable changes in the 2026 Plan in certain transactions and events affecting our common stock, such as stock dividends, stock splits, stock repurchases, acquisitions, consolidations and other corporate transactions.

acquisitions, consolidations and other corporate transactions. In addition, in the event of a change of control or reciprocal transactions with our stockholders known as “equity restructurings,” the plan administrator may make equitable adjustments to the 2026 Plan and outstanding awards. In the event of a change

defined in the 2026 Plan), to the extent that the surviving entity declines to continue, convey or replace outstanding awards, then all such awards will become fully vested and exercisable in connection with the transaction. Upon or in anticipation of a change in control, the plan administrator may cause outstanding awards to terminate at a specified time in the future and give the participant the right to receive such awards during a period of time determined by the plan administrator in its sole discretion. Award agreements may provide for additional accelerated vesting and payment provisions.

Repricing. Our board of directors may, without approval of the stockholders, reduce the exercise price of any stock option or SAR, or cancel any stock option or SAR in exchange for cash, other awards or stock options or SARs with an exercise price per share that is less than the exercise price per share of the original stock options or SARs.

Plan Amendment and Termination. Our board of directors may amend or terminate the 2026 Plan at any time; however, no amendment, other than an amendment that increases the number of shares available for awards under the 2026 Plan, may materially and adversely affect an award outstanding under the 2026 Plan without the consent of the affected participant, and stockholder approval will be obtained for any amendment to the extent necessary to comply with applicable laws. The 2026 Plan will remain in effect until the 10th anniversary of the effective date of the 2026 Plan, unless earlier terminated. No awards may be granted under the 2026 Plan after its termination.

Foreign Participants, Claw-back Provisions, Transferability and Participant Payments. The plan administrator may modify award terms, establish subplans and/or adjust other terms and conditions of awards, subject to the share limits described above, in order to facilitate grants of awards subject to applicable and/or stock exchange rules of countries outside of the U.S. All awards will be subject to a clawback policy as set forth in such clawback policy or the applicable award agreement. Awards granted under the 2026 Plan are generally non-transferrable, except by will or the laws of descent and distribution, and subject to the plan administrator's consent, pursuant to a domestic relations order, and are generally exercisable only by the participant. With regard to tax withholding, exercise price and purchase price obligations, in connection with awards under the 2026 Plan, the plan administrator may, in its discretion, accept payment by check, shares of our common stock that meet specified conditions, a "market sell order" or other form of consideration as it deems suitable.

Employee Stock Purchase Plan

In connection with this offering, we intend to adopt, and our stockholders will approve, the 2026 Inc. Employee Stock Purchase Plan, or the ESPP. The material terms of the ESPP are summarized below.

The ESPP is comprised of two distinct components in order to provide increased flexibility in granting stock options to purchase shares under the ESPP. Specifically, the ESPP authorizes (i) the grant of options to employees that are intended to qualify for favorable U.S. federal tax treatment under Section 423 of the Code, if eligible under applicable law (the "Section 423 Component"), and (ii) the grant of options to employees not intended to be tax-qualified under Section 423 of the Code to facilitate participation for employees located within and outside of the U.S. who are not eligible for or otherwise do not benefit from the U.S. federal tax treatment and to provide flexibility to comply with non-U.S. law and other conditions (the "Non-Section 423 Component"). Where permitted under applicable law, we expect the Non-Section 423 Component will generally be operated and administered on terms and conditions similar to the Section 423 Component.

Shares Available; Administration

The initial share reserve under the ESPP will equal 2% of the number of fully-diluted shares of our Class A and Class B common stock outstanding (assuming all shares reserved under the ESPP have been issued) as of immediately following the completion of this offering (which is expected to be approximately 10,000 shares, assuming an initial public offering price of \$ 12.00 per share, which is the midpoint of the price range set forth on the cover page of this prospectus).

In addition, the number of shares available for issuance under the ESPP will be annually on the first day of each calendar year beginning January 1, 2027 and ending on and including January 1, 2028, equal to the lesser of (i) 0.5% of the aggregate number of shares of our Class A and Class B common stock outstanding on the final day of the immediately preceding calendar year and (ii) such small number of shares as is determined by our board of directors. In no event will more than 50,000,000 shares of our Class A common stock be available for issuance under the ESPP.

Our board of directors or a committee designated by our board of directors will have the authority to interpret the terms of the ESPP and determine eligibility of participants. The compensation committee will be the administrator of the ESPP.

Eligibility

The plan administrator may designate certain of our subsidiaries as participating subsidiaries in the ESPP and may change these designations from time to time. Employees of our company and our designated subsidiaries are eligible to participate in the ESPP if they meet the requirements under the ESPP established from time to time by the plan administrator. An employee participating in the Section 423 Component may not be granted rights to purchase stock under the ESPP if such employee, immediately after the grant, would own (directly or through attribution) more than 10% of the total combined voting power or value of all classes of our common stock, or would be possessing 5% or more of the total combined voting power or value of all classes of our common stock. Eligible employees become participants in the ESPP by enrolling and authorizing payroll deductions by the deadline established by the plan administrator prior to the relevant offering period. Directors who are not employees, as well as consultants, are not eligible to participate. Employees may choose to not participate, or are not eligible to participate at the start of an offering period but become eligible thereafter, may enroll in any subsequent offering period.

Participation in an Offering

Stock will be offered under the ESPP during offering periods. The length of offering periods under the ESPP will be determined by the plan administrator and may be up to 27 months long. Employees' payroll deductions will be used to purchase shares on each purchase date during an offering period. The number of purchase periods within, and purchase dates during, each offering period will be established by the plan administrator. Offering periods under the ESPP will commence when determined by the plan administrator. The plan administrator may, in its discretion, modify the terms of future offering periods. In jurisdictions where participation in the ESPP through payroll deductions is prohibited, the plan administrator may provide that an eligible employee may elect to participate through contribution to a participant's account under the ESPP in a form acceptable to the plan administrator in lieu of payroll deductions.

The ESPP will permit participants to purchase our common stock through payroll deductions up to 15% of their eligible compensation, which will include a participant's gross base salary for salary, excluding one-time bonuses, periodic bonuses, sales commissions, expense reimbursements, fringe benefits, and other special payments. However, the plan administrator may change the definition of eligible compensation and the maximum amount of possible payroll deductions, with respect to any offering period under the ESPP. The plan administrator will establish a maximum number of shares that may be purchased by a participant during any offering period or purchase period, which, in the absence of a contrary determination, will be 10,000 shares for an offering period and/or a purchase period. In addition, no employee will be permitted to accrue the right to purchase stock under the Section 423 Component at a rate that would result in the purchase of more than \$25,000 worth of shares during any calendar year during which such a purchase right is outstanding. The purchase price will be on the fair market value per share of our common stock as of the first day of the offering period.

On the first trading day of each offering period, each participant automatically will be granted an option to purchase shares of our common stock. The option will be exercised on the applicable purchase date(s) during the offering period, to the extent of the payroll deductions accumulated during the offering period or purchase period. The purchase price of the shares, in the absence of a contrary determination by the plan administrator, will be 85% of the lower of the fair market value of our common stock on the first trading day of the offering period or on the applicable purchase date, which will be the final trading day of the applicable purchase period.

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Participants may voluntarily end their participation in the ESPP at any time at least two weeks before the end of the applicable offering period or, if earlier, purchase period (or such longer or shorter period as specified by the plan administrator), and will be paid their accrued payroll deductions that have not been used to purchase shares of common stock. Participation ends automatically upon a participant's termination of employment.

Transferability

A participant may not transfer rights granted under the ESPP other than by will, the laws of the state of distribution or as otherwise provided in the ESPP.

Certain Transactions

In the event of certain transactions or events affecting our common stock, such as any stock split, stock dividend, other distribution, change in control, reorganization, merger, consolidation or other corporate transaction, the plan administrator will make equitable adjustments to the ESPP and outstanding rights. In the event of the foregoing transactions or events or certain significant transactions, including a change in control, the plan administrator may provide for (i) either the replacement of outstanding rights with new rights or property or termination of outstanding rights in exchange for cash, (ii) the acquisition or substitution of outstanding rights by the successor or survivor corporation or parent or subsidiary, (iii) the adjustment in the number and type of shares of stock subject to outstanding rights, (iv) the use of participants' accumulated payroll deductions to purchase stock on a new purchase date prior to the next scheduled purchase date and termination of any rights under ongoing offering periods or (v) the termination of all outstanding rights. Under the ESPP, a change in control has the same definition as given in the 2026 Plan.

Plan Amendment; Termination

The plan administrator may amend, suspend or terminate the ESPP at any time. However, the approval of any amendment to the ESPP must be obtained for any amendment which increases the number of shares or changes the type of shares that may be sold pursuant to rights under the ESPP, or for any corporations or classes of corporations whose employees are eligible to participate in the ESPP, as required to comply with applicable law.

Director Compensation

During the fiscal year ended December 31, 2025, our non-employee directors' compensation were Dr. Kam Ghaffarian, Christopher F. Ginther, Gregory J. Goff, Kathleen M. Hynes, and Michael J. Wallace, who we refer to as our non-employee directors in this section.

2025 Director Compensation Table

The following table sets forth information for 2025 regarding the compensation awarded or paid to the non-employee directors.

Name	Fees Earned or Paid in Cash (\$) ⁽¹⁾	Stock Awards (\$) ⁽¹⁾
Dr. Kam Ghaffarian	400,000	
Christopher F. Ginther ⁽²⁾	80,212	465,690
Gregory J. Goff	95,000	465,690
Kathleen W. Hyle	100,000	384,440
Michael J. Wallace	80,000	242,090

- (1) Amounts reflect the aggregate grant date fair value of profits interests granted during the year ended December 31, 2025 computed in accordance with FASB ASC Topic 718, Compensation—Stock-Based Compensation and does not correspond to the actual economic value that may be received by the employee directors from these awards. We provide information regarding the assumptions used in the calculation of the aggregate grant date fair value of profits interests granted during the year ended December 31, 2025 in the following table:

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calculate the value of all profits interests made to non-employee directors in Note 12 – Compensation Expense to the financial statements included in this prospectus.

- (2) The amounts for Mr. Ginther originally denoted in local currency (CAD) have been converted to USD using an average exchange rate for December 31, 2025 of 1 USD to 1.37 CAD.

The following table summarizes the number of outstanding unvested Incentive Units held by each non-employee director, as of December 31, 2025.

Name	Stock Awards
Dr. Kam Ghaffarian	
Christopher F. Ginther	
Gregory J. Goff	
Kathleen W. Hyle	
Michael J. Wallace	

Non-Employee Director Compensation Program

In connection with this offering, we intend to approve and implement a compensation program for our non-employee directors (each, an “Eligible Director”). The Director Compensation Program will include annual cash retainer fees and long-term equity awards. The material terms of the Director Compensation Program are described below.

The Director Compensation Program consists of the following components:

Cash Compensation:

- Annual Retainer: \$80,000
- Annual Committee Chair Retainer:
 - Audit and Risk: \$25,000
 - Compensation: \$20,000
 - Nominating and Governance: \$15,000
- Non-Executive Chair: \$50,000

The annual cash retainers will be paid in quarterly installments in arrears. Annual cash retainer will be pro-rated for any partial calendar quarter of service.

Equity Compensation:

- **2026 Awards:** Our board intends to approve the grant of RSU awards pursuant to the 2026 Plan to our non-employee directors, which grants will become effective in connection with the closing of this offering (the “2026 Awards”). The dollar-denominated value of each award will be determined solely with respect to the Non-Executive Chair of the Board as of such Annual Meeting. The aggregate dollar-denominated value of these awards will be \$150,000. The aggregate number of shares of our Class A common stock that will be subject to the 2026 Awards will be determined based on the initial public offering price per share of our common stock in this offering.

The 2026 Awards will vest in full on the date of the annual meeting of stockholders to be held in 2026, or, if earlier, the first anniversary of the closing of this offering, subject to continued service and the applicable vesting date.

- **Annual Award:** An Eligible Director who is serving on our board of directors as of the date of the annual meeting of stockholders (beginning with calendar year 2027) automatically shall be granted, on the date of such annual meeting, an RSU award with an aggregate value of \$150,000. The number of RSUs subject to the award will be determined by dividing the value of the award by the price per share of our Class A common stock on the applicable grant date.

Each Annual Award will vest in full on the earlier to occur of the first anniversary of the grant date or the date of the next annual meeting following the grant date, subject to continued service.

In addition, each equity award granted to an Eligible Director under the Director Compensation Program will vest in full immediately prior to the occurrence of a “change in control” (as defined in the 2026 Plan). Compensation under the Director Compensation Program will be subject to the annual non-employee director compensation set forth in the 2026 Plan.

CERTAIN RELATIONSHIPS AND RELATED PARTY TRANSACTIONS

The following includes a summary of transactions since January 1, 2024 and any current transactions, to which we were or are to be a participant, in which (i) the amount involved exceeds \$120,000; and (ii) any of our directors, executive officers, or holders of more than 5% stock, or any affiliate or member of the immediate family of the foregoing persons or entities have a direct or indirect material interest, other than compensation and other arrangements described under “Executive and Director Compensation.”

We believe the terms obtained or consideration that we paid or received, as applicable, with the transactions described below were comparable to terms available or the amounts that we paid or receive, as applicable, in arm’s-length transactions.

Transactions

In connection with the Transactions, we will engage in certain transactions with our directors, executive officers and other persons and entities which are or will become holders of our voting securities upon the consummation of the Transactions. These transactions are described in our “Organizational Structure.”

XERC LLC Agreement

In connection with the closing of this offering, XERC will amend and restate its limited liability company agreement by adopting the XERC LLC Agreement. The XERC LLC Agreement will govern the issuance and ownership of the post-offering equity of XERC and (ii) admit X-energy as a member of XERC.

Appointment as Managing Member. Under the XERC LLC Agreement, X-Energy, Inc. is the managing member of XERC. As the managing member, X-Energy, Inc. is able to control the day-to-day business affairs and decision-making of XERC without the approval of any other members. Such, X-Energy, Inc., through its officers and directors, will be responsible for all operational and administrative decisions of XERC and the day-to-day management of XERC’s business. Pursuant to the terms of the XERC LLC Agreement, X-Energy, Inc. cannot be removed as the managing member by other members.

Compensation. X-Energy, Inc. is not entitled to compensation for its services as managing member. X-Energy, Inc. is entitled to reimbursement by XERC for fees and expenses incurred on behalf of XERC, including all expenses associated with the Transactions and maintaining its corporate existence.

Capitalization. The XERC LLC Agreement provides for a single class of Common Units. All Common Units shall have identical rights and privileges in all respects. Each Common Unit holder is entitled to a pro rata share of the net profits and net losses and distributions of XERC.

Distributions. The XERC LLC Agreement requires “Tax Distributions,” as that term is defined in the XERC LLC Agreement, to be made by XERC to X-Energy, Inc. and to its “Members,” as defined in the XERC LLC Agreement. Tax Distributions shall be made quarterly to X-Energy, Inc. and each Member based on the greater of (i) their allocable share of the taxable income of XERC multiplied by the tax rate that will be determined by X-Energy, Inc. and (ii) the pro rata amount of tax distributions to the Members holding the same class of equity. The tax rate used to determine tax distributions shall be the federal corporate tax rate regardless of the actual final tax liability of any such member. The XERC LLC Agreement also requires that distributions to be made by XERC to its members on a pro rata basis out of “distributable cash.”

the amount of cash that may be distributed by XERC to its Members in accordance with the agreements.

Common Unit Redemption Right. The XERC LLC Agreement provides a redemption right to Members (other than X-Energy, Inc. and its subsidiaries) and option holders (in connection with the exercise of an XERC Option, as such term is defined in the XERC LLC Agreement), which entitles them to have their Common Unit redeemed, in whole or in part, at the election of each such person, for a number of shares of X-Energy, Inc. Class A common stock on a one-to-one basis or, to the extent cash is available from a contemporaneous public offering or private sale of X-Energy, Inc. Class A common stock, by X-Energy, Inc.,

cash (in each case, subject to the terms and restrictions set forth in the XERC LLC Agreement). Alternatively, X-Energy, Inc. may instead authorize a cash payment equal to a volume weighted average of the market prices of one share of X-Energy, Inc. Class A common stock for each unit redeemed, subject to customary adjustments, including for stock splits, stock dividends and similar events affecting the market price of X-Energy, Inc. Class A common stock. If X-Energy, Inc. decides to make a cash payment, the Member may elect to rescind its redemption request within a specified time period. Upon the exercise of this right, the redeeming member will surrender its units for cancellation. The XERC LLC Agreement provides that X-Energy, Inc. contribute cash or shares of X-Energy, Inc. Class A common stock to XERC to satisfy the redemption for an amount of units that will be issued to us equal to the number of units redeemed from XERC. XERC will then distribute the cash or shares of X-Energy, Inc. Class A common stock to such Member to complete the redemption. In the event of such election by a Member, X-Energy, Inc. may, at its discretion, effect a direct exchange of cash or X-Energy, Inc. Class A common stock for such units in lieu of cash. Whether by redemption or exchange, we are obligated to ensure that at all times the number of Common Units that X-Energy, Inc. own equals the number of shares of X-Energy, Inc. Class A common stock issued by X-Energy, Inc. (subject to certain exceptions for treasury shares and shares of certain convertible or exchangeable securities). Shares of X-Energy, Inc. Class B common stock will be cancelled on a one-to-one basis if we, at the election of a Member, redeem or exchange units of a Member pursuant to the terms of the XERC LLC Agreement.

Issuance of Equity-based Compensation. X-Energy, Inc. may implement equity compensation plans and any actions taken under such equity compensation plans (such as the grant or exercise of options to acquire shares of X-Energy, Inc. Class A common stock), whether taken with respect to or by X-Energy, Inc. or other service provider of X-Energy, Inc., XERC or its subsidiaries, in a manner determined by X-Energy, Inc., in accordance with the initial implementation guidelines attached to the XERC LLC Agreement. The XERC LLC Agreement may be amended from time to time. X-Energy, Inc. may amend the XERC LLC Agreement (including the initial implementation guidelines attached thereto) as necessary or advisable in its sole discretion in connection with the adoption, implementation, modification or termination of an equity compensation plan. In the event of such an amendment, XERC will provide notice of such amendment to the Members. XERC is expressly authorized to issue units (i) in accordance with the terms of any equity compensation plan, and (ii) in an amount equal to the number of shares of Class A common stock issued pursuant to any equity compensation plans, without any further act, approval or vote of any Member or any other Person.

Maintenance of One-to-One Ratios. X-Energy, Inc.'s Amended and Restated Certificate of Incorporation and the XERC LLC Agreement will require that X-Energy, Inc. and XERC, respectively, at all times maintain (i) a one-to-one ratio between the number of Common Units owned by X-Energy, Inc. directly or indirectly, and the aggregate number of outstanding shares of X-Energy, Inc. Class A common stock, and (ii) a one-to-one ratio between the number of Common Units owned by each Member of XERC (other than X-Energy, Inc. and its subsidiaries), directly or indirectly, and the aggregate number of outstanding shares of X-Energy, Inc. Class B common stock owned by such Member.

Transfer Restrictions. The XERC LLC Agreement generally does not permit transfers of Units, by Members, subject to limited exceptions. Any transferee of Common Units must execute the XERC LLC Agreement and any other agreements executed by the holders of Common Units and related to the Common Units.

Dissolution. The XERC LLC Agreement provides that the decision of X-Energy, Inc. to dissolve, with the approval of a majority of the equity interests (including, but not limited to, Common Units) outstanding (excluding all units held directly or indirectly by X-Energy, Inc.) will be required to dissolve XERC. In addition to a voluntary dissolution, XERC will be dissolved under Section 18-801 of the DGCL because all members withdraw/resign (unless XERC is continued without dissolution) or pursuant to Section 18-802 of the DGCL by operation of law, including entry of a court order for judicial dissolution.

Confidentiality. Each Member (other than X-Energy, Inc.) agrees to hold confidential information in confidence and may not disclose or use such information except as otherwise authorized in writing by X-Energy, Inc. This obligation excludes information that (i) is, or becomes, generally known to the public other than as a direct or indirect result of a disclosure by such Member or its representatives; (ii) is, or becomes, available to such Member from a source other than X-Energy, Inc., XERC or their respective representatives; (iii) is approved for release by written authorization of the executive officer, chief financial officer or general counsel of X-Energy, Inc. or any other officer or employee of X-Energy, Inc.; or

(iv) is or becomes independently developed by such Member or its respective representatives without reference to the confidential information.

Indemnification and Exculpation. The XERC LLC Agreement provides for indemnification of expenses, liabilities and losses (including attorneys' fees, judgments, fines, excise taxes and other expenses) reasonably incurred or suffered by reason of the fact that such person is or was a Member of XERC or thereof or is or was serving as manager or a director, officer, employee, advisor, attorney, accountant or other agent or representative of the manager, the Company Representative (as such term is defined in the XERC LLC Agreement), or a director, manager, officer, employee, advisor, attorney, accountant or other agent or representative of XERC or is or was serving at the request of XERC as a manager, officer, director, principal, member, employee, advisor, attorney, accountant or other agent or representative of XERC; *provided, however*, that no indemnified person shall be indemnified for any expenses, liabilities or losses suffered that are attributable to such indemnified person's or its affiliates' fraud, willful misconduct or knowing violation of law or for any present or future breaches of any representations, warranties or covenants by such indemnified person or its affiliates contained in the XERC LLC Agreement or any agreements with XERC.

Amendments. The XERC LLC Agreement may be amended or modified (including by merger, consolidation or other business combination to which XERC is a party) upon the written consent of X-Energy, Inc. together with the prior written consent of the holders of a majority of the equity interests (including, but not limited to, Common Units) then outstanding (excluding all units held directly or indirectly by X-Energy, Inc.); provided, that no alteration, modification or amendment shall be made until written notice has been provided to the Members. Notwithstanding the foregoing, no amendment of any of the terms and conditions of the XERC LLC Agreement that expressly require the approval of a specified percentage of certain persons may be made without obtaining the consent of the requisite specified percentage of such persons who are entitled to approve or take action on such matter. Notwithstanding the foregoing, no alteration, modification or amendment may be made to any of the terms and conditions of the XERC LLC Agreement that would (A) reduce the amounts distributable to a Member in a manner not pro rata with respect to all Members, (B) modify the limited liability of any Member or

liabilities of such Member hereunder, (C) otherwise materially and adversely affect a holder in a manner materially disproportionate to any other holder of units or remove a right or privilege of any Member (other than amendments, modifications and waivers necessary to implement the terms of the Charter, permitting substitution or admission of Members) or (D) alter or change any rights, preferences or privileges of any units in a manner that is different or prejudicial relative to any other units in the same class of unit or materially and adversely affect the rights of any Member, in each case without the written consent of such Member or holder of units.

Tax Receivable Agreement

As a result of our organizational structure, we expect to obtain:

- Existing Basis;
- Basis Adjustments;
- Blocker Tax Attributes; and
- Interest Deductions.

The parties intend to treat each redemption or exchange of Common Units pursuant to the Tax Receivable Agreement as our direct purchase of Common Units from a Continuing Equity Owner for our tax purposes, for income and other applicable tax purposes, regardless of whether such Common Units are surrendered to the Continuing Equity Owner to XERC for redemption, or, to the extent there is cash available, in connection with a contemporaneous public offering or private sale of our Common Stock by us and we so agree, or directly to us. Any Existing Basis, Basis Adjustments, Blocker Tax Attributes and Interest Deductions will have the effect of reducing the amount of taxes that we would otherwise pay in the future to the extent of such tax savings. The Existing Basis, Basis Adjustments, Blocker Tax Attributes and Interest Deductions will also have the effect of decreasing gains (or increasing losses) on future dispositions of certain assets to the extent tax basis is reduced to those assets.

In connection with the closing of this offering, we will enter into a Tax Receivable Agreement with XERC and the TRA Holders. The Tax Receivable Agreement will provide for the payment by us to the TRA Holders of 85% of the amount of cash tax savings, if any, that we actually realize (or in some cases, that we are deemed

to realize) as a result of the Basis Adjustments, Existing Basis Blocker Tax Attributes and Interest Deductions, including those resulting from payments pursuant to the Tax Receivable Agreement. We and its applicable subsidiaries will have an election under Section 754 of the U.S. Internal Revenue Code of 1986, as amended (the “Code”), in effect for each taxable year in which a redemption or exchange of Common Units for our Class A common stock or cash occurs. Assuming no material changes in applicable tax law and that we earn sufficient taxable income to realize all tax benefits that are subject to the Tax Receivable Agreement, we expect that the tax savings associated with the (i) Basis Adjustments, (ii) Existing Basis, (iii) Blocker Tax Attributes and (iv) Interest Deductions would aggregate to approximately \$ 1.5 million over 5 years from the date of this offering based on a \$ 1.50 per share of our Class A common stock and assuming all future redemptions or exchanges would occur at the price of this offering at the same assumed price per share. Under such scenario, assuming future payments are made on the due date (with extension) of each relevant U.S. federal income tax return, we would be able to pay approximately 85% of such amount, or approximately \$ 1.275 million, over the 5 years from the date of this offering, and we would benefit from the remaining 15% of the tax benefits. The Tax Receivable Agreement payments are not conditioned upon any continued ownership interest in the company or us by any TRA Holder. The rights of each TRA Holder under the Tax Receivable Agreement

assignable regardless of whether the underlying Common Units are also assigned. In general, TRA Holders' rights under the Tax Receivable Agreement may not be assigned, sold, pledged, transferred or otherwise alienated to any person, other than certain permitted transferees, without our consent and without becoming a party to the Tax Receivable Agreement and agreeing to succeed to the applicable obligations and interest therein.

The actual Basis Adjustments, Existing Basis, Blocker Tax Attributes and Interest Deductions will vary depending on as any amounts paid to the TRA Holders under the Tax Receivable Agreement, will vary depending on a number of factors, including:

- the price of our Class A common stock at the time of redemptions or exchanges — Basis Adjustments and Blocker Tax Attributes, as well as any related increase in any tax deductions, will be directly related to the price of our Class A common stock at the time of each redemption or exchange;
- the timing of any subsequent redemptions or exchanges — for instance, the amount of tax deductions will vary depending on the fair value, which may fluctuate over time, of the tangible and intangible or amortizable assets of XERC and certain of its direct and indirect subsidiaries at the time of the redemption, exchange or distribution (or deemed distribution) as well as the amount of the existing tax basis at the time of such redemption, exchange or distribution (or deemed distribution);
- the extent to which such redemptions or exchanges are taxable — if a redemption or exchange is not taxable for any reason, certain of the increased tax deductions will not be available;
- the extent to which such Basis Adjustments and Blocker Tax Attributes are immediately deductible — we may be permitted to immediately expense a portion of the Basis Adjustments and Blocker Tax Attributes attributable to a redemption or exchange, which could significantly accelerate the realization of our realization of the associated tax benefits. Under the XERC LLC Agreement, the decision as to whether to immediately expense such Basis Adjustments and Blocker Tax Attributes will be made in our sole discretion; and
- the amount and timing of our income — the Tax Receivable Agreement generally will require us to pay 80% of the amount of cash tax savings as and when such cash tax savings are realized under the terms of the Tax Receivable Agreement. If we do not have sufficient cash flow or income to realize any of the applicable tax benefits, we generally will not be required to do so in circumstances requiring an early termination payment or in the event of a change of control or Material Breach requiring the use of certain calculation assumptions) to make payments under the Tax Receivable Agreement for that taxable year because no tax benefits will have been realized. However, any tax benefits that do not result in realized tax benefits in a given year may generate tax attributes that may be used to generate tax benefits in previous or subsequent taxable years. The use of any such tax attributes will result in payments under the Tax Receivable Agreement.

For purposes of the Tax Receivable Agreement, cash savings in income taxes will be determined by comparing our actual income tax liability to the amount of such taxes that we would have been required to pay had there been no Basis Adjustments, Existing Basis, Blocker Tax Attributes and Interest Deductions.

provided that, for purposes of determining cash savings with respect to state and local income taxes, the assumed tax rate will be used. The Tax Receivable Agreement will generally apply to our taxable years, beginning with the first taxable year ending after the completion of this offering and continuing for a maximum term for the Tax Receivable Agreement, although, as discussed further below, the Tax Receivable Agreement may be terminated by us pursuant to an early termination procedure that requires us to elect, to pay the TRA Holders an agreed upon amount equal to the estimated present value of

payments to be made under the agreement (calculated based on certain assumptions, including rates and use of the Basis Adjustments, Existing Basis, Blocker Tax Attributes and Interest Ded

The payment obligations under the Tax Receivable Agreement are obligations of us and our subsidiaries. Although the actual timing and amount of any payments that may be made under the Tax Receivable Agreement will vary, it is expected that the payments that we may be required to make to the TRA Holders could be substantial. Any payments made by us to the TRA Holders under the Tax Receivable Agreement will generally reduce the amount of overall cash flow that might have otherwise been available to us to the extent that we are unable to make payments under the Tax Receivable Agreement for any taxable year. Unpaid amounts generally will be deferred and will accrue interest until paid by us; provided, that nonpayment for a specified period may constitute a Material Breach under the Tax Receivable Agreement and, therefore, the future payments under the Tax Receivable Agreement for each taxable year following such Material Breach would be calculated utilizing certain deemed exchanges and valuation assumptions, which future payment could be substantial and in excess of the tax benefits we realize. We may fund our ordinary course payments under the Tax Receivable Agreement from cash flow from our operations, our subsidiaries, available cash or available borrowings under our existing credit facilities and other agreements.

Decisions made by us in the course of running our business, such as with respect to acquisitions, sales, other forms of business combinations or other changes in control, may influence the timing and amount of payments that we are required to make to a TRA Holder under the Tax Receivable Agreement. For example, the earlier disposition of assets following an exchange or acquisition transaction may accelerate payments under the Tax Receivable Agreement and increase the present value of such payments.

The Tax Receivable Agreement provides that if we elect an early termination of the Tax Receivable Agreement and if we further elect to make a single, lump sum payment, then our obligations under the Tax Receivable Agreement would accelerate and become payable, based on certain assumptions, including an assumption that we would have sufficient taxable income to fully use all potential future tax benefits that are subject to the Tax Receivable Agreement. In those circumstances, TRA Holders would be deemed to exchange any remaining outstanding Common Units for our Class A common stock and the TRA Holders generally would be entitled to payments under the Tax Receivable Agreement resulting from such deemed exchanges. We may elect to completely terminate the Tax Receivable Agreement early only with the written approval of a majority of our board of directors” (within the meaning of the rules of the Nasdaq). The amount due and payable under the Tax Receivable Agreement (i.e., the single lump sum payment option) is based on the present value (at a discount rate of SOFR plus 100 basis points) of projected future tax benefits that are based on certain assumptions, including an assumption that we would have sufficient taxable income to fully use all potential future tax benefits that are subject to the Tax Receivable Agreement.

Based on such assumptions, if we were to exercise our termination right immediately upon the consummation of this offering and elect to make a single lump sum payment, the aggregate amount of our termination payments would be approximately \$ million. Importantly, upon a change of control or a Material Breach, the Tax Receivable Agreement will not terminate nor will a single, accelerated payment be due. If we commit a Material Breach under the Tax Receivable Agreement, experience a change of control (as defined in the Tax Receivable Agreement) or elect an early termination of the Tax Receivable Agreement and do not elect the single, lump sum payment obligation, our (or our successors') future payments under the Tax Receivable Agreement for each taxable year after any such event will be calculated utilizing certain valuation assumptions, including that (i) any Common Units that have been exchanged are deemed exchanged for the market value of the shares of our Class A common stock at the time of the change of control and (ii) X-Energy, Inc. will have sufficient taxable income to fully utilize the tax attributes covered by the Tax Receivable Agreement. These future payments will be paid to the TRA Holders due in the taxable year in which the underlying tax attributes are deemed utilized by us (or our successors) according to the valuation assumptions.

As a result of the foregoing, we could, in the case of an early termination, be required to make an immediate cash payment, and in each case, any cash payments may possibly be significantly less than the actual realization, if any, of such future cash tax savings. We also could be required to make cash payments to the TRA Holders that are greater than 85% of the actual cash tax savings we ultimately realize in respect of the tax benefits that are subject to the Tax Receivable Agreement. In these situations, our obligations under the Tax Receivable Agreement could have a substantial negative impact on our liquidity and could have the effect of deferring or preventing certain mergers, asset sales, other form reorganizations, combinations, or other changes of control. There can be no assurance that we will be able to satisfy our obligations under the Tax Receivable Agreement.

Payments under the Tax Receivable Agreement will be based on the tax reporting positions we determine, which are complex and factual in nature, and the IRS or another taxing authority may challenge all or any part of the Basis Adjustments, Existing Basis Blocker Tax Attributes or Interest Deductions, as well as other tax positions that we take, and a court may sustain such a challenge. We will not be required to make any cash payments previously made to the TRA Holders pursuant to the Tax Receivable Agreement if any tax benefits initially claimed by us are subsequently challenged by a taxing authority and are disallowed. Instead, any excess cash payments made by us to a TRA Holder will be netted against future cash payments, if any, we might otherwise be required to make under the terms of the Tax Receivable Agreement to such TRA Holders. However, a challenge to any tax benefits initially claimed by us may arise for a number of years following the initial time of such payment or, even if challenged, an excess cash payment may be greater than the amount of future cash payments, if any, we might otherwise be required to make under the terms of the Tax Receivable Agreement and, as a result, there may not be sufficient future cash payments from which to net against. As a result, it is possible that we could make cash payments under the Tax Receivable Agreement that are substantially greater than 85% of our actual cash tax savings.

We will have full responsibility for, and sole discretion over, all our and XERC's tax matters, including the filing and amendment of all tax returns and claims for refund and defense of all tax controversies, and certain participation and approval rights held by certain TRA Holders. If the outcome of any challenge to all or part of the Basis Adjustments, Existing Basis, Blocker Tax Attributes, Interest Deduction, or other tax benefits we claim would reasonably be expected to materially affect a TRA Holder's rights or obligations under the Tax Receivable Agreement, then we will not be permitted to settle such challenge without the consent (not to be unreasonably withheld or delayed) of certain TRA Holders. The interests of such TRA Holders in any such challenge may differ from or conflict with our and our investors' interests. We will not allow TRA Holders to exercise their consent rights relating to any such challenge in a manner that would be inconsistent with our and our investors' interests.

Under the Tax Receivable Agreement, we are required to provide each TRA Holder with a copy of our interest in the Tax Receivable Agreement and to which a tax benefit or detriment is attributable. We will provide a schedule showing the calculation of payments that are due under the Tax Receivable Agreement to each taxable year with respect to which a payment obligation to such holder arises within the period of 45 and fifty (150) days after filing our U.S. federal income tax return for such taxable year. The amount of such payments will be based upon the advice of our tax advisors. Payments are generally due under the Tax Receivable Agreement within a specified period of time following the filing of our tax return for the taxable year with respect to which the payment obligation arises, although interest on such payments will begin to accrue at a rate of SOFR plus 100 basis points from the due date (without extensions) of such tax return. If payments that may be made under the Tax Receivable Agreement will continue to accrue interest at a rate of SOFR plus 500 basis points until such payments are made, including any late payments that we may subsequently make because we did not have enough available cash to satisfy our payment obligations at the time at which they originally arose.

Registration Rights Agreement

We intend to enter into a registration rights agreement with certain X-energy Members (“Registration Rights Agreement”) in connection with this offering. The Registration Rights Agreement provides certain X-energy Members and certain of their affiliates with certain demand registration rights, including shelf registration rights, in respect of any shares of our Class A common stock held by them, subject to certain conditions. In addition, in the event that we register additional shares of Class A common stock for sale to the public following the completion of this offering, we will be required to include such registration to certain X-energy Members and certain of their affiliates, and, subject to certain limitations, include shares of Class A common stock held by them in such registration. The agreement also includes customary indemnification provisions in favor of certain X-energy Members and certain of their affiliates and related parties against certain losses and liabilities (including reasonable costs of attorneys’ fees and legal expenses) arising out of or based upon any filing or other disclosure made by us in connection with securities laws relating to any such registration.

Historical Transactions with Affiliates

In the normal course of business, we enter into transactions with related parties in which certain of our affiliates hold financial interests, which are described in more detail below.

Our Relationship with Intuitive Machines

Kamal Ghaffarian, the founder and executive chairman of X-energy, is a co-founder and executive chairman of Intuitive Machines, Inc. (“Intuitive Machines”). In 2021, X-energy and Intuitive Machines formed a joint venture, IX, LLC, to pursue nuclear space propulsion and surface power systems in support of future space exploration goals. Intuitive Machines currently holds 510 membership unit interests in IX, LLC, representing 51% of the issued and outstanding membership unit interests of IX, LLC. X-energy currently holds 490 membership unit interests of IX, LLC, representing 49% of the issued and outstanding membership unit interests of IX, LLC. During the years ended December 31, 2025 and 2024, the total costs associated with this joint venture were \$0.8 million and \$0.2 million, direct costs were \$0.1 million and \$0.1 million, and general and administrative expenses were \$0.2 million and \$0.1 million, respectively. As of the date of this prospectus, X-energy and Intuitive Machines are in the process of dissolving IX, LLC. The parties expect the dissolution of IX, LLC to be completed by June 30, 2026.

Our Relationship with IBX

Kamal Ghaffarian, the founder and executive chairman of X-energy, is a co-founder and executive member of management of IBX, LLC (“IBX”), an investment firm. X-energy relies on IBX for a variety of general and administrative services in the day-to-day operation of its business. These expenses include, among others, fees for the provision of capital raise and proposal services. As such, expenses incurred in relation to IBX are incurred in the normal course of business and amounts are settled under normal commercial terms. For the fiscal years ended December 31, 2025 and 2024, these costs were \$0.2 million and \$0.8 million, respectively.

Our Relationship with IBX Thompson

Kamal Ghaffarian, the founder and executive chairman of X-energy, is a co-founder and executive member of management of IBX 801 Thompson Realty, LLC (“IBX Thompson”). The Company has entered into an agreement with IBX Thompson for general and administrative services. For the fiscal years ended December 31, 2025 and 2024, these costs were \$0.4 million and \$1.2 million, respectively.

Our Professional Services Agreement with Kamal Ghaffarian

In January 2020, X-energy entered into a professional services agreement with Kamal C founder and executive chairman, pursuant to which Mr. Ghaffarian agreed to provide consulting and advising services to X-energy for a period of three years beginning January fixed fee of \$0.08 million per quarter. Beginning January 2022, the fee was increased to \$0.1 quarter. The agreement shall continue in effect until the first of the following to occur: (a) X-e Mr. Ghaffarian that the services have been completed, (b) the sum of all awarded task or reached and has not

been increased in accordance with the agreement or (c) the agreement has been terminated i with its terms. X-energy extended this agreement for an additional one year terms on January and 2025. For the fiscal years ended December 31, 2025 and 2024, the Company incurred \$0. \$0.4 million, respectively, of consulting fees in connection with this agreement. The C Dr. Ghaffarian plan to terminate this agreement after the offering.

Our Agreement with Aerotherm

Martin Van Staden, our Senior Vice President and Chief Technology Officer, is the presi and principal shareholder of Aerotherm, Inc. (“Aerotherm”). X-energy entered into a contra consulting services from Aerotherm. During the years ended December 31, 2025 and 2024, incurred subcontracting costs to Aerotherm, Inc. in the amount of \$0.4 million and respectively, for these services.

Our Agreement with Hatch

Hatch Associates Consultants, Inc (“Hatch”) is a unitholder of XERC. The Company en agreement to receive support services related to its U.S. and international engineering and activities, including services related to the DOE ARDP from Hatch. The services include, but a to, project management, quality assurance management, cost control management and management. During the years ended December 31, 2025, and 2024, the Company incurred c in the amount of \$4.4 million and \$7.4 million, respectively, for these services. As of Decem \$4.5 million due to Hatch Associates Consultants, Inc. was included within our financial state to related parties.

Our Agreement with Zachry Nuclear

Zachry Nuclear Engineering, Inc (“Zachry Nuclear”) is a unitholder of XERC. The Com into an agreement to receive support services related to the DOE ARDP from Zachry Nuclear include, but are not limited to, site design and pre-construction activities, such as safety analysis support, engineering management and oversight, and system design services. Dur ended December 31, 2025 and 2024, the Company incurred costs to Zachry Nuclear of a amount and \$1.0 million, respectively, for these services. As of December 31, 2024, an imma due to Zachry Nuclear was included within our financial statements as due to related parties.

Our Agreement with Burns & McDonnell

Burns & McDonnell Engineering Company, Inc (“Burns & McDonnell”) is a unitholder o Company entered into an agreement to receive support services related to the DOE ARDP f McDonnell. The services include, but are not limited to, preliminary design support and pre support. During the years ended December 31, 2025 and 2024, the Company incurred cost McDonnell in the amount of \$1.1 million and \$2.8 million, respectively, for these ser

December 31, 2024, \$0.1 million due to Burns & McDonnell was included within our financial statements as due to related parties.

Our Relationship with Curtiss-Wright Corporation

Curtiss-Wright Corporation is a unitholder of XERC. During the years ended December 31, 2025 and 2024, the Company incurred subcontracting costs to Curtiss-Wright Corporation, Inc. in the amount of \$11.6 million and \$4.2 million, respectively, related to the DOE ARDP. As of December 31, 2025 and 2024, \$0.5 million was due to Curtiss-Wright Corporation, Inc. and was included within our financial statements as due to related parties.

Our Relationship with The Dow Chemical Company

Dow is a unitholder and warrant holder of XERC. During 2022, the Company entered into a binding letter of intent with Dow to develop a small modular reactor at one of Dow's U.S. Gulf Coast sites. In the first quarter of 2023, the Company entered into a joint development agreement with Dow. During 2023, the Company entered into a master project development agreement and a commercial agreement with Dow. During the years ended December 31, 2025 and 2024, the Company incurred costs to Dow in the amount of \$8.6 million and \$5.8 million, respectively. As of December 31, 2025 and 2024, \$6.2 million and \$2.3 million due to Dow was included within our financial statements as due to related parties. For the years ended December 31, 2025 and 2024, the revenues associated with these contracts are \$6.9 million and \$26.8 million, respectively. As of December 31, 2025 and 2024, \$4.6 million and \$15.8 million due to Dow was included within our financial statements as due from related parties, respectively.

Our Relationship with Amazon

Amazon.com NV Investment Holdings LLC ("Amazon") is a unitholder and warrant holder of XERC. In 2024, the Company entered into a project with Amazon to provide design and engineering services. During the years ended December 31, 2025 and 2024, the Company recorded \$0.5 million and \$0.6 million within revenue associated with this project, respectively. As of December 31, 2025 and 2024, \$0.5 million and \$0.6 million associated with deferred revenue for this project was included within our financial statements as due from related parties, respectively.

Our Relationship with Centrica

Centrica is the largest provider of energy services across the U.K. and owns a 20% stake in a fleet of operating nuclear reactors in the country. In September 2025, X-energy and Centrica signed a Joint Development Agreement (JDA) dedicated to building and operating Xe-100 reactors in the U.K. Securing appropriate permissions and licenses, the first electricity generation is expected in the mid-2030s.

Our Agreement with Desbuild Construction

Nick Ghaffarian, the brother of Kam Ghaffarian, is the owner of Desbuild Construction ("Desbuild Construction"). During 2022, X-energy engaged Desbuild Construction for construction services for X-energy's training facility in Frederick, Maryland (the "Frederick Facility"). The services included, but were not limited to, demolition and design services for the build out of the Frederick Facility. In addition, in January 2023, X-energy entered into another agreement with Desbuild Construction to which Desbuild Construction agreed to remodel the Frederick Facility for a fixed fee of \$1.5 million.

energy makes payments in installments to Desbuild Construction upon completion of the work specified in the agreement. For the fiscal years ended December 31, 2025 and 2024, the Company incurred costs of \$0.6 million and \$0.7 million, respectively, associated with these services as leasehold improvements, property and equipment, net. Refer to Note 5 — Government Grants for additional information regarding grants received for these costs.

Our Relationship with Chris Ginther

Christopher F. Ginther, a member of X-energy's board of directors, was an Executive Vice President of OPG. In 2022, X-energy and OPG entered into a framework agreement to work exclusively with each other with respect to advanced nuclear power industrial applications in Ontario, Canada, and to develop and advance the Xe-100 as the nuclear technology of choice for industrial applications through OPG will be the operator of any Xe-100 facilities that are deployed under this agreement. OPG is an equity investor in X-energy.

We also have a project management consulting agreement with Outpost Consultants, in which OPG is an owner.

Director and Officer Indemnification and Insurance

Prior to the consummation of this offering, we intend to enter into separate indemnification agreements with each of our directors and executive officers. We have also purchased directors' and officers' liability insurance. See "Description of Capital Stock — Limitations on Liability and Indemnification of Directors."

Policies and Procedures for Related Persons Transactions

The Board of X-Energy, Inc. will adopt a formal written related person transaction policy effective upon the Closing. Such policy will provide that X-energy's executive officers, directors, nominees, any member of the immediate family of any of the foregoing persons, or a security holder, shall not, without the approval of X-energy's audit committee, subject to certain exceptions, enter into a "related person transaction" (as defined under Item 404 of Regulation S-K) without the approval of X-energy's audit committee, subject to certain exceptions.

PRINCIPAL STOCKHOLDERS

The following table sets forth information regarding the beneficial ownership of our Class A common stock and Class B common stock by (1) each person known to us to beneficially own more than 1% of our voting securities, (2) each of our directors, (3) each of our named executive officers and (4) all of our named executive officers as a group.

The number of shares of Class A common stock and Class B common stock outstanding as of the date of beneficial ownership before and after this offering are based on (i) the number of shares of Class A Units to be issued and outstanding immediately prior to and after the consummation of this offering, giving effect to the Transactions and (ii) an assumed initial public offering price of \$10.00 (the midpoint of the estimated price range set forth on the cover page of this prospectus). See “Capital Structure.”

Beneficial ownership of Class A common stock and Class B common stock is determined in accordance with the rules of the SEC. In accordance with the rules of the SEC, beneficial ownership includes investment power with respect to securities and includes shares issuable pursuant to exchangeable securities rights that are exercisable within 60 days of the date of this prospectus.

To our knowledge, except as indicated in the footnotes to this table and pursuant to applicable community property laws, the persons named in the table have sole voting and investment power with respect to all shares of common stock.

Name of Beneficial Owner ⁽¹⁾	Class A common stock Beneficially Owned						Class B Beneficially Owned	
	Beneficially Owned Prior to the Offering		After the Offering Assuming Underwriters' Option is Not Exercised		After the Offering Assuming Underwriters' Option is Exercised in Full		Beneficially Owned Prior to the Offering	
	Number	%	Number	%	Number	%	Number	%
Greater than 5% Stockholders				%		%		
Named Executive Officers:								
J. Clay Sell								
Directors:								
Dr. Kamal Ghaffarian								
Christopher F. Ginther								
Gregory J. Goff								
Kathleen W. Hyle								
David Kaplan								
Allyson Satin								
Edward Sonnenschein								
Michael J. Wallace								
Directors and executive officers as a group (14 persons)				%		%		

* Less than 1% of common stock outstanding.

(1) Unless otherwise indicated in the below, the address of each of the individuals named above is:

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DESCRIPTION OF CAPITAL STOCK

The following is a description of the material terms of, and is qualified in its entirety by, the company's charter, articles of association, and restated certificate of incorporation and amended and restated bylaws, each of which will be filed with the SEC upon the consummation of this offering, the forms of which are filed as exhibits to the registration statement, of which this prospectus is a part.

Our purpose is to engage in any lawful act or activity for which corporations may now or hereafter be organized under the DGCL. Upon consummation of this offering, our authorized capital stock will consist of shares of common stock, par value \$0.0001 per share, and shares of preferred stock. Following the completion of this offering, there are expected to be outstanding shares of common stock.

Common Stock

Class A common stock

Voting rights. Each holder of Class A common stock is entitled to one vote for each share of Class A common stock held of record in person, virtually or by proxy on all matters submitted to the holders of Class A common stock, whether voting separately as a class or otherwise.

Dividend rights. Subject to applicable law and the rights and preferences of any other outstanding series of Preferred Stock or any class or series of stock having a preference over or the right to participate with the Class A common stock with respect to the payment of dividends, holders of Class A common stock, as such, shall be entitled to the payment of dividends on the Class A common stock and if declared by the Board in accordance with applicable law.

The payment of future dividends on the shares of Class A common stock will depend on the condition of the Company after the completion of this offering, and subject to the discretion of the Board. The Company does not guarantee that cash dividends will be declared. The ability of the Company to declare dividends may be limited by the terms and conditions of other financing and other agreements entered into by or on behalf of the Company from time to time.

Rights upon liquidation. In the event of liquidation, dissolution or winding up of the affairs of the Company, whether voluntary or involuntary, after payment or provision for payment of the debts and other liabilities of the Company after making provisions for preferential and other amounts, if any, to which the holders of Preferred Stock or any class or series of stock having a preference over or the right to participate with the Class A common stock with respect to payments in liquidation shall be entitled, the remaining assets and funds of the Company for distribution shall be divided among and paid ratably to the holders of all outstanding shares of Class A common stock in proportion to the number of shares held by each such stockholder.

Other rights. The holders of Class A common stock have no pre-emptive or conversion or subscription rights. There are no redemption or sinking fund provisions applicable to the Class A common stock. The rights, preferences and privileges of holders of the Class A common stock will be subject to the rights of the holders of any shares of the Preferred Stock that may issue in the future.

Class B common stock

Voting rights. Each holder of Class B common stock is entitled to one vote for each share of Class B common stock held of record in person, virtually or by proxy on all matters submitted to the holders of Class B common stock, whether voting separately as a class or otherwise.

Dividend rights. Other than in connection with a dividend declared by the Board in connection with a “poison pill” or similar stockholder rights plan, dividends shall not be declared or paid on the shares of Class B common stock and the holders of shares of Class B common stock shall have no right to receive dividends in respect of such shares of Class B common stock.

Rights upon liquidation. Each holder of shares of Class B common stock shall be entitled to receive \$0.0001 per share of Class B common stock owned of record by such holder on the record date for distribution, and upon receiving such amount, the holders of shares of Class B common stock, in their capacity as such, shall not be entitled to receive any other assets or funds of the Company.

Permitted Transfers. From and after the effectiveness of the Proposed Certificate of Incorporation (the “Charter Effective Time”), shares of Class B common stock may be issued only to, and registered in the name of, the X-energy Members, their respective successors and assigns and their respective

transferees (the X-energy Members, together with all such subsequent successors, assigns and transferees, collectively, the “Permitted Class B Owners”), and the aggregate number of shares of common stock at any time registered in the name of each such Permitted Class B Owner must not exceed the aggregate number of X-energy Common Units held of record at such time by such Permitted Class B Owner under the XERC LLC Agreement. A Permitted Class B Owner may transfer or assign such Class B common stock (or any legal or beneficial interest in such shares) (directly or indirectly, in whole or in part, and by operation of law) only to a permitted transferee of such holder or to a non-permitted transferee, with the prior approval in advance and in writing by X-energy, and only if such holder also simultaneously transfers to the non-permitted transferee, as applicable, in compliance with the XERC LLC Agreement. Permitted transfers include transfers pursuant to certain redemption or direct exchange scenarios, a transfer by a Permitted Class B Holder to the Company or any of its subsidiaries, or to an affiliate of such Permitted Class B Holder, in each case subject to and in compliance with the XERC LLC Agreement.

Cancellation of Class B Common Stock. A holder of Class B common stock may surrender such Class B common stock to X-energy for cancellation for no consideration at any time. Shares of Class B common stock are automatically transferred to X-Energy, Inc. upon the redemption or exchange of such Common Units pursuant to the terms of the XERC LLC Agreement and will be canceled and the shares will be reissued. Following the surrender or other acquisition of any shares of Class B common stock, whether by or for X-energy, X-energy will take all actions necessary to cancel and retire such shares and the shares shall not be re-issued by X-energy.

Permitted Transfers. The Board (including a majority of the directors who are disinterested in the transaction with respect to the relevant transaction serving on the Board at such time) may, to the extent permitted by law, from time to time establish, modify, amend or rescind, by bylaw or otherwise, regulations and restrictions that are not inconsistent with the provisions described for determining whether any transfer or acquisition of Class B common stock would violate the restrictions described and for the orderly administration and implementation of the provisions of the Proposed Certificate of Incorporation.

Preferred Stock

The Proposed Certificate of Incorporation will authorize the board of directors of to establish one or more series of Preferred Stock. Unless required by law or any stock exchange, the authority to issue Preferred Stock will be available for issuance without further action by the holders of Common Stock.

The Board has the discretion to determine the powers, preferences and relative, participating, and other special rights, including voting rights, dividend rights, conversion rights, redemption rights, and liquidation preferences, of each series of Preferred Stock. The issuance of Preferred Stock may have the effect of delaying, deferring or preventing a change in control of without further action by the holders of Common Stock. Additionally, the issuance of Preferred Stock may adversely affect the holders of the Common Stock by restricting dividends on the Common Stock, diluting the voting power of the Common Stock, and subordinating the liquidation rights of the Common Stock. As a result of these or other factors, the issuance of Preferred Stock could have an adverse impact on the market price of the Common Stock.

Anti-Takeover Effects of Our Amended and Restated Certificate of Incorporation and Amended and Restated Bylaws and Certain Provisions of Delaware Law

The provisions of the Proposed Certificate of Incorporation, the Proposed Bylaws and the Proposed Restated Certificate of Incorporation and Restated Bylaws summarized below may have an anti-takeover effect and may delay, defer or prevent a takeover.

attempt that you might consider in your best interest, including an attempt that might result in the payment of a premium over the market price for your shares of Class A common stock.

The Proposed Certificate of Incorporation and Proposed Bylaws contain certain provisions intended to enhance the likelihood of continuity and stability in the composition of the Board of Directors. These provisions may have the effect of delaying, deferring or preventing a future takeover or change in control of the company if such takeover or change in control is approved by such board of directors.

These provisions include:

- *Authorized but Unissued Capital Stock.* Our authorized but unissued shares of Common Stock and Preferred Stock will be available for future issuance without stockholder approval. These shares may be utilized for a variety of corporate purposes, including future public offerings, raising additional capital, corporate acquisitions and employee benefit plans. The existence of a large number of authorized but unissued shares of Common Stock and Preferred Stock could render more difficult or more costly an attempt to obtain control of a majority of common stock by means of a proxy contest, tender offer, merger or otherwise.
- *Director Designees; Classes of Directors.* Pursuant to the Proposed Certificate of Incorporation, the directors of X-energy will be divided into three classes, with each class serving staggered terms. The existence of a classified board of directors could discourage a third party from making a tender offer or otherwise attempting to obtain control of us as it is more difficult and time consuming for stockholders to replace a majority of the directors on a classified board of directors.
- *No Cumulative Voting for Directors.* The DGCL provides that stockholders are not permitted to cumulate votes in the election of directors unless a corporation's certificate of incorporation provides otherwise. The Proposed Certificate of Incorporation does not provide for cumulative voting. As a result, the holders of shares of Common Stock representing a majority of the voting power of the outstanding shares of our capital stock of will be able to elect all of the directors for election.
- *Quorum.* The Proposed Bylaws will provide that at any meeting of the Board, a majority of the number of directors then in office constitutes a quorum for the transaction of business.
- *Action by Written Consent.* The Proposed Certificate of Incorporation provides that any action required or permitted to be taken by the stockholders of must be effected at a duly called meeting or special meeting of the stockholders of (and may not be taken by consent of the stockholders of a meeting). In addition to the foregoing, any action required or permitted to be taken by the holders of any series of Preferred Stock, voting separately as a series or separately as a class, or by the holders of one or more other such series, may be taken without a meeting, without prior notice and without a vote, to the extent expressly so provided by the applicable certificate of designation for that series of Preferred Stock, if a consent or consents in writing, setting forth the action so taken, shall be signed by the holders of outstanding shares of the relevant series of Preferred Stock representing at least the minimum number of votes that would be necessary to authorize or take such action at a meeting at which all shares entitled to vote thereon were present and voted and shall be filed in accordance with the applicable provisions of the DGCL.
- *Special Meetings of Stockholders.* Subject to the special rights of the holders of one or more series of Preferred Stock, special meetings of the stockholders of may be called, for any purpose, at any time only by or at the direction of (i) the board of directors, the chairman of the board of directors, the chief executive officer or president, and (ii) special meetings of the stockholders of X-energy may not be called by the stockholders of X-energy or any other person.
- *Advance Notice Procedures.* The Proposed Bylaws establish an advance notice procedure for stockholder proposals to be brought before an annual meeting of our stockholders and for stockholder nominations of persons for election to the board of directors of X-energy. Stockholders must submit proposals or nominations before an annual or special meeting of stockholders. Stockholders at an annual meeting must submit proposals or nominations before the meeting. Stockholders are not able to consider proposals or nominations specified in the notice of meeting or brought forward at a meeting.

meeting by or at the direction of the board of directors of or by a stockholder who was of record on the record date for the meeting, who is entitled to vote at the meeting and the secretary

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of X-energy timely written notice, in proper form, of the stockholder's intention business or nomination before the meeting. Although the Proposed Bylaws do not give directors the power to approve or disapprove stockholder nominations of candidates regarding other business to be conducted at a special or annual meeting, as applicable, Bylaws may have the effect of precluding the conduct of certain business at a meeting procedures are not followed or may discourage or deter a potential acquirer from solicitation of proxies to elect its own slate of directors or otherwise attempting to obtain X-energy.

Limitations on Liability and Indemnification of Officers and Directors

The DGCL authorizes corporations to limit or eliminate the personal liability of directors to corporations and their stockholders for monetary damages for breaches of directors' fiduciary duties, subject to certain exceptions. The Proposed Certificate of Incorporation provision that eliminates the personal liability of directors and officers for monetary damages for breach of fiduciary duty as a director or officer, except to the extent such exemption from liability thereof is not permitted under the DGCL. The effect of these provisions is to eliminate the personal liability of X-energy and its stockholders, through stockholders' derivative suits on X-energy's behalf, for monetary damages from a director or officer for breach of fiduciary duty as a director or officer for breaches resulting from grossly negligent behavior. However, exculpation does not apply to a director or officer if the director or officer has acted in bad faith, knowingly or intentionally violated the law, authorized illegal dividends or Redemptions or derived an improper benefit from such person's position as a director or officer.

The Proposed Bylaws provide that X-energy must indemnify and advance expenses to directors and officers to the fullest extent authorized by the DGCL. X-energy is also expressly authorizing directors' and officers' liability insurance providing indemnification for directors, officers and employees for some liabilities. X-energy believes that these indemnification and advancement provisions and insurance are useful to attract and retain qualified directors and executive officers.

The limitation of liability, indemnification and advancement provisions in the Proposed Certificate of Incorporation and the Proposed Bylaws may discourage stockholders from bringing a lawsuit against directors and officers for breach of their fiduciary duty. These provisions also may have the effect of reducing the likelihood of derivative litigation against directors and officers, even though such litigation, if successful, might otherwise benefit X-energy and its stockholders. Your investment may be affected to the extent X-energy pays the costs of settlement and damage awards against directors and officers pursuant to these indemnification provisions. X-energy believes that these provisions, along with directors' and officers' liability insurance and any indemnity agreements that may be entered into are necessary to attract and retain qualified and experienced directors and officers.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted under the Proposed Certificate of Incorporation and the Proposed Bylaws for directors, officers and controlling persons pursuant to the foregoing provisions, or otherwise, you are hereby advised that in the opinion of the SEC such indemnification is against public policy as expressed in the Securities Act and is, therefore, unenforceable.

There is currently no pending material litigation or proceeding involving any of our directors, officers or employees for which indemnification is sought.

Transfer Agent and Registrar

The Transfer Agent and registrar for the shares of Common Stock will be _____.

Listing

We intend to apply to have our common stock listed on Nasdaq under the symbol “XE.”

SHARES ELIGIBLE FOR FUTURE SALE

Prior to this offering, there has been no public market for shares of our Class A common stock. We cannot predict the effect, if any, future sales of shares of Class A common stock, or the amount of a future sale of shares of Class A common stock, will have on the market price of shares of our Class A common stock prevailing from time to time. Future sales of substantial amounts of our Class A common stock in the public market (including shares of Class A common stock issuable upon redemption of Common Units of our Continuing Equity Owners) or the perception that such sales may adversely affect market prices of our Class A common stock prevailing from time to time may impair our future ability to raise capital through the sale of our equity or equity-related securities at the time and price that we deem appropriate. Furthermore, there may be sales of substantial amounts of our Class A common stock in the public market after the existing legal and contractual restrictions lapse, which may adversely affect the prevailing market price and our ability to raise equity capital in the future. *Factors — Risks Related to the Offering and Ownership of our Class A common stock — Sales of a substantial amount of our Class A common stock in the public markets by our Continuing Equity Owners or security holders may cause the price of our Class A common stock to decline.”*

Upon completion of this offering, we will have a total of _____ shares of our Class A common stock outstanding. Of the outstanding shares, the shares sold in this offering (or shares if the underwriter exercises in full their over-allotment option) will be freely tradable without restriction or further registration under the Securities Act, except that any shares held by our affiliates, as that term is defined under the Securities Act, including our directors, executive officers and other affiliates (including our existing owners) will be sold only in compliance with the limitations described below.

In addition, each Common Unit held by our Continuing Equity Owners will be redeemable at the election of each Continuing Equity Owner, for, at our election (determined solely by our board of directors (within the meaning of Nasdaq rules) who are disinterested), newly issued shares of our Class A common stock on a one-for-one basis or a cash payment equal to a volume weighted average market price of one share of Class A common stock for each Common Unit so redeemed, in each case, in accordance with the terms of the XERC LLC Agreement; provided that, at our election (determined solely by our independent directors (within the meaning of Nasdaq rules) who are disinterested), we may elect to deliver in exchange by X-Energy, Inc. of such Class A common stock or such cash, as applicable, for such Common Units. The Continuing Equity Owners may, subject to certain exceptions, exercise such redemption rights as long as their Common Units remain outstanding. See “Certain Relationships and Related Transactions — XERC LLC Agreement.” Upon consummation of the Transactions, our Continuing Equity Owners will hold _____ Common Units, all of which will be exchangeable for shares of our Class A common stock.

common stock. The shares of Class A common stock we issue upon such exchanges would be “restricted securities” as defined in Rule 144 unless we register such issuances.

Lock-up Agreements

We, our officers, and directors, and holders of substantially all of our shares of capital stock and securities convertible into or exchangeable for shares of our capital stock outstanding upon completion of this offering have agreed or will agree that, without the prior written consent of J.P. Morgan LLC, we and they will not, subject to certain exceptions, during the period ending 180 days after the date of this prospectus:

(i) offer, sell, contract to sell, pledge, grant any option to purchase, lend or otherwise dispose of any shares of our Class A common stock, or any options or warrants to purchase any shares of our Class A common stock, or any securities convertible into, or exchangeable for, or that entitle the holder to receive, shares of our Class A common stock; or

(ii) engage in any hedging or other transaction or arrangement (including, without limitation, a short sale or the purchase or sale of, or entry into, any put or call option, or combination thereof, or forward, swap or any other derivative transaction or instrument, however described or designated, which is designed to, or which reasonably could be expected to lead to, or result in, a sale, loan, or other disposition of shares of our Class A common stock or any securities convertible into, or exercisable or exchangeable for, shares of our Class A common stock);

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for shares of our Class A common stock, whether any transaction described above is to be made by the delivery of our Class A common stock or such other securities, in cash or otherwise.

Upon the expiration of the applicable lock-up periods, substantially all of the shares subject to the lock-up restrictions will become eligible for sale, subject to the limitations discussed under the heading “Underwriting” for additional information.

In addition to the restrictions contained in the lock-up agreements described above, we have entered into agreements with certain security holders that contain market standoff provisions imposing restrictions on the ability of such security holders to offer, sell, or transfer our equity securities for a period of 180 days following the date of this prospectus.

Rule 144

In general, under Rule 144, as currently in effect, once we have been subject to public filing and reporting requirements for at least 90 days, a person (or persons whose shares are aggregated for purposes of Rule 144) who is deemed to be or have been one of our affiliates for purposes of the Securities Act at any time during the 12 months preceding a sale and who has beneficially owned the shares proposed to be sold for at least one year, including the holding period of any prior owner other than an affiliate, is entitled to sell such shares without complying with the manner of sale, volume limitation or notice provisions of Rule 144, provided that such person is in compliance with the public information requirements of Rule 144. If such a person has not been an affiliate, then such person is entitled to sell such shares without complying with any of the provisions of Rule 144.

In general, under Rule 144, as currently in effect, our affiliates or persons selling shares of our Class A common stock on behalf of our affiliates, who have met the six month holding period for purposes of Rule 144, and whose ownership of “restricted shares” of our Class A common stock, are entitled to sell upon the expiration of the six month holding period.

lock-up agreements described above, within any three-month period beginning 90 days after the date of the offering, a number of shares that does not exceed the greater of:

- 1% of the number of shares of our Class A common stock then outstanding, which number shall be determined as of approximately 90 days after this offering; or
- the average reported weekly trading volume of our Class A common stock on Nasdaq during the 90 calendar weeks preceding the filing of a notice on Form 144 with respect to such sale.

Sales under Rule 144 by our affiliates or persons selling shares on behalf of our affiliates shall be subject to certain manner of sale provisions and notice requirements and to the availability of public information about us. The sale of these shares, or the perception that sales will be made, could adversely affect the price of our Class A common stock after this offering because a great supply of shares would or would be perceived to be, available for sale in the public market.

We are unable to estimate the number of shares that will be sold under Rule 144 since this depends on the market price for our Class A common stock, the personal circumstances of the stockholders, and other factors.

Rule 701

In general, under Rule 701 as currently in effect, any of our employees, directors, officers, or advisors who received shares of our Class A common stock from us in connection with a stock or option plan or other written agreement before the effective date of this offering are entitled to sell such shares 90 days after the effective date of this offering in reliance on Rule 144, in the case of our employees, without having to comply with the holding period requirements of Rule 144 and, in the case of our affiliates, without having to comply with the public information, holding period, volume limitations, and filing requirements of Rule 144.

Registration Rights

See “Certain Relationships and Related Party Transactions — Registration Rights Agreements” for a description of these registration rights. If the sale of the shares is registered, the shares will be freely tradable without restriction under the Securities Act and a large number of shares may be sold into the public market.

of these shares is registered, the shares will be freely tradable without restriction under the Securities Act and a large number of shares may be sold into the public market.

Registration Statements on Form S-8

We intend to file with the SEC a registration statement on Form S-8 under the Securities Act covering the offer and sale of all shares of Class A common stock reserved for issuance under our 2014 ESPP.

Such registration statement is expected to be filed and become effective as soon as practicable after the completion of this offering. Accordingly, shares registered under such registration statement will be available for sale in the open market following its effective date, subject to certain Rule 144 requirements applicable to affiliates, vesting restrictions, and the lock-up agreements and market standstill obligations described above, if applicable.

MATERIAL U.S. FEDERAL INCOME TAX CONSEQUENCES TO NON-U.S. HOLDERS OF A COMMON STOCK

The following discussion is a summary of the material U.S. federal income tax consequences to Non-U.S. Holders (as defined below) of the purchase, ownership, and disposition of our Class A common stock issued pursuant to this offering, but does not purport to be a complete analysis of all potential U.S. federal income tax consequences. The effects of other U.S. federal tax laws, such as estate and gift tax laws, and any applicable state and local non-U.S. tax laws are not discussed. This discussion is based on the Code, Treasury Regulations, and other authorities promulgated thereunder, judicial decisions, and published rulings and administrative pronouncements of the IRS, in each case in effect as of the date hereof. These authorities may change or be subject to differing interpretations. Any such change or differing interpretation may be applied retroactively in a manner that could adversely affect a Non-U.S. Holder. We have not sought and will not seek any rulings from the IRS regarding the matters discussed below. There can be no assurance the IRS or a court will not take a position to that discussed below regarding the tax consequences of the purchase, ownership, and disposition of our Class A common stock.

This discussion is limited to Non-U.S. Holders that hold our Class A common stock as a “passive foreign investor” within the meaning of Section 1221 of the Code (generally, property held for investment). This discussion does not address all U.S. federal income tax consequences relevant to a Non-U.S. Holder in all circumstances, including the impact of the Medicare contribution tax on net investment income, the alternative minimum tax. In addition, it does not address consequences relevant to Non-U.S. Holders subject to special rules, including, without limitation:

- U.S. expatriates and former citizens or long-term residents of the U.S.;
- persons holding our Class A common stock as part of a hedge, straddle, or other non-investment strategy or as part of a conversion transaction or other integrated investment;
- holders of our X-energy Warrants, or any other types of our warrants;
- banks, insurance companies, and other financial institutions;
- brokers, dealers, or traders in securities;
- “controlled foreign corporations,” “foreign controlled foreign corporations,” “passive foreign investment companies,” and corporations that accumulate earnings to avoid U.S. federal income tax;
- partnerships or other entities or arrangements treated as partnerships for U.S. federal income tax purposes (and investors therein);
- tax-exempt organizations or governmental organizations;
- persons deemed to sell our Class A common stock under the constructive sale provision of the Code;
- persons who hold or receive our Class A common stock pursuant to the exercise of a stock option or otherwise as compensation;
- tax-qualified retirement plans;
- “qualified foreign pension funds” as defined in Section 897(l)(2) of the Code and entities with interests of which are held by qualified foreign pension funds; and
- persons subject to special tax accounting rules as a result of any item of gross income with respect to the stock being taken into account in an applicable financial statement.

If an entity or arrangement treated as a partnership for U.S. federal income tax purposes, Class A common stock, the tax treatment of a partner in the partnership will depend on the partner, the activities of the partnership, and certain determinations made at the partner level. Partnerships holding our Class A common stock and the partners in such partnerships should consult their tax advisors regarding the U.S. federal income tax consequences to them.

THIS DISCUSSION IS FOR INFORMATIONAL PURPOSES ONLY AND IS NOT TAX ADVICE. INVESTORS SHOULD CONSULT THEIR TAX ADVISORS WITH RESPECT TO THE APPLICATION OF THE U.S. FEDERAL INCOME TAX LAWS TO THEIR PARTICULAR SITUATIONS AS WELL AS THE TAX CONSEQUENCES OF THE PURCHASE, OWNERSHIP, AND

DISPOSITION OF OUR CLASS A COMMON STOCK ARISING UNDER THE U.S. FEDERAL INCOME TAX LAWS OR UNDER THE LAWS OF ANY STATE, LOCAL, OR NON-UNITED STATES JURISDICTION OR UNDER ANY APPLICABLE INCOME TAX TREATY.

Definition of a Non-U.S. Holder

For purposes of this discussion, a “Non-U.S. Holder” is any beneficial owner of our Class A common stock that is neither a “U.S. person” nor an entity treated as a partnership for U.S. federal income tax purposes. A U.S. person is any person that, for U.S. federal income tax purposes, is or is treated as being one of the following:

- an individual who is a citizen or resident of the U.S.;
- a corporation created or organized under the laws of the U.S., any state thereof, or the District of Columbia;
- an estate, the income of which is subject to U.S. federal income tax regardless of its source;
- a trust that (1) is subject to the primary supervision of a U.S. court and the control of one or more “United States persons” (within the meaning of Section 7701(a)(30) of the Code), or (2) has an election in effect to be treated as a U.S. person for U.S. federal income tax purposes.

Distributions

As described in the section entitled “Dividend Policy,” we do not anticipate declaring or paying dividends to holders of our Class A common stock in the foreseeable future. However, if we make distributions of cash or property on our Class A common stock, such distributions will constitute dividends for U.S. federal income tax purposes to the extent paid from our current or accumulated earnings and profits, as determined under U.S. federal income tax principles. Amounts not treated as dividends for U.S. federal income tax purposes will constitute a return of capital and first be applied against and reduce the U.S. Holder’s adjusted tax basis in its Class A common stock, but not below zero. Any excess will be treated as capital gain and will be treated as described below under “— Sale or Other Taxable Dispositions.”

Subject to the discussion below on effectively connected income, dividends paid to a Non-U.S. Holder will be subject to U.S. federal withholding tax at a rate of 30% of the gross amount of the dividends, or such lower rate specified by an applicable income tax treaty, provided the Non-U.S. Holder furnishes to us an IRS Form W-8BEN or W-8BEN-E (or other applicable documentation) certifying qualification for the reduced treaty rate). A Non-U.S. Holder that does not timely furnish the required documentation, but for which a reduced treaty rate may obtain a refund of any excess amounts withheld by timely filing an appropriate claim for refund with the IRS. Non-U.S. Holders should consult their tax advisors regarding their entitlement to benefits under any applicable income tax treaty.

If dividends paid to a Non-U.S. Holder are effectively connected with the Non-U.S. Holder's conduct of a trade or business within the U.S. (and, if required by an applicable income tax treaty, the Non-U.S. Holder maintains a permanent establishment in the U.S. to which such dividends are attributable), the Non-U.S. Holder will be exempt from the U.S. federal withholding tax described above. To claim this exemption, the Non-U.S. Holder must furnish to the applicable withholding agent a valid IRS Form W-8EPC that certifies that the dividends are effectively connected with the Non-U.S. Holder's conduct of a trade or business within the U.S.

Any such effectively connected dividends will be subject to U.S. federal income tax on a net income basis at the regular rates. A Non-U.S. Holder that is a corporation also may be subject to a branch profits tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on such effectively connected dividends, as adjusted for certain items. Non-U.S. Holders should consult their tax advisors regarding any applicable tax treaties that may provide for different rules.

Sale or Other Taxable Disposition

A Non-U.S. Holder will not be subject to U.S. federal income tax on any gain realized upon the sale or other taxable disposition of our Class A common stock unless:

- the gain is effectively connected with the Non-U.S. Holder's conduct of a trade or business within the U.S. (and, if required by an applicable income tax treaty, the Non-U.S. Holder maintains a permanent establishment in the U.S. to which such gain is attributable);

- the Non-U.S. Holder is a nonresident alien individual present in the U.S. for 183 days or more during the taxable year of the disposition and certain other requirements are met; or
- our Class A common stock constitutes a U.S. real property interest ("USRPI") by reason of our being a U.S. real property holding corporation ("USRPHC") for U.S. federal income tax purposes.

Gain described in the first bullet point above generally will be subject to U.S. federal income tax on a net income basis at the regular rates. A Non-U.S. Holder that is a corporation also may be subject to branch profits tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on such effectively connected gain, as adjusted for certain items.

A Non-U.S. Holder described in the second bullet point above will be subject to U.S. federal income tax at a rate of 30% (or such lower rate specified by an applicable income tax treaty) on gain realized upon the sale or other taxable disposition of our Class A common stock, which may be offset by capital losses of the Non-U.S. Holder (even though the individual is not considered a resident alien) provided the Non-U.S. Holder has timely filed U.S. federal income tax returns with respect to such gain.

With respect to the third bullet point above, we believe we currently are not, and do not expect to become, a USRPHC. Because the determination of whether we are a USRPHC depends, in part, on the fair market value of our USRPIs relative to the fair market value of our non-U.S. real property interests and other business assets, there can be no assurance we currently are not a USRPHC or will not become one in the future. Even if we are or were to become a USRPHC, gain arising from the sale or other taxable disposition of our Class A common stock by a Non-U.S. Holder will not be subject to U.S. federal income tax if our Class A common stock is "regularly traded," as defined by applicable Treasury Regulations, on an established securities market and such Non-U.S. Holder owned, actually and constructively, at least 10% of our Class A common stock throughout the shorter of the five-year period ending on the date of the sale or other taxable disposition or the Non-U.S. Holder's holding period.

Non-U.S. Holders should consult their tax advisors regarding potentially applicable income tax rules that may provide for different rules.

Information Reporting and Backup Withholding

Payments of dividends on our Class A common stock will not be subject to backup withholding provided the applicable withholding agent does not have actual knowledge or reason to know that the holder is a U.S. person and the holder either certifies its non-U.S. status, such as by furnishing a valid W-8BEN, W-8BEN-E, or W-8ECI, or otherwise establishes an exemption. However, information reporting is required to be filed with the IRS in connection with any distributions on our Class A common stock to the Non-U.S. Holder, regardless of whether such distributions constitute dividends or whether backup withholding is actually withheld. In addition, proceeds of the sale or other taxable disposition of our Class A common stock within the U.S. or conducted through certain U.S.-related brokers generally will not be subject to backup withholding or information reporting if the applicable withholding agent receives the certification described above and does not have actual knowledge or reason to know that such holder is a U.S. person and the holder otherwise establishes an exemption. Proceeds of a disposition of our Class A common stock conducted through a non-U.S. office of a non-U.S. broker generally will not be subject to backup withholding or information reporting.

Copies of information returns that are filed with the IRS may also be made available to the Non-U.S. Holder pursuant to the provisions of an applicable treaty or agreement to the tax authorities of the country in which the Non-U.S. Holder resides or is established.

Backup withholding is not an additional tax. Any amounts withheld under the backup withholding rules may be allowed as a refund or a credit against a Non-U.S. Holder's U.S. federal income tax liability, provided the required information is timely furnished to the IRS.

Additional Withholding Tax on Payments Made to Foreign Accounts

Withholding taxes may be imposed under Sections 1471 to 1474 of the Code (such Section 1471 referred to as the Foreign Account Tax Compliance Act, or "FATCA") on certain types of payments made to

non-U.S. financial institutions and certain other non-U.S. entities. Specifically, a 30% withholding tax may be imposed on dividends on, or (subject to the proposed Treasury Regulations discussed below) on the proceeds from the sale or other disposition of, our Class A common stock paid to a "foreign financial institution" or a "non-financial foreign entity" (each as defined in the Code), unless (1) the foreign financial institution undertakes certain diligence and reporting obligations, (2) the non-financial foreign entity certifies it does not have any "substantial United States owners" (as defined in the Code) and provides identifying information regarding each substantial U.S. owner, or (3) the foreign financial institution or non-financial foreign entity otherwise qualifies for an exemption from these rules. If the payee is a foreign financial institution and is subject to the diligence and reporting requirements in (1) above, it may enter into an agreement with the U.S. Department of the Treasury requiring, among other things, that it agree to identify accounts held by certain "specified United States persons" or "United States owned entities" (each as defined in the Code), annually report certain information about such accounts, and not withhold 30% on certain payments to non-compliant foreign financial institutions and certain non-U.S. holders. Foreign financial institutions located in jurisdictions that have an intergovernmental agreement with the U.S. governing FATCA may be subject to different rules.

Under the applicable Treasury Regulations and administrative guidance, withholding tax generally applies to payments of dividends on our Class A common stock. While withholding tax

would have applied also to payments of gross proceeds from the sale or other disposition of stock. On January 1, 2019, proposed Treasury Regulations eliminate FATCA withholding on payments of gross proceeds entirely. Taxpayers generally may rely on these proposed Treasury Regulations until the Regulations are issued.

Prospective investors should consult their tax advisors regarding the potential application of FATCA withholding under FATCA to their investment in our Class A common stock.

UNDERWRITING

We and the underwriters named below have entered into an underwriting agreement with respect to the offering of shares of Class A common stock described in this prospectus. J.P. Morgan Securities LLC, Morgan Stanley & Co. LLC, Jefferies LLC, and Moelis & Company LLC are acting as the representative underwriters. Subject to the terms and conditions of the underwriting agreement, each underwriter severally agreed to purchase, at the public offering price less the underwriting discounts and commissions set forth on the cover page of this prospectus, the number of shares of Class A common stock specified in the following table.

Name	Number
J.P. Morgan Securities LLC	
Morgan Stanley & Co. LLC	
Jefferies LLC	
Moelis & Company LLC	
Cantor Fitzgerald & Co.	
Guggenheim Securities, LLC	
Nomura Securities International, Inc. ⁽¹⁾	
TD Securities (USA) LLC	
UBS Securities LLC	
WR Securities, LLC ⁽¹⁾	
Total	

- (1) “Wolfe | Nomura Alliance” is the marketing name used by Wolfe Research Securities International, Inc. in connection with certain equity capital markets activities jointly by the firms. Both Nomura Securities International, Inc. and WR Securities, LLC are underwriters in the offering described herein. In addition, WR Securities, LLC and its affiliates may provide sales support services, investor feedback, investor education, and independent equity research services in connection with this offering.

The underwriters are committed to purchase all the shares of Class A common stock of us that they purchase any shares of Class A common stock. The underwriting agreement also provides that, in the event of an underwriter default, the purchase commitments of non-defaulting underwriters may also be increased. The offering may be terminated.

Shares of Class A common stock sold by the underwriters to the public will initially be at the initial public offering price set forth on the cover of this prospectus. Any shares of Class A common stock sold by the underwriters to securities dealers may be sold at a discount of up to \$ _____ per share below the initial public offering price. After the initial offering of the shares of Class A common stock, the representatives may change the offering price and the other selling terms. The sale of the shares of Class A common stock by the underwriters is subject to receipt and acceptance by us. We reserve to the underwriters’ right to reject any order in whole or in part. Sales of any shares of Class A common stock made outside of the U.S. may be made by affiliates of the underwriters.

To the extent that the underwriters sell more than _____ shares of Class A common stock in the offering, the underwriters have the option to purchase, exercisable within 30 days from the date of this prospectus, up to an additional _____ shares of Class A common stock from us at the public offering price, less the underwriting discounts and commissions. If any shares of Class A common stock are sold by the underwriters with this option to purchase additional shares of Class A common stock, the underwriters will purchase shares of Class A common stock in approximately the same proportion as shown in the table above. If any additional shares of Class A common stock are purchased, the underwriters will purchase additional shares on the same terms as those on which the shares are being offered.

The underwriting fee is equal to the public offering price per share less the amount payable to the underwriters to us per share. The following table shows the per-share and total underwriting discounts and commissions.

commissions to be paid to the underwriters assuming both no exercise and full exercise of the option to purchase additional shares of Class A common stock.

	Without option to purchase additional shares exercise
Per Share	\$
Total	\$

We estimate that the total expenses of this offering, including registration, filing and printing fees and legal and accounting expenses, but excluding the underwriting discounts and will be approximately \$. We have agreed to reimburse the underwriters for certain expenses in connection with this offering in amount not to exceed \$.

A prospectus in electronic format may be made available on the web sites maintained by underwriters, or selling group members, if any, participating in the offering. The underwriters allocate a number of shares of Class A common stock to underwriters and selling group members through their online brokerage account holders. Internet distributions will be allocated by the representative underwriters and selling group members that may make Internet distributions on the same basis as the allocations.

We have agreed that we will not (i) offer, pledge, sell, contract to sell, sell any option or contract to purchase, purchase any option or contract to sell, grant any option, right or warrant to purchase or otherwise transfer or dispose of, directly or indirectly, or submit to, or file with, the Securities and Exchange Commission a registration statement under the Securities Act relating to, any shares of our common stock or securities convertible into or exercisable or exchangeable for any shares of our common stock, or (ii) disclose the intention to make any offer, sale, pledge, loan, disposition or filing, or (ii) enter into or other arrangement that transfers all or a portion of the economic consequences associated with the ownership of any shares of common stock or any such other securities (regardless of whether such transactions are to be settled by the delivery of shares of common stock or such other securities or otherwise), in each case without the prior written consent of J.P. Morgan Securities LLC for a period of 30 days after the date of this prospectus, other than the shares of our common stock to be sold in the offering.

The restrictions on our actions, as described above, do not apply to certain transactions, including: (i) the issuance of shares of common stock or securities convertible into or exercisable for shares of common stock pursuant to the conversion or exchange of convertible or exchangeable securities or the exercise of warrants or options (including net exercise) or the settlement of restricted stock units (including net settlement), in each case outstanding on the date of the underwriting agreement described in this prospectus; (ii) grants of stock options, stock awards, restricted stock, RSUs, or other equity awards and the issuance of shares of our common stock or securities convertible into or exercisable or exchangeable for shares of our common stock (whether upon the exercise of stock options or otherwise) to our officers, directors, advisors, or consultants pursuant to the terms of an equity compensation plan in effect on the closing of this offering and described in this prospectus, provided that such recipients are bound by a lock-up agreement with the underwriters; or (iii) our filing of any registration statement under the Securities Act relating to securities granted or to be granted pursuant to any plan in effect on the date of the underwriting agreement and described in this prospectus or any assumed benefit plan pursuant to an acquisition or strategic transaction.

Our directors, officers, and certain of our stockholders (such persons, the “lock-up parties”), have entered into lock-up agreements with the underwriters prior to the commencement of this offering. Pursuant to which each lock-up party, with limited exceptions, for a period of 180 days after the date of this prospectus (such period, the “restricted period”), may not (and may not cause any of their direct or indirect affiliates to), without the prior written consent of J.P. Morgan Securities LLC, (1) offer, sell, contract to sell, sell any option or contract to purchase, purchase any option or contract to sell, grant any option, right or warrant to purchase, lend or otherwise transfer or dispose of, directly or indirectly, any shares of our common stock or any securities convertible into or exercisable or exchangeable for shares of common stock (including, without

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limitation, common stock or such other securities which may be deemed to be beneficially owned by the lock-up parties in accordance with the rules and regulations of the SEC and securities which may be received upon exercise of a stock option or warrant (collectively with the common stock, the “lock-up securities”). No person or entity (1) enter into any hedging, swap or other agreement or transaction that transfers, in whole or in part, the economic consequences of ownership of the lock-up securities, whether any such transaction is described in clause (1) or (2) above is to be settled by delivery of lock-up securities, in cash or otherwise, in response to any demand for, or exercise any right with respect to, the registration of any lock-up securities; (2) (4) publicly disclose the intention to do any of the foregoing. Such persons or entities have acknowledged that these undertakings preclude them from engaging in any hedging or other trading arrangements (including, without limitation, any short sale or the purchase or sale of, or entry into, any put or call option, or combination thereof, forward, swap or any other derivative transaction or arrangement, however described or defined) designed or intended, or which could reasonably be expected to result in, a sale or disposition or transfer (by any person or entity, whether or not a signatory to this agreement) of any economic consequences of ownership, in whole or in part, directly or indirectly, of the lock-up securities, whether any such transaction or arrangement (or instrument provided for in the offering) would be settled by delivery of lock-up securities, in cash or otherwise.

The restrictions described in the immediately preceding paragraph and contained in the lock-up agreements between the underwriters and the lock-up parties do not apply, subject in certain cases to the following conditions, to certain transactions, including (a) transfers of lock-up securities: (i) as bona fide gifts for bona fide estate planning purposes, (ii) by will or intestacy, (iii) to any trust for the direct or indirect benefit of the lock-up party or any immediate family member, (iv) to a partnership, limited liability company or other entity of which the lock-up party and its immediate family members are the legal and beneficial owners of all of the outstanding equity securities or similar interests, (v) to a nominee or custodian of the lock-up party or to an entity to whom a disposition or transfer would be permissible under clauses (i) through (iv), (vi) to any subsidiary of a corporation, partnership, limited liability company, trust or other business entity, (A) to any subsidiary of a corporation, partnership, limited liability company, trust or other business entity that is an affiliate of the lock-up party, or to any investment fund or other entity controlling, controlled by, managing or acting in concert with the lock-up party or its affiliates or (B) as part of a distribution to the lock-up party or its shareholders of the lock-up party; (vii) by operation of law, (viii) to us from an employee of the lock-up party upon disability or termination of employment of such employee, (ix) as part of a sale of lock-up securities to us acquired in open market transactions after the completion of this offering, (x) to us in connection with the vesting, settlement or exercise of restricted stock units, options, warrants or other rights to purchase shares of our common stock (including “net” or “cashless” exercise), including for the payment of dividends, interest and tax and remittance payments, or (xi) pursuant to a bona fide third-party tender offer or business combination or consolidation or other similar transaction approved by our board of directors and made to all holders of our common stock involving a change in control, provided that if such transaction is not completed, all such lock-up securities would remain subject to the restrictions in the immediately preceding paragraph; (b) exercise of options, warrants or settlement of RSUs or other equity awards, or the exercise of warrants granted pursuant to plans or arrangements described in this prospectus, provided that any lock-up securities received upon such exercise, vesting or settlement would be subject to restrictions similar to those in the immediately preceding paragraph; (c) the exercise of outstanding preferred stock, warrants to acquire preferred stock, or convertible securities which may be converted into our common stock or warrants to acquire shares of our common stock, provided that any common stock or warrant received upon such conversion would be subject to restrictions similar to those in the immediately preceding paragraph; and (d) the establishment by lock-up parties of trading plans under Rule 10b5-1 of the Exchange Act, provided that such plan does not provide for the transfer of lock-up securities during the restricted period.

J.P. Morgan Securities LLC, in its sole discretion, may release the securities subject to any lock-up agreements with the underwriters described above, in whole or in part at any time.

We have agreed to indemnify the underwriters against certain liabilities, including liabilities under the Securities Act of 1933.

We intend to apply to list our Class A common stock on Nasdaq under the symbol “XE.”

In connection with this offering, the underwriters may engage in stabilizing transactions that involve making bids for, purchasing and selling shares of Class A common stock in the open market for the purpose of preventing or retarding a decline in the market price of the Class A common stock following the offering is in

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progress. These stabilizing transactions may include making short sales of Class A common stock. Such a short sale involves the sale by the underwriters of a greater number of shares of Class A common stock than the underwriters are required to purchase in this offering, and purchasing shares of Class A common stock on the open market to cover positions created by short sales. Short sales may be “covered” shorts, which are short positions in an amount not greater than the underwriters’ option to purchase additional shares of Class A common stock as referred to above, or may be “naked” shorts, which are short positions in excess of that amount. The underwriters may close out any covered short position either by exercising their option to purchase additional shares of Class A common stock, in whole or in part, or by purchasing shares of Class A common stock in the open market. In making this determination, the underwriters will consider, among other things, the price of shares of Class A common stock available for purchase in the open market and the price at which the underwriters may purchase shares of Class A common stock through the exercise of their option to purchase additional shares of Class A common stock. A naked short position is more likely to be created if the underwriters are concerned that there may be downward pressure on the price of the Class A common stock in the open market that could adversely affect investors who purchase in this offering. If the underwriters create a naked short position, they will purchase shares of Class A common stock in the open market to cover the position.

The underwriters have advised us that, pursuant to Regulation M of the Securities Act of 1933, they may also engage in other activities that stabilize, maintain or otherwise affect the price of our Class A common stock, including the imposition of penalty bids. This means that if the representatives of the underwriters purchase Class A common stock in the open market in stabilizing transactions or in connection with short sales, the representatives can require the underwriters that sold those shares of Class A common stock as part of this offering to repay the underwriting discount received by them.

These activities may have the effect of raising or maintaining the market price of the Class A common stock or preventing or retarding a decline in the market price of the Class A common stock, and the price of the Class A common stock may be higher than the price that otherwise might exist in the open market. If the underwriters commence these activities, they may discontinue them at any time. The underwriters may carry out these transactions on Nasdaq, in the over the counter market or otherwise.

Prior to this offering, there has been no public market for our Class A common stock. The offering price will be determined by negotiations between us and the representatives. In determining the initial public offering price, we and the representatives of the underwriters expect to consider a number of factors including:

- the information set forth in this prospectus and otherwise available to the representative;
- our prospects and the history and prospects for the industry in which we compete;
- an assessment of our management;
- our prospects for future earnings;

- the general condition of the securities markets at the time of this offering;
- the recent market prices of, and demand for, publicly traded common stock of generally comparable companies; and
- other factors deemed relevant by the underwriters and us.

Neither we nor the underwriters can assure investors that an active trading market will develop for the shares of Class A common stock, or that the shares of Class A common stock will trade in the market at or above the initial public offering price.

The underwriters and their respective affiliates are full service financial institutions engaged in a variety of activities, which may include sales and trading, commercial and investment banking, advisory services, asset management, investment research, principal investment, hedging, market making, brokerage services, and other financial and non-financial activities and services. Certain of the underwriters and their respective affiliates have provided, and may in the future provide, a variety of these services to the issuer and to other clients or entities with relationships with the issuer, for which they received or will receive customary fees and expenses. In addition, from time to time, certain of the underwriters and their affiliates may execute transactions for their

own account or the account of customers, and hold on behalf of themselves or their customers, long or short positions in our debt or equity securities or loans, and may do so in the future.

Other than in the U.S., no action has been taken by us or the underwriters that would prevent the offering of the securities offered by this prospectus in any jurisdiction where action for the offering is required. The securities offered by this prospectus may not be offered or sold, directly or indirectly, in any jurisdiction by this prospectus or any other offering material or advertisements in connection with the offer and sale of such securities be distributed or published in any jurisdiction, except under circumstances that comply with the applicable rules and regulations of that jurisdiction. Persons into whose possession this prospectus comes are advised to inform themselves about and to observe any restrictions on the offering and the distribution of this prospectus. This prospectus does not constitute an offer or a solicitation of an offer to buy any securities offered by this prospectus in any jurisdiction in which such offer or a solicitation is unlawful.

NOTICE TO PROSPECTIVE INVESTORS IN THE EUROPEAN ECONOMIC AREA

In relation to each Member State of the European Economic Area (each a “Relevant State”) have been offered or will be offered pursuant to the offering to the public in that Relevant State the publication of a prospectus in relation to the shares which has been approved by the competent authority in that Relevant State or, where appropriate, approved in another Relevant State and notified to the authority in that Relevant State, all in accordance with the Prospectus Regulation, except that shares may be made to the public in that Relevant State at any time under the following exemptions from the Prospectus Regulation:

- (a) to any legal entity which is a qualified investor as defined under Article 2 of the Prospectus Regulation;
- (b) to fewer than 150 natural or legal persons (other than qualified investors as defined under Article 2 of the Prospectus Regulation), subject to obtaining the prior consent of the underwriters; or
- (c) in any other circumstances falling within Article 1(4) of the Prospectus Regulation,

provided that no such offer of shares shall require us or any underwriter to publish a prospectus pursuant to Article 3 of the Prospectus Regulation or supplement a prospectus pursuant to Article 23 of the Prospectus Regulation, and each person who initially acquires any shares or to whom any offer is made will be deemed to have represented, acknowledged and agreed to and with each of the underwriters and the Company that such person is a “qualified investor” within the meaning of Article 2(e) of the Prospectus Regulation. In the event that shares being offered to a financial intermediary as that term is used in the Prospectus Regulation, the financial intermediary will be deemed to have represented, acknowledged and agreed that the shares acquired by it in the offer have not been acquired on a non-discretionary basis on behalf of, or for the account of, persons who have been acquired with a view to their offer or resale to, persons in circumstances which may give rise to an offer of any shares to the public other than their offer or resale in a Relevant State to qualified investors as so defined or in circumstances in which the prior consent of the underwriters have been obtained in relation to such proposed offer or resale.

For the purposes of this provision, the expression an “offer to the public” in relation to the shares in a Relevant State means the communication in any form and by any means of sufficient information to enable investors to make an investment decision in terms of the offer and any shares to be offered so as to enable an investor to decide to purchase or subscribe for any shares, and the expression “Prospectus Regulation” means Regulation (EU) 2017/1129.

NOTICE TO PROSPECTIVE INVESTORS IN THE UNITED KINGDOM

No shares have been offered or will be offered pursuant to the offering to the public in the United Kingdom the publication of a prospectus in relation to the shares which has been approved by the Financial Conduct Authority, except that the shares may be offered to the public in the U.K. at any time:

- (a) to any legal entity which is a qualified investor as defined under Article 2 of the U.K. Prospectus Regulation;
- (b) to fewer than 150 natural or legal persons (other than qualified investors as defined under Article 2 of the U.K. Prospectus Regulation), subject to obtaining the prior consent of underwriters in relation to the offer; or

(c) in any other circumstances falling within Section 86 of the Financial Services and Markets Act 2000 (“FSMA”);

provided that no such offer of the shares shall require the Company or any underwriter to publish a prospectus pursuant to Section 85 of the FSMA or supplement a prospectus pursuant to Article 17 of the U.K. Prospectus Regulation. For the purposes of this provision, the expression an “offer to the public” in relation to the shares in the U.K. means the communication in any form and by any means of information on the terms of the offer and any shares to be offered so as to enable an investor to make a purchase or subscribe for any shares and the expression “U.K. Prospectus Regulation” means the U.K. Prospectus Regulation (EU) 2017/1129 as it forms part of domestic law by virtue of the European Union (Withdrawal) Act 2018.

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In addition, in the U.K., this document is being distributed only to, and is directed only at, persons to whom an offer subsequently made may only be directed at persons who are “qualified investors” (as defined in the U.K. Prospectus Regulation) (i) who have professional experience in matters relating to investments (as defined within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended) (the “Order”) and/or (ii) who are high net worth companies (or persons to whom it may be lawfully communicated) falling within Article 49(2)(a) to (d) of the Order (all such persons being referred to as “relevant persons”) or otherwise in circumstances which have not resulted in the offer being a result in an offer to the public of the shares in the U.K. within the meaning of the FSMA.

Any person in the U.K. that is not a relevant person should not act or rely on the information contained in this document or use it as a basis for taking any action. In the U.K., any investment or investment decision that this document relates to may be made or taken exclusively by relevant persons.

NOTICE TO PROSPECTIVE INVESTORS IN CANADA

The shares may be sold only to purchasers purchasing, or deemed to be purchasing, as primary purchasers, accredited investors, as defined in National Instrument 45-106 Prospectus Exemptions or subsection 73(1) of the Securities Act (Ontario), and are permitted clients, as defined in National Instrument 33-109 Registration Requirements, Exemptions and Ongoing Registrant Obligations. Any resale of the shares may be made in accordance with an exemption from, or in a transaction not subject to, the prospectus requirements of applicable securities laws.

Securities legislation in certain provinces or territories of Canada may provide a purchaser with remedies for rescission or damages if this prospectus (including any amendment thereto) contains a misrepresentation, provided that the remedies for rescission or damages are exercised by the purchaser within the time limit prescribed by the securities legislation of the purchaser’s province or territory. The purchaser should refer to any applicable provisions of the securities legislation of the purchaser’s province or territory for particulars of these rights or consult with a legal advisor.

Pursuant to section 3A.3 of National Instrument 33-105 Underwriting Conflicts (NI 33-105), the underwriters are not required to comply with the disclosure requirements of NI 33-105 relating to underwriter conflicts of interest in connection with this offering.

NOTICE TO PROSPECTIVE INVESTORS IN SWITZERLAND

This prospectus does not constitute an offer to the public or a solicitation to purchase or subscribe for shares. No shares have been offered or will be offered to the public in Switzerland, except insofar as shares may be made to the public in Switzerland at any time under the following exemption from the Swiss Financial Services Act (“FinSA”):

- (a) to any person which is a professional client as defined under the FinSA;
 - (b) to fewer than 500 persons (other than professional clients as defined under the FinSA) obtaining the prior consent of J.P. Morgan Securities LLC, Morgan Stanley & Co. LLC, Jefferies & Company LLC or Moelis & Company LLC for any such offer; or
 - (c) in any other circumstances falling within Article 36 FinSA in connection with Art. 36 Swiss Financial Services Ordinance,
- provided that no such offer of shares shall require the Company or any investment bank to file a prospectus pursuant to Article 35 FinSA.

The shares have not been and will not be listed or admitted to trading on a trading venue in Switzerland.

Neither this document nor any other offering or marketing material relating to the shares or the prospectus as such term is understood pursuant to the FinSA and neither this document nor any other offering or marketing material relating to the shares may be publicly distributed or otherwise made available in Switzerland.

NOTICE TO PROSPECTIVE INVESTORS IN THE DUBAI INTERNATIONAL FINANCIAL CENTRE (DIFC)

This document relates to an Exempt Offer in accordance with the Markets Law, DIFC Law No. 1 of 2012, as amended. This document is intended for distribution only to persons of a type specified in the Markets Law, DIFC Law No. 1 of 2012, as amended. It must not be delivered to, or relied on by, any person who is not a person of a type specified in the Markets Law, DIFC Law No. 1 of 2012, as amended. The Dubai Financial Services Authority (DFSA) has no responsibility for reviewing or approving documents in connection with Exempt Offers. The DFSA has not approved this prospectus and has not taken steps to verify the information set forth herein and has no responsibility for this document. The securities to which this document relates may be illiquid and/or subject to restrictions on their sale. Prospective purchasers of the securities offered should conduct their own due diligence on the securities. If you do not understand the contents of this document you should consult an authorized financial adviser.

In relation to its use in the DIFC, this document is strictly private and confidential and is not to be distributed to a limited number of investors and must not be provided to any person other than the intended recipient, and may not be reproduced or used for any other purpose. The interests in the securities offered may be offered or sold directly or indirectly to the public in the DIFC.

NOTICE TO PROSPECTIVE INVESTORS IN THE UNITED ARAB EMIRATES

The shares have not been, and are not being, publicly offered, sold, promoted or advertised in the United Arab Emirates (including the Dubai International Financial Centre) other than in compliance with the laws of the United Arab Emirates (and the Dubai International Financial Centre) governing the offering and sale of securities. Further, this prospectus does not constitute a public offer of securities in the United Arab Emirates (including the Dubai International Financial Centre) and is not intended to be used in connection with an offer. This prospectus has not been approved by or filed with the Central Bank of the United Arab Emirates or the Securities and Commodities Authority, Financial Services Regulatory Authority (FSRA) or the Dubai Financial Services Authority (DFSA).

NOTICE TO PROSPECTIVE INVESTORS IN AUSTRALIA

This prospectus:

- does not constitute a disclosure document or a prospectus under Chapter 6D.2 of the Act 2001 (Cth) (the “Corporations Act”);
- has not been, and will not be, lodged with the Australian Securities and Investments (“ASIC”), as a disclosure document for the purposes of the Corporations Act and does not include the information required of a disclosure document for the purposes of the Corporations Act and
- may only be provided in Australia to select investors who are able to demonstrate that they fall within one or more of the categories of investors, available under section 708 of the Corporations Act (“Exempt Investors”).

The shares may not be directly or indirectly offered for subscription or purchased or sold in Australia. Invitations to subscribe for or buy the shares may be issued, and no draft or definitive offer memorandum, advertisement or other offering material relating to any shares may be distributed in Australia, except where disclosure to investors is not required under Chapter 6D of the Corporations Act. By submitting a request for the shares, you represent and warrant to us that you are an Exempt Investor.

As any offer of shares under this document will be made without disclosure in Australia, Chapter 6D.2 of the Corporations Act, the offer of those securities for resale in Australia without disclosure may, under section 707 of the Corporations Act, require disclosure to investors under Chapter 6D.2 of the Corporations Act if the exemption in section 708 applies to that resale. By applying for the shares, you undertake that you will not, for a period of 12 months from the date of issue of the shares, offer, transfer or otherwise alienate those shares to investors in Australia except in circumstances where disclosure to investors is not required under Chapter 6D.2 of the Corporations Act or where a compliance document is prepared and lodged with ASIC.

NOTICE TO PROSPECTIVE INVESTORS IN JAPAN

The shares have not been and will not be registered pursuant to Article 4, Paragraph 1 of the Financial Instruments and Exchange Act. Accordingly, none of the shares nor any interest therein may be sold, directly or indirectly, in Japan or to, or for the benefit of, any “resident” of Japan (which herein means any person resident in Japan, including any corporation or other entity organized under the laws of Japan), or to others for re-offering or resale, directly or indirectly, in Japan or to or for the benefit of a resident of Japan, except pursuant to an exemption from the registration requirements of, and in compliance with, the Financial Instruments and Exchange Act and any other applicable law and ministerial guidelines of Japan in effect at the relevant time.

NOTICE TO PROSPECTIVE INVESTORS IN HONG KONG

The shares have not been offered or sold and will not be offered or sold in Hong Kong, by any document, other than (a) to “professional investors” as defined in the Securities and Futures Ordinance (Cap. 571 of the Laws of Hong Kong) (the “SFO”) of Hong Kong and any rules made thereunder, or (b) in other circumstances which do not result in the document being a “prospectus” as defined in the Securities and Futures (Winding Up and Miscellaneous Provisions) Ordinance (Cap. 32) of Hong Kong (the “CO”) or which would constitute an offer to the public within the meaning of the CO. No advertisement, invitation or other document relating to the shares has been or may be issued or has been or may be in the possession of a person for the purposes of issue, whether in Hong Kong or elsewhere, which is directed at, or the contents of which are likely to be accessed or read by, the public of Hong Kong (except if permitted to do so under

laws of Hong Kong) other than with respect to shares which are or are intended to be disposed of to persons outside Hong Kong or only to “professional investors” as defined in the SFO and any rules made thereunder.

NOTICE TO PROSPECTIVE INVESTORS IN SINGAPORE

Each of J.P. Morgan Securities LLC, Morgan Stanley & Co. LLC, Jefferies LLC and Moelis & Company LLC and the Company has acknowledged that this prospectus has not been registered with the Monetary Authority of Singapore. Accordingly, each of J.P. Morgan Securities LLC, Morgan Stanley & Co. LLC, Jefferies LLC and Moelis & Company LLC, and the Company have not and agreed that it has not offered or sold any shares or caused the shares to be made the subject of an invitation for subscription or purchase and will not offer or sell any shares or cause the shares to be the subject of an invitation for subscription or purchase, and has not circulated or distributed, nor will it circulate or distribute, this prospectus or any other document or material in connection with the offer, sale, or invitation for subscription or purchase, of the shares, whether directly or indirectly, to any person in Singapore other than:

- (a) to an institutional investor (as defined in Section 4A of the Securities and Futures Act (Chapter 289) of Singapore, as modified or amended from time to time (the “SFA”)) pursuant to Section 275 of the SFA;
- (b) to a relevant person (as defined in Section 275(2) of the SFA) pursuant to Section 275 of the SFA, or any person pursuant to Section 275(1A) of the SFA, and in accordance with the conditions set out in Section 275 of the SFA; or
- (c) otherwise, pursuant to, and in accordance with the conditions of, any other applicable provision of the SFA.

Where the shares are subscribed or purchased under Section 275 of the SFA by a relevant person, which is:

- (a) a corporation (which is not an accredited investor (as defined in Section 4A of the SFA)) the business of which is to hold investments and the entire share capital of which is owned by one or more individuals, each of whom is an accredited investor; or
 - (b) a trust (where the trustee is not an accredited investor) whose sole purpose is to hold investments and each beneficiary of the trust is an individual who is an accredited investor,
- and the shares are securities or securities-based derivatives contracts (each term as defined in Section 2(1) of the SFA) in which the corporation or the beneficiaries’ rights and interest (howsoever described) in that trust are not transferred

within six months after that corporation or that trust has acquired the shares pursuant to an offer made pursuant to Section 275 of the SFA except:

- (i) to an institutional investor or to a relevant person, or to any person arising from a transfer referred to in Section 275(1A) or Section 276(4)(c)(ii) of the SFA;
- (ii) where no consideration is or will be given for the transfer;
- (iii) where the transfer is by operation of law;

- (iv) as specified in Section 276(7) of the SFA; or
- (v) as specified in Regulation 37A of the Securities and Futures (Offers of (Securities and Securities-based Derivatives Contracts) Regulations 2018.

LEGAL MATTERS

Latham & Watkins LLP, New York, New York, which has acted as our counsel in connection with this offering, will pass upon the validity of the issuance of the shares of our Class A common stock.

this prospectus. Certain legal matters in connection with this offering will be passed to the underwriters by Skadden, Arps, Slate, Meagher & Flom LLP.

EXPERTS

The consolidated financial statements of X-Energy Reactor Company, LLC at December 31, 2024, and for the two years in the period ended December 31, 2025, appearing in this Preliminary Registration Statement have been audited by Ernst & Young LLP, independent registered public accounting firm, as set forth in their report thereon appearing elsewhere herein, and are included in reliance on the report given on the authority of such firm as experts in accounting and auditing.

The financial statements of X-Energy, Inc. as of December 31, 2025 appearing in this Preliminary Registration Statement have been audited by Ernst & Young LLP, independent registered public accounting firm, as set forth in their report thereon appearing elsewhere herein, and is included in reliance on the report given on the authority of such firm as experts in accounting and auditing.

CHANGE IN AUDITOR

On May 2, 2025, the Company notified Grant Thornton LLP (“GT”) of its intent to replace GT as the Company’s certifying accountant. On June 9, 2025, the Company appointed Ernst & Young LLP (“EY”) as its new independent registered public accounting firm. The reports of GT on the Company’s consolidated financial statements for the fiscal years ended December 31, 2024 and 2023 did not contain a disclaimer of opinion or a disclaimer of opinion, and were not qualified or modified as to uncertainty, audit scope, or accounting principles. During the years ended December 31, 2024 and 2023, and the subsequent interim period through May 2, 2025, there were (i) no disagreements between the Company and GT concerning the application of accounting principles or practices, financial statement disclosure or auditing scope or procedure, which, if not resolved to GT’s satisfaction, would have caused GT to make reference to the subject of the disagreement in connection with its report for such years, and (ii) no “reportable events” as defined in Item 304(a)(1)(v) of Regulation S-K for such years and subsequent interim period through May 2, 2025, except for GT’s communication of the material weakness discussed in the Risk Factors section of this Preliminary Registration Statement. The Company discussed such reportable events with GT, and the Company has authorized GT to respond to the inquiries of EY concerning such reportable events. In connection with the filing of this Preliminary Registration Statement, EY reaudited the Company’s financial statements for the year ended December 31, 2024 in accordance with the standards of the Public Company Accounting Oversight Board (United States) (“PCAOB”). EY’s audit report for the fiscal year ended December 31, 2024 is included herein. Accordingly, GT’s previous audit report on the fiscal year ended December 31, 2024 is not included in this Preliminary Registration Statement. For the years ended December 31, 2024 and 2023, and the subsequent interim period through May 2, 2025, neither the Company nor anyone on its behalf has consulted EY with respect to either (i) the application of accounting principles to a specified transaction, either completed or proposed, or the type of audit opinion that might be rendered on the Company’s consolidated financial statements or the effectiveness of the Company’s internal control over financial reporting, where either a written report or oral advice was provided to the Company, or (ii) that EY concluded was an important factor considered by the Company in reaching a decision regarding an accounting, auditing or financial reporting issue; or (ii) any matter that was either the subject of a disagreement (as defined in Item 304(a)(1)(iv) of Regulation S-K and related instructions) or a reportable event (as defined in Item 304(a)(1)(v) of Regulation S-K). The Company has provided GT with a copy of this Current Report.

WHERE YOU CAN FIND MORE INFORMATION

We have filed with the SEC a registration statement on Form S-1 (including the exhibits, amendments thereto) under the Securities Act, with respect to our Class A common stock of Xcel Energy Inc. This prospectus does not contain all of the information set forth in the registration statement, including the exhibits and schedules thereto. For further information with respect to the Class A common stock, hereby, we refer you to the registration statement and the exhibits and schedules filed therewith. The information contained in this prospectus as to the contents of any contract, agreement or other document and the material terms of such contract, agreement or other document and are not necessarily complete. With respect to each of these contracts, agreements or other documents filed as an exhibit to the registration statement, reference is made to the exhibits for a more complete description of the matter in which the SEC maintains a website that contains reports, proxy and information statements and other documents regarding registrants that file electronically with the SEC. The address of the SEC's website is www.sec.gov. A copy of the registration statement, of which this prospectus forms a part, and the exhibits and schedules thereto may be downloaded from the SEC's website.

As a result of this offering, we will become subject to full information reporting requirements under the Exchange Act and will file with or furnish to the SEC periodic reports and other information. We will furnish our shareholders with annual reports containing our audited financial statements prepared in accordance with GAAP and certified by an independent public accounting firm. We also intend to make available to our shareholders quarterly reports containing our unaudited interim financial statements for the first three fiscal quarters of each fiscal year. Our website is located at www.x-energy.com. From the completion of this offering, we intend to make our periodic reports and other information furnished to the SEC available, free of charge, through our website, as soon as reasonably practicable after those reports and other information are electronically filed with or furnished to the SEC. The information contained on our website or linked therein or otherwise connected thereto does not constitute part of this prospectus incorporated by reference into this prospectus or the registration statement of which this prospectus is a part.

INDEX TO FINANCIAL STATEMENTS

X-Energy, Inc.**Audited Financial Statements as of December 31, 2025:**[Report of Ernst & Young LLP \(PCAOB ID: 42\)](#)[Balance Sheet as of December 31, 2025](#)[Notes to Balance Sheet](#)**X-Energy Reactor Company, LLC****Audited Financial Statements for the Years Ended December 31, 2025 and 2024:**[Report of Ernst & Young LLP \(PCAOB ID: 42\)](#)[Consolidated Balance Sheets as of December 31, 2025 and 2024.](#)[Consolidated Statements of Operations and Comprehensive Loss for the Years Ended December 31, 2025 and 2024](#)[Consolidated Statements of Changes in Members' Deficit and Mezzanine Equity for the Years Ended December 31, 2025 and 2024](#)[Consolidated Statements of Cash Flows for the Years Ended December 31, 2025 and 2024](#)[Notes to Consolidated Financial Statements](#)

Report of Independent Registered Public Accounting Firm

To the Board of Directors of X-Energy, Inc.

Opinion on the Financial Statements

We have audited the accompanying balance sheet of X-Energy, Inc. (the “Company”) as of December 31, 2025, and the related notes (collectively referred to as the “financial statements”) in our opinion, the financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2025, in accordance with U.S. generally accepted accounting principles.

Basis for Opinion

These financial statements are the responsibility of the Company’s management. Our responsibility is to express an opinion on the Company’s financial statements based on our audit. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB). We are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audit in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. The Company is not required to have, and we were not engaged to perform, an audit of its internal control over financial reporting. As part of our audit, we are required to obtain an understanding of internal control over financial reporting but not for the purpose of expressing an opinion on the effectiveness of the Company’s internal control over financial reporting. Accordingly, we express no such opinion.

Our audit included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those risks. The procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audit also included evaluating the accounting principles used and the estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audit provides a reasonable basis for our opinion.

/s/ Ernst & Young LLP

We have served as the Company’s auditor since 2025.

Tysons, Virginia
March 20, 2026

X-ENERGY, INC.
BALANCE SHEET

Assets**Current assets**

Cash and cash equivalents

Total assets**Shareholder's Equity**

Common stock, par value \$0.0001 per share, 1,000 authorized and 0 issued or outstanding

Additional paid-in capital

Total shareholder's equity**Total liabilities, and shareholder's equity**

X-ENERGY, INC.**NOTES TO FINANCIAL STATEMENTS****NOTE 1 — ORGANIZATION****Company Overview**

X-Energy, Inc. (the “Company”) was incorporated in Delaware on September 18, 2025. With its incorporation, the Company authorized 1,000 shares of common stock for \$0.0001 per share and issued 0 shares. The Company was formed for the purpose of completing a public offering of common stock in connection with the Reorganization transactions (the “Reorganization”) in order to conduct the business of the X-Energy Reactor Company, LLC, which is a related party, as a publicly-traded entity. The Company had no operations from its incorporation on September 18, 2025, and had no activities since the date of incorporation and therefore has omitted presenting statements of operations, changes in shareholder’s equity and cash flows.

Following the successful completion of the Reorganization and this offering, the Company will be a public holding company and its sole asset will be equity interests in X-Energy Reactor Company, LLC. The Company will be the general partner of X-Energy Reactor Company, LLC.

NOTE 2 — SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES

The accompanying financial statements of the Company have been prepared in accordance with accounting principles generally accepted in the United States of America (“U.S. GAAP”).

NOTE 3 — SHAREHOLDER’S EQUITY

The Company is authorized to issue 1,000 shares of common stock (par value \$0.0001 per share).

NOTE 4 — COMMITMENTS AND CONTINGENCIES

In the ordinary course of business, the Company may be subject to various legal, regulatory, and administrative proceedings. There are currently no such proceedings to which the Company is a party.

In the normal course of business, the Company may enter into contracts that contain indemnification clauses. The Company’s maximum exposure under these arrangements cannot be determined as these indemnities relate to future claims that may be made against the Company that have not yet occurred. However, the Company has not had prior claims or losses pursuant to these arrangements and expects the risk of loss to be remote.

Report of Independent Registered Public Accounting Firm

To the Members and the Board of Directors of X-Energy Reactor Company, LLC

Opinion on the Financial Statements

We have audited the accompanying consolidated balance sheets of X-Energy Reactor Company, LLC (the Company) as of December 31, 2025 and 2024, the related consolidated statements of operations, comprehensive loss, changes in members' deficit and mezzanine equity and cash flows for the two years in the period ended December 31, 2025, and the related notes (collectively referred to as the "consolidated financial statements"). In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company at December 31, 2025 and 2024, and the results of its operations and its cash flows for each of the two years in the period ended December 31, 2025, in conformity with U.S. generally accepted accounting principles.

Basis for Opinion

These financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on the Company's financial statements based on our audits. We are a registered public accounting firm registered with the Public Company Accounting Oversight Board (United States) and are required to be independent with respect to the Company in accordance with the securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the financial statements are free of material misstatement, whether due to error or fraud. The Company is not required to have, and we were not engaged to perform, an audit of its internal control over financial reporting. As part of our audits, we are required to obtain an understanding of internal control over financial reporting but not for the purpose of expressing an opinion on the effectiveness of the Company's internal control over financial reporting. Accordingly, we express no such opinion.

Our audits included performing procedures to assess the risks of material misstatement of the financial statements, whether due to error or fraud, and performing procedures that respond to those assessed risks. The procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the financial statements. Our audits also included evaluating the accounting principles used and estimates made by management, as well as evaluating the overall presentation of the financial statements. We believe that our audits provide a reasonable basis for our opinion.

/s/ Ernst & Young LLP

We have served as the Company's auditor since 2025.

Tysons, Virginia
March 20, 2026

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X-ENERGY REACTOR COMPANY, LLC
CONSOLIDATED BALANCE SHEETS
(in thousands, except units)

ASSETS**Current assets**

Cash and cash equivalents
Short-term investments
Accounts receivable
Unbilled receivables and contract assets
Prepaid and other current assets
Due from related parties

Total current assets

Long-term investments
Restricted cash
Property and equipment, net
Operating lease right-of-use assets
Other long-term assets

Total assets**LIABILITIES, MEZZANINE EQUITY, AND MEMBERS' DEFICIT****Current liabilities**

Accounts payable
Accrued liabilities
Due to related parties
Short-term borrowings

Total current liabilities

Long-term deferred revenue
Long-term operating lease liabilities
Warrant liabilities

Total liabilities**Mezzanine equity**

Class A Common Units: 367,055,779 units authorized, 3,128,026 units issued and outstanding as of December 31, 2025, and 328,688,824 units authorized, 3,128,026 units issued and outstanding as of and December 31, 2024

Class B Common Units: 41,149,242 units authorized, 16,838,205 units issued and outstanding as of December 31, 2025, and 34,043,242 units authorized, 13,960,705 units issued and outstanding as of December 31, 2024

Series A redeemable convertible preferred units: 90,625,588 units authorized, issued and outstanding as of December 31, 2025, and 2024; liquidation preference of \$52,146 as of December 31, 2025 and 2024

Series A-1 redeemable convertible preferred units: 8,808,351 units authorized, issued and outstanding as of December 31, 2025, and 2024; liquidation preference of \$67,250 as of December 31, 2025 and 2024

Series B redeemable convertible preferred units: 11,643,171 units authorized, issued and outstanding as of December 31, 2025, and 2024; liquidation preference of \$120,214 as of December 31, 2025 and \$117,030 as of December 31, 2024

Series C redeemable convertible preferred units: 41,418,916 units authorized; 39,963,592 units issued and outstanding as of December 31, 2025, and 65,185,243 units authorized; 37,093,420 units issued and outstanding as of December 31, 2024; liquidation preference of \$305,114 as of December 31, 2025 and \$283,201 as of December 31, 2024

Series C-1 redeemable convertible preferred units: 162,246,180 units authorized; 107,908,114 units issued and outstanding as of December 31, 2025, and 148,122,321 units authorized, 99,672,593 units issued and outstanding as of December 31, 2024; liquidation preference of \$874,999 as of December 31, 2025 and \$808,220 as of December 31, 2024

Series D redeemable convertible preferred units: 48,154,955 units authorized, issued and outstanding as of December 31, 2025, and zero units authorized, issued and outstanding as of December 31, 2024; liquidation preference of \$700,000 as of December 31, 2025 and \$0 as of December 31, 2024

Total mezzanine equity

Accumulated deficit
Accumulated other comprehensive income (loss)
Additional paid-in capital

Total members' deficit**Total liabilities, mezzanine equity, and members' deficit**

X-ENERGY REACTOR COMPANY, LLC
CONSOLIDATED STATEMENTS OF OPERATIONS AND COMPREHENSIVE LOSS
(in thousands)

	Year Ended 2025
Services revenue ⁽¹⁾	\$ 94,26
Grant income	14,83
Total revenues and grant income	\$ 109,09
Operating expenses	
Direct costs	161,36
Selling, general and administrative	116,31
Research and development	1,70
Total operating expenses	279,39
Operating loss	(170,29)
Other income (expense)	
Interest expense	(47)
Interest income	20,29
Other income (expense), net	(239,30)
Total other expense, net	(219,48)
Net loss	\$(389,77)
Other comprehensive loss	
Foreign currency translation adjustment	(88)
Reclassification of OCI for conversion of C-1 and C-2 Notes	—
Changes in fair value of liabilities under fair value option attributable to changes in instrument-specific credit risk	76
Other comprehensive loss	(12)
Comprehensive loss	\$(389,90)

(1) Includes related party revenue of \$6,943 and \$27,555 for the years ended December 31, 2024 and 2025, respectively.

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X-ENERGY REACTOR COMPANY, LLC
CONSOLIDATED STATEMENTS OF CHANGES IN MEMBERS' DEFICIT AND MEZZANINE
(in thousands, except units)

	Class A Common Units		Class B Common Units		Series A Preferred Units		Series A-1 Preferred Units		Series B Preferred Units		Series C Preferred Units	
	Units	Amount	Units	Amount	Units	Amount	Units	Amount	Units	Amount	Units	Amount
	Accumulated Deficit	Accumulated Other Comprehensive Income (Loss)	Additional Paid-In Capital	Total Members' Deficit	Units	Amount	Units	Amount	Units	Amount	Units	Amount
Balance, January 1, 2024	\$ (716,769)	\$ 879	\$ —	\$ (715,890)	3,128,026	\$ 1,800	10,139,818	\$ 74,250	90,625,588	\$218,408	8,808,351	\$ 21,477
Unit-based compensation	(3,838)	—	—	(3,838)	—	—	694,000	4,988	—	—	—	—
Issuance of Class B Units in conjunction with the issuance and modification of debt	—	—	—	—	—	—	2,369,752	9,061	—	—	—	—
Issuance of Class B Units in conjunction with the conversion of C-1 Notes to Series C Preferred Units	—	—	—	—	—	—	757,135	2,560	—	—	—	—
Conversion of Series C-1 Notes into Series C Preferred Units	—	—	—	—	—	—	—	—	—	—	—	—
Exercise of the October 2022 Warrants	—	—	—	—	—	—	—	—	—	—	—	—
Conversion of C-2 Notes into Series C-Preferred Units	—	—	—	—	—	—	—	—	—	—	—	—
Issuance of Series C-1 Preferred Units, net of issuance costs	—	—	—	—	—	—	—	—	—	—	—	—
Conversion of debt into Series C-1 Preferred Units, net of issuance costs	—	—	—	—	—	—	—	—	—	—	—	—
Net Loss	(125,960)	—	—	(125,960)	—	—	—	—	—	—	—	—
Foreign Currency Translation Adjustment	—	474	—	474	—	—	—	—	—	—	—	—
Changes in fair value of liabilities under fair value option attributable to changes in instrument specific credit risk	—	(6,220)	—	(6,220)	—	—	—	—	—	—	—	—
Reclassification of OCI for conversion of C-1 and C-2 Notes	—	4,873	—	4,873	—	—	—	—	—	—	—	—
Balance, December 31, 2024	\$ (846,567)	\$ 6	—	\$ (846,561)	3,128,026	\$ 1,800	13,960,705	\$ 90,859	90,625,588	\$218,408	8,808,351	\$ 21,477
Unit-based compensation	—	—	12,167	12,167	—	—	2,877,500	2,494	—	—	—	—
Issuance of Series C-1 Preferred Units, net of issuance costs	—	—	—	—	—	—	—	—	—	—	—	—
Conversion of C-2 Notes Issuance of Series D Preferred Units, net of issuance costs	—	—	—	—	—	—	—	—	—	—	—	—
Net Loss	(389,778)	—	—	(389,778)	—	—	—	—	—	—	—	—
Foreign Currency Translation Adjustment	—	(888)	—	(888)	—	—	—	—	—	—	—	—
Changes in fair value of liabilities under fair value option attributable to changes in instrument specific credit risk	—	765	—	765	—	—	—	—	—	—	—	—
Balance, December 31, 2025	\$ (1,236,345)	\$ (117)	\$12,167	\$ (1,224,295)	3,128,026	\$ 1,800	16,838,205	\$ 93,353	90,625,588	\$218,408	8,808,351	\$ 21,477

X-ENERGY REACTOR COMPANY, LLC
CONSOLIDATED STATEMENTS OF CASH FLOWS
(in thousands)

	Year Ended
	2025
Cash flows from operating activities:	
Net loss	\$(389,77)
Adjustments to reconcile net loss to net cash used in operating activities:	
Depreciation and amortization	1,39
Unit-based compensation	14,13
Loss on extinguishment of debt	—
(Gain) loss on conversion of C-1 & C-2 Notes	4,02
Non-cash selling, general, and administrative expenses	—
Payable in-kind interest	—
Mark-to-market loss (gain) on warrant liabilities	223,53
Mark-to-market loss on C-1 & C-2 Notes	13,01
Mark-to-market gain on derivative liabilities	—
Mark-to-market gain on 2024 Financial Instrument	—
Reclassification of OCI for conversion of C-1 & C-2 Notes	—
Amortization of deferred financing costs, debt discount, and other	97
Accretion and amortization on investments	(27)
Changes in operating assets and liabilities:	
Accounts receivable and unbilled receivables	(24,80)
Prepaid and other current assets	(4,02)
Due from related parties	11,39
Operating lease right-of use assets	2,43
Accounts payable and accrued liabilities	8,73
Long-term deferred revenue	15,15
Payment of payable-in-kind interest	—
Other long-term assets	(18,18)
Due to related parties	(4,25)
Operating lease liabilities	(3,33)
Net cash used in operating activities	\$(149,86)
Cash flows from investing activities:	
Capital expenditures	(117,23)
Reimbursement of capital expenditures under government grant	54,83
Purchase of investments	(565,94)
Net cash used in investing activities	\$(628,34)
Cash flows from financing activities:	
Repayments on bridge loans	—
Borrowings on bridge loans	—
Repayments on lines of credit	—
Borrowings from lines of credit	—
Payments of mezzanine equity issuance costs	(25,27)
Payment of debt issuance costs	(50)
Payment of deferred transaction costs	(1,01)
Proceeds from issuance of Preferred Units	752,92
Net cash provided by financing activities	\$ 726,13
Net effect of exchange rate	10
Net increase (decrease) in cash, cash equivalents, and restricted cash	(51,97)

Cash, cash equivalents and restricted cash at beginning of period	514,60
Cash, cash equivalents and restricted cash at end of period	\$ 462,63

X-ENERGY REACTOR COMPANY, LLC
NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

NOTE 1 — ORGANIZATION AND NATURE OF BUSINESS

Company Overview

X-Energy Reactor Company, LLC (“X-energy” or the “Company” or “its”) is a Delaware limited liability company formed on December 14, 2018, and is the successor for financial reporting purposes to X-energy, LLC, a Maryland limited liability company founded in 2009. The Company is headquartered in Rockville, Maryland. The Company is a developer of advanced small modular nuclear reactor technology for clean energy generation.

NOTE 2 — SUMMARY OF SIGNIFICANT ACCOUNTING POLICIES AND RECENT ACCOUNTING PRONOUNCEMENTS

Basis of Presentation

The consolidated financial statements for X-Energy Reactor Company, LLC as of and for the years ended December 31, 2025 and 2024, include the accounts of the Company’s wholly owned and controlled subsidiaries. The consolidated financial statements have been prepared pursuant to the accounting principles generally accepted in the United States of America (“U.S. GAAP”). All intercompany transactions have been eliminated in consolidation.

Use of Estimates

The preparation of the Company’s financial statements in conformity with U.S. GAAP requires the use of estimates and assumptions that affect the reported amounts of assets and liabilities and the disclosure of contingent assets and liabilities as of the date of the financial statements and the reported revenues and expenses for the reporting period. Some of the more significant estimates include the valuation of unit-based compensation, preferred units, profits interests units (“PIUs”), and warrants. Due to the inherent uncertainty involved in making estimates, actual results could differ from those estimates. Such differences could have a material effect on the financial condition and results of operations in future periods.

The Company bases its estimates and assumptions on historical experience and other factors, including the current economic environment and various other judgments that it believes to be reasonable under the circumstances. The Company adjusts such estimates and assumptions when facts and circumstances change. Changes in those estimates resulting from continuing changes in the economic environment will be reflected in the financial statements in future periods.

Certain Significant Risks and Uncertainties

Financial instruments that potentially subject the Company to concentrations of credit risk consist principally of cash and cash equivalents, accounts receivable, and short and long-term investments. In the nature, all such financial instruments involve risks, including the credit risk of nonperformance by counterparties. The Company generally does not require collateral to support the obligations of counterparties and cash levels held at banks may be in excess of federally insured limits. To limit its exposure to credit loss by maintaining its cash and cash equivalents and investments in highly rated financial institutions and investing in US Government securities and high credit quality debt. Further, the Company’s revenue and credit relationships are primarily concentrated within the United States Government which represents a concentration risk but a low credit risk.

For the years ended December 31, 2025 and 2024, two customers, the United States Government and The Dow Chemical Company (“Dow”) accounted for 87% and 6%, and 74% and 22%, respectively, of the Company’s total revenue and grant income. Refer to Note 3—Revenue Recognition for the disaggregation of revenue and grant income by customer for the years ended December 31, 2025 and 2024.

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As of December 31, 2025 and 2024, the Company's receivable balances related to services provided to the United States Department of Energy ("DoE") and United States Department of Defense ("DoD") are as follows (in thousands). Refer to Note 14 — Related Party Transactions for more information on receivable balances with Dow.

	December 31, 2025	December 31, 2024
Accounts Receivable		
DoE	\$ 23,628	\$ 23,628
Unbilled Receivables and Contract Assets		
DoE	\$ 39,644	\$ 39,644
DoD	1,091	1,091

Cash and Cash Equivalents

Cash and cash equivalents consist primarily of cash deposits, cash held in financial institutions, and short-term investments purchased with an original maturity of three months or less. The carrying amount of cash and cash equivalents approximates fair value because of the short-term nature of these investments. The Company maintains its cash in bank deposit accounts with high credit quality financial institutions. At times, may exceed the federally insured limit. The Company does not believe it is exposed to a significant credit risk regarding these deposits.

The components of cash, cash equivalents, and restricted cash as of December 31, 2025 and 2024 are as follows (in thousands):

	December 31, 2025	December 31, 2024
Cash and cash equivalents	\$458,932	\$458,932
Restricted cash	3,698	3,698
Cash, cash equivalents, and restricted cash as presented in the Statement of Cash Flows	\$462,630	\$462,630

Investments

The Company classifies all debt investments as held to maturity and are reported at amortized cost, net of any allowance for credit losses. The Company does not intend to sell the investments and it is more likely than not that the Company will be required to sell the investments before their maturity. Interest income, including amortization of premiums and accretion of discounts, is included in interest income in the consolidated statements of operations.

Held to maturity investments are classified within Level 1 of the fair value hierarchy for investments that are valued using inputs at quoted prices for identical assets in active markets. For those investments that are valued using inputs other than quoted prices in active markets, the Company uses observable inputs for assets that are observable either directly or indirectly.

Accounts Receivable

Accounts receivable are presented at the invoiced receivable amounts, less any allowance for potential expected uncollectible amounts, and do not bear interest. The Company estimates credit losses based on the credit worthiness of each customer, historical collections experience, looking information, adverse situations that may affect a customer's ability to pay, and both macro and macroeconomic factors. The Company grants uncollateralized credit in the form of accounts receivable to its customers. The Company's receivables are heavily concentrated from the United States, specifically the DoE. Historically, the Company has not experienced any significant credit-risk and management does not believe that any government agencies represent a significant credit-risk. Accordingly, historical write-offs of accounts receivable have not occurred and as of December 31, 2025 and 2024, no reserve was recorded.

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Customer payments on contracts are typically due within 30 days of billing, depending on the terms of the contract.

Prepaid Costs

Prepaid costs, primarily consisting of prepaid service fees, software and other general expenses, amounted to \$7.5 million and \$2.9 million as of December 31, 2025 and 2024, respectively. Vendors' deposits to vendors for long-lead materials. As of December 31, 2025, the Company made \$1.5 million deposits related to long-lead materials, which is included in other long-term assets on the consolidated balance sheets.

Transaction and Financing Costs

The Company has recorded deferred financing costs incurred in conjunction with its debt issuances that are held at amortized cost. The Company amortizes deferred financing costs over the remaining term of the related debt and records the amortization within interest expense in the consolidated statements of operations.

In connection with the issuance and amendment of debt, the Company incurred debt issuance costs in the amount of \$0.5 million and \$2.9 million during the years ended December 31, 2025 and 2024, respectively. The debt issuance costs are treated as a debt discount and are amortized over the term of the related loan. During the years ended December 31, 2025 and 2024, \$0.5 million and \$4.8 million, respectively, of amortization expense related to the discount created from debt issuance costs was recorded to Interest expense in the consolidated statements of operations.

The Company records transaction costs incurred in conjunction with the issuance of preferred equity, including the reduction of mezzanine equity on the consolidated balance sheets. For the years ended December 31, 2025 and 2024, the Company incurred \$2.2 million and \$11.1 million, respectively, in issuance costs associated with the issuance of the Series C-1 Preferred Units. Additionally, during the year ended December 31, 2025, the Company incurred \$22.4 million in issuance costs associated with the issuance of the Series D-1 Preferred Units.

The Company capitalizes qualified legal, accounting, and other direct and incremental costs incurred in connection with the potential public offering transaction and began incurring such costs in 2025. These costs amounted to \$3.9 million as of December 31, 2025 and are included in Prepaid and other current assets on the consolidated balance sheet.

Property and Equipment, Net

Property and equipment, net is stated at cost, less accumulated depreciation and amortization, and less accumulated impairments. Depreciation and amortization is recognized using the straight-line method over the estimated useful life of the asset.

the following estimated useful lives of assets and is recorded within selling, general, and administrative expenses on the consolidated statements of operations and comprehensive loss:

Asset	Useful Life
Equipment and materials	5 – 35 years
Computer equipment and software	3 – 5 years
Office furniture and fixtures	2 – 7 years
Leasehold improvements	Shorter of lease term or useful life

The costs of maintenance and repairs that do not materially prolong the useful life of an asset are expensed as incurred. Asset betterments are capitalized. Costs for property and equipment not ready for use at the time they are placed into service, including advance payments for materials and equipment and payments made in connection with contractual progress milestones, are capitalized as construction-in-progress and will be depreciated in accordance with the above guidelines once placed into service.

Impairment of Long-Lived Assets

Long-lived assets are reviewed for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset or group of assets may not be recoverable. Such impairment losses are recognized in circumstances may

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include significant decreases in the market price of the asset, significant adverse changes in the economic environment or legal factors, accumulation of costs significantly in excess of the amount originally estimated, significant operating losses, the acquisition or construction of the asset, current period cash flow or operating losses compared to the historical trend, a history of losses or a forecast of continuing losses associated with the use of the asset, or an expectation that the asset will more likely than not be sold or disposed of significantly before the end of its estimated useful life. If an impairment indicator is present, the Company evaluates recoverability by comparing the carrying amount of the asset to its future undiscounted net cash flows. If the carrying amount of a long-lived asset is not recoverable, it is impaired, and the impairment loss is measured as the difference between the carrying value of the asset and its fair value. No impairment losses were recognized in the years ended December 31, 2025 and 2024.

Leases

The Company leases facilities including office space under non-cancelable lease contracts with terms greater than 12 months and the Company has classified these leases as operating leases. The Company does not have any material finance leases. When determining lease term, the Company considers all options to extend or terminate the lease when it is reasonably certain the Company will exercise such options.

The Company elected to account for lease and non-lease components related to office space as a single lease component. Leases with an initial term of 12 months or less are not included within the long-lived assets (“ROU”) assets and lease liabilities recognized on the consolidated balance sheets, and instead, lease payments for those short-term leases are recognized on a straight-line basis over the lease term.

Lease ROU assets are calculated as the net present value of future payments plus any capitalized costs less any tenant improvements or lease incentives. Lease liabilities are calculated as the present value of future payments. For leases that do not provide an implicit interest rate, the Company uses the incremental borrowing rate to calculate the lease liability. Upon lease modification, the Company remeasures the ROU asset and lease liability as of the modification date.

Lease expense for minimum operating lease payments is recognized on a straight-line basis over the lease term. Any variable non-lease components are not included within the lease ROU asset or liability on the consolidated balance sheets and instead are reflected as an expense in the period in which they occur.

Deferred Revenue

The Company recognizes a contract liability, referred to as deferred revenue in its financial statements, when the Company receives a payment from customers, or has an unconditional right to consideration, for services that have yet to be performed. Refer to Note 3 — Revenue Recognition for further discussion on deferred revenue balances.

Financial Instruments and Fair Value Measurements

The Company estimates fair value based on assumptions that active market participants would use in pricing an asset or liability in the principal or most advantageous market. When considering the most appropriate participant assumptions in fair value measurements, the following fair value hierarchy is used, with the highest priority given to observable inputs and the lowest priority given to unobservable inputs. Fair value measurements are categorized according to the hierarchy criteria below based on the lowest level of input that is significant to the overall fair value measurement of the instrument:

- Level 1 inputs: Quoted prices (unadjusted) in active markets for identical assets or liabilities that the reporting entity can access at the measurement date;
- Level 2 inputs: Inputs other than quoted prices included within Level 1 that are observable for the asset or liability, either directly or indirectly; and
- Level 3 inputs: Unobservable inputs for the asset or liability. These are used to measure fair value to the extent those observable inputs are not available, thereby allowing for situations in which the Company's judgment is minimal, if any, market activity for the asset or liability at the measurement date.

The carrying amount reflected in the Company's consolidated balance sheets for accounts receivable and accounts payable approximate fair value due to the short-term maturities of these instruments.

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Note 13 — Fair Value Measurements for the financial assets and liabilities that are recorded at fair value on a recurring basis within the consolidated balance sheets.

Loss Contingencies

The Company accrues liabilities for loss contingencies when it is probable that a loss has been incurred on account of investigations, litigation, disputes, or claims related to its business activities and the amount of loss is reasonably estimable. When the amount of loss cannot be reasonably estimated, the Company discloses contingent liabilities when there is a reasonable possibility that a loss or additional loss has been incurred. The Company's future earnings could be affected by changes in the assessment of the probability of a loss or changes in the estimates related to such matters.

Revenue and Cost Recognition

For the years ended December 31, 2025 and 2024, the Company generated all of its revenue from contracts with customers, a substantial portion of which was generated from contracts with the U.S. Government. A majority of the Company's contracts with the U.S. Government are generally subject to Federal Acquisition Regulation and are priced based on estimated costs of providing the services.

The Company accounts for a contract when the parties have approved the contract and are to perform on it, the rights of each party and the payment terms are identified, the contract has commercial substance, and collection of substantially all of the consideration is probable.

The Company evaluates if its contracts are in the scope of Accounting Standards Codification (“ASC”) 606, Revenue from Contracts with Customers, or other guidance or a combination. If a contract is partially in the scope of other guidance, the Company separates and allocates the arrangement to those components in accordance with ASC 606 unless the other guidance provides its own specific allocation guidance.

At contract inception, the Company determines whether the services to be provided are to be accounted for as a single performance obligation or as multiple performance obligations. This evaluation requires professional judgment and may impact the timing and pattern of revenue recognition.

The Company’s contracts may include variable consideration, such as adjustments to price based on performance or other contractual terms. Variable consideration is estimated at contract inception and is updated throughout the contract term as additional information becomes available. The Company includes variable consideration in the transaction price only to the extent that it is probable that a significant reversal of cumulative revenue recognized will not occur when the uncertainty associated with the variable consideration is resolved.

The Company generally recognizes revenue over time throughout the performance period as the customer simultaneously receives and consumes the benefits provided on services-based arrangements. The Company satisfies its performance obligation as services are rendered. An exception is used for cost-based contracts, based on the cost of services which correspond directly with the Company’s performance completed to date. For fixed-fee contracts, the Company applies the cost-to-cost method — specifically the cost-to-cost approach — where revenue is recognized in proportion to costs incurred, reflecting progress towards complete satisfaction of the performance obligation.

Contract modifications are reviewed to determine whether they should be accounted for as a modification of the original performance obligation or as a separate contract. When a contract modification changes the scope or price and the additional performance obligations are at their standalone selling price, the original contract is terminated and the Company accounts for the change prospectively when the new services to be transferred are distinct from those already provided. When the contract modification includes services that are not distinct from those already provided, the Company records a cumulative adjustment to revenue based on a remeasurement of progress towards the complete satisfaction of the not yet fully satisfied performance obligation.

The Company utilizes other parties in the performance of some services. Based on the Company’s evaluation using a control model, the Company determined that in all of its performance obligations, the Company is the principal.

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as a principal rather than an agent within its revenue arrangements. Revenue and the associated costs are both reported on a gross basis within the consolidated statements of operations and comprehensive income.

Government Grants

Under the Company’s accounting policy for government grants, which is consistent with International Accounting Standard 20 (“IAS 20”) Accounting for Government Grants and Government Assistance, the Company recognizes government grants when there is reasonable assurance that the Company will comply with the conditions attached to the grant arrangement and the funds will be received. This assessment is performed for each grant as of each reporting period. Amounts received are recognized as income over the period in which the Company incurs the related costs.

advance of the Company incurring eligible expenses are recorded as deferred grant income with deferred revenue until the eligible expense is incurred.

For grants related to assets, the Company has elected to record such grants as a deduction of the carrying value of the asset. For grants related to income, the Company has elected to record such grants as income separately from revenues on the Consolidated Statements of Operations and Comprehensive Income. The goods and services provided for under the grant are consistent with the operating activities of the Company. The expenses related to grant income are recorded on a gross basis as direct costs.

Grant income and direct costs related to activities of third parties including subcontractors and grantees are recorded on a gross basis when the Company has the power to redirect the funds to a different party, or retain those funds itself, in order to meet the conditions of the grant. The costs considered include, but are not limited to, whether the Company is responsible for and has discretion over the decision whether to engage third parties, which third parties to engage and on what terms.

Direct Costs

Direct costs on the consolidated statements of operations and comprehensive loss include costs that are directly attributable to providing services under contracts with customers and grants related to operations, such as direct labor, direct materials and subcontracting costs. Indirect costs are allocated to direct costs in the same manner as such costs are defined in disclosure statements under U.S. Government Cost Accounting Standards.

Selling, General and Administrative Expense

Selling, general and administrative expenses consist of human capital related expenses for executive and non-executive management involved in general corporate functions; rent relating to the Company's office space; professional fees; and other general corporate costs.

Unit-Based Compensation

Unit-based compensation represents costs related to unit-based awards granted to executive and non-executive management members of the Board of Directors. The Company recognizes unit-based compensation, including restricted stock, using the accelerated attribution method, based upon the estimated fair value of awards on the grant date and the vesting schedule of the classified awards. Forfeitures are accounted for as they occur. The recognition period for these awards begins on the date at either the applicable service inception date or grant date and continues throughout the required service period. Due to the contingently redeemable nature of Class B Common Units (inclusive of Class B-1 Common Units and Class B-2 Common Units, as discussed in Note 11 — Member Units and Preferred Units and Note 12 — Unit-Based Compensation Expense), the Company classifies these units as mezzanine equity and records the redemption value as of the grant date for vested awards within mezzanine equity on the consolidated balance sheets.

Research and Development

The Company conducts research and development activities related to the development and improvement of technologies pertaining to nuclear reactor and fuel design engineering. The costs for conducting the research and development primarily include equipment, material, and labor. All research and development costs of research and development are expensed in the period incurred.

Other income (expense), net for the years ended December 31, 2025 and 2024 consists of (in thousands):

	Year Ended Dec
	2025
Foreign currency transaction gain (loss)	\$ 924
Mark-to-market loss on C-1 & C-2 Notes ⁽¹⁾	(13,015)
Gain (loss) on conversion of C-1 & C-2 Notes ⁽¹⁾	(4,023)
Mark-to-market gain (loss) on warrant liabilities ⁽¹⁾	(223,532)
Mark-to-market gain on embedded derivatives ⁽¹⁾	—
Mark-to-market gain on 2024 Financial Instrument ⁽¹⁾	—
Reclassification of OCI for conversion of C-1 & C-2 Notes ⁽¹⁾	—
Loss on extinguishment of debt ⁽¹⁾	—
Other income (expense)	345
Total other income (expense), net	\$ (239,301)

(1) Refer to Note 7 — Debt

Foreign Currency Translation

The functional currency of the Company’s foreign operations is the reported local currency. Currency adjustments result from translating the Company’s foreign subsidiaries’ financial statements into U.S. dollars. The balance sheet accounts of the Company’s foreign subsidiaries are translated into U.S. dollars using the exchange rate in effect at the balance sheet date. Revenue and expense accounts are translated using average exchange rates for each month during the fiscal year. The resulting gains or losses are recorded as a component of accumulated other comprehensive income (loss), net of tax. The foreign currency gains and losses are not material for the periods presented.

Income Taxes

The Company is a limited liability company which is classified as a partnership for U.S. federal income tax purposes. Accordingly, the Company’s U.S. operations are not subject to income tax in the U.S. The Company’s owners separately account for their pro rata share of the Company’s income, deductions, losses and credits annually. The Company’s foreign operations are subject to income tax in the foreign jurisdictions in which they operate. Income tax expense and benefit were immaterial for the years ended December 31, 2025 and 2024.

On July 4, 2025, the One Big Beautiful Bill Act (“OBBBA”) was signed into law. The legislation is a sweeping tax and spending law which makes permanent many provisions of the 2017 Tax Cuts and Jobs Act, while introducing new tax policies and restructuring others. OBBBA does not currently have a material impact to the Company.

Other Comprehensive Income (Loss)

Other comprehensive income (loss) refers to revenues, expenses, gains, and losses that are recognized in GAAP but are excluded from net income. The components of other comprehensive income (loss) consist of foreign currency translation adjustments, the reclassification of other comprehensive income for the conversion of the C-1 and C-2 Notes, and the change in fair value of financial liabilities accounted for pursuant to the fair value option attributable to instrument-specific credit risk.

The components of accumulated other comprehensive income for the years ended December 31, 2023 and 2024 are as follows (in thousands):

	Changes in fair value of liabilities under fair value option attributable to changes in instrument specific credit risk	Foreign currency translation adjustments
Accumulated other comprehensive income (loss), balance at January 1, 2024	\$ 582	\$ 297
Other comprehensive income (loss)	(6,220)	474
Reclassification of OCI for conversion of C-1 and C-2 Notes	4,873	—
Accumulated other comprehensive income (loss), balance at December 31, 2024	\$ (765)	\$ 771
Other comprehensive income (loss)	765	(888)
Accumulated other comprehensive income (loss), balance at December 31, 2025	\$ —	\$ (117)

Accounting Standards Not Yet Adopted

In December 2023, the FASB issued ASU No. 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures* (“ASU 2023-09”), to enhance the transparency and decision usefulness of income tax disclosures. ASU 2023-09 is effective for annual periods beginning after December 15, 2024, on a prospective basis and early adoption is permitted. The Company is currently evaluating the impact of the adoption of this standard on its disclosures.

In March 2024, the FASB issued ASU No. 2024-01, *Compensation — Stock Compensation: Application of Profits Interest and Similar Awards* (“ASU 2024-01”), which clarifies the scope of the standard for profits interest awards and similar awards and provides illustrative examples intended to increase diversity in practice in determining whether such awards should be accounted for under Topic 718. For public business entities, ASU 2024-01 is effective for annual reporting periods, including interim periods within those annual periods, beginning after December 15, 2024; for all other entities, the ASU is effective for annual reporting periods, including interim periods within those annual periods, beginning after December 15, 2025; early adoption is permitted for all entities. The Company is currently evaluating the impact of the adoption of this standard on its consolidated financial statements and related disclosures.

In November 2024, the FASB issued ASU No. 2024-03, *Income Statement — Reporting Comprehensive Income — Expense Disaggregation Disclosures (Subtopic 220-40): Disaggregation of Income Statement Expenses* (“ASU 2024-03”), and in January 2025, the FASB issued ASU No. 2025-01, *Income Statement — Reporting Comprehensive Income — Expense Disaggregation Disclosures (Subtopic 220-40): Clarification of Effective Date*, which clarified the effective date of ASU 2024-03. ASU 2024-03 requires public business entities to provide enhanced disclosures about certain categories of expenses in the notes to the financial statements. The guidance is intended to improve the transparency and decision usefulness of the information provided to financial statement users. The ASU is effective for fiscal years beginning after December 15, 2026, and interim periods within fiscal years beginning after December 15, 2026; early adoption is permitted. The Company is currently evaluating the impact of this standard on its disclosures.

In November 2024, the FASB issued ASU No. 2024-04, *Debt with Conversion and Other Options (Subtopic 470-20): Induced Conversions of Convertible Debt Instruments* (“ASU 2024-04”), which clarifies the assessment of whether certain settlements of convertible debt instruments should be accounted for as conversions. The ASU is effective for fiscal years beginning after December 15, 2026, and interim periods within fiscal years beginning after December 15, 2026; early adoption is permitted. The Company is currently evaluating the impact of this standard on its disclosures.

inducement conversion or extinguishment of convertible debt. ASU 2024-04 is effective for annual periods beginning after December 15, 2025 on either a prospective or retrospective basis and early adoption is permitted. The Company's outstanding convertible debt matured in September 2025, prior to the Company's planned adoption date. Therefore, the Company does not expect the adoption of this standard to have a material impact on its consolidated financial statements and related disclosures.

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In May 2025, the FASB issued ASU No. 2025-04, *Compensation — Stock Compensation and Revenue from Contracts with Customers (Topic 606): Clarifications to Share-Based Compensation Payable to a Customer* ("ASU 2025-04"). ASU 2025-04 refines key aspects of the guidance, including the definition of performance condition as well as the measurement requirements and the accounting for forfeitures. ASU 2025-04 is effective for annual periods beginning after December 15, 2025, on a modified retrospective or a retrospective basis and early adoption is permitted. The Company is currently evaluating the impact of the adoption of this standard on its consolidated financial statements and related disclosures.

In July 2025, the FASB issued ASU No. 2025-05, *CECL Practical Expedient for Accounting for Expected Credit Losses on Certain Current Accounts Receivable and Contract Assets* ("ASU 2025-05"), which provides an optional practical expedient for measuring expected credit losses on certain current accounts receivable and contract assets arising from transactions accounted for under Topic 606. ASU 2025-05 is effective for annual reporting periods beginning after December 15, 2025, and interim reporting periods within those annual periods. The Company is currently evaluating whether it will elect the practical expedient and the impact of the adoption on its consolidated financial statements and related disclosures.

In December 2025, the FASB issued ASU No. 2025-10, *Government Grants (Topic 832) — Guidance for Government Grants Received by Business Entities* ("ASU 2025-10"), which establishes consistent guidance for the recognition, measurement, and disclosure of government grants received by business entities. The ASU addresses both monetary and certain nonmonetary government grants and income tax credits. Annual disclosure requirements regarding the nature, terms, and accounting policies related to government grants and income tax credits are required. ASU 2025-10 is effective for annual periods beginning after December 15, 2029 on either a prospective approach, modified retrospective approach, or retrospective approach and early adoption is permitted. The Company is currently evaluating the impact of the adoption of this standard on its consolidated financial statements and related disclosures.

NOTE 3 — REVENUE AND GRANT INCOME RECOGNITION

The Company recognizes revenue under contracts with customers to provide nuclear reactor design engineering services in the areas of research and development, systems engineering, and construction development. The Company's revenues are generally derived from contract services performed for the U.S. Government and commercial entities. All revenues for the years ended December 31, 2025 and 2024 were recognized over time.

There are two main types of contracts: cost-based contracts and cost-plus fixed fee contracts. Consideration in the Company's contracts is paid in the form of incremental incurred cost reimbursement over time upon the delivery of the Company's invoices.

The allowability of certain costs under government contracts is subject to audit by the U.S. Government. Certain indirect costs are charged to contracts using provisional or estimated indirect rates, which are subject to later revision based on actual costs incurred and subject to government audits of the Company's costs. The government has performed audits of the Company's costs. Refer to Note 15 — Commitments and Contingencies, for further discussion of these cost audits.

Advanced Reactor Demonstrator Program

During the year ended December 31, 2021, the Company was named an awardee under the Department of Energy's Advanced Reactor Demonstrator Program ("ARDP"). The objective of the program is to accelerate the development and demonstration of advanced nuclear reactor technologies, designs that improve safety, efficiency, and economic viability for commercial deployment. Under the agreement, the Company will develop the Xe-100 demonstrator reactor and the TRISO-X core fabrication facility through a public-private cost-share partnership. The DoE will obtain certain intellectual property ("IP") developed under ARDP but will not obtain ownership of the reactor or fuel facility. The DoE will fund a portion of the direct and indirect costs incurred for development and development costs to develop the IP and the costs of developing and constructing both the fuel facility and Xe-100 demonstrator reactor. The program provides for an approximate reimbursement of up to \$2.4 billion of eligible costs (\$1.2 billion reimbursement) through 2025. Funding is subject to future government appropriations which may not occur, and the government may cancel the contract at any time without incurring a substantive

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penalty. Dow has been named a subawardee under the ARDP in connection with their involvement in the development and construction of the demonstrator reactor. The Company is constructing and operating the fuel facility.

The Company determined that the ARDP is partially in the scope of ASC 606 and partially outside the scope of other guidance as the DoE only receives benefit from the development of the IP and does not receive benefit from construction of the demonstrator reactor or the fuel facility. The Company has considered the consideration between revenue and grant components of the ARDP on the basis of the stand-alone fair value price of each of the components. The Company has analogized to IAS 20 to account for the portion of the ARDP that is not within the scope of ASC 606 and deemed to be a government grant. The revenue components consist of an asset grant and an income grant. The Company recognizes grants as revenue when the Company incurs the eligible costs. Refer to Note 2 — Summary of Significant Accounting Policies, Recent Accounting Pronouncements and Note 5 — Government Grants for Property and Equipment.

Under the Company's separate agreement with Dow, certain costs incurred by Dow for the demonstrator reactor are funded by the Company using funds obtained through the ARDP. The Company has recognized the funds received from the DoE and the costs incurred by Dow on a gross basis as revenue and income and direct costs, respectively. The Company has recognized \$6.2 million and \$5.6 million as revenue and income and \$5.3 million and \$4.6 million as direct costs during the years ended December 31, 2024 and 2023, respectively related to costs incurred by Dow for the demonstrator reactor. The Company also recognizes revenue from Dow as a customer for certain site specific engineering, permitting, and other services related to the demonstrator reactor. Refer to Note 14 — Related Party Transactions.

Disaggregated Revenues and Grant Income

A summary of revenues and grant income by customer type is as follows for the years ended December 31 (in thousands):

	Year Ended December 31, 2025
Customer type:	
DoE ⁽¹⁾	\$ 89,396
DoD	5,444
Commercial	14,258
Total revenues and grant income	\$109,098

- (1) For the year ended December 31, 2025, \$74.6 million is classified as services revenue and \$14.8 million is classified as grant income. For the year ended December 31, 2024, \$74.6 million is classified as services revenue and \$36.2 million is classified as grant income.

A summary of revenues and grant income by contract type is as follows for the year ended December 31 (in thousands):

	Year Ended December 31, 2025
Contract type:	
Cost-based ⁽¹⁾	\$ 96,340
Fixed fee	2,358
Cost plus fixed fee	5,308
Time & materials	5,092
Total revenues and grant income	\$109,098

- (1) For the year ended December 31, 2025, \$81.5 million is classified as services revenue and \$14.8 million is classified as grant income. For the year ended December 31, 2024, \$81.5 million is classified as services revenue and \$36.2 million is classified as grant income.

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Revenues and Grant Income by Geographic Location

The Company has revenues and grant income in the following countries for the year ended December 31 (in thousands):

	Year Ended December 31, 2025
Customer Location:	
United States	\$108,584
Canada	69
United Kingdom	445
Total Revenues and Grant Income	\$109,098

Contract Balances

The timing of revenue recognition, billings, and cash collections results in billed accounts receivable and unbilled receivables (amounts billable where the right to consideration is unconditional), contract liabilities, and contract assets.

which certain conditions must be satisfied before the right to bill is obtained), and contract assets (customer advances and deposits) on the consolidated balance sheets. Amounts are billed as work progresses in accordance with agreed-upon contractual terms, either at periodic intervals (e.g., biweekly or monthly) or upon achievement of contractual milestones. Generally, billing occurs subsequent to revenue recognition, resulting in unbilled receivables. However, the Company sometimes receives advances or deposits from customers, or bills customers where it has an unconditional right to consideration before revenue is recognized, resulting in contract liabilities. These liabilities are reported on the consolidated balance sheet on a contract-by-contract basis at the end of each reporting period.

Contract liabilities as of December 31, 2025 and 2024 include deferred revenue. As of December 31, 2025, the Company has deferred revenue of \$15.3 million, of which \$15.2 million is recorded in short-term deferred revenue and \$0.1 million is recorded in accrued liabilities. Long-term deferred revenue attributable to related parties is \$2.4 million. During the year ended December 31, 2025, \$1.1 million of revenue was recognized that had previously been deferred. As of December 31, 2024, the Company has deferred revenue of \$1.1 million, of which \$0.3 million is recorded within accrued liabilities and \$0.8 million is recorded within due to related parties, and long-term deferred revenue of zero.

Contract assets represent revenue recognized that exceeds the amount billed to the customer. The Company excludes amounts billable where the right to consideration is solely subject to the passage of time. As of December 31, 2025, the Company has contract assets of \$7.5 million, which is recorded within unbilled receivables and contract assets and is not associated with related parties. As of December 31, 2024, the Company has contract assets of \$9.5 million, of which \$7.6 million is recorded within unbilled receivables and contract assets and \$1.9 million is recorded within due from related parties.

Remaining Performance Obligation

Remaining performance obligations represent non-cancelable contracted revenue that has not yet been recognized and will be recognized as revenue in future periods. The Company has elected the optional exemption for the disclosure of remaining performance obligations as the Company's contracts are predominantly short-term or less in duration.

NOTE 4—PROPERTY AND EQUIPMENT, NET

Property and equipment, net consists of the following (in thousands):

	December 31, 2025
Equipment and materials	\$ 2,335
Computer equipment and software	3,760
Office furniture and fixtures	355
Leasehold improvements	2,708
Land	1,697
Construction-in-progress	42,839
Property and equipment at cost ⁽¹⁾	53,694
Accumulated depreciation	(3,589)
Property and equipment, net	\$ 50,105

(1) Net of reimbursement from the United States Government.

All of the Company's property and equipment is located within the United States.

NOTE 5 — GOVERNMENT GRANTS FOR PROPERTY AND EQUIPMENT

During the years ended December 31, 2025 and 2024, the Company received reimbursements from the DoE under the ARDP related to construction of its nuclear fuel fabrication facility. During the year ended December 31, 2025, the Company also received reimbursements from the DoE under the ARDP related to the purchase of a building for its reactor testing facility. The Company recognizes the grant from the time in which the costs are incurred in compliance with the conditions of the grant, which was \$10.0 million and \$7.7 million for the years ended December 31, 2025 and 2024, respectively.

NOTE 6 — ACCRUED LIABILITIES

The following table sets forth the components of accrued liabilities (in thousands):

	December 31, 2025
Operating lease liabilities	\$ 2,344
Accrued payroll and related expenses	21,735
Accrued subcontractor costs	18,160
Accrued financing costs	—
Accrued transaction costs	2,926
Incurred cost audits reserve (Note 15)	1,068
Accrued professional fees	3,423
Accrued liabilities – other	1,561
Total accrued liabilities	\$ 51,217

NOTE 7 — DEBT

The Company's outstanding debt as of December 31, 2024 was attributable to the C-2 Notes (defined below) and was classified as short-term. The fair value of the C-2 Notes outstanding as of December 31, 2024 equates to the carrying value as the Company elected to apply the fair value measurement of the C-2 Notes. No debt was outstanding at December 31, 2025.

Live Oak Credit Facility

On June 14, 2021, the Company and one of its subsidiaries entered into a revolving credit facility with Live Oak Bank (the “Live Oak Credit Facility”) with maximum borrowings of up to \$15.0 million based on an asset-based borrowing base based on eligible accounts receivable, net of lender reserves. The Company incurred lender and third-party fees and a 0.25% commitment fee on the unused portion of the revolving commitment; such fees were immaterial for the years ended December 31, 2025 and 2024.

The obligations under the Live Oak Credit Facility were guaranteed by the Company and its wholly owned domestic subsidiaries and were secured by a first priority security interest in substantially all of the Company’s and such subsidiaries’ equipment, accounts receivable, investment property, intellectual property, intangibles (and related proceeds), as well as 100% of the common equity interests of the Company and its domestic subsidiaries. The facility could be prepaid without premium or penalty, and prepayments were required in certain customary circumstances.

Borrowings under the Live Oak Credit Facility bore interest at a floating rate indexed to the Prime Rate plus 1.0%, subject to a 4% floor. Interest expense associated with the facility was \$0.4 million for the year ended December 31, 2024.

The facility was amended during 2024 to extend its maturity to October 31, 2024, at which time it matured with no outstanding borrowings following repayment of the outstanding principal in full. The Company reestablished the facility on May 9, 2025 with an expiration date of December 31, 2025. There were no draws during the year ended December 31, 2025, and the facility matured on December 31, 2025 with no outstanding borrowings.

C-1 Convertible Notes

In 2022, the Company issued convertible promissory notes (“C-1 Notes”) with an aggregate principal amount of \$57.4 million and a 7.00% annual interest rate. The C-1 Notes contained embedded derivatives. Absent the election of the fair value option, would have been bifurcated and accounted for as a liability. Accordingly, the Company elected the fair value option and classified the C-1 Notes as a liability. The C-1 Notes were measured at fair value, remeasuring them at each reporting period with changes in fair value recorded in other comprehensive income (except for the portion attributable to instrument-specific credit risk, which was recorded in earnings). The C-1 Notes were converted into 5,957,135 Class B Common Units.

On December 5, 2023, \$37.4 million principal was automatically converted into 5,957,135 Class B Common Units at a discounted price per unit of \$7.94 in connection with a qualified equity financing.

On March 29, 2024, the remaining \$20.0 million principal was converted into 3,210,000 Class B Common Units. In connection with this conversion, the Company issued 757,135 Class B Common Units and entered into a Letter Agreement with a related party investor creating a contingent reimbursement obligation (the “2024 Financial Instrument”) whereby the Company would reimburse the investor for the payments made to former C-1 note holders if certain conditions related to a future financing were met. The 2024 Financial Instrument was accounted for as a liability remeasured at each reporting period with changes in fair value recorded in other income (expense), net. The March 2024 conversion was accounted for as an induced conversion, resulting in recognition of an inducement expense of \$14.8 million, net of an expense, net which included the value of the Class B Common units and 2024 Financial Instrument. Additionally, warrants previously issued in connection with the C-1 Notes were exercised into 1,000,000 Series C Preferred Units.

C-2 Convertible Notes

In 2022 and 2023, the Company issued convertible notes payable in an aggregate principal amount of \$28.0 million and \$85.0 million (“C-2 Notes”), respectively, of which \$70.0 million of the C-2 Notes were issued to related parties. The C-2 Notes were due on September 30, 2025 and accrued 10.0% interest.

kind interest annually. The C-2 Notes provided holders with conversion rights into equity securities under certain conditions, including upon an IPO or at the holder's election after August 4, 2023.

The Series C-2 Convertible Notes contained embedded derivatives which, absent the election of the fair value option, would have been bifurcated and accounted for at fair value. Accordingly, the fair value option was elected. The Company classified the C-2 Notes as a liability at fair value and remeasured them at fair value to

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their fair value at each reporting period, with the portion of the change in fair value attributable to the instrument-specific credit risk recorded within other comprehensive income (loss), and the portion of the change in fair value recorded within other income (expense), net. As the fair value option was elected, the issuance costs associated with the C-2 Notes were expensed in the period incurred. Refer to Note 3, Fair Value Measurements for further information on the remeasurement of the C-2 Notes.

On October 11, 2024, certain of the C-2 Notes with an aggregate principal balance of \$18.4 million were converted into 16,960,021 Series C Preferred Units, which was accounted for as a debt extinguishment. Such extinguishment resulted in a gain on debt extinguishment of \$17.6 million for the year ended December 31, 2024, which was recorded in other income (expense), net in the consolidated statements of operations. Further, since the C-2 Notes were historically accounted for by applying the fair value option in accordance with ASC 825-10, the Company included in net loss the \$6.1 million cumulative loss previously recorded in other comprehensive income (loss) for the extinguished C-2 Notes that resulted from the instrument-specific credit risk.

As of December 31, 2024, the outstanding principal, inclusive of payable-in-kind interest, on the C-2 Notes was \$18.4 million and the fair value was \$18.5 million, resulting in a cumulative loss of \$0.1 million.

On September 30, 2025, the outstanding principal and unpaid accrued interest on the C-2 Notes were converted into 2,870,172 Series C Preferred Units, resulting in the extinguishment of the C-2 Notes. Therefore, as of December 31, 2025, there was no outstanding principal on the C-2 Notes. The cumulative loss recognized on the C-2 Notes was \$4.0 million for the year ended December 31, 2025, which was recorded in other income (expense), net in the consolidated statements of operations.

2023 Bridge Loan

On October 4, 2023, the Company entered into a credit agreement (the "2023 Bridge Loan Agreement") with a related party, Ares Acquisition Holdings, LP for a \$10.0 million term loan. In addition, subject to the terms of the agreement of Ares Acquisition Holdings, LP, additional draws of up to \$10.0 million per calendar quarter were available to the Company and the Company made additional draws of \$14.2 million in 2023.

The annual interest rate on the 2023 Bridge Loan was 12.0%. For the year ended December 31, 2023, interest expense of \$3.0 million with the 2023 Bridge Loan was recorded within interest expense in the consolidated statements of operations. The interest was payable in-kind and increased the principal amount of the 2023 Bridge Loan.

During 2024, the Company entered into a series of amendments to the 2023 Bridge Loan Agreement. Among other things, extended the maturity date to March 26, 2025. The Company concluded that the amendments to the 2023 Bridge Loan constituted, for accounting purposes, debt extinguishment. The Company recognized debt extinguishment losses of \$6.8 million. The Company drew a total of \$25.8 million under the 2023 Bridge Loan during 2024, including draws made in conjunction with amendments. In connection with the draws, the Company issued 1.7 million Class B Common Units to Ares Acquisition Holdings, LP. Class B Common Units have the same rights as the Class B Profits Interests discussed in Note 3, Fair Value Measurements. The Class B Common Units were recorded in the mezzanine equity section of the balance sheet as Based Compensation Expense. The Class B Common Units were recorded in the mezzanine equity section of the balance sheet as Based Compensation Expense.

of the consolidated balance sheets either at fair value, for Class B Common Units issued in conjunction with the debt extinguishments, or at relative fair value, for Class B Common Units issued in conjunction with draws.

On October 11, 2024, in accordance with the 2023 Bridge Loan's stated terms, the Company settled the outstanding principal and interest on the 2023 Bridge Loan of \$53.5 million, and the 2023 Bridge Loan was settled. The settlement was accounted for as an extinguishment of the 2023 Bridge Loan. As a result, the Company recorded a loss on debt extinguishment of \$0.1 million for the year ended December 31, 2024. Other income (expense), net in the consolidated statements of operations and no balance was recorded at December 31, 2024.

The Company assessed the terms of the 2023 Bridge Loan in order to identify embedded derivatives and determine if any such features should be bifurcated and accounted for separately as derivatives. The Company determined that the event of default interest rate adjustment feature, event of default interest rate feature, and contingent additional interest feature required bifurcation and separate accounting for the embedded liabilities. Accordingly, the bifurcated embedded derivatives were recorded at fair value upon issuance and each relevant

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draw and were subsequently remeasured at the end of each reporting period. The bifurcated embedded derivatives that were created upon new draws resulted in a discount on the 2023 Bridge Loan, which was subsequently amortized over the life of the 2023 Bridge Loan through periodic charges to interest expense.

During the year ended December 31, 2024, the Company recorded \$4.2 million in interest expense related to the amortization of the discount created by embedded derivatives. Additionally, the Company recorded embedded derivatives, with a total fair value of \$2.7 million, were recorded as a part of the amortization expense included in the loss on extinguishment. The Company determined the fair value of the embedded derivatives by considering the probability of the settlement of the feature, its settlement value, and other factors. Refer to Note 13 — Fair Value Measurements for further information.

2024 Convertible Note

On September 26, 2024, the Company entered into a convertible note with Amazon Investment Holdings LLC in the principal amount of \$20.0 million (the "2024 Convertible Note"). The 2024 Convertible Note was outstanding, the debt accrued paid-in-kind interest of 12% per annum, compounded quarterly. The interest expense recognized related to the 2024 Convertible Note was immaterial during the year ended December 31, 2024. The 2024 Convertible Note was convertible at the option of the creditor into Series C-1 Preferred Units from the issuance date until the occurrence of a Qualified Financing, which is defined in the 2024 Convertible Note's terms as the Company's completion of a financing round of Series C-1 Preferred Units with total proceeds of at least \$100.0 million on or before the maturity date. The maturity date of the 2024 Convertible Note was March 26, 2025, or upon the receipt of a third-party investment of at least aggregate net cash proceeds of at least \$100.0 million, if such an investment occurs prior to the maturity date.

On October 11, 2024, in conjunction with the Series C-1 Preferred Units financing discussed in Note 12, in accordance with the 2024 Convertible Note's stated terms, the outstanding principal and interest on the 2024 Convertible Note of \$20.1 million was converted into 3,097,477 Series C-1 Preferred Units (the "2024 Convertible Note Conversion"). Since the 2024 Convertible Note was converted via the creditor's exercise of its unit-settled redemption right, the conversion was accounted for as a debt extinguishment. The balance remained outstanding at December 31, 2024. The fair value of the Series C-1 Preferred Units

approximated the value of the extinguished debt plus interest, and therefore, no gain or loss during the year ended December 31, 2024.

2024 Bridge Loan

On September 26, 2024, the Company entered into a bridge loan with Escape2, LLC, an entity affiliated with an investor (the “2024 Bridge Loan”), in the amount of \$3.8 million. While outstanding, the 2024 Bridge Loan accrued paid-in-kind interest of 12% per annum, compounded quarterly. The maturity date of the 2024 Bridge Loan was March 26, 2025, or upon the receipt of a third-party investment with aggregate proceeds of at least \$100.0 million, if such an investment occurred prior to the maturity date. In connection with the issuance of this debt, the Company issued 124,430 Class B Common Units to Ghaffarian Enterprises, LLC, which is controlled by the same related party investor as Escape 2, LLC. The Class B Common Units have the same rights as the Class B Profits Interests discussed in Note 12 — Compensation Expense. The Class B Common Units were recorded in the mezzanine equity section of the consolidated balance sheets and included in the discount of the 2024 Bridge Loan.

On October 11, 2024, in accordance with its stated terms, the 2024 Bridge Loan was redeemed, and the Company paid the outstanding principal and interest associated with the 2024 Bridge Loan of \$3.8 million. Since the 2024 Bridge Loan was initially recognized at its relative fair value, the result of the concurrent issuance of the Class B Common Units to Ghaffarian Enterprises, LLC and the amount at settlement was not equal to the cash paid by the Company. As a result, the debt was extinguished at its carrying value, the cash payment was credited, and a loss on extinguishment was recorded for the difference. The Company recorded an immaterial loss on debt extinguishment during the year ended December 31, 2024 in other income (expense), net in the consolidated statements of operations. The balance remained outstanding at December 31, 2024.

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Bank of New York Credit Facility

On July 28, 2020, the Company executed a credit agreement with Pershing LLC, an affiliate of Bank of New York Mellon, in the form of a revolving credit facility (the “Bank of New York Credit Facility”). The facility was subject to the guarantee by Ghaffarian Enterprises, LLC and Ghaffarian Enterprises, LLC and related party investors.

During the year ended December 31, 2024, the Company entered into Credit Support Subrogation Agreements (the “2024 Credit Support Fee Agreements”) with GM Enterprises, LLC and Ghaffarian Enterprises, LLC, entities affiliated with an investor, which increased the availability of the Bank of New York Credit Facility to \$20.0 million and extended the maturity of the credit facility to March 26, 2025. In conjunction with the 2024 Credit Support Fee Agreements, the Company issued 562,483 Class B Common Units to GM Enterprises, LLC and Ghaffarian Enterprises, LLC a monthly 12% credit support fee to the Company in kind. Pursuant to the terms of the 2024 Credit Support Fee Agreements, the Company paid credit support fees and issued 562,483 Class B Common Units to GM Enterprises, LLC and Ghaffarian Enterprises, LLC. Refer to Note 14 — Related Party Transactions for further information on the credit support fee arrangement. The Class B Common Units have the same rights as the Class B Profits Interests discussed in Note 12 — Compensation Expense. The Class B Common Units were recorded in the mezzanine equity section of the consolidated balance sheets and included in the discount of the Bank of New York Credit Facility.

On October 11, 2024, in accordance with the facility’s stated terms, the Company redeemed the outstanding principal and interest associated with the Bank of New York Credit Facility with cash of \$20.2 million, and the 2024 Credit Support Fee Agreements were terminated. The Bank of New York Credit Facility was extinguished at its carrying value, and a loss on extinguishment was recorded for the difference.

Facility did not have an outstanding balance as of December 31, 2024, and the facility mat balance on March 26, 2025.

NOTE 8—LEASES

The Company leases office spaces, which have initial operating lease terms of three to Some leases have options to extend the lease term, ranging from six months to six years. Certain early termination options, which the Company is not reasonably certain to exercise.

During 2024, the Company modified certain existing leases to extend lease terms, w ranging from approximately six months to two years.

During the year ended December 31, 2025, the Company entered into a lease agreement office space, with a lease term of twelve years commencing in August 2025. The lease agreem \$3.9 million refundable security deposit, of which \$3.5 million is classified as restricted consolidated balance sheets, which is not considered a lease payment and is excluded from the of the right-of-use (“ROU”) asset and lease liability. The lease agreement also includes \$17 lease incentives, of which the lessor has provided a \$14.7 million tenant improvement allowa treated as a lease incentive for lessee-owned improvements. The lease incentives reduce the asset and lease liability recognized at lease commencement as the Company is reasonably cert the lease incentives.

Additionally, during the year ended December 31, 2025, the Company amended exist leases to extend lease terms, increase the leased premises, and to exercise early termination op the year ended December 31, 2025, the Company paid early termination fees totaling \$2.8 milli

Lease balances as of December 31, 2025 and 2024 are as follows (in thousands):

	Decemb
	2025
Operating lease ROU assets	\$22,696
Current portion of operating lease liabilities	2,344
Long-term portion of operating lease liabilities	20,887
Total operating lease liabilities	<u>\$23,231</u>

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The current portion of operating lease liabilities is reflected within accrued liabi consolidated balance sheets.

Operating lease cost totaled \$6.5 million and \$4.4 million for the years ended December 2024, respectively, and was recognized on a straight-line basis over the lease term. Variable le and short-term lease expenses were immaterial for the years ended December 31, 2025 and 202

Future minimum lease payments under operating leases as of December 31, 2025 are a thousands):

2026
2027
2028
2029
2030
Thereafter
Total minimum lease payments
Less: lease incentives
Less: amounts representing interest or imputed interest
Present value of lease obligations

Supplemental cash flow information relating to the Company’s leases is as follows (in thousands):

	December 31, 2025
Operating cash flows used in operating leases	\$ 8,461
Non-cash items:	
Change to ROU asset and lease liability due to lease modifications and reassessments	(1,616)
ROU assets recorded under new operating leases	15,748

The weighted average remaining lease term and discount rates for operating leases are as follows:

	Year Ended December 31, 2025
Weighted average remaining lease term	9.3 years
Weighted average discount rate	10.0%

NOTE 9 — PROFIT SHARING PLAN

The Company maintains a defined contribution profit sharing plan (the “Plan”), which includes a deferral arrangement, under the provisions of Section 401(k) of the Internal Revenue Code. Employees are eligible to participate in the Plan on the date of employment. Company matches 100% of employee contributions up to 5% of annual compensation. The Company contributed \$4.4 million and \$4.1 million to the Plan during the years ended December 31, 2025 and 2024, respectively.

NOTE 10 — INCOME TAX

X-energy is a limited liability company which is classified as a partnership for U.S. federal income tax purposes. Accordingly, the Company’s U.S. operations are not subject to income tax in the U.S. The Company’s owners separately account for their pro rata share of the Company’s income, deductions,

losses and credits annually. The Company's foreign operations are subject to income taxes in the jurisdictions in which they operate.

The following table sets forth the components of income (loss) before income taxes recorded in the consolidated statements of operations and comprehensive loss (in thousands):

	Year Ended December 31,
	2025
U.S.	\$(389,611)
Foreign	(167)
Total	<u><u>\$(389,778)</u></u>

No income tax (expense) benefit was recorded for the years ended December 31, 2025 and 2024.

A reconciliation of income taxes computed at the U.S. federal statutory income tax rate to the Company's income tax (expense) benefit was as follows:

	December 31,
	2025
U.S. federal statutory tax rate	21.0%
Domestic income not subject to income tax	(21.0%)
Foreign rate differential	0.0%
Prior year adjustments	0.0%
Other	0.0%
Valuation allowance	(0.0%)
	<u>0.0%</u>

The Company's effective tax rate for the years ended December 31, 2025 and 2024 was 0.0%.

The components of deferred tax assets and liabilities are as follows (in thousands):

	December 31,
	2025
Deferred tax assets:	
Property and equipment	\$ 0
Net operating losses	4,053
Total deferred tax assets	4,053
Valuation allowance	(4,053)
Total deferred tax assets net of valuation allowance	0
Deferred tax liabilities:	
Foreign exchange	0
Total deferred tax liabilities	0
Net deferred tax assets/(liabilities)	<u><u>\$ —</u></u>

The Company's net deferred tax assets are comprised primarily of net operating loss carryforwards. The ultimate realization of the net deferred tax assets is dependent upon the generation of future taxable income sufficient to utilize the deferred tax assets on the Company's income tax returns. The Company has determined that its net deferred tax assets are not more likely than not going to be realized.

Company's three- year cumulative loss position and the generation of future taxable income. The Company is subject to taxation in the United States and various foreign jurisdictions. Tax years 2023 and forward are open for examination in the United States, tax years 2023 and forward are open for examination in the United Kingdom, and all years are open in Canada. Considering this and the Company recognized a

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full valuation allowance of \$4.1 million and \$3.9 million as of December 31, 2025 and 2024, respectively. The valuation allowance increased by \$0.2 million during the year ended December 31, 2025 and decreased by \$2.1 million during the year ended December 31, 2024.

As of December 31, 2025 and 2024, the Company had unused Canadian net operating losses of \$7.5 million and \$7.3 million, respectively, which begin to expire in 2041. As of December 31, 2024, the Company had unused United Kingdom net operating losses of \$8.2 million and \$8.1 million, respectively, which can be carried forward indefinitely.

The Company had no liability for uncertain tax positions and had no interest or penalties related to unrecognized tax benefits as of December 31, 2025 and 2024.

NOTE 11 — MEMBER UNITS AND PREFERRED UNITS

Common Units

The holders of Class A Common Units are entitled to one vote for each unit of Class A Common Unit held at all meetings of stockholders. The voting, dividend, and liquidation rights of the holders of Class A Common Units are subject to and qualified by the rights, powers, and privileges of the holders of Class B Common Units set forth in the original or amended Certificate of Incorporation. The Class A Common Units have not been classified as mezzanine equity as they are redeemable for cash upon the occurrence of certain events that are considered outside of the Company's control.

The Class B Common Units have the same rights and preferences as the Class A Common Units, except that provided however, the Class B Common Units shall not have voting rights and are subject to a 10% ownership interest threshold as may be set forth in grant agreements under the Company's unit-based compensation plan discussed in Note 12 — Unit-Based Compensation Expense or certain credit agreements discussed in Note 7 — Debt.

During the year ended December 31, 2024, the Company issued 3,126,887 Class B-2 Common Units in conjunction with the issuance and modification of debt and in conjunction with the conversion of convertible Notes as discussed in Note 7 — Debt. The Class B-2 Common Units are classified as mezzanine equity as they are redeemable for cash upon the occurrence of certain events that are considered outside of the Company's control.

Series A Preferred Units

The Series A Preferred Units were issued to a related party in 2023 in conjunction with the Company's Preferred financing. The units were recognized at fair market value on the issuance date.

Series A-1 Preferred Units

The Series A-1 Preferred Units were issued in 2023 together with the Series C Preferred Units and Series B-2 Common Units in exchange for a cash investment of \$50.0 million. The Series A-1 Preferred Units have been initially recognized at their relative fair market value.

Series B Preferred Units

The Series B Preferred Units were issued prior to 2023. The Series B Preferred Units were initially recognized at fair market value.

Series C Preferred Units

During 2023, the Company issued 16,340,900 Series C Preferred Units through a financing round, a party investment (resulting in a loss on issuance), and the conversion of C-1 outstanding notes.

During the year ended December 31, 2024, 20,752,520 Series C Preferred Units were issued in connection with the C-1 Conversion, the C-2 Note Conversion, and the exercise of the C-1 Warrants. The Series C Preferred Units issued as a result of these transactions were recognized at fair market value.

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On September 30, 2025, the Company's C-2 Note was converted into 2,870,172 Series C Preferred Units. The Series C Preferred Units issued as a result of this transaction were recognized at fair market value.

Refer to Note 7 — Debt for further discussion on these convertible note conversions and the exercise.

Series C-1 Preferred Units

During 2024, the Company completed a financing round of Series C-1 Preferred Units with the receipt of gross cash proceeds of approximately \$626.5 million from various investors. In this financing round, the Company issued 99,672,593 Series C-1 Preferred Units. The Series C-1 Preferred Units issued through these transactions were recorded at fair value.

In connection with the issuance of the Series C-1 Preferred Units, one investor and potential customer was issued a warrant on Series C-1 Preferred Units (the "2024 Warrant") for no consideration. The 2024 Warrant is required to be classified as a liability and measured at fair value on an ongoing basis pursuant to ASC 480 since the underlying Series C-1 Preferred Units are redeemable upon the occurrence of certain events that are considered outside of the Company's control. The Warrant may be exercised by the holder at any time from the issuance date until the 18-month term of the issuance date and entitles the holder to purchase up to 40,214,207 Series C-1 Preferred Units at an exercise price of approximately \$7.46 per unit. As the 2024 Warrant was issued for no consideration and did not meet the criteria for capitalization of a contract asset, the issuance of the 2024 Warrant of \$55.3 million was expensed upon its issuance. The expense is reflected in selling, general, and administrative expenses within the consolidated statement of operations for the year ended December 31, 2024.

On January 24, 2025, the Company completed additional financing rounds of Series C-1 Preferred Units, issuing 8,235,521 Series C-1 Preferred Units and receiving gross cash proceeds of approximately \$53.4 million from various investors. On October 3, 2025, the Company issued a warrant to a potential customer to purchase up to 14,123,859 Series C-1 Preferred Units at an exercise price of \$6.00 per unit.

(the “2025 Warrant”). See Note 14 -Related Party Transactions for the details of the 2025 Series C-1 Preferred Units issued through these transactions were recorded at fair value.

Series D Preferred Units

On November 21, 2025, the Company completed a financing round of Series D Preferred 48,154,955 Series D Preferred Units and receiving gross cash proceeds of approximately \$7. The Series D Preferred Units issued through these transactions were recorded at fair market value.

Preferred Units

Together, the Series A Preferred Units, the Series A-1 Preferred Units, the Series B Preferred Units, the Series C Preferred Units, the Series C-1 Preferred Units, and the Series D Preferred Units shall be referred to as the “Preferred Units” for the purpose of this Note. Preferred Units have been issued in several series, each of such series consisting of such number of units and to have such terms, rights, preferences, and the qualifications and limitation with respect thereto, as stated or expressed in the original or amended Certificate of Incorporation. Specifics regarding the conversion features and other terms associated with, and the balance sheet classification of, the Preferred Units outstanding are as follows:

Optional Conversion Feature

Each Preferred Unit is convertible, at the option of the holder thereof, at any time, and upon the payment of additional consideration by the holder thereof, into an equal number of Class A Common Units as is determined by dividing the Original Issue Price, initially (i) \$0.5754 per Unit for each Series A Preferred Unit, (ii) \$7.6348 for each Series A-1 Preferred Unit, (iii) \$9.1162 per Unit for each Series B Preferred Unit, (iv) \$7.6348 per Unit for each Series C Preferred Unit, (v) \$6.487 per Unit for each Series C-1 Preferred Unit, and (vi) \$14.53 per Unit for each Series D Preferred Unit, for each preferred unit by the conversion price, initially equal to the applicable original issue price per unit of each series of Preferred Units in effect at the time of conversion.

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the time of conversion. Outstanding Preferred Units are therefore convertible into the following number of Class A Common Units as of December 31, 2025: 90,625,588 relating to Series A Preferred Units, 8,808,351 relating to Series A-1 Preferred Units, 11,643,171 relating to Series B Preferred Units, 39,963,592 relating to Series C Preferred Units, 107,908,114 relating to Series C-1 Preferred Units, and 48,154,955 related to Series D Preferred Units. Conversion may be effected at any time at the discretion of the holder.

Automatic Conversion Feature

In the event of (i) a Qualified IPO, with the exception of the Series D Preferred Units, or (ii) the approval of the requisite holders, the Preferred Units will automatically be converted into an equal number of Class A Common Units at the applicable conversion ratio.

For Series D Preferred Units specifically, if the Company undertakes a Qualified IPO, the conversion price of the Series D Preferred Units will be adjusted, if necessary, to be equal to the lesser of (a) the price of the Series D Preferred Units as of the date of the IPO, or (b) the price per Class A Common Unit, or the price per the equivalent common unit, of the Company or its corporate successor, in either scenario offered to the public in such IPO; or (b) the then-current conversion price of the Series D Preferred Units prior to such adjustment.

Voting Rights

Each of the holders of Class A Common Units, Series A Preferred Units, Series B Preferred Units, Series C Preferred Units, Series C-1 Preferred Units, and Series D Preferred Units are entitled to elect members of the Board of Directors in whatever manner necessary to ensure that the size of the Board of Directors of the Company is set and remains at nine directors who serve as the managers of the Company. The Series A-1 Preferred Units and Class B Common units do not vote on the Board of Directors.

Cumulative Preferred Return

The Preferred Units have an annual 3% cumulative preferred return on the original issue price, beginning on the date such Preferred Units were issued. As of December 31, 2025, the cumulative preferred return not reflected in the consolidated balance sheet for Series A, A-1, B, C, C-1, and D Preferred Units was \$7.0 million, \$4.2 million, \$14.1 million, \$14.2 million, \$24.4 million, and \$2.2 million, respectively. As of December 31, 2024, the accumulated preferred return not reflected in the consolidated balance sheet for Series A, A-1, B, C, and C-1 Preferred Units was \$5.4 million, \$2.2 million, \$5.5 million, and \$3.5 million, respectively. In the event of a deemed liquidation, the cumulative preferred return is payable solely for the Series B Preferred Units and therefore has been incorporated into the Series B Preferred Units liquidation preference amount presented on the balance sheet. The cumulative preferred return on the remaining units is subject to discretion of the Board of Directors.

Liquidation Preference

In the event of any liquidation, dissolution, or winding up of the Company, or a deemed liquidation event, including a merger or consolidation, or a sale or other disposition of all or substantially all of the Company's assets, the holders shall receive distributions in an amount up to their liquidation preference. See the consolidated balance sheet for the liquidation preference amounts. The preference of distributions shall be paid first to both Series D Preferred Unit and Series C-1 Preferred Unit holders *pari passu*, second to Series A Preferred Unit holders, third to Series B Preferred Units holders, fourth to Series A and Series C Preferred Units holders, and fifth to Class B-1 Profits Interest Units, until members have received distributions equal to approximately \$9.5 million. After the payment of the full liquidation preference, the redeemable convertible preferred stock and profits interests, the Company's remaining assets, if any, available for distribution, if any, would be distributed ratably to the holders of Class A Common Units.

Classification

A redeemable equity security is to be classified as temporary or mezzanine equity if it is not redeemable upon the occurrence of an event that is not solely within the control of the issuer.

The Preferred Units and Common Units are redeemable for cash upon the occurrence of a liquidation event. As of December 31, 2025 and 2024, a deemed liquidation event is not probable, and the occurrence of such an event is considered to be outside of the Company's control. Accordingly, the Preferred Units and Common Units are considered conditionally redeemable. The occurrence of an event that is not solely within the control of the issuer and, therefore, the Company classified the Preferred Units and Common Units as mezzanine equity in the consolidated balance sheet as of December 31, 2025 and 2024. The mezzanine equity amount related to Common Units is based on the grant date redemption value, or the modification date redemption value, if applicable. The difference between the grant date fair value and the redemption value for vested

presented in permanent equity. As of December 31, 2025 and December 31, 2024, \$74.9 million and \$72.4 million, respectively, has been presented within mezzanine equity, and the excess of redemption value over grant date fair value for vested awards has been recorded within additional paid in capital and the accumulated deficit for those years, respectively. Additionally, the Company does not believe that the occurrence of contingent events and the redemption of the Class B Common Units is probable to occur.

NOTE 12 — UNIT-BASED COMPENSATION EXPENSE

Class B-1 Profits Interest Units

On September 12, 2022, the Company issued service-vesting Class B-1 PIUs in exchange for the Class B-1 Options previously granted as unit-based compensation. The Class B-1 PIUs contain a catch-up provision payable upon a deemed liquidation event to compensate holders for the loss in appreciation from the strike price of the options to the profits interest participation threshold. Generally, Class B-1 PIUs vest 25% on the first vesting date, and an additional 25% on each anniversary of the vesting date until the units are fully vested. The Company has the right to repurchase vested units at fair value upon termination, however the Company does not have plans to exercise this right.

The following table summarizes the activity related to the Company's Class B-1 Profits Interest Units for the years ended December 31, 2025 and 2024:

	Number of Units	Weighted Average Grant Date Fair Value
Unvested Class B-1 Units at January 1, 2024	231,250	
Vested	(231,250)	
Forfeited	—	
Unvested Class B-1 Units at December 31, 2024	—	
Vested	—	
Forfeited	—	
Unvested Class B-1 Units at December 31, 2025	—	

Issuance of Class B-2 Profits Interest Units

Also on September 12, 2022, the Company issued service-vesting Class B-2 PIUs to certain employees. The Class B-2 PIUs have identical rights and participations as the Class B-1 PIUs with the exception of the catch-up distribution. Class B-2 PIUs vest 25% on the first vesting date, and an additional 25% on each anniversary of the vesting date until the units are fully vested. The Company has the right to repurchase vested units at fair value upon termination, however the Company does not have plans to exercise this right.

In May and June of 2025, under the Company's 2022 long-term incentive plan, the Company issued 9,794,000 additional Class B-2 PIUs, with an aggregate fair value of \$21.4 million, with some units vesting upon the grant, and the remaining vesting 25% on the first vesting date, and an additional 25% on each anniversary of the vesting date until the units are fully vested, and a two year sale restriction in addition to existing equity transfer restrictions in the Limited Liability Agreement. The fair value of the awards is based on a Black-Scholes model and requires significant judgements and use of assumptions, particularly with regard to time to exit and volatility. Volatility is determined by reference to the volatility of several publicly traded companies in the same industry.

traded companies that are similar to the Company in its industry sector. The expected time to grant date, is 2.9 years. This period reflects the anticipated duration before the units can be sold. The significant inputs into the valuation of these PIUs are:

Significant Inputs

Risk-free rate

Equity volatility

The risk-free rate utilized was based on term-matched U.S. Treasury Constant Maturity selected volatility rate reflected the expected variability in the value of the new Class B Common volatility estimate is derived from equity class volatilities derived from the option pricing dividends are expected to be paid on the new Class B Common PIUs during the restriction period

In July and August 2025, under the Company's 2022 long-term incentive plan, the Company additional 471,000 Class B-2 PIUs, with an aggregate fair value of \$1.6 million, with some vesting upon the grant. Additionally, in December 2025, the Company granted 8,230,995 Class B-2 PIUs with an aggregate fair value of \$25.2 million. These PIUs have a two year sale restriction in existing equity transfer restrictions in the Limited Liability Agreement. The fair value of the award on a Black-Scholes model and requires significant judgements and use of estimates, particularly to time to exit and volatility. Volatility is determined by reference to the actual volatility of similar traded companies that are similar to the Company in its industry sector. The expected time to grant date, is 2.2 years. This period reflects the anticipated duration before the units can be sold. The significant inputs into the valuation of these PIUs are:

Significant Inputs

Risk-free rate

Equity volatility

The risk-free rate utilized was based on term-matched U.S. Treasury Constant Maturity selected volatility rate reflected the expected variability in the value of the new Class B Common volatility estimate is derived from equity class volatilities derived from the option pricing dividends are expected to be paid on the new Class B Common PIUs during the restriction period

The following table summarizes the activity related to the Company's Class B-2 Profits Interests for the years ended December 31, 2025 and 2024:

	Number of Units	Weighted Average Date
Unvested Class B-2 Units at January 1, 2024	1,181,500	\$
Vested	(464,000)	
Forfeited	(34,500)	
Unvested Class B-2 Units at December 31, 2024	683,000	\$
Granted	18,495,995	
Vested	(2,877,500)	
Forfeited	(685,500)	
Unvested Class B-2 Units at December 31, 2025	15,615,995	\$

Unit-Based Compensation

For the year ended December 31, 2025, unit-based compensation expense of \$11.5 million was classified as selling, general, and administrative expense, \$2.6 million was classified as direct costs, and \$0.5 million was capitalized into property and equipment as a part of the Company's construction project. For the year ended December 31, 2024, unit-based compensation expense of \$1.0 million was classified as selling, general, and administrative expense and \$0.1 million was classified as direct costs. The total fair value of units

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that vested during the years ended December 31, 2025, and 2024 was \$6.7 million and \$10.7 million, respectively. As of December 31, 2025, the Company had \$32.7 million of unrecognized compensation that is expected to be recognized over a weighted average period of 2.3 years.

NOTE 13 — FAIR VALUE MEASUREMENTS

The liabilities recorded at fair value and measured on a recurring basis are all Level 3 measurements as of December 31, 2025 and 2024.

Rollforward of Level 3 Measurements

A roll forward of the Level 3 measurements as of December 31, 2025 and 2024 is as follows:

	June 2022 Warrants	October 2022 Warrants	December 2022 Warrants	C-1 Notes	C-2 Notes	2024 Warrants	2025 Financial Instruments
Beginning balance as of January 1, 2024	\$ 3,082	\$ 4,450	\$ 912	\$ 25,901	\$ 121,240	\$ —	\$ —
Exercises/Settlements	—	(5,175)	—	—	—	—	—
Conversions	—	—	—	(27,138)	(116,800)	—	—
Issuances	—	—	—	—	—	55,252	12,000
Change in fair value recognized in other income (expense), net	554	725	482	1,237	7,877	(9,648)	(12,000)
Change in fair value attributable to instrument- specific credit risk recognized in other comprehensive income ⁽¹⁾	—	—	—	—	6,220	—	—
Ending balance as of December 31, 2024 .	\$ 3,636	\$ —	\$ 1,394	\$ —	\$ 18,537	\$ 45,604	\$ —
Beginning Balance as of January 1, 2025	3,636	—	1,394	—	18,537	45,604	—
Conversions	—	—	—	—	(30,787)	—	—
Change in fair value recognized in other income (expense), net	7,140	—	(1,394)	—	13,015	217,786	—
Change in fair value attributable to instrument- specific credit risk recognized in other comprehensive income ⁽¹⁾	—	—	—	—	(765)	—	—
Ending Balance as of December 31, 2025.	\$10,776	\$ —	\$ —	\$ —	\$ —	\$263,390	\$ —

- (1) As the fair value option was elected for the C-1 Notes and C-2 Notes, the change attributable to instrument-specific credit risk associated with each instrument is recorded in comprehensive income (loss), and the remaining change in fair value is recorded within (expense), net on the consolidated statements of operations and comprehensive loss. The instrument-specific credit risk is estimated based on the option adjusted spread of companies and the option adjusted spread of the US high yield index for companies with ratings.

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Valuation Techniques and Significant Unobservable Inputs

A significant increase or decrease in any of the significant unobservable inputs used in measurement of the warrant liabilities or C-2 Notes could result in a significantly higher or lower measurement. The quantitative inputs and assumptions used for items categorized in Level 3 fair value hierarchy as of December 31, 2025 and December 31, 2024 are summarized below:

2024 Warrants

The fair value of the 2024 Warrants is based on an option pricing method of allocation. The inputs, including significant unobservable inputs, used in the recurring Level 3 fair value measurement of the 2024 Warrants as of December 31, 2025 and 2024 are as follows:

Significant Inputs	December 31, 2025
Expected term (years)	0.3
Equity volatility	80.6% – 94.0%
Risk-free rate	3.6%

June 2022 Warrants

In conjunction with a guarantee to the Bank of New York Credit Facility, the Company issued \$10.0 million of warrants (the “June 2022 Warrants”) to Ghaffarian Enterprises on June 30, 2022. The fair value of the June 2022 Warrants is based on an option pricing method of allocation. The significant inputs, including significant unobservable inputs, used in the recurring Level 3 fair value measurement of the June 2022 Warrants as of December 31, 2025 and 2024, are as follows:

Significant Inputs	December 31, 2025
Expected term (years)	0.3 – 2.2
Equity volatility	94.0 – 101.8%
Risk-free rate	3.5 – 3.6%

December 2022 Warrants

On December 5, 2022, the Company issued 444,444 warrants (the “December 2022 Warrants”) to purchase common units of the Company. The December 2022 Warrants have an exercise price of \$1.00 per unit and are not exercisable until 36 months following the issuance of the C-2 Notes, subject to the achievement of certain business requirements. The December 2022 Warrants are freestanding

instruments and are liability classified as of December 31, 2024. As of December 31, 2024, the December 2022 Warrants expired as business requirements of the holder were not met.

The fair value is determined based on an option pricing method of allocation using the strike price per the warrant agreement, and an estimate of the per unit value of the warrants. The significant inputs, including significant unobservable inputs, used in the recurring Level 3 fair value measurement of the December 2022 Warrants as of December 31, 2024 are as follows:

Significant Inputs
Expected term (years)
Equity volatility
Discount for lack of marketability
Risk-free rate

C-2 Notes

As of December 31, 2025, all C-2 Notes were converted into Series C Preferred Units. Refer to Note 7 — Debt for further information.

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The fair value of the C-2 Notes is based on the binomial lattice model, which is considered a Level 3 fair value measurement. The significant inputs, including significant unobservable inputs, used in the recurring Level 3 fair value measurements of the C-2 Notes as of December 31, 2024 are as follows:

Significant Inputs
Discount rate
Credit spread
Equity volatility
Risk-free rate

Held-to-Maturity Securities — Amortized Cost and Fair Value

The estimated fair values of the Company’s financial instruments that are not measured at fair value on a recurring basis, categorized based upon the fair value hierarchy, at December 31, 2025 are as follows:

	Carrying Value	Fair Value Level 1
December 31, 2025		
Cash equivalents		
Money market fund	\$152,951	\$152,951
Commercial paper and certificates of deposit	45,430	—
Corporate bonds	36,021	—
Foreign issuer debt securities	1,191	—
Total in cash and cash equivalents	\$235,593	\$152,951
Short term investments		
Corporate bonds	\$163,331	\$ —
Government treasury bills	88,154	88,206
Commercial paper and certificates of deposit	35,717	—
Foreign issuer debt securities	17,706	—
Total in short-term investments	\$304,908	\$ 88,206
Long term investments		
Corporate bonds	\$208,419	\$ —
Government treasury bills	43,468	43,540
Foreign issuer debt securities	9,571	—
Total in long-term investments	\$261,458	\$ 43,540

The following table summarizes the carrying values and fair values of the Company's debt securities held to maturity at December 31, 2025. The following table summarizes the carrying values and fair values of the Company's debt securities held to maturity at December 31, 2025. The following table summarizes the carrying values and fair values of the Company's debt securities held to maturity at December 31, 2025. (in thousands):

	Amortized Cost Basis	Allowance for Credit Losses	Net carrying amount	Gross Unrealized Gains	Gross Unrealized Losses
U.S. Government securities	\$131,623	\$ —	\$131,623	\$ 123	\$ —
Corporate securities	407,466	—	407,466	105	(28)
Foreign securities	27,277	—	27,277	5	(1)
Total held to maturity securities	\$566,366	\$ —	\$566,366	\$ 233	\$ (29)

The Company did not hold held to maturity securities during the year ended December 31, 2024.

The amortized cost and estimated fair value of held to maturity securities at December 31, 2025, by contractual maturity are shown below (in thousands):

	Amortized Cost Basis
Due in less than one year	\$304,908
Due after one year through five years	261,458
Total	\$566,366

Expected maturities may differ from contractual maturities because borrowers have the right to prepay obligations with or without call or prepayment penalties.

NOTE 14—RELATED PARTY TRANSACTIONS

In the ordinary course of business, the Company enters into transactions with various related parties.

Related Party Warrants

On October 3, 2025, agreements with a related party customer were modified through the 2025 Warrant and the Company accounted for such modification in accordance with ASC 606. The 2025 Warrant may be exercised by the holder when all of the Xe-100 units associated with the agreements are considered fully operational (the “Vesting Event”). The exercise period ends at the earliest of (i) one year from the Vesting Event, (ii) event of non-compliance and (iii) a consummation of an initial public offering, a change of control or a listing event of the Company, to the extent it occurs after vesting. The Vesting Event is considered a performance condition in accordance with ASC 718, as the Vesting Event is a target defined solely by reference to the Company’s own operations. As the 2025 Warrant was issued for consideration from a related party customer and not in exchange for a distinct good or service that the customer transfers to the Company, the Company accounts for the 2025 Warrant as consideration payable to a customer under ASC 606. The Company will recognize the Warrant as a decrease to the transaction price, measured based on the grant date fair value when the performance condition is deemed probable. The grant date fair value was determined to be \$162.7 million.

Moreover, the Company determined that the 2025 Warrant will be classified as an equity instrument on the criteria of ASC 480 and ASC 718. When the performance condition is deemed probable, the Company will recognize the 2025 Warrant as a temporary equity classified instrument on its balance sheet, measured based on its grant date redemption value. Upon the performance condition becoming probable, any difference between the 2025 Warrant’s grant date fair value and its grant date fair value shall be presented in permanent equity.

As of December 31, 2025, the 2025 Warrant was not deemed probable of vesting, and the Company did not recognize any amount related to the 2025 Warrant. The Company will assess the probability that the 2025 Warrant will vest at the end of every reporting period.

In connection with the issuance of the Series C-1 Preferred Units, one investor and one related party customer was issued a warrant on Series C-1 Preferred Units (the “2024 Warrant”) for consideration. The 2024 Warrant is required to be classified as a liability and measured at fair value on an ongoing basis pursuant to ASC 480 since the underlying Series C-1 Preferred Units are redeemable at the option of the Company upon the occurrence of certain events that are considered outside of the Company’s control. The 2024 Warrant may be exercised by the holder at any time from the issuance date until the 18-month period following the issuance date and entitles the holder to purchase up to 40,214,207 Series C-1 Preferred Units at an exercise price of approximately \$7.46 per unit. As the 2024 Warrant was issued for consideration and did not meet the criteria for capitalization of a contract asset, the issuance of the 2024 Warrant of \$55.3 million was expensed upon its issuance. The expense is reflected in selling, general, and administrative expenses within the consolidated statement of operations for the year ended December 31, 2024.

Related Party Revenue

Dow is a unitholder and warrant holder of X-Energy. During the year ended December 31, 2024, the Company entered into the Master Project Development Agreement (“MPDA”) with Dow, which

continuation of the project under the previously existing joint development agreement with into during the year ended December 31, 2023, whereby the MPDA and joint development together are considered the “Dow Agreements.”

Additionally, during the year ended December 31, 2024, the Company entered into a separate related party to provide design and engineering services.

The table below summarizes related party revenues that are reflected within revenue in the statement of operations and comprehensive loss for the years ended December 31, 2025 (in thousands):

	Year Ended Dec 2025	
Revenue associated with Dow	\$ 6,943	\$
Revenue associated with design and engineering services	—	\$
Total	\$ 6,943	\$

Related Party Expenses

The Company enters into various arrangements with related party vendors. These primarily include subcontracting services with a related party investor, and general and administrative services, as well as credit support fees, with an affiliate of the Chairman of the Board of Directors of the Company.

The subcontracting services primarily pertain to support services related to ARDP, and the administrative services include rent for office space, consulting services, and other administrative services. The credit support fee relates to the Bank of New York Credit Facility above in Note 7 — Debt. The table below summarizes the expenses with related parties that are reflected within the consolidated statements of operations and comprehensive loss the years ended December 31, 2024 (in thousands):

	Year Ended Dec 2025	
Subcontracting services ⁽¹⁾	\$ 29,982	\$
General and administrative services ⁽²⁾	1,490	\$
Credit support fee ⁽³⁾	—	\$
Total	\$ 31,472	\$

- (1) Expenses relating to subcontracting services are reflected within direct costs in the consolidated statements of operations and comprehensive loss.
- (2) Expenses related to general and administrative services are reflected within selling, general and administrative expenses in the consolidated statements of operations and comprehensive loss.
- (3) Expenses related to credit support fee are reflected within interest expense in the consolidated statements of operations and comprehensive loss.

Due to/from Related Parties

As of December 31, 2025 and 2024, the following balances were recorded within due to/from related parties (in thousands):

	Decem 2025
Due from related parties ⁽¹⁾	\$ 4,580
Due to related parties	4,225
Short-term deferred revenue	
Total due to related parties	\$ 4,225
Long-term deferred revenue	\$ 2,353

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- (1) As of December 31, 2025 and 2024, \$4.5 million and \$15.8 million, respectively, is related to parties which \$0.7 million and \$7.7 million, respectively, is unbilled.

Refer to Note 7 — Debt for disclosures related to debt from related parties.

Upon closing of Series D Preferred Units and resulting change in capital ownership, the Company reevaluated its related parties. Accordingly, certain entities were previously considered related parties but that the Company has historically conducted business with are no longer considered related parties as defined in ASC 850 beginning on November 21, 2025. The Company has disclosed transactions with these entities prior to November 21, 2025 with these entities as related party transactions.

NOTE 15 — COMMITMENTS AND CONTINGENCIES

Incurred Cost Audits

The Company predominantly performs contract services for the DoE. The Company's costs for such contracts with the DoE are subject to final incurred cost audits prior to the close out of such contracts as specified in such contracts. Billings under this contract are based on provisional rates that include a percentage of allowable overhead, and general and administrative expenses not exceeding certain limits. The Company is subject to review by the government on an annual basis. When final determination and approval of allowable rates have been made, billings may be adjusted.

The Company is currently undergoing audits of the costs incurred from January 1, 2023 to December 31, 2024. As of December 31, 2025, the Company has \$1.1 million in contract-related liabilities for its estimate of potential refunds to customers recorded in accrued liabilities on the consolidated balance sheet. As of December 31, 2024, the impact of any cost audit findings were immaterial.

Investigations and Litigation

From time to time the Company may become involved in legal proceedings or be subject to claims arising in the ordinary course of its business. The Company is not, to the best of its knowledge, a party to any legal proceedings that, if determined adverse to the Company, would individually or together have a material adverse effect on its business, operating results, financial condition, or

Unconditional Purchase Obligations

The Company has entered into certain agreements in which the Company is committed to purchase goods or services, primarily related to supply agreements for graphite components.

Year Ended December 31,	Un- F O
2026	\$
2027	
2028	
2029	
2030 and thereafter	
Total	\$ =

As of December 31, 2025, there were no unconditional purchase obligations required to be

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NOTE 16 — SUPPLEMENTAL CASH FLOW INFORMATION

Supplemental cash flow disclosures are as follows (in thousands):

	Year Ended Dec
	2025
Cash paid for interest	\$ — \$
Non-cash Investing and Financing Activities:	
Conversion of C-1 and C-2 notes to Series C Preferred Units	30,787
Conversion of 2024 Convertible Note to Series C-1 Preferred Units	—
Valuation of derivative liability	—
Class B common units related to the issuance and conversion of debt	—
Deferred transaction costs	3,039
Government grant reimbursement receivable for purchase of property and equipment	20,631
Property and equipment included in accounts payable and accrued expenses	4,490

NOTE 17 — SEGMENTS

Under the guidelines of ASC 280, which require companies to present financial information by operating segments, products and services, geographic regions, and significant customers, the Chief Executive Officer serves as the chief operating decision maker (“CODM”). The CODM evaluates financial performance, allocates resources, and makes strategic decisions based on the consolidated financial performance of the Company, rather than evaluating discrete lines of business. Accordingly, management has determined that the Company comprises a single reportable segment.

In making resource allocation and assessing operational outcomes, the CODM relies on operating income (loss) as the primary metric both during the annual planning process and for evaluating performance on a quarterly basis. The primary expenses presented to the CODM and incorporated in segment performance are operating expenses; direct costs, selling, general and administrative costs, and research and development costs — as well as the non-operating expenses; interest expense, interest income,

income (expense). The consolidated statements of operations for the years ended December 2024 present these significant segment expenses for the Company's single operating segment. The consolidated balance sheets as of December 31, 2025 and 2024 likewise display the assets and liabilities attributable to this single segment.

NOTE 18 — SUBSEQUENT EVENTS

In preparing these financial statements, the Company has evaluated subsequent events to determine if any require recognition or disclosure.

2024 Warrant

In March 2026, the Company amended the 2024 Warrant to permit the holder to elect the date of exercise of the 2024 Warrant at an earlier date than provided by the 2024 Warrant's original terms. A warrant holder exercised the warrant on March 18, 2026 and received 19,576,222 Series C Units.

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Shares

X•ENERGY

X-Energy, Inc.

Class A Common Stock

PRELIMINARY PROSPECTUS

, 2026

Joint Bookrunning Managers

J.P. Morgan Morgan Stanley Jefferies Moelis & Company
Cantor UBS Investment Bank TD Securities
Guggenheim Securities Wolfe | Nomura Alliance

Until , 2026 (the 25th day after the date of this prospectus), all dealers effecting transactions in these securities, whether or not participating in this offering, may be required to deliver a prospectus in addition to a dealer's obligation to deliver a prospectus when acting as an underwriter and with respect to an unsold allotment or subscription.

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PART II

INFORMATION NOT REQUIRED IN THE PROSPECTUS

Item 13. Other expenses of issuance and distribution.

The following table sets forth all fees and expenses, other than the underwriting commissions payable solely by X-Energy Reactor Company, LLC in connection with the offering of the securities being registered. All amounts shown are estimated except for the SEC registration fee, the FINRA fee and the stock exchange listing fee.

	Amount
SEC registration fee	\$
FINRA filing fee	\$
Stock exchange listing fee	\$
Printing expenses	\$
Accounting fees and expenses	\$
Transfer agent and registrar fees	\$
Legal fees and expenses	\$

Item 14. Indemnification of directors and officers.

Section 145 of the General Corporation Law of the State of Delaware provides that a corporation may indemnify a director, officer, employee, or agent of the corporation, or a person who served the corporation in any capacity, for the amount of the reasonable expenses incurred by or for such person in connection with the request of the corporation for another corporation, partnership, joint venture, trust or other enterprise, or in connection with any legal proceedings in which he or she is involved in his or her capacity as a director, officer, employee, or agent of the corporation, or in his or her capacity as a director, officer, employee, or agent of any related capacities against expenses (including attorneys' fees), judgments, fines and amounts paid in settlement actually and reasonably incurred by the person in connection with an action, suit or proceeding in which he or she is or may be a party or is threatened to be made a party to any threatened, pending or actual action, suit or proceeding by reason of such position, if such person acted in good faith and in the reasonably believed to be in or not opposed to the best interests of the corporation, and, in connection with any action or proceeding, had no reasonable cause to believe his conduct was unlawful, except that in the case of actions brought by or in the right of the corporation, no indemnification shall be made with respect to any claim, issue or matter as to which such person shall have been adjudged to be liable to the corporation, unless and only to the extent that the Court of Chancery or other adjudicating court determines that, in the adjudication of liability but in view of all of the circumstances of the case, such person was in fact reasonably entitled to indemnity for such expenses which the Court of Chancery or such other court or judge shall deem proper.

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or has agreed to become, a director or officer, or is or was serving, or has agreed to serve, at our request as a director, officer, partner, employee or trustee of, or in a similar capacity with, another corporation, partnership, joint venture, trust or other enterprise (all such persons being referred to as an “Indemnified Person”) or by reason of any action alleged to have been taken or omitted in such capacity, against such Indemnified Person (including attorneys’ fees), judgments, fines and amounts paid in settlement actually and reasonably incurred in connection with such action, suit or proceeding and any appeal therefrom, if such Indemnified Person acted in good faith and in a manner he or she reasonably believed to be in, or not opposed to, our best interests, and, with respect to any criminal action or proceeding, he or she had no reasonable grounds to believe his or her conduct was unlawful. Our certificate of incorporation and bylaws provides that we will indemnify any Indemnified Person who was or is a party to an action or suit by or in the right of us or our corporation, or judgment in our favor by reason of the fact that the Indemnified Person is or was, or has agreed to become, a director or officer, or is or was serving, or has agreed to serve, at our request as a director, officer, partner, employee or trustee of, or in a similar capacity with, another corporation, partnership, joint venture, trust or other enterprise, or by reason of any action alleged to have been taken or omitted in such capacity, against all expenses (including attorneys’ fees) and, to the extent permitted by law, amounts paid in settlement actually and reasonably incurred in connection with such action, suit or proceeding, and any appeal therefrom, if the Indemnified Person acted in good faith and in a manner he or she reasonably believed to be in, or not opposed to, our best interests, except that no indemnification shall be made with respect to any such action or suit, or judgment or settlement, or issue or matter as to which such person shall have been adjudged to be liable to us, unless the court or arbitrator determines that, despite such adjudication but in view of all of the circumstances, he or she should nevertheless receive indemnification of such expenses. Notwithstanding the foregoing, to the extent that any Indemnified Person has been successful, on the merits or otherwise, he or she will be indemnified by us against all expenses (including attorneys’ fees) actually and reasonably incurred in connection therewith. Expenses actually and reasonably advanced to an Indemnified Person under certain circumstances.

We maintain a general liability insurance policy that covers certain liabilities of directors and officers of our corporation arising out of claims based on acts or omissions in their capacities as directors and officers.

Item 15. Recent sales of unregistered securities.

C-2 Convertible Notes

In 2022 and 2023, XERC issued convertible notes payable in an aggregate principal amount of \$28.0 million and \$85.0 million (“C-2 Notes”), respectively, of which \$70.0 million of the C-2 Notes were issued to related parties. The C-2 Notes were scheduled to mature on September 30, 2025 and to pay interest of payable-in-kind interest annually.

On October 11, 2024, certain of the C-2 Notes with an aggregate principal balance of \$18.4 million were converted into 16,960,021 Series C Preferred Units, which was accounted for as a debt extinguishment. As of December 31, 2024, the outstanding principal, inclusive of payable-in-kind interest, on the C-2 Notes was \$18.4 million and the fair value was \$18.5 million, resulting in a cumulative loss of \$0.1 million.

In 2024, the C-2 Notes contained embedded derivatives which, absent the election of the cash settlement option, would have been bifurcated and accounted for at fair value. Accordingly, the fair value adjustments were elected for the C-2 Notes.

On September 30, 2025, the outstanding principal and unpaid accrued interest on the C-2 Notes were converted into 2,870,172 Series C Preferred Units, resulting in the extinguishment of the C-2 Notes. Therefore, as of September 30, 2025, there was no outstanding principal on the C-2 Notes.

2024 Convertible Note

On September 26, 2024, the Company entered into a convertible note with Amazon Investment Holdings LLC in the principal amount of \$20.0 million (the “2024 Convertible Note”). As of September 30, 2024, the debt accrued paid-in-kind interest of 12% per annum, compounded quarterly. The expense recognized related to the 2024 Convertible Note was immaterial through September 30, 2024. On October 11, 2024, in conjunction with the Series C-1 Preferred Units financing discussed above, and in accordance with the 2024 Convertible Note’s stated terms, the outstanding principal and interest on the 2024 Convertible Note of \$20.1 million was converted into 3,097,477 Series C-1 Preferred Units (the “2024 Convertible Note Conversion”).

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2024 Warrant Exercise

In March 2026, the Company amended the 2024 Warrant to permit the holder to elect the exercise of the 2024 Warrant at an earlier date than provided by the 2024 Warrant's original terms. The warrant holder exercised the warrant on March 18, 2026 and received 19,576,222 Series C Preferred Units.

2025 Warrant Issuance

On October 3, 2025, XERC issued the 2025 Warrant to a related party to purchase up to 10,000,000 Series C-1 Preferred Units at an exercise price of \$6.4870 per unit.

Series D Equity Issuance

On November 21, 2025 and November 25, 2025, XERC issued 48,134,320 and 20,000,000 Series D Preferred Units, respectively, and received gross cash proceeds of approximately \$700.0 million. Specifically, if the Company undertakes a Qualified IPO, the conversion price of the Series D Preferred Units will be adjusted, if necessary, to be equal to the lesser of: (a) seventy percent (75%) of the price per Class A Common Unit, or the price per the equivalent common share of the Company's corporate successor following a conversion of the Company to a C corporation; or (b) the then-current conversion price of the Series D Preferred Units prior to such adjustment.

December Profits Interest Unit ("PIU") Issuance

In December 2025, XERC issued 8,230,995 additional Class B-2 PIUs to certain employees.

In connection with the formation transactions described herein and pursuant to the Reorganization Transactions that will be completed prior to the closing of this offering, we will issue 10,000,000 shares of our Class B common stock, representing an aggregate 10% non-economic interest in the Company, to the Continuing Equity Owners. Such issuance will not involve any underwriters, discounts or commissions or a public offering, and such issuance will be exempt from the registration requirements pursuant to Section 4(a)(2) of the Securities Act.

Item 16. Exhibits.

(a) Exhibits

The following documents are filed as exhibits to this registration statement:

Exhibit Number	Description
*1.1	Form of Underwriting Agreement.
3.1	Certificate of Incorporation of X-Energy, Inc.
3.2	Bylaws of X-Energy, Inc.
*3.3	Form of Amended and Restated Bylaws of X-Energy, Inc.
4.1	Specimen Common Stock Certificate of X-Energy, Inc.
*5.1	Form of Opinion of Latham & Watkins LLP as to the legality of the securities being registered.
10.1†	Form of X-Energy, Inc. 2026 Equity Incentive Plan.
10.2†	Form Restricted Stock Unit Award Agreement under X-Energy, Inc. 2026 Equity Incentive Plan.
10.3†	Form Option Agreement under X-Energy, Inc. 2026 Equity Incentive Plan.
10.4†	Form of X-Energy, Inc. 2026 Employee Stock Purchase Plan.
10.5†	Offer Letter, dated January 15, 2019, by and between X-Energy, LLC and J. Clay Sell.
10.6†	Offer Letter, dated November 20, 2025, by and between X-Energy, LLC and Daniel Groves.
10.7†	Offer Letter, dated September 22, 2025, by and between X-Energy Canada, Inc. and Daniel Groves.
10.8†	X-Energy, Inc. Non-Employee Director Compensation Program
10.9	Form of Indemnification Agreement.

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Exhibit Number	Description
*10.10	Form of Eighth Amended and Restated Limited Liability Company Agreement of X-Energy, Inc. LLC.
*10.11	Form of Lock-Up Agreement.
*10.12	Form of Master Reorganization Agreement.
*10.13	Form of Tax Receivable Agreement.
10.14	Assistance Agreement with the U.S. Department of Energy dated February 2, 2021 (portions of the agreement have been redacted).
*21.1	List of subsidiaries of X-Energy, Inc.
23.1	Consent of Ernst & Young LLP, independent registered public accounting firm to X-Energy, Inc. LLC.
23.2	Consent of Ernst & Young LLP, independent registered public accounting firm to X-Energy, Inc. LLC.
*23.3	Consent of Latham & Watkins LLP (included as part of Exhibit 5.1 hereto).
24.1	Power of Attorney (included on the signature page of the initial filing of this Registration Statement).
*99.1	Consent of Director Nominee
107	Calculation of Filing Fee Table.

* To be filed by amendment.

† Management contract or compensatory plan or arrangement.

Pursuant to Item 601(b)(10)(iv) of Regulation S-K, portions of this exhibit (indicated by #) have been omitted as the registrant has determined that (i) the omitted information is not material and (ii) the omitted information is the type that the registrant treats as private or confidential. The registrant agrees to furnish supplementally an unredacted copy of the exhibit to the SEC upon its request.

(b) Financial Statement Schedules

See the index to the financial statements included on page F-1 for a list of the financial statements included in this registration statement.

Item 17. Undertakings.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted by law, the registrant understands that its officers and controlling persons of the registrant pursuant to the foregoing provisions, or any other person, has been advised that in the opinion of the SEC such indemnification is against public policy as expressed in the Securities Act and is, therefore, unenforceable. In the event that a claim for indemnification against such liabilities (other than the payment by the registrant of expenses incurred or paid by an officer or controlling person of the registrant in the successful defense of any action, suit or proceeding) is asserted by such director, officer or controlling person in connection with the securities being registered, the registrant will, unless in the opinion of its counsel the matter has been settled by controlling authority, submit to a court of appropriate jurisdiction the question whether such indemnification by the registrant is against public policy as expressed in the Securities Act and will be governed by the final adjudication of that court.

The undersigned registrant hereby undertakes that:

- (1) For purposes of determining any liability under the Securities Act, the information contained in the form of prospectus filed as part of this registration statement in reliance upon Rule 433, and the information contained in a form of prospectus filed by the registrant pursuant to Rule 424(b) or 433, shall be deemed to be part of this registration statement from the time it was declared effective.
- (2) For the purpose of determining any liability under the Securities Act, each amendment that contains a form of prospectus shall be deemed to be a new registration statement relating to the securities offered therein, and the offering of such securities at that time shall be deemed to be the initial bona fide offering thereof.

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Pursuant to the requirements of the Securities Act of 1933, the registrant has duly caused this registration statement to be signed on its behalf by the undersigned, thereunto duly authorized, at Rockville, State of Maryland, on March 20, 2026.

X-Energy, Inc.

By: /s/ J. Clay Sell

Name: J. Clay Sell

Title: Chief Executive Officer

POWER OF ATTORNEY

Each person whose signature appears below appoints J. Clay Sell and Steven Miller and any of whom may act without the joinder of the other, as their true and lawful attorneys-in-fact, with full power of substitution and re-substitution, for them and in their name, place and stead, in all capacities, to sign any and all amendments (including post-effective amendments) to this registration statement and any registration statement (including any amendment thereto) for this offering, to file the same effective upon filing pursuant to Rule 462(b) under the Securities Act of 1933, and to file the same with the SEC, exhibits thereto, and all other documents in connection therewith, with the Securities and Exchange Commission, granting unto said attorneys-in-fact and agents, and each of them, full power and authority to do and perform each and every act and thing requisite and necessary to be done in connection

fully to all intents and purposes as they might or could do in person, hereby ratifying and confirming the acts of said attorneys-in-fact and agents, or any of them, or their substitute and substitutes, may lawfully cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Act of 1933, this registration statement has been signed by the following persons in the capacities indicated on March 20, 2026.

Signature	Title
<u>/s/ J. Clay Sell</u> J. Clay Sell	Chief Executive Officer (Principal Executive Officer)
<u>/s/ Daniel Gross</u> Daniel Gross	Chief Financial Officer (Principal Financial Officer)
<u>/s/ Laura Garcia</u> Laura Garcia	Chief Accounting Officer (Principal Accounting Officer)
<u>/s/ Kamal Ghaffarian</u> Kamal Ghaffarian	Director
<u>/s/ Edward Sonnenschein</u> Edward Sonnenschein	Director
<u>/s/ Michael J. Wallace</u> Michael J. Wallace	Director
<u>/s/ Kathleen W. Hyle</u> Kathleen W. Hyle	Director

Signature	Title
<u>/s/ Christopher F. Ginther</u> Christopher F. Ginther	Director
<u>/s/ David B. Kaplan</u> David B. Kaplan	Director
<u>/s/ Allyson Satin</u> Allyson Satin	Director
<u>/s/ Gregory Goff</u> Gregory Goff	Director